Lectures on Optics

"To Understand the universe we must understand the light"

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LECTURES ON OPTICS

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Abstract

The field of optics is a captivating and continually evolving branch of science that plays an indispensable role in our modern world. "Advances in Optics: Exploring the Boundaries of Light and Vision" presents a comprehensive overview of the latest developments, theories, and applications in optics, spanning from fundamental principles to cutting-edge technologies.

This book delves into the fundamental properties of light, offering a clear and concise explanation of wave-particle duality, the nature of electromagnetic waves, and the behavior of light in various media. It explores the principles of geometric optics, including reflection, refraction, and the formation of images, while also addressing wave optics phenomena such as interference, diffraction, and polarization.

In addition to the foundational aspects, "Advances in Optics" goes beyond traditional optics to cover emerging topics and technologies. Readers will find detailed discussions on advanced optical materials, including metamaterials and photonic crystals, and their revolutionary applications in the manipulation of light for imaging, communication, and sensing. Quantum optics, an exciting frontier in optical science, is also explored, with a focus on quantum entanglement, quantum cryptography, and quantum computing.

Moreover, this book highlights the practical applications of optics in various fields, including medicine, telecommunications, astronomy, and environmental monitoring. It provides insights into state-of-the-art optical instruments and techniques, such as lasers, optical fibers, and adaptive optics systems, and their pivotal roles in advancing technology and scientific discovery.

"Advances in Optics" is designed to cater to a broad readership, from students and educators seeking a comprehensive textbook to researchers and professionals looking for the latest developments in the field. Its accessible style, accompanied by illustrative examples and diagrams, ensures that both newcomers and experts can appreciate the wonders of optics and its ever-expanding frontiers.

As we venture further into the 21st century, the exploration of light and vision promises to yield groundbreaking discoveries and transformative technologies. "Advances in Optics" serves as an invaluable resource for anyone curious about the principles, innovations, and limitless potential of optics in shaping our future.

Chapter 1

The loving Story of Optics

"The light is the God's shadow"
– Saint Agustin

When we study physics beyond doing long calculations, formulate complicated theories or doing extremely elaborated experiments in an advanced laboratory there is a really important fact many people ignore, the *story of physics* if we want to study physics we must have understand where was born the physics and more important, how was it done. Here we are going to check the story of physics from the geometrical optics, to corpuscular optics to wave optics, to electromagnetic optics and the most recent the quantum optics.

1.1 Optics from Euclid to Huygens

Here I'm going to copy and paste one of the best articles I've read about the story of optics (Herzberger, 1966).

If I were endowed with dictatorial powers, I would require everyone receiving a degree in a scientific subject to know its history and to have read the classical papers relating to it. Historical knowledge is important because it stimulates creative thinking. The man who first struggled with an idea, trying to find a law, looked at the situation with different eyes than do we who accept the law as a matter of course. He considered alternatives to the law, and different interpretations, and some of these alternatives and interpretations may still be stimulating and worth thinking about.

I was fortunate during the eight years of my affiliation with the Zeiss works at the beginning of my career to have access to their extensive library; it contained the originals, or at least translations, of nearly all the writings in optics from Euclid to contemporary times. Over thirty years ago, I wrote a historical survey describing the content of important papers on geometrical optics. This task was immensely facilitated by the extensive bibliography in the third edition of Czapski-Eppenstein,I to which I had contributed my small part as a work student at the firm of Zeiss. Lately, I have looked through most of these papers again, as far as they were available to me, and have discovered many important details that I had not understood at that time. I should like to share these with you. I emphasize that my interest is not basically historical. I am not interested so much in questions of priority and originality of documents. My object is to entice you to read and to enjoy the old scriptures, and maybe take along an idea or two that might be of interest.

1.1.1 Ancient times

I am sure that the Egyptians and, even more, the Babylonians, with their great interest in astronomy, must have had some knowledge of optical laws. But the first important mention of one of the phenomena of refraction I found was in Plato's Republic.' Plato(427-347 B.C.) quotes Socrates as saying (Jowett's translation): " . . . the same object appears straight when looked at out of the water, and crooked when in the water; and the concave becomes convex, owing to the illusion about colours to which the sight is liable. Thus every sort of confusion is revealed within us; and this is that weakness of the human mind on which the art of conjuring and of deceiving by light and shadow and other ingenious devices imposes, having an effect upon us like magic.¨ Our senses give an erroneous picture of reality; they are untrustworthy; alone can lead us to truth pure thinking. This idea persisted for two millennia under the influence of the misinterpreted teachings of Aristotle.

How we see is a question that long occupied the Greek philosophers, and, through them, the Neo-Platonists and the Scholastics. (Indeed, it cannot be said that the question is settled, and in recent times Ronchi has set forth some cogent ideas.) Even before Plato there was speculation in the writings of the Greek philosophers about the process of seeing. Most of the Greek thinkers believed that the eye sends out rays to the object, and Aristotle (384-322 B.C.) seems to have been the first to question this belief. If, says Aristotle, the eye sends out fiery rays that stream from it as from a lantern, why can we not see in the dark? His explanation, that the eye contains a transparent substance and that seeing is associated with the motion of something between the object and the eye, seems quite modern.

After the conquest of Greece by Alexander, Alexandria (in Egypt) and Sicily became the strongholds of Greek culture. The first great textbook of optics was written by Euclid of Alexandria, who lived about 300 B.C. And is better known for his Elements of Geometry. This famous work is still must reading for every scientist and is available in several editions. When King Archelaos once complained about the difficulty of learning mathematics and asked whether there was not an easier way, Euclid's answer is supposed to have been: "There is no royal route to geometry¨. When one of his pupils asked him what of material value could be gained by learning geometry, he told a slave, "Give the man threepence and tell him to go away¨.

Under Euclid's name we have two books of interest in the present connection: Optics and Catoptrics. The first deals with the laws of vision and perspective and the second with reflection. An English translation of the first was published in the Journal of the Optical Society of America twenty years ago. 7 It is a striking fact that one of the definitions in this book contains an idea that could be used to develop many problems of diffraction theory.

I quote some of Euclid's definitions, slightly paraphrased from the Optical Society of America translation: "Let it be assumed that lines drawn directly from the eye pass through a space of great extent [not infinite]; that the form of this space is the mantle of a cone with its apex in the eye and its base at the limits of our vision; that those things upon which the cone falls are seen and those things upon which the cone does not fall are not seen; and that things seen through several cones simultaneously appear to be more distinct.¨ From this, Euclid concludes, among other things, that nothing can be seen in its entirety. The same objects located nearby are seen more distinctly than if they were at a distance. Every object that is visible has a certain limit of distance. If that is reached-that is, if the object lies entirely inside the mantle of the cone of vision-it is no longer seen. These are the axioms from which he derives the laws of perspective.

In Catoptrics the law of reflection is stated, namely, that incoming and outgoing rays form the same angle with the surface normal. From this he derives the exact position of the image behind a plane mirror and the principle that left- and right-hand sides are interchanged. He generalizes these laws for spherical mirrors, both concave and convex, describing virtual images as well as stating that concave mirrors can give real images. As the position of the image is formed by reflection at curved surfaces, he suggests the point of intersection of the ray with the axis (shades of my definition of diapoint!). He shows the existence of a focal point for spherical mirrors, although he errs with respect to its position. One of his theorems says that an eye at the center of a spherical mirror sees only itself. Another states that you can light a fire at the focus of a concave mirror. He states that straight lines appear convex in a convex mirror. One can see the same object through many plane mirrors if they are positioned along the sides of a regular polygon with two more sides than there are mirrors.

According to Roman historians, the greatest mathematician of ancient times, Archimedes (287-212 B.C.), is supposed to have used mirrors to destroy the Roman fleet besieging Syracuse. Like most scientists, I have doubted this story, but a personal experience made me doubt my doubts.

Two years ago I had an optical accident. My wife had me get a large bottle of distilled water, and I left it in my auto in front of our laboratory. Driving home that evening I smelled smoke and stopped at a garage. After thoroughly investigating all the mechanical details and finding them in order, the mechanics discovered that smoke was coming out of the upholstery. The cylindrical bottle had acted like a lens and the sun had made a hole in the upholstery and ignited the stuffing inside. After this personal experience, I am somewhat more reluctant to doubt the story of Archimedes' mirrors.

The next person I should like to discuss, Hero of Alexandria, lived sometime between about 150 B.C. and A.D. 250. We have two outside dates for this philosopher: he mentions Hipparchos, who flourished between 146 B.C. and 126 B.C., but the mathematician Pappus (ca. A.D. 300) follows Hero's treatment in his own writings on mechanics. Be this as it may, Hero's Catoptrics is one of the most interesting books on optics. He is not content with Euclid's statement that light travels in straight lines, nor is he content with the form of the reflection law. He looks for a reason for both laws and derives it from the assumption that light chooses the shortest possible path between two points. We know that this minimum principle is not quite correct: the path is, in general, only an extremum. It may or may not be significant that Hero's book illustrates the concept with a picture of a convex mirror, for which the minimum principle is correct-not a concave mirror, for which it can be wrong. We shall find the same trick worked later by Fermat, who extended Hero's law to derive the refraction law. However, Fermat was called on the carpet for this by Abbé Mersenne, one of his contemporaries.

Hero of Alexandria-who, by the way, in his Mechanics described the so-called seven simple machines known to us-not only discussed spherical mirrors in his Catoptrics but, as an early practical experimentalist, also showed what can be done with cylindrical mirrors. If you go into an amusement parlor and see yourself in mirrors of all kinds (I especially like those that elongate the height and diminish the girth), you know the proprietor has read Hero of Alexandria. If you see a play in a tanagra theater, where the actors appear on the stage, by the skillful use of mirrors, as beautifully dressed dwarfs, you know that Hero of Alexandria wrote the prescription. If you want to see around the corner or look at a person without letting him see you, Hero of Alexandria tells you how to do it with mirrors.

Ptolemy of Alexandria, who lived about A.D. 150, is best known for his Almagest, but he was also the first to make a serious attempt to study the law of refraction. He wrote a five-part volume entitled Optics, several copies of which have come down to us. This work, of which the first part has been lost (although we have a description of its contents), can be summarized as follows.

The first part deals with the relations between the eye and light, and the second with the conditions of visibility: size, form, color, and motion of bodies. Ptolemy explains why we usually see only one image of an object, although we have two eyes, and describes the conditions under which we see more than one image. The third part deals with the laws of reflection by plane and convex mirrors. Ptolemy gives probably the first attempt at an explanation of why the moon and the sun appear larger at the horizon than at the zenith a question that is still being discussed.

In the fifth part, Ptolemy tries to find a law for refraction. First he observes that for small incident angles the angles of incidence and of refraction are proportional to each other. He also remarks that, in transition from a thinner to a denser medium, the refracted ray comes earer to the perpendicular, the opposite being true for the reverse direction of travel. He uses an ingenious contraption to measure the incident and the refracted angles for the transition from air into water, air into glass, and water into glass. This instrument consists of a circular disk divided into 3600 with a colored marker in the center and two movable markers on the periphery. You put the instrument into water, line up the three markers, and read the two positions, one giving the incident angle and the other the refracted angle. Ptolemy gives refraction tables in 100 intervals. My friend, the late Hans Boegehold, studied hese tables carefully and found that the values for 500 and 600 incident angles are surprisingly accurate, but that the other angles were obviously calculated by a second-degree interpolation formula. (The first differences are exactly 1 30'.14) Ptolemy applies the same method for the transition of light from air into glass and from water into glass. He also discusses atmospheric refraction, the cause of which he assumes to be that the densities of air and of ether vary with height. He points out that only for a star in the zenith do the true and the apparent positions coincide.

1.1.2 The arabain period

After the fall of Greece, scientific progress ceased on the continent of Europe for nearly a century and a half, but Alexandria remained the center of scientific discovery. When the Arabs took over, they brought their scientific inheritance and knowledge of mathematics and the mathematical sciences into northern Africa and Spain. It is remarkable that, at a time when magic and mysticism and misinterpretation of Aristotelian doctrines prevented scientific progress over most of Europe, the Arabs held high the torch of science in their countries.

The next big contribution to our science was by Ibn al-Haitham, or Alhazen of Cairo, whose date is frequently set around 1100, although some sources tell us he lived from 965 to 1038. Alhazen's Optics" consists of seven parts. The first gives an anatomical description of the eye, which, he maintains, contains three liquids separated by four membranes. The names which he gives to them the watery medium, the crystalline medium, and the glassy medium are still used today. The phenomenon of binocular vision is explained by assuming that the pictures from the two eyes are transported to one visual nerve. Alhazen discusses the pupil of the eye and speaks of a pyramid of rays (rather than a cone) coming from the object to the eye.

The second part, like the corresponding book by Ptolemy, deals with the properties of bodies, and he distinguishes no fewer than twenty-two different properties.

The third part deals with optical illusions. He draws attention to the illusions arising from a false interpretation of a picture by the human mind and imagination. The fourth, fifth, and sixth parts deal with problems of reflection and mirrors. One of the problems studied in great detail has since been called after Alhazen: it consists in finding the point on a mirror where the incident and the reflected rays meet. Alhazen corrects errors of the Greeks about the position of the images in spherical, conical, and cylindrical mirrors. He also emphasizes the fact that the incident and reflected rays lie in the same plane.

Alhazen's seventh part deals with refraction. He proposes methods for measuring incident and refracted rays through different media, but he gives no tables. He knows that a glass sphere magnifies. He explains the apparently large size of the sun at the horizon by the assumption that we compare the sun with objects of known size lying along the line of sight. He investigates binocular vision and makes a good guess at the height of the atmosphere. He breaks with the idea of the Greeks that light rays originate at the eye and go to the objects, supporting his claim by the circumstance that sunlight can hurt the eye and that we can see afterimages.

Other manuscripts show that Alhazen used the pinhole to view solar eclipses and that he knew how the mage becomes blurred and finally becomes circular as the pinhole increases in size. He did not write as though this were a discovery of his own, and Aristotle had noted that the spot formed by the rays from the uneclipsed sun is circular when the rays pass through a rectangular hole in wickerwork if the hole is small, but becomes rectangular when the hole is large. However, Alhazen's is the first known unequivocal description of the camera obscura. Alhazen's works were of enormous influence in the development of optics.

1.1.3 The middle ages

The knowledge of the Arabs was transmitted to Europe via Spain and Sicily and set in motion the intellectual revival that culminated in the Renaissance. The first of the European scientists to play a role in optics was Vitello, a Polish mathematician who lived and studied in Italy about 1270. His book in ten parts is probably the most voluminous optical work ever written. It contains about five hundred closely printed folio pages. Unfortunately, he is more famous for his errors than for his contributions.

He reissued the tables of Ptolemy and added tables for the passage of light from water into air, from glass into air, and from glass into water. It is unfortunate that he never tells us how he arrived at his values, for they include impossible data, being in contradiction with the then unknown phenomenon of total reflection.

It was this error that later prevented Kepler from finding the correct form of the refraction law. Nevertheless, Vitello made two notable contributions. He explained the colors of the rainbow as originating from refraction and reflection by the water drops, and he made the proposal to collect the light of the sun into a point by using parabolic mirrors.

The Englishman Roger Bacon (1215-1294) discovered the spherical aberration of a mirror; he further stated that rays forming a cone around the axis in object space also form a cone around the axis in image space. From the magnifying property of a sphere, he speculated that it might be possible to invent instruments in which a man would look like a mountain and a small group like a big army. However, since he did not say how this could be achieved, it is doubtful whether he should be called (as he was by Sir John Herschel) the inventor of the telescope. His writings are so sketchy and in definite that it is impossible to attribute specific inventions to him or to reach any conclusions about what he actually knew.

1.1.4 The renaissance

Coming to the Renaissance, we must mention Leonardo da Vinci (1452-1519). His notebooks show that he studied the eye anatomically and made a model of it, although he supposed the rays to cross twice in it to form an upright image. 18 , 9 Immediately afterward he described the camera obscura but correctly showed the image inverted!20 He described, with sketches, a Rumford-type photometer and a machine for grinding concave mirrors. Unfortunately for his optical reputation and the development of optics, he never organized his notes into publishable form, so they were practically unknown until Venturi published them in 1797.

About half a century later came another Italian we should mention, Francesco Maurolico (1494-1577), who solved one of the problems that had bothered previous thinkers: that the image of the sun is round even if formed by a square pinhole. He improved our knowledge of vision by suggesting that the crystalline lens of the eye works like a double-convex lens. This enabled him to attribute myopia and hyperopia to an eye whose lens has a larger or smaller curvature than is proper for the length of the eyeball. He emphasized for the first time that the rays from an object point envelop a surface that is now called the caustic surface and that this surface is the place of the highest light concentration. And he explained the colors of the rainbow by multiple reflection in the water drops. All this information was concisely organized by 1567 into a work entitled Photismi de lumine , although it was not published for some thirty years (1611) after the author's death, and then only as a consequence of the interest in optics engendered by Galileo's telescopic discoveries.

In Italy, under the influence of the philosophy of the time, scientific progress and magic were mixed in a way that led to interesting reading. The development of mathematical methods, however, served to gradually separate speculation and imagination from the solid basis of observation.

One of the most colorful books of the period was the Magia naturalis of Giovanni Battista della Porta (ca. 1538 or 1543 -1615). In his youth, della Porta had written a four-part volume entitled De miraculis rerum naturalium (1558) and, subsequently, many books and plays, including De refractione (Fig. 6) in nine parts in 1593. The Magia naturalis, which was an expansion of the earlier De iraculis into twenty volumes, appeared in 1589. It became so famous that it was soon translated from Latin into the vernacular languages Italian, French, Spanish, and English. It is a strange medley. Only the seventeenth part concerns optics, but some of the others are also worth noting for our amusement if nothing else. I mention the titles of a few of the twenty parts, to give you an idea of their contents. The first is entitled, "The Causes of the Miraculous. It deals with astrology, sympathy and antipathy, and the influence of place and time on events. The second deals with the origin of animals. He quotes many authorities who agree that snakes are made from the marrow of men and the hair of women. The third deals with garden plants; the fourth, with how to become wealthy by good housekeeping; the fifth, with how to make precious stones artificially;

and the ninth, with make-up and other ways of improving the beauty of women (as if this were necessary!).

Finally, the seventeenth part deals with focal mirrors and other mirrors and explains what one can do with them. This book exhibits progress in the understanding of optical phenomena. In it are discussed the same tricks that Hero had dealt with 1400 years earlier, namely, how to use mirrors to obtain desired effects. However, we find here the first exact theory of multiple mirrors, the first complete description of the camera obscura, with pinhole or lens, and a discussion of combinations of positive and negative lenses designed to improve vision. Porta compares the eye and its pupil to the camera obscura and its stop. (Leonardo had anticipated him here, but less definitely and accurately.) At the end of this century also falls the invention of the telescope and the microscope.

Eyeglasses had apparently been invented toward the end of the thirteenth century, but they seem not to have attracted the attention of writers on optics until the practical Leonardo discussed methods of making them. Some lens grinder-more likely, several independentlymust have discovered the principle of the Galilean telescope by accident, and certain passages in Leonardo's notebooks a hundred years before Galileo can be interpreted as instructions for making such a telescope. Be that as it may, a Dutchman, Zacharias Jansen (1588-1632), was said by his son to have made a telescope in 1604 according to the model of an Italian dated 1590, and soon telescopes became well known in Holland. Galileo Galilei (1564-1642) heard of them and, apparently independently, made an instrument that he exhibited in 1609, and in his lifetime made over a hundred. The Dutch instruments were so imperfect that a magnifying power of more than 3X was impractical. Galileo made instruments with a useful power of approximately 30 X, an order of magnitude greater. He was by no means the inventor of the telescope or even the first scientist to describe it; Vavilov reminds us that Leonardo may have anticipated him and that Porta described, three months after Galileo's demonstration, an instrument he had made independently. But Galileo is outstanding on two counts: he revolutionized man's ideas of the uni- verse by his applications of the telescope and, by improving it by an order of magnitude, Ronchi points out that Galileo is entitled to be considered the father of the art of fine optics.

Moreover, from telescope to microscope is a short optical step, and by 1624 Galileo was using lenses of short focal length and thus making precursors of the modern compound microscope. 9 A few decades thereafter, the Dutchman Antonj van Leeuwenhoek (1632-1723) improved the quality of simple lenses used for magnifiers or simple microscopes, and in a letter of 1674 he began the descriptions of the discoveries for which he became justly famous.24 Meanwhile, Robert Hooke (1635-1703) in England also improved the quality of lenses in order to make the compound microscope practical, and in 1665 the Royal Society published his Micrographia.

Aside from publications, which were in an embryonic state, the seventeenth century brought a lively exchange of ideas and thoughts between scientists, mostly friendly but sometimes polemical and frequently expressed in code to conceal them from the uninitiated. It is therefore necessary to study not only the books of these scientists but also their correspondence (cf. Leeuwenhoek's letters quoted by Dobell).

The same period witnessed the development of scientific knowledge to the system that is still important.Indeed, scientific development in the seventeenth century was greater than in any century before or since. Scientific giants like Kepler, Descartes, Pascal, Fermat, Newton, Leibniz, and Huygens appeared and left their marks in all fields of science, including optics. Let us sketch some of their contributions to geometrical optics.

Johannes Kepler (1571-1630) wrote two books on optics: Supplements to Vitelo in 1604 and Dioptrice in 1611. The object of the first is to find the law of refraction. Kepler nearly succeeds, though he does not look for a relation between the incident angle and the refracting angle but between the incident angle and the deviation. He asks himself what form a surface should have to refract a parallel beam of light exactly into a point image. He studies the hyperboloid, which, as we know, fulfills the condition; but he rejects it because it disagrees with the values that Vitello gave for the relation between the angles of the incident ray and the refracted ray. If only the Kepler of 1611 and the Dioptrice, in which he discovered the law of total reflection, had reread his book of 1604, he would have seen that the data of Vitello had been erroneous and thus he might have found the correct form of the law of refraction.

Kepler investigated the function of the eye more profoundly than his predecessors. He discovered that the eye is not at rest but moves around a point inside it, and he investigated the importance of this motion on our vision. He drew attention to the fact that the image on the retina is inverted. All this is in his Supplements to Vitello.

His Dioptrice , although a relatively small book, was of importance to the development of optics. In it, Kepler assumes that for small angles the angle of refraction is proportional to the angle of incidence. From this assumption, he develops the laws of firstorder optics for thin lenses and systems of thin lenses with finite separations. He describes, among other things, the Keplerian telescope, in which two positive lenses are placed so that the back focus of the first is at the front focus of the second, and compares it with the form named after Galileo, consisting of a positive and a negative lens placed so that the focus of the former s at the virtual focus of the latter. He gives the focal length correctly for all forms of lenses. His book contains many practical proposals for lens design and is highly recommended to those who are interested in these problems.

Rene Descartes (1569-1650) must be considered as one of the discoverers of the correct law of refraction, for he published the sine law in his Dioptrique in 1637. After his death, he was violently attacked by Isaac Voss, who accused him of stealing the law from Willebrord Snel* (1591-1626). Descartes had indeed visited Leyden in 1630, where Snell, a professor of physics, had written a book on optics containing the refraction law. Although the book had never appeared, Voss, in his posthumous attack, claimed that Descartes had seen it on his visit after Snell's death, and Voss quoted Snell's proof.

The question of priority has been much discussed in the literature. A recent study has been published by Boegehold, who refers to a paper by P. Kramer. Boegehold found that Descartes, in some letters written in 1627, had suggested making plano-convex lenses with hyperbolic surfaces. Since such a lens would give a sharp point image for an infinite object if and only if one assumes the sine law, it is fair to conclude that Descartes knew of the law three years before his visit to Leyden in 1630. It is strange that Voss should have years after waited for twenty-five years-twelve Descartes's death-to make his accusation. Credit for discovering the law of refraction should therefore be divided between Descartes and Snel, with a fair share going to Johannes Kepler, who failed to find the sine law only because he trusted the tables of Vitello.

Descartes's Dioptrique is beautifully written. It tries to explain the law of refraction by means of the theory of elasticity. A light ray is compared to the path of a ball, first being bounced off the ground, then being slowed down as it goes into a denser medium. Descartes was the first to suggest determining the velocity of light from the position of the moon at an eclipse.

He also made experiments with human and animal eyes. If the eye is removed and put in the opening of a window in a darkened room, one can see the images of distant objects on the retina, and by pressing the eye a little, one can see the image of near-by objects, showing that the eye can accommodate.

In the eighth chapter of his book, Descartes investigates surfaces that image a point sharply-the socalled Cartesian surfaces, of which the conic sections constitute a special case. He proposes machines for grinding lenses and demonstrates the colors of the rainbow with small glass spheres filled with water.

But much of Descartes's reasoning in connection with the refraction law was faulty and was rejected by the great mathematician Pierre de Fermat (1601-1665), who pointed out that water would have to be less dense than air to fit Descartes's explanation. Fermat returned to the formulation of the refraction law given by Hero, deriving it from the assumption that light has different velocities in different media and takes the quickest path between any two points. Like Hero, he illustrates this idea with a drawing of a convex surface, for which the minimum principle is correct, rather than a concave surface, for which it can be wrong. But he was called on the carpet by Abbé Mersenne, as was mentioned earlier, one of the great scientists and letter writers of his day. Fermat's answer is a classic example of double talk: he says that the path would be a minimum if the surface were a plane, and the incoming ray sees so little of the surface that it cannot distinguish whether the surface is plane or curved!

1.1.5 Newton and Huygens

Sir Isaac Newton (1642-1727), in his Lectiones opticae of 1669-1671 (published posthumously), showed how to calculate the spherical aberration of a plano-convex lens for an infinite object point. This book contains the formula for locating the image, still known by his name, in terms of the distances of the object and the image from the focal points. It also contains a thorough investigation of reflection in prisms, including the position of minimum deviation. It even goes into some aberrations, in particular, image errors as functions of color, and it compares the size of the color aberration with the spherical aberration of a given lens and prism.

Although Newton's best known work in optics is his Opticcs , I should like to draw your attention especially to his beautiful paper on color, in which he describes the separation of white light into the colors of the spectrum by means of prisms. This charming paper should be read by every student of optics, especially since the writer not only gives his final conclusions but describes many false attempts at interpreting his experiments and the refutation of these attempts.

Newton was misled into believing that all materials have the same relative dispersion, since he found this to be true for such different media as glass and water. The conclusion that he drew-that telescopes should not contain refracting surfaces-was erroneous. But it led him to study mirror telescopes and, with his teacher Gregory, he contributed much to the development of these instruments.

Christian Huygens (1629-1695) wrote two books on optics: Dioptrical in 1653, published posthumously , and Traite de la lumiere , written in 1678 but not published for twelve years. The first book shows enormous advances with respect to the understanding of optical image formation. The second proposes a new and revolutionary theory of light, deriving refraction and reflection laws from a completely new principle.

This principle, set forth earlier in the Dioptrica, is a starting point from which all the laws of Gaussian optics can be derived for lens systems with finite thicknesses. It is expressed as follows: if we interchange object and eye in an optical system, the object will have the same apparent size and orientation as before. The law is known to us in the form that the product of lateral magnification and angular magnification is constant in an optical system, and equal to the ratio of the refractive indices.

Huygens found, before Newton, the law that, for a single surface or for a single lens, the image distance is proportional to the object distance when the distances are measured from the respective principal foci. He knew of the aplanatic surface. He was acquainted with the important property of the optical center of a lens (the point which divides the distance between the vertices in the ratio of the radii), namely, that a ray through it emerges parallel to its original direction. He had an approximate formula for the spherical aberration of a thick lens that enables the form for minimum spherical aberration to be determined.

In his Traite de la lumiere, which should be required reading for every physicist and is now readily available, Huygens derives optical laws from the fact that the light emerging from an object point P forms a wave surface S at any given time t. The light rays are then the normals to this surface and are not straight unless the velocity is constant throughout the medium.

The surface can also be constructed in the following way. At an arbitrary time , light from the point source P forms a wave surface S. (assumed to be spherical in the figure). Considering each point of S, as a source sending out a new wave, we see that the envelope of all the waves at the time gives the original wave surface S.

With the help of this concept, Huygens derives the straight paths of light in a medium of constant velocity. He derives not only the reflection and refraction laws but also the law of refraction in the atmosphere and in doubly refracting crystals. He clearly defines the caustic as the evolute of the rays and investigates it after reflection and refraction, in addition to investigating the degenerate form of the wave surfaces in the points of the caustic. He proves Malus' law: that rays form a normal system for a single reflection and refraction.

It should be emphasized, however, that this formulation is mathematically identical with ray optics, as Hamilton showed in his paper of 1831. The assumption that light has a finite wavelength and that it can be described as a sine wave going along a ray is foreign to Huygens. However, once this theory was established, the methods of Huygens enabled Fresnel to derive all the laws of diffraction in a most elegant manner.

Chapter 2

Geometrical Optics

The Euclidean Geometry exists thanks to light. Euclid wanted to undestand the light rays, so he discovered the plane geometry.

"Music is the arithmetic of sounds as optics is the geometry of light."
– Claude Debussy

2.1 The postulates of geometrical optics and the dark Room

We already know the history of the geometrical optics, well before starting the study of this theory we must take back the contributions of Al Azhen and talk about the **postulates of geometrical physics** which are

1. The speed of light is finite.
2. The light propagates in a straight line.
3. The light rays obey the Euclid's postulates of plane geometry.
   (a) A straight line segment can be drawn joining any two points.
   (b) A finite straight line (line segment) can be extended continuously in a straight line indefinitely.
   (c) A circle can be drawn with any point as its center and any distance as its radius.
   (d) All right angles are equal to one another, regardless of their orientation.
   (e) If a straight line falling on two straight lines makes the interior angles on the same side less than two right angles, the two lines, if extended indefinitely, will eventually meet on that same side.
4. In the matter the light's speed is less than in vacuum.

Now, before starting to use these postulates we must understand the physical meaning of them. The first postulate, **The speed of light is finite** is fundamental, it is incredibly important.

How could Hero of Alexandria understand that the speed of light must be finite? To be honest, I don't really know his motivation but, we could get an idea about why this should be like that from simple phenomenological reasoning without the use a mathematics or advanced physics. *Lets imagine a world where the speed of light is infinite*

How are going to be the nights in such world? It should be a complete chaos. Every star in the sky is a source of light, a source of infinite rays which each one travels at an infinite speed. There wont be any night, all the day is going to be full of light and even every event in the universe is going to be seen simultaneously, the big bang, the birth and death of a star. *Each past and present event is going to be seen at the same time.*

Other effect that only can be explained in a universe with light with finite speed. Imagine you're standing in the rain, which is falling vertically. If you start running, the rain hits you in the face. This happens because your movement changes the apparent direction of the rain. An astronomer, James Bradley, observed that the stars didn't appear to be exactly in their calculated positions. To see a star directly overhead, he had to tilt his telescope slightly in the direction of Earth's movement around the Sun, the light from the star "falls" toward Earth at a finite speed. Earth is moving. For the light beam to travel from the telescope lens to the eyepiece (and for us to see it), Earth has moved a little. Therefore, you have to point the telescope at a slight angle toward where Earth will be when the light reaches the eyepiece, not where it is at the moment it enters the lens.

If the speed of light were infinite, this effect wouldn't exist. The rain (light) would fall so fast that your (Earth's) movement would be irrelevant. The fact that we have to "aim ahead" of the target is irrefutable proof that light travels at a finite speed.

There are many other consequences but lets continue with all the other postulates. Now lets talk about **The light propagates in a straight line**. This does have sense because it is what is seen in any normal day, all the source of light can be seen as sources of infinite many rays which propagates at a finite speed. With tools purely geometrical there is no way to give a proof of this, however, from a wave theory for light, Huygens was able to prove that the rays are, in fact, the normal vectors of the wavefront are the rays, and, from a quantum perspective, Richard Feynman proved that the rays are the superposition of the ”watches” that represent the probability amplitudes.

Often we see that light is not an straight line, for example, an easy experiment consists on a translucid bowl full of salted water, we're going to see that the ray presents a deflection, we're going to see that this curves can be generated as a composition of infinitely many straight refractions.

Because we've postulated that the light travels in straight lines, so we have to give a theory for the propagation so **The light rays obey the Euclid's postulates of plane geometry**. A point is going to be a source, a place where two rays meet, the place where two media separates, etc.

The extension of a line segment to a continuously straight line is a way to make or represent ray tracing. The circle postulate talks about the sphere that propagates alongside all the rays (which behave as the radius). The right angles are just a geometrical formalism which does not have a dramatic optical interpretation.

The last postulate is really important, from a physical point of view it tell us that all the rays are going to be propagate on a completely plane space, so we could ask ourselves if we want to study the light's behavior under the gravitational effects of compact objects we must change the fifth Euclid's postulate, and the short answer is yup, we have to. Under gravitational effects the differential geometry is a better tool to study the light-type observers on the curved space-time.

So, we have many arguments that supports the idea the light's speed must be finite, and if it is finite there should be a way to measure it, right? The answer is yes, and here we're going to use the postulates to follow the Fizeau's ideas on how to measure the light's speed.

In 1849, Hippolyte Fizeau performed a groundbreaking experiment that provided the first non-astronomical measurement of the speed of light. His ingenious setup transformed the problem of measuring an extremely short time interval into a problem of measuring a rotation speed, which could be done with much greater precision. Imagine a very fast runner who starts from a point A, runs to a point B and immediately returns to A. To measure the runner's speed, you could place a door at A that opens and closes at regular intervals. If you open the door just when the runner leaves, then close it and open it again periodically, you might catch a glimpse of him when he returns if the timing is right. But if the runner is extremely fast, the door might be closed when he gets back. This is exactly the idea Fizeau used with light, replacing the door with a rapidly rotating toothed wheel and the runner with a light pulse.

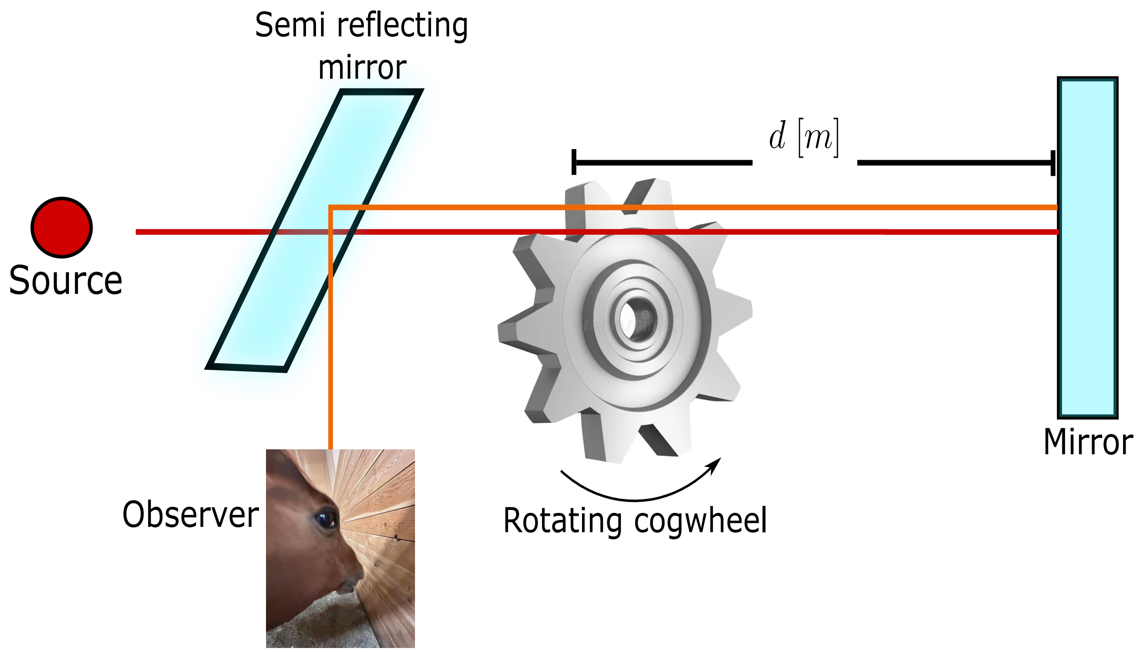


Figure 2.1: Schematic representation of the Fizeau's experiment pretended to measure the light's speed.

Fizeau set up his apparatus on two hills in the outskirts of Paris, with the light source and observer on one hill (Montmartre) and a mirror on the other (Mont Valérien), about 8.633 kilometers away. A beam of light from a lamp was directed toward a partially reflecting mirror (a beam splitter) that sent part of the light toward a toothed wheel. The wheel could be spun at a controlled speed and had a large number of teeth and gaps – typically 720 teeth, meaning 720 gaps as well. After passing through a gap, the light traveled to the distant mirror, was reflected, and returned along the same path. On its way back, it would again have to pass through a gap in the wheel to reach the observer's eye via the beam splitter.

If the wheel were stationary, the returning light would pass through the same gap and be seen. But as the wheel began to rotate, the situation changed. When the wheel turned slowly, the gap through which the light had originally passed would have moved only slightly during the light's round trip, so most of the returning light still made it through. As the speed increased, the gap shifted more, and the light began to be partially blocked by the edge of a tooth. At a certain critical speed, the gap moved exactly so that by the time the light returned, the adjacent tooth had taken its place, completely eclipsing the light. The observer would see the light disappear for the first time. By knowing the rotation speed at which this first extinction occurred, the number of teeth, and the distance to the mirror, Fizeau could deduce the time it took for light to travel to the mirror and back.

Let us analyze the situation mathematically. In the figure 2.1 let d be the distance from the wheel to the distant mirror. The total path for the light is $2d$. The wheel has N teeth and consequently

N gaps, so the angular width of one tooth (or one gap) is π/N radians (since a full circle has 2π radians and there are $2N$ segments – teeth and gaps – each of equal angular size). For the first extinction, the time required for the light to make the round trip must equal the time needed for the wheel to rotate by the angle corresponding to one tooth (i.e., to replace a gap with a tooth). If the wheel rotates with angular velocity ω (radians per second), the time to turn through an angle π/N is

$$t_{\text{rot}} = \frac{\pi/N}{\omega}. \tag{2.1}$$

The light travel time is

$$t_{\text{light}} = \frac{2d}{c}, \tag{2.2}$$

and equation these gives us

$$\frac{2d}{c} = \frac{\pi}{N\omega}. \tag{2.3}$$

But $\omega = 2\pi f$ where f is the rotation frequency in revolutions per second. Substituting,

$$\frac{2d}{c} = \frac{\pi}{N \cdot 2\pi f} = \frac{1}{2Nf}. \tag{2.4}$$

Solving for c yields

$$c = 4dNf. \tag{2.5}$$

Fizeau used $d = 8.633 \times 10^3$ m, $N = 720$, and observed the first extinction at $f \approx 12.6$ revolutions per second. Plugging in these numbers,

$$c = 4 \times 8.633 \times 10^3 \times 720 \times 12.6 \approx 3.13 \times 10^8 \text{ m/s}, \tag{2.6}$$

a value about 5% higher than the modern accepted value of 2.99792458×10^8 m/s, but remarkably accurate for the time. The discrepancy arose mainly from uncertainties in the distance measurement and the precise determination of the extinction condition. Fizeau's experiment was a triumph of experimental physics, demonstrating that light travels at a finite speed and providing a method that would be refined by later researchers, most notably Léon Foucault, who replaced the toothed wheel with a rotating mirror to achieve even greater precision. The conceptual beauty of Fizeau's approach lies in its translation of an infinitesimal time interval (on the order of 1/20000 of a second) into a measurable mechanical rotation, a perfect illustration of the deep connection between optics and mechanics.

Now we know the speed of light is finit, so lets use these postulates to understand one of the biggest contribution of Al Azhen, the dark room, see figure 2.2.

Before develop mathematically the dark room we're going to set an important convention

- The light propagates from left to right along the z direction.
- A distance measured in the direction of light is positive and a distance measured against the direction of light is negative.

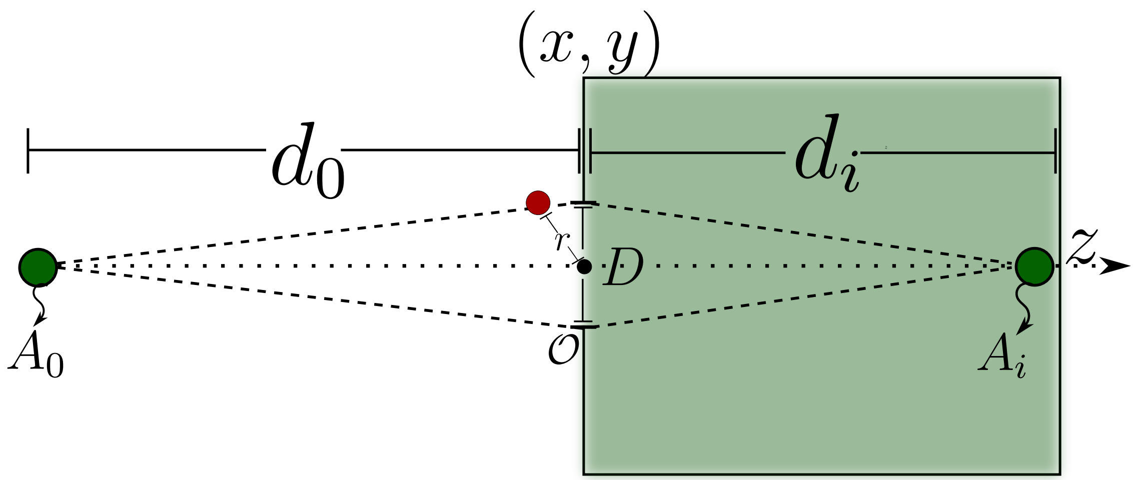


Figure 2.2: Graphical representation of a generic dark room which is a box of an arbitrary material which has a side opened and while it satisfies the Rayleigh's inequality i will form an inverted image.

Once defined the convention lets do it. A source in the point A_o is at a distance d_o respect to the opening O of the dark room, (d_o is in fact negative because it is measure against the light direction from the point O), an image if formed at a distance d_i from O thus we must find out a relation between these parameters and the diameter D of the DR. First we're going to calculate the distances $\overline{A_oOA_i}$, $\overline{A_oRA_i}$ where R is the radius of the opening, therefore

$$\overline{A_oOA_i} = d_i - d_o \;\; d_i = \overline{OA_i} \;\; d_0 = \overline{OA_o}, \tag{2.7}$$

$$\overline{A_oRA_i} = \left[d_i^2 + x^2 + y^2\right]^{1/2} - \left[d_o^2 + x^2 + y^2\right]^{1/2}, \tag{2.8}$$

having on count our convention implies that $\sqrt{d_o^2} = -d_o$ so

$$\overline{A_oRA_i} = d_i \left[1 + \frac{r^2}{d_i^2}\right]^{1/2} - d_o \left[1 + \frac{r^2}{d_o^2}\right]^{1/2}. \tag{2.9}$$

Historically it was observed in the dark room that the image that was formed depended on the diameter of the opening and the distance of the object to view, it was observed that the image was formed for far objects, but how far is far? It is relative to the radius of the opening, then the ratios must satisfy that

$$\frac{r^2}{d_o^2} \ll 1\,,\;\; \frac{r^2}{d_i^2} \ll 1, \tag{2.10}$$

using this and using the Taylor's theorem for a polynomial expansion of the function $\sqrt{1+x}$ then

$$\overline{A_oRA_i} \approx d_i\left[1+\frac{1}{2}\frac{r^2}{d_i^2}+\mathcal{O}(\delta^2)\right]-d_o\left[1+\frac{1}{2}\frac{r^2}{d_o^2}+\mathcal{O}(\delta^2)\right], \tag{2.11}$$

$$\overline{A_oRA_i} \approx (d_i-d_o)+\frac{r^2}{2}\left(\frac{1}{d_i}-\frac{1}{d_o}\right)=\overline{A_oOA_i}+\frac{r^2}{2}\left(\frac{1}{d_i}-\frac{1}{d_o}\right) \tag{2.12}$$

with this we can express this in terms the difference of the light's path and the diameter of the dark room

$$\triangle L=\frac{r^2}{2}\left(\frac{1}{d_i}-\frac{1}{d_o}\right)=\frac{D^2}{8}\left(\frac{1}{d_i}-\frac{1}{d_o}\right). \tag{2.13}$$

In the equation 2.13 lies the hearth of the dark room, it was found by Rayleigh a certain relation between the light's wavelength and the light's path difference between the upper and lower part of the opening of the dark room, this relation (which will be discussed physically in a moment) is:

$$\triangle L<\frac{\lambda}{4}. \tag{2.14}$$

Using this relation we will find the maximum diamater for the dark room

$$\frac{D^2}{8}\left(\frac{1}{d_i}-\frac{1}{d_o}\right)<\frac{\lambda}{4}, \tag{2.15}$$

$$D_{max}=\left(\frac{2\lambda d_i d_o}{d_o-d_i}\right)^{1/2}. \tag{2.16}$$

In the real life there is no monochromatic light, it is forbidden by the laws of quantum mechanics, we can have quasi-monochromatic light but it is only in advanced laboratories so for a polychromatic source in the equation 2.16 we should do the calculations with the maximum wavelength that conforms the beam the origin of this expression have many origins which converges to the same phenomena. If we want to form an image the electromagnetic field coming from the object must arrive at the same time at the image point, if there is some kind of phase difference there will be aberrations as astigmatism or chromatic aberration (even if it is not a lens we can see it from the equation 2.16) from a quantum point of view this is related to an interpretation of the *size* of a photon (Santos et al., 2018).

Now we're going to analyze how does the light interact (from a simple and geometrical point of view) with transparent media. Suppose a source radiates from a point A and the ray light arrives at B, see figure 2.3.

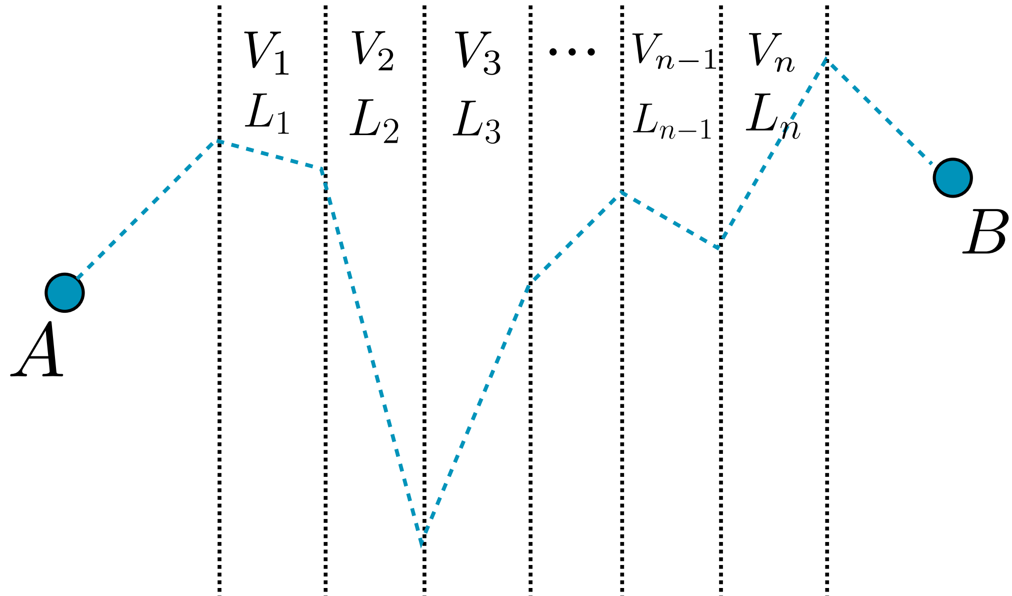


Figure 2.3: A schematic representation of a ray beam from A to B which goes through n different transparent media and in each one it had a different speed and traveled different length.

There are n different transparent media each one hast a different thickness and the light in each one is different, therefore the time that takes the light to go from A to B is

$$t_{AB} = \sum_{i=1}^{n} \frac{L_i}{v_i} = \sum_{i=1}^{n} \frac{L_i C}{v_i} \frac{1}{C} \ [s]. \tag{2.17}$$

Then we define the refraction index in the ith medium as the relation between the speed of light in vacuum and the speed of light in the ith medium. *It is a scalar in linear and homogeneous and isotropic media but mostly it is a tensor.*

$$t_{AB} = \frac{1}{C} \sum_{i=1}^{n} n_i L_i = \frac{L_{AB}}{C} \ [s], \tag{2.18}$$

Definition 1 *(Optical path): The optical path L_{AB} is the distance that must travel in vacuum the light to have the same time to go from a point A to a point B across n different media each one characterized by a different (or equal) refraction index.*

The optical path has a length greater than the geometrical path of the light, that is why it is called optical path. This can be easily generalized to continuous media using a Lebesgue's integral (or Riemann integral if it is well behaved)

$$L_{AB} = \int_A^B n(S) ds \; [m]. \tag{2.19}$$

Before going further we must have to set some definitions.

Definition 2 *(Phase): The phase is the relative angular displacement the ray has.*

Definition 3 *(Wavefront): The geometrical space where the phase of the light is constant is called the wavefront of the light.*

2.2 The Snell-Descartes law and the Fermat's principle

As we wad discussed in the historical chapter it was Heron of Alexandria who realizes that in light reflection it travels the least distance and Al Azhen proposes a maximum time, it was Fermat who formalizes it as a principle the **Fermat's principle**

Principle 1 *(Fermat's principle): The light travels a trajectory such that the optical path is stationary*

$$\delta L = 0. \tag{2.20}$$

The Fermat's principle can be "deduced¨ from the Hamilton principle or even more from a variational principle, lets give a look to it, lets look for an stationary curve for the time the light travels

$$t = \int_{t_0}^{t_f} dt = \int_A^B \frac{1}{C} dL = \int_{\tau_0}^{\tau_f} L d\tau, \tag{2.21}$$

from here we see that, in a mathematical way, if we want to find the relation between the trajectory of the light and the time it takes to go from A to B we will have to solve the Euler-Lagrange equation. Now lets use the Fermat's principle to see how the light interacts with a diopter (a refracting surface) in a vector way, known as the *vectorial Snell-Descartes law.*

Definition 4 *(Diopter): A Diopter is a surface $\Sigma \subset \mathbb{R}^2$ such that it separates two media with different refraction index.*

In the figure 2.4 we see two media separated by a transparent surface (diopter) with refractive index n_o,n_1, respectively, a source A emits a beam which reaches the diopter in I and it is refracted to B, from A there is an unitary vector $\hat{u}_o$ in the direction $\overline{AI}$ and the observer O analyzes a variation of A as A' and the incident point on the diopter is I', then lets analyze it vectorially

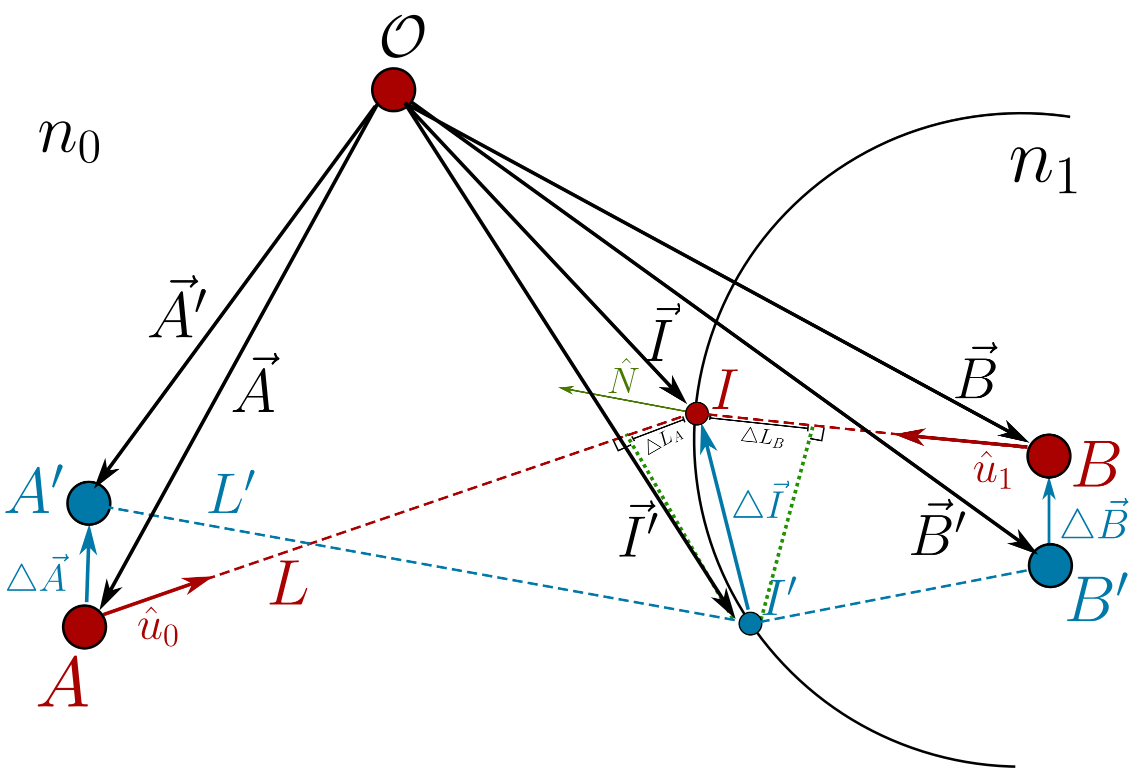


Figure 2.4: two media separated by a diopter which an observer O measures the path the light will follow from A to B and proposes a displacement of the source A to A' to see how must be the light path.

$$L = \left|\left|\vec{I} - \vec{A}\right|\right|, \tag{2.22}$$

$$\triangle\vec{I} = \vec{I}' - \vec{I}, \tag{2.23}$$

$$\triangle\vec{A} = \vec{A}' - \vec{A}, \tag{2.24}$$

with this we can calculate the path difference $L - L'$ vectorially as a proyection of the unitary vector in the direction of L given by $\hat{u}_o$ along $\triangle\vec{I} - \triangle\vec{A}$

$$\triangle L_A = \frac{\left(\vec{I} - \vec{A}\right) \cdot \left(\triangle\vec{I} - \triangle\vec{A}\right)}{\left|\left|\vec{I} - \vec{A}\right|\right|} = \hat{u}_o \cdot \left(\triangle\vec{I} - \triangle\vec{A}\right), \tag{2.25}$$

now for the optical path

$$\mathcal{L}_A = n_o \triangle L = n_o \hat{u}_o \left(\triangle\vec{I} - \triangle\vec{A}\right), \tag{2.26}$$

doing the same for the right side of the diopter but for the point B we then have got the total optical path difference as

$$\triangle\mathcal{L} = n_o\hat{u}_o \cdot \left(\triangle\vec{I} - \triangle\vec{A}\right) + n_1\hat{u}_1 \cdot \left(\triangle\vec{B} - \triangle\vec{I}\right) \tag{2.27}$$

because the points A and B are fixed then $\triangle\vec{A} = 0$ and $\triangle\vec{B} = 0$ thus

$$\triangle\mathcal{L} = (n_o\hat{u}_o - n_1\hat{u}_1) \cdot \triangle\vec{I}. \tag{2.28}$$

If the Fermat's principle is satisfied the equation 2.28 implies straightfully that $\triangle\vec{I}$ is a tangent vector to the diopter this implies then

$$n_o\hat{u}_o - n_1\hat{u}_1 = k\hat{N}. \tag{2.29}$$

The equation 2.29 is known as the vectorial Snell-Descartes law, from this equation we can arrive to the law of reflection and refraction to do it we only have to do the cross product of the equation 2.29

$$\hat{N} \times (n_0\hat{u}_0 - n_1\hat{u}_1) = k\hat{N} \times \hat{N} = 0, \tag{2.30}$$

$$n_0 sin(\theta_0) = n_1 sin(\theta_1). \tag{2.31}$$

2.2.1 The critical angle, and the refraction by a prism

We had deduced the vectorial Snell-Descartes law, which is a strong law and we are going to apply it to some physical situations, but before doing that we are going to talk briefly about where is the physical regime where this law can work on.

We are working under the postulates of the geometrical optics, but lets analyze a bit the second one, *the light propagates in a straight line* we haven't say anything about the light's shape, if we say it propagates in a straight line we can image light rays as circular cylinders but what about the radius of this ray? Well, in fact there are more shapes we can imagine of the wavefront and the beam's profile, but we are going to talk about it in the chapter of electromagnetic optics, for now keep with the cylinder image.

If we make this ray to inside over a perfectly polished mirror as shown in 2.5 then we can see an prove easily experimentally the Snell-Descartes law will be fulfilled but what happen if we have not got a polished surface but an extremely irregular surface? we will have reflections in all the points over the surface as shown in the figure 2.6.

The pehonomena shown in the figure 2.6 is, often, called *diffuse refraction* but it is a kind of macroscopic effect of the Snell-Descartes law but there are other behaviors of the refraction index

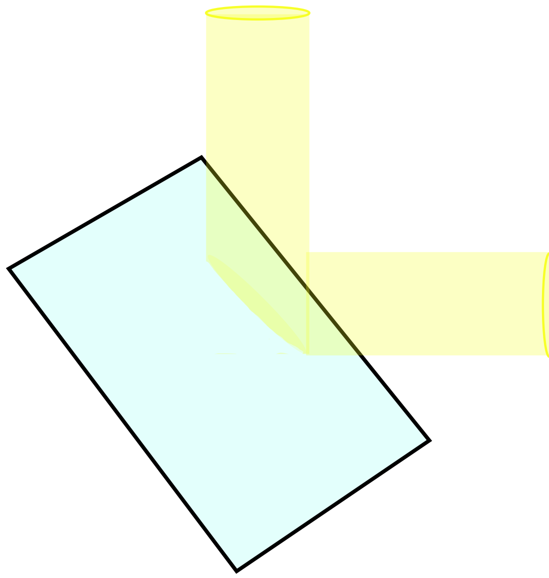


Figure 2.5: Representation of a cylindrical ray light being reflected by a perfectly polished mirror.

which give a particular behavior to the SD law, which are related to the electromagnetic properties of the matter, in fact, the refraction indez n is a complex quantity, which its real part is related to the refractive properties of the material and the imaginary part is related to absorption an losses in the energy but we will work on this later.

The are some other special materials where the equations 2.29 has an amazing behavior. The meta materials which has a negative refraction index, but how can we interpret this? It is easy to see in the equation 2.29 that a variable changes of $n \to -n$ implies a physical effect of $\theta \to -\theta$ this means, the rays will be reflected or refracted to the other side!

Well then, lets come back to our well defined Behavior, where the SD law 2.29 will be fulfilled. Can we find a condition between two material such that there is no refraction? It is worth to ask about it because in nature the reflection and refraction happen at the same time both simultaneously because a translucent surface is not perfectly transparent and a refractive surface is not perfectly reflective this is related to the Fresnel's coefficients but this will be hard work of the chapter of electromagnetic optics, for now have on mind reflection and refraction happen at the same time. To try to kill the refraction we have to study two cases between two media.

1. $n_1 > n_2$
2. $n_1 < n_1$

For the first case then we have from 2.29

$$n_1 sin\theta_1 = n_2 sin\theta_2, \tag{2.32}$$

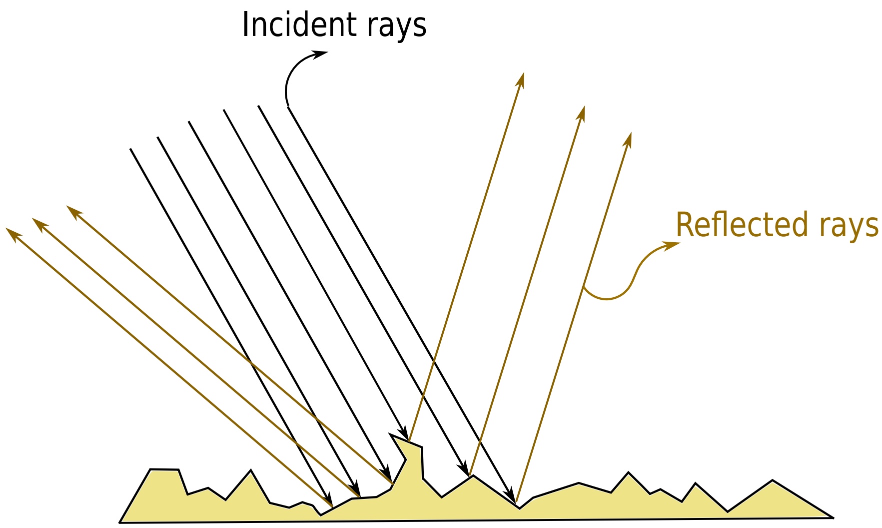


Figure 2.6: Representation of a cylindrical bunch of rays being reflected by an inhomogeneus surface.

$$\frac{n_1}{n_2} sin\theta_1 = sin\theta_2. \tag{2.33}$$

If $\theta_1 = 0$ then $\theta_2 = 0$ thus if the incident ray is perfectly parallel to the normal vector to surface there wont be any refracted ray, only a completely reflected in the same plane as the incident but with opposite direction, this is said to get a phase shift of π. If the ray's incidence angle is perpendicular to the normal vector to surface $\theta_1 = \pi/2$ then we will have

$$\frac{n_1}{n_2} = sin\theta_2, \tag{2.34}$$

but $n_1 > n_2$ then $n_1/n_2 > 1$ which has no sense with the right hand side of the equation 2.34 because $0 \leq sin\theta_2 \leq 1$ therefore for $\theta_1 = \pi/2$ there is no refracted ray this is illustrated in the figure 2.7.

Now, we have to consider and study the case where $n_1 < n_2$ the obvious case is when $\theta_1 = 0$ therefore $\theta_2 = 0$ and there wont be any refracted ray, but the interesting thing comes when $\theta_1 = \pi/2$ then again we have got the same equation 2.34 but here is the main difference, now we've got $n_1 < n_2$ or $n_1/n_2 < 1$ therefore there exists this angle and it is called the *total reflection angle* or *critical angle* θ_c this is illustrated in the figure 2.8.

2.3 The Descartes's ovoid and diopters

From here on we are going to start to study the theory of image formation where the concepts of **object** and **image** are really important because there are real and virtual objects and image, this

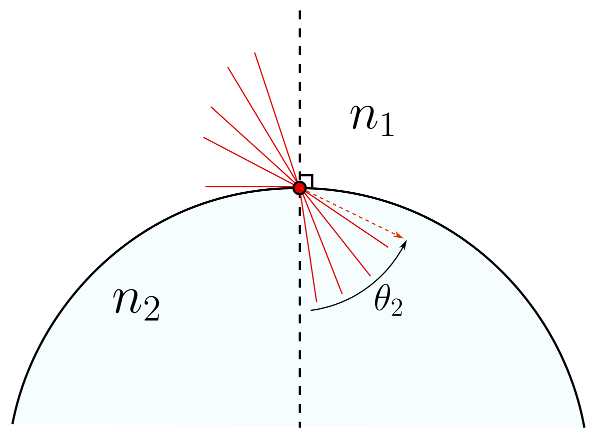


Figure 2.7: Representation of all the incident and refracted rays from a media with refraction index n_1 to another with refraction index n_2 such that $n_1 > n_2$. The angle θ_2 here shown wont never reach a value equal to $asin\left(\frac{n_1}{n_2}\right)$.

depends on the relation between these points ant the diopter, in fact, something is called virtual if we put a photo detector in that point and we wont measure any field if an object is to the left of a diopter is called real and if it is to the right is called virtual, in the same order of ideas if the image is to the left of the diopter it is virtual and if it is to the right is real.

Before starting we have to give some fundamental definitions

Definition 5 *(Optical system) an optical system is a set of translucid media (diopters) and reflective media (catopters) and we say that the* ***system is centered*** *if all the diopters are revolution surfaces and shared the same symmetry axis.*

The systems conformed only by refractive surfaces all called ***dioptric systems*** *and the system conformed only by reflective surfaces are called* ***catoptric systems*** *and the systems formed by these two kind of surfaces are called* ***catadioptrics systems****.*

Definition 6 *(Real and virtual points) Given a optical system $\mathcal{S}$ we say that an object point A_o is real if it is to the left and outside the optical system and we way that the object point is virtual if it is to the right and inside the optical system.*

In the same way, we say that an image point A_i is real if it is to the right and outside the optical system, and we way that the image point is virtual if it is inside the optical system.

To understand better these two definitions lets follow the figures 2.9 there we can see two examples of optical systems. In the first one we can see an object point located to the left and outside of the system $\mathcal{S}$ so it is a real object point, and the point A_i is located to the right and outside the optical system, therefore it is a real image point. In the second optical system we can see that the same

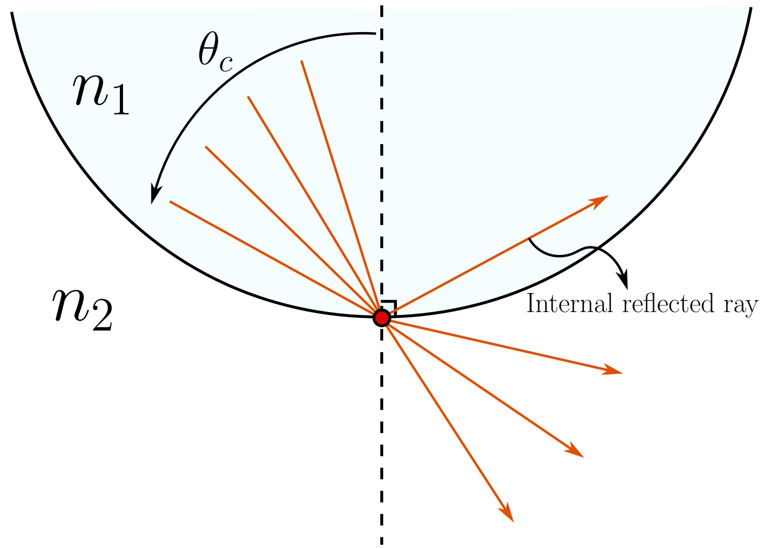


Figure 2.8: Representation of all the incident and refracted rays from a media with refraction index n_1 to another with refraction index n_2 such that $n_1 < n_2$. The angle θ_c here separates the rays that will be refracted , (angles less than θ_c), and the rays that will be totally reflected internally.

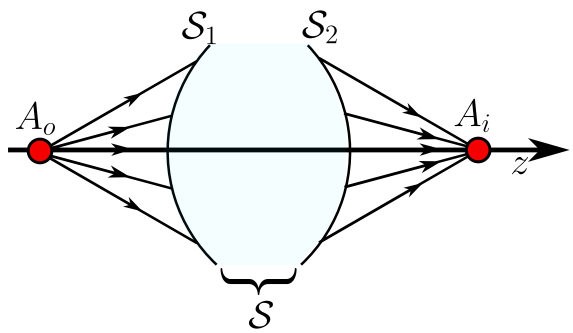
real object point and that the point A_i is located inside the optical system, therefore it is a virtual image point for A_o.

In the figure 2.10 we can see two diopter $\mathcal{S}_1, \mathcal{S}_2$ which forms the optical system $\mathcal{S}$ where the point A_o is located inside the optical system, then it behaves as a virtual object point and A_i is to the right of A_o and also inside the optical systen, therefore it is a virtual image point.

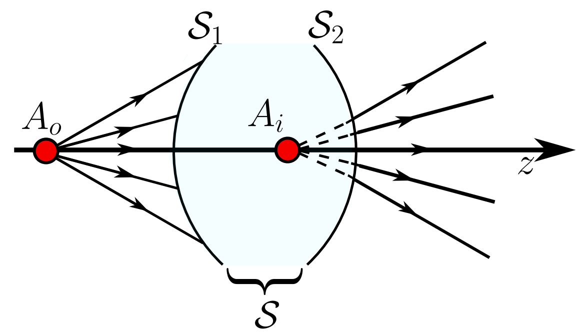
In the optical system 2.11 we can see three optical systems $\mathcal{S}_1, \mathcal{S}_2, \mathcal{S}_3$ and a configuration of many points. The first one, A_o is completely to the left and outside any optical system, so it is a real object point, and the point A' is inside the optical system $\mathcal{S}_1$ so it is the **the virtual image of the real object** A_o but, as we can see, A' is to the left and outside the optical system $\mathcal{S}_2$ so it is a real object point for the rest of the total optical system. The point A'' is inside the optical system $\mathcal{S}_3$ so it is the virtual image of the real object A', and in the very last there is the point A_i which is completely to the right and outside any optical system, so it is the real image point of the virtual object point A''.

Definition 7 *(Stigmatism) From a* ***geometrical point of view*** *we say that an optical system $\mathcal{S}$ is* ***stigmatic*** *if for any given point source, all the rays that pass through $\mathcal{S}$ converge to a point source. In other words, a optical system $\mathcal{S}$ is stigmatic if for any object point there is one unique image point.*

We can also, define the stigmatism from a wavefront perspective. An rigorously stigmatic system is one that transforms an spherical wavefront $\mathcal{A}$ to a new spherical wavefront $\mathcal{A}'$ whose curvature centers, A_o and A_i are the rigorously stigmatic points (see figure 2.12). Thus, the spherical wavefront $\mathcal{A}'$ is the image of the spherical wavefront $\mathcal{A}$.



(a) Optical system $\mathcal{S}$ formed by two diopters $\mathcal{S}_1, \mathcal{S}_2$ where the point A_o is a real object and A_i is a real image point.



(b) Optical system $\mathcal{S}$ formed by two diopters $\mathcal{S}_1, \mathcal{S}_2$ where the point A_o is a real object and A_i is a virtual image point.

Figure 2.9: Representation of two optical systems where appear real object points, real and virtual image points.

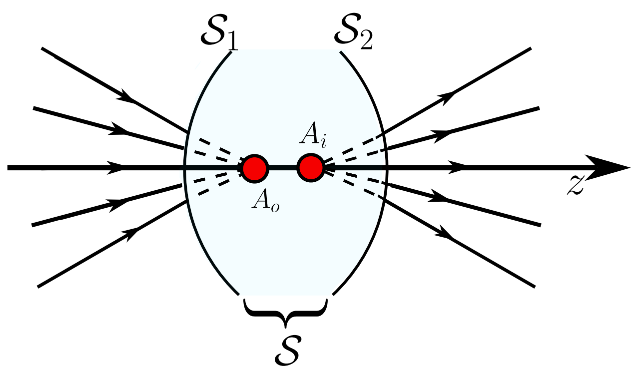


Figure 2.10: Optical system $\mathcal{S}$ formed by two diopters $\mathcal{S}_1, \mathcal{S}_2$ where the point A_o is a virtual object and A_i is a virtual image point.

Understanding that an image-forming optical system is one that maps each point source of an object's electromagnetic field to the point sources of the image's electromagnetic field, corresponding to a homothety between these fields, the interest arises in studying what these optical systems are and what their characteristics are. In general, these systems can be composed of multiple elements, but initially we will study the simplest configuration for two reasons:

1. because it most easily allows us to understand the principles of these systems.
2. because later, more complex systems can be understood as compositions of these.

These elementary systems are made up of a single optical surface, either refractive or reflective, which is capable of perfectly focusing all the rays emitted by a point source into a perfect point image. In geometric optics, this means that all the rays travel the same optical path length. Initially, this

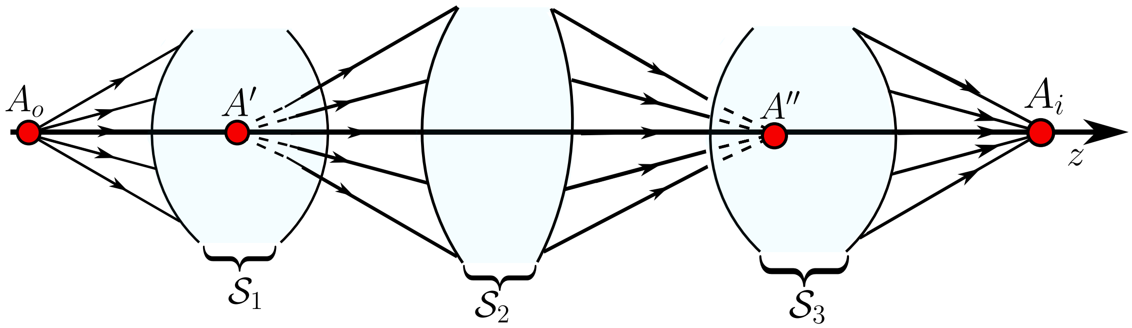


Figure 2.11: Optical system $\mathcal{S}$ formed by three optical systems $\mathcal{S}_1, \mathcal{S}_2, \mathcal{S}_3$ where the point A_o is a real object point, A' is a virtual image point for $\mathcal{S}_1$, but a real object point for $\mathcal{S}_2, \mathcal{S}_3$, A'' is a virtual image for the real object A' and A_i is the real image of the virtual object A''.

property associated with a single pair of points (stigmatism) will be studied, and then the conditions under which this property becomes invariant to the translation of these points, which is known as aplanetism, will be studied.

Now we are going to develope the *rigurous stigmatism* thes are system capable of making perfect images, this means from a point source there is a perfect point image. Let a diopter which separates two media with refraction index n_o and n_i to the left of the diopter there is an object A_o at a point $[0, 0, z_o]$ the field from A_o reaches a point I, measured from the diopter's vertex v, from the point I the beam is refracted and forms an image A_i at a point $[0, 0, z_i]$ then we are going to find the relations between all these parameters to make all the rays from A_o arrive at A_i making a perfect image. From the Fermat's principle we have got

$$n_0 l_0 + n_i l_i = cte \tag{2.35}$$

the, as we can see in the figure 2.13 we then have

$$-n_0[x^2 + y^2 + (z - z_o)^2]^{1/2} + n_i[x^2 + y^2 + (z - z_i)^2]^{1/2} = cte \tag{2.36}$$

and, remembering our sign convention $\sqrt{z_o^2} = -zo$, $z_i^2 = z_i$ and to find the constant we can see that if we want to make all the rays from A_o arrive to A_i then the rays that travel parallel to z satisfy also the Fermat's principle, thus

$$n_i z_i - n_o z_o = cte, \tag{2.37}$$

$$n_i[x^2 + y^2 + (z - z_i)^2]^{1/2} - n_0[x^2 + y^2 + (z - z_o)^2]^{1/2} = n_i z_i - n_o z_o, \tag{2.38}$$

now, we are going to do the following change of variables given by

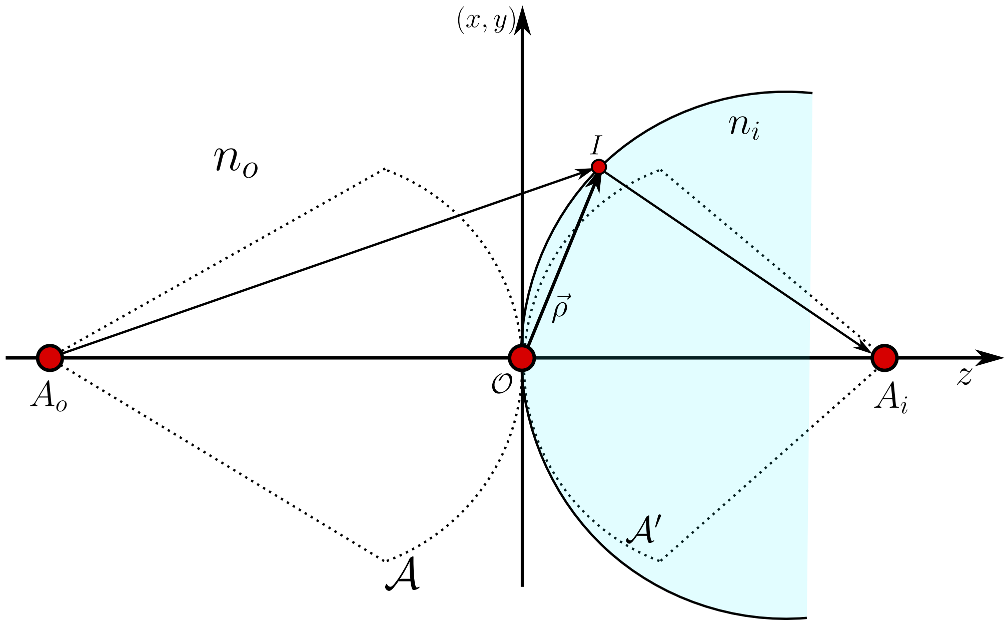


Figure 2.12: The stigmatic system transform the incident wavefront $\mathcal{A}$ to another new spherical wavefront $\mathcal{A}'$.

$$\rho^2 = x^2 + y^2 + z^2, \tag{2.39}$$

which represents the square of the vector from the vertex to the interface shown in 2.13, so, having this on count, the equation becomes

$$n_i \left[\rho^2 + z_i^2 - 2zz_i\right]^{1/2} - n_0 \left[\rho^2 + z_0^2 - 2zz_0\right]^{1/2} = n_i z_i - n_0 z_0. \tag{2.40}$$

Now, the main goal is to transform the equation 2.40 into a more explicit and useful algebraic equation, so, the first thing we're going to do is to square both side of the equation 2.40 and then we'll have

$$\begin{aligned}&n_o^2 \left[\rho^2 + z_o^2 - 2zz_o\right] + n_i^2 \left[\rho^2 + z_i - 2zz_i\right] - 2n_o n_1 \left[\rho^2 + z_o^2 - 2zz_o\right]^{1/2} \left[\rho^2 + z_i^2 - 2zz_i\right]^{1/2}\\ &= (n_i z_i - n_o z_o)^2\end{aligned}$$

Now, we're going to re-order all the equation for convenience as follows

$$\begin{aligned}&n_o^2 \left[\rho^2 + z_o^2 - 2zz_o\right] + n_i^2 \left[\rho^2 + z_i - 2zz_i\right] - (n_i z_i - n_o z_o)^2\\ &= 2n_o n_1 \left[\rho^2 + z_o^2 - 2zz_o\right]^{1/2} \left[\rho^2 + z_i^2 - 2zz_i\right]^{1/2},\end{aligned}$$

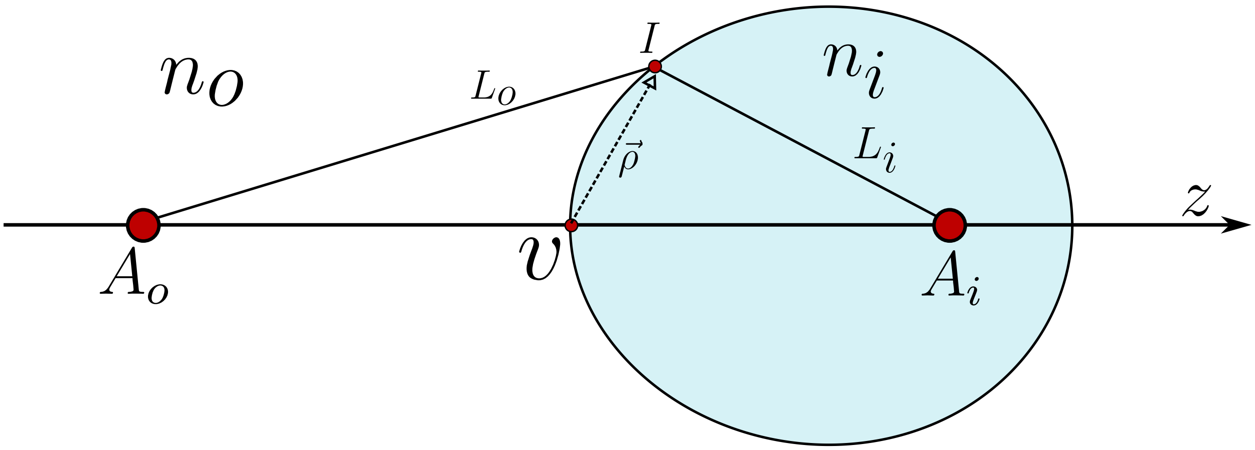


Figure 2.13: Scheme representation of the development of a rigorously stigmatic system, from a point source A_o at a position $[0, 0, z_o]$ the diopter with a vertice v forms a perfect image A_i at $[0, 0, z_i]$.

on the left hand side we have got the following

$$\begin{aligned}
&n_o^2\left[\rho^2+z_o^2-2zz_o\right]+n_i^2\left[\rho^2+z_i-2zz_i\right]-\left(n_iz_i-n_oz_o\right)^2=\\
&n_o^2\left[\rho^2+z_o^2-2zz_o\right]+n_i^2\left[\rho^2+z_i-2zz_i\right]-\left(n_i^2z_i^2+n_o^2z_o^2-2n_on_iz_oz_i\right)=\\
&n_o^2\left[\rho^2-2zz_o\right]+n_i^2\left[\rho^2-2zz_i\right]+2n_on_iz_oz_i,
\end{aligned}$$

where we just only simplified the common terms. Having this on count, the equation reduces to

$$n_o^2\left[\rho^2-2zz_o\right]+n_i^2\left[\rho^2-2zz_i\right]+2n_on_iz_oz_i=2n_on_1\left[\rho^2+z_o^2-2zz_o\right]^{1/2}\left[\rho^2+z_i^2-2zz_i\right]^{1/2}. \quad (2.41)$$

Now, we must square again, the whole equation as follows

$$\begin{aligned}
&\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]^2+\left(2n_on_iz_oz_i\right)^2+4n_on_iz_oz_i\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]=\\
&4n_o^2n_i^2\left(\rho^2+z_o^2-2zz_o\right)\left(\rho^2+z_i^2-2zz_i\right),
\end{aligned}$$

now we are going to expand the product on the right hand side in the following way

$$\begin{aligned}
&\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]^2+\left(2n_on_iz_oz_i\right)^2+4n_on_iz_oz_i\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]=\\
&4n_o^2n_i^2\left(\rho^2-2zz_o\right)\left(\rho^2-2zz_i\right)+4n_o^2n_i^2\left[z_o^2\left(\rho^2-2zz_i\right)+z_i^2\left(\rho^2-2zz_o\right)+z_o^2z_i^2\right].
\end{aligned}$$

Now, we must do a simplification, to do this, we must expand the first term of the left hand side like this

$$\begin{aligned}
&\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]^2\\
&=\left[n_o^2\left(\rho^2-2zz_o\right)\right]^2+\left[n_i^2\left(\rho^2-2zz_i\right)\right]^2+2n_o^2n_i^2\left(\rho^2-2zz_o\right)\left(\rho^2-2zz_i\right),
\end{aligned}$$

so the first term of the left hand side have on common in its expansion the same term than the expansion of the term on the right hand side, so, we could simplify then and look at the binomial is the same but with a minus sign instead of a plus sign, thus

$$\left[n_o^2\left(\rho^2-2zz_o\right)-n_i^2\left(\rho^2-2zz_i\right)\right]^2+\left(2n_on_iz_oz_i\right)^2+4n_on_iz_oz_i\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]=$$
$$4n_o^2n_i^2\left[z_o^2\left(\rho^2-2zz_i\right)+z_i^2\left(\rho^2-2zz_o\right)+z_o^2z_i^2\right].$$

Now, we are going to forget, momentarily the first term of the left hand side and focus on all the rest of the equations because there are going to be many important simplifications. Then, having this on count we'll have the following:

$$\begin{aligned}&\left(2n_on_iz_oz_i\right)^2+4n_on_iz_oz_i\left[n_o^2\left(\rho^2-2zz_o\right)+n_i^2\left(\rho^2-2zz_i\right)\right]\\&-4n_o^2n_i^2\left[z_o^2\left(\rho^2-2zz_i\right)+z_i^2\left(\rho^2-2zz_o\right)+z_o^2z_i^2\right]=0\end{aligned}$$

$$\begin{aligned}&4n_on_i\left[z_oz_in_o^2\left(\rho^2-2zz_o\right)+z_oz_in_i^2\left(\rho^2-2zz_i\right)-n_on_iz_o^2\left(\rho^2-2zz_i\right)-n_on_iz_i^2\left(\rho^2-2zz_o\right)\right.\\&\left.-n_on_izo^2z_i^2+n_on_iz_o^2z_i^2\right],\end{aligned}$$

now, we have to factorize the common factors

$$\begin{aligned}&4n_on_i\left[\left(\rho^2-2zz_o\right)\left(z_oz_in_o^2-n_on_iz_i^2\right)+\left(\rho^2-2zz_i\right)\left(z_oz_in_i^2-n_on_iz_o^2\right)\right]=\\&4n_on_i\left[z_in_o\left(\rho 2-2zz_o\right)\left(z_on_o-n_iz_i\right)+z_on_i\left(\rho^2-2zz_i\right)\left(z_in_i-z_on_o\right)\right]=\\&4n_on_i\left[n_iz_i-n_oz_o\right]\left[n_iz_o\left(\rho^2-2zz_i\right)-n_oz_i\left(\rho^2-2zz_o\right)\right]=\\&4n_on_i\left[n_iz_i-n_oz_o\right]\left[\rho^2\left(n_iz_o-n_oz_i\right)+2zz_iz_o\left(n_o-n_i\right)\right].\end{aligned}$$

So, joining this result with the first term we will have the following

$$\left[n_o^2\left(\rho^2-2zz_o\right)-n_i^2\left(\rho^2-2zz_i\right)\right]^2+4n_on_i\left[n_iz_i-n_oz_o\right]\left[\rho^2\left(n_iz_o-n_oz_i\right)+2zz_iz_o\left(n_o-n_i\right)\right]=0,$$

so, the last things we have to do are is re organize in the following way

$$\left[\left(n_i^2-n_o^2\right)\rho^2-2z\left(n_i^2z_i-n_o^2z_o\right)\right]^2-4n_in_o\left[n_iz_i-n_oz_o\right]\left[\left(n_oz_i-n_iz_o\right)\rho^2+2zz_oz_i\left(n_i-n_o\right)\right]=0. \tag{2.42}$$

The equation 2.42 is the algebraic expression for the Descartes Ovoid the rigorously stigmatic surface we wanted.

2.4 The spherical diopter

Now, we are going to see that the Descartes Ovoid 2.42 contains the information of all the other diopters known. So, the first one we are going to deduce is the spherical diopter this is obtained in a vicinity of the vertex shown in the figure 2.13. Therefore, to deduce this particular case, we are going to take the equation 2.42 and rewrite in the following convenient way

$$\left[\left(n_o^2-n_i^2\right)\rho^2+2zz_oz_i\left(\frac{n_i^2}{z_o}-\frac{n_o^2}{z_i}\right)\right]^2-4n_in_o\left[n_iz_i-n_oz_o\right]\left[\left(n_oz_i-z_iz_o\right)\rho^2+2zz_oz_i\left(n_i-n_o\right)\right]=0. \tag{2.43}$$

So, the main idea is to construct the sphere that is located in the vertex v. To do this, we must see that the squared term in 2.43 looks very similar to the right factor term in the second term, so, we want to equate them and then simplify, which leads us to the following equations

$$n_o^2-n_i^2=c\left(n_oz_i-n_iz_o\right), \tag{2.44}$$

$$\frac{n_i^2}{z_o}-\frac{n_o^2}{z_i}=c\left(n_i-n_o\right) \tag{2.45}$$

where c is a constant we must calculate. The second thing we are going to do, is to re-order the equation 2.44 in the following way

$$n_o^2-cz_in_o=n_i^2-cz_on_i, \tag{2.46}$$

and if we re-order the equation 2.45 we'll have

$$\begin{aligned}z_in_i^2-z_on_o^2&=cz_oz_i(n_i-n_o),\\ z_in_i^2-cz_oz_in_i&=z_on_o^2-cz_oz_in_o,\\ z_o\left(n_o^2-cz_in_o\right)&=z_i\left(n_i^2-cz_on_i\right).\end{aligned} \tag{2.47}$$

So if we use 2.46 on 2.47 we have got the following two equations

$$\left(z_o-z_i\right)\left(n_i^2-cz_on_i\right)=0, \tag{2.48}$$

$$\left(z_o-z_i\right)\left(n_o^2-cz_in_o\right)=0, \tag{2.49}$$

using the equations 2.48 and 2.49 we will have

$$z_o=\frac{n_i}{c}, \qquad z_i=\frac{n_0}{c}, \tag{2.50}$$

and using the equation 2.44 we will have

$$z_0=z_i \qquad z_{0,i}=\frac{n_o+n_i}{c}. \tag{2.51}$$

Now, before going further analyzing this results, we have to use the equations 2.44, 2.45 on 2.43

$$\begin{aligned}&\left[c\left(n_o z_i - n_i z_o\right)\rho^2 + 2zz_oz_ic(n_i - n_o)\right]^2\\&-4n_in_o\left[n_iz_i - n_oz_o\right]\left[\left(n_oz_i - z_iz_o\right)\rho^2 + 2zz_oz_i\left(n_o - n_o\right)\right] = 0,\\&\left[\left(n_oz_i - n_iz_o\right)c\rho^2 + 2zz_oz_ic(n_i - n_o)\right]\\&\times\left[\left(n_oz_i - n_iz_o\right)\rho^2 + 2zz_oz_i(n_i - n_o) - 4n_in_o\left(n_iz_i - n_oz_o\right)\right] = 0,\end{aligned}$$

and by the fundamental theorem of Algebra, we have got the equation for the spherical diopter

$$\left(n_oz_i - n_iz_o\right)\rho^2 + 2zz_oz_i(n_i - n_o) = 0. \tag{2.52}$$

2.4.1 The Young-Weirstrass points

From the case 2.50 we have that

$$n_iz_i - n_oz_o = 0,$$

and using the equation 2.52 we will have

$$\rho^2\left(n_o^2 - n_i^2\right) + 2czz_on_i(n_i - n_o) = 0, \tag{2.53}$$

now, dividing by $n_o^2 - n_i^2$ and then we must complete squares as follows

$$x^2 + y^2 + z^2 + \frac{2czz_oz_i}{n_i + n_o} = x^2 + y^2 + z^2 + \frac{2czz_oz_i}{n_i + n_o} + \left(\frac{cz_oz_i}{n_i + n_o}\right)^2 = \left(\frac{cz_oz_i}{n_i + n_o}\right)^2,$$

$$x^2 + y^2 + \left(z - c\frac{z_oz_i}{n_o + n_i}\right)^2 = c^2\frac{z_o^2z_i^2}{(n_i + n_o)^2}, \tag{2.54}$$

which corresponds to the equation of an sphere whose radius is given by

$$R = c\frac{z_oz_i}{n_o + n_i} \longrightarrow x^2 + y^2 + (z - R)^2 = R^2, \tag{2.55}$$

and using 2.50 $z_o = n_i/c,\ z_i = n_o/c$ we will have

$$R = c\frac{z_in_i}{c(n_o + n_i)} \longrightarrow z_i = R\left(1 + \frac{n_o}{n_i}\right), \tag{2.56}$$

$$R = c\frac{z_on_o}{c(n_o + n_i)} \longrightarrow z_i = R\left(1 + \frac{n_i}{n_o}\right) \tag{2.57}$$

which are known as **Young-Weirstrass** points or the aplanetic points. One of these points is inside and the other outside the sphere as is shown in 2.14.

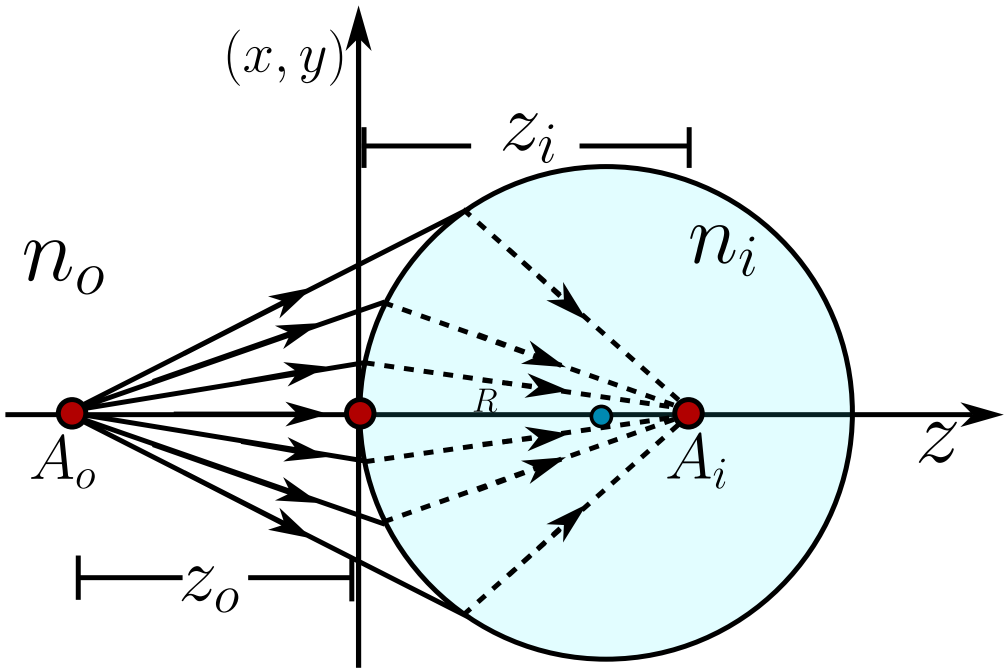


Figure 2.14: The spherical Diopter for the first case, where the object point is real and it is outside the sphere but the image point is virtual and inside the diopter.

2.4.2 The second spherical diopter

So, for the second case we have 2.51 and the equation 2.52 then let z_c given by

$$z_c = z_o = z_i = \frac{n_i + n_o}{c},$$

then we will have

$$\rho^2 (n_o z_c - n_i z_c) + 2 z z_c^2 (n_i - n_o) = 0, \tag{2.58}$$
$$\rho^2 - 2 z z_c = 0, \tag{2.59}$$

again we must complete squares in the following way

$$x^2 + y^2 + z^2 - 2 z z_c + z_c^2 = z_c^2, \tag{2.60}$$
$$x^2 + y^2 + (z - z_c)^2 = z_c^2. \tag{2.61}$$

The equation 2.61 is an spherical diopter where the object point is on the center and the image point is also on the center.

2.5 Objects and images to the infinity

Some other cases arise when either the object or the image is at infinity, which is a convenient condition when the object is very far away, for example, in telescopes.

But even in microscopy, systems are designed to form the image at infinity, since under these conditions, an eyepiece facilitates visual accommodation for observation.

For an object in the infinity we divide by z_o^2 and take the limit $z_o \to -\infty$ in 2.42 in the following way

$$\begin{aligned}
&\lim_{z_o \to -\infty} \left[\frac{\rho^2 \left(n_o^2 - n_i^2\right)}{z_o} + 2z\frac{n_o^2 z_o + n_i^2 z_i}{z_o} \right]^2 - \lim_{z_o \to -\infty} \frac{4n_o n_i \left(n_o z_o + n_i z_i\right)}{z_o} \left[\frac{\rho^2 \left(n_o z_i + n_i z_o\right)}{z_o} \right. \\
&\left. - \frac{2zz_o z_i (n_i - n_o)}{z_o} \right] = 0, \\
&(2zn_o^2)^2 - 4n_o 2n_i^2 \left[\rho^2 n_i - 2zz_i(n_i - n_o)\right] = 0, \\
&4z^2 n_o^4 - 4n_o^2 n_i^2 \rho^2 + 8zz_i n_o^2 n_i^2 - 8zz_i n_o^3 n_i = 0, \qquad \text{Dividing all the equation by } 4n_o^2, \\
&z^2 \left(n_o^2 - n_i^2\right) + 2zn_i z_i (n_i - n_o) - n_i(x^2 + y^2) = 0, \\
&[n_o - n_i] \left[z^2 \left(n_o + n_i\right) - 2zn_i z_i\right] - n_i(x^2 + y^2) = 0.
\end{aligned}$$

Now, to complete the algebraic expression we must multiply all the equation by $(n_o + n_i)$ and complete the squares as follows

$$\begin{aligned}
&(n_o - n_i) \left[(n_o + n_i)(n_o + n_i)z^2 - 2zn_i z_i (n_o + n_i)\right] - n_i^2 (n_o + n_i)(x^2 + y^2) = 0, \\
&(n_o - n_i) \left[(n_o + n_i)^2 z^2 - 2zn_i z_i (n_o + n_i) + n_i^2 z_i^2 - n_i^2 (n_o + n_i)(x^2 + y^2)\right] = (n_o - n_i) n_i^2 z_i^2 \ , \\
&(n_o - n_i) \left[(n_o + n_i)z - n_i z_i\right]^2 - n_i^2 (n_o + n_i)(x^2 + y^2) = (n_o - n_i) n_i^2 z_i^2
\end{aligned}$$

and dividing all by $(n_o - n_i)n_i^2 z_i^2$ we will have

$$\left(\frac{n_i + n_o}{n_i z_i}\right)^2 \left[z - \frac{n_i z_i}{n_i + n_o}\right]^2 + \frac{n_i + n_o}{z_i^2 (n_i - n_o)} (x^2 + y^2) = 1. \tag{2.62}$$

The result 2.62 represents a conical section of revolution around the z-axis from this equation we are going to obtain the Ellipsoid and the hyperboloid.

2.5.1 Revolution Ellipsoid

if $n_i > n_o$ we are going to define the following scalars

$$a = \left(\frac{n_i - n_o}{n_i + n_o} \right)^{1/2} |z_i|, \qquad b = \left(\frac{n_i}{n_i + n_o} \right) |z_i|, \tag{2.63}$$

and notice that $b > a$ so we will have **prolate ellipsoid** (see figure 2.15 given by

$$\frac{1}{b^2}(z - b)^2 + \frac{1}{a^2}(x^2 + y^2) = 1. \tag{2.64}$$

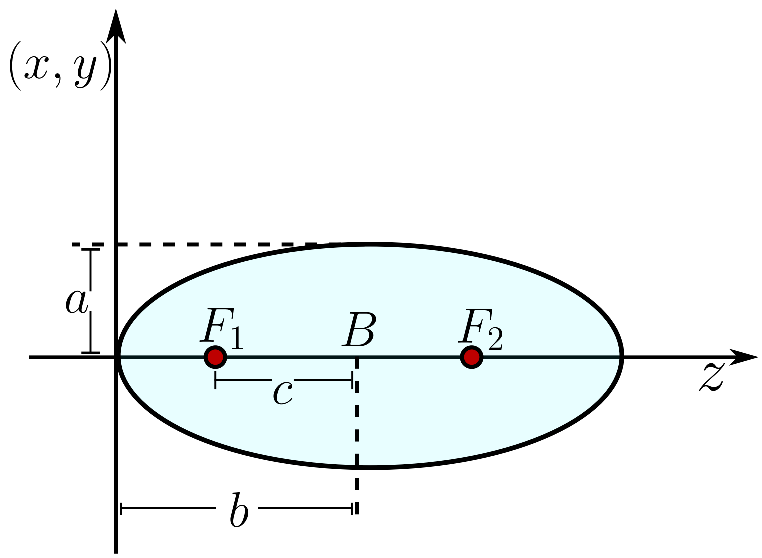


Figure 2.15: Prolate Ellipsoid centered and $z = B$.

We have got two cases, when $z_i > 0$ and $z_i < 0$. For both cases what is next is to study the eccentricity of the ellipse and the foci of the ellipsoids.

When $z_i > 0$ lets define the segment $c = \overline{F_1 B}$ such that the eccentricty is given by

$$\epsilon = \frac{c}{b} = \frac{\sqrt{b^2 - a^2}}{b} = \sqrt{1 - \frac{a^2}{b^2}} = \sqrt{\frac{n_o^2}{n_i^2}} = \frac{n_o}{n_i} < 1, \tag{2.65}$$

therefore the foci F_1 is centered to a distance

$$f_1 = b - c = b - \epsilon b = (1 - \epsilon) b = \frac{n_i - n_o}{n_i + n_o} z_i, \tag{2.66}$$

and the second foci is at

$$f_2 = b + c = b + \epsilon b = (1 + \epsilon) b = z_i. \tag{2.67}$$

This means, that for this first case that all the rays that comes parallel to the z-axis from infinity converges to the second Foci of the ellipsoid.

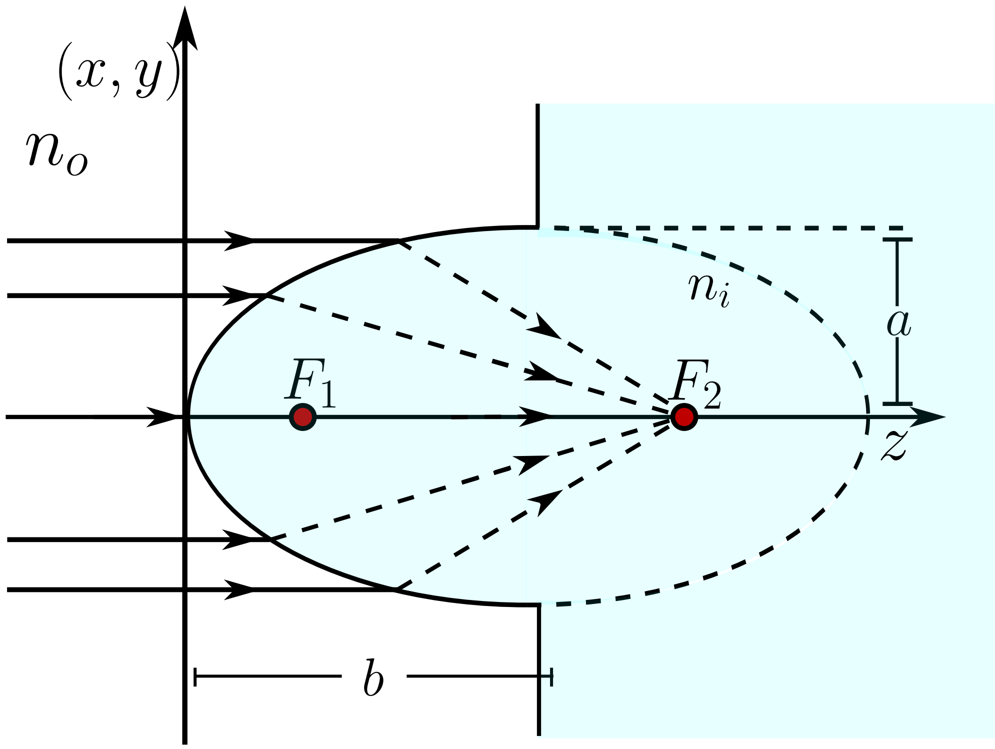


Figure 2.16: Ellipsoid diopter when $n_i > n_o$.

If $z_i < 0$ by the convention we have that $\sqrt{z_i^2} = -z_i$ and we define

$$b = -\frac{n_i}{n_i + n_o} z_i, \tag{2.68}$$

then the conical section equation 2.62 becomes

$$\frac{1}{b^2}(z + b)^2 + \frac{1}{a^2}(x^2 + y^2) = 1. \tag{2.69}$$

Again, we define $c = \overline{F_1 B}$ so the eccentricty is given by

$$\epsilon = \frac{c}{b} \frac{\sqrt{b^2 - a^2}}{b} = \sqrt{1 - \frac{a^2}{b^2}} = \frac{n_o}{n_i} < 1,$$

and the foci is now going to be to a distance

$$f_1 = -b - c = -b - \epsilon b = -b(1 + \epsilon) = z_i, \tag{2.70}$$

and the second one is at

$$f_2 = -b + c = -b + \epsilon b = (\epsilon - 1)b = \frac{n_i - n_o}{n_i + n_o} z_i. \tag{2.71}$$

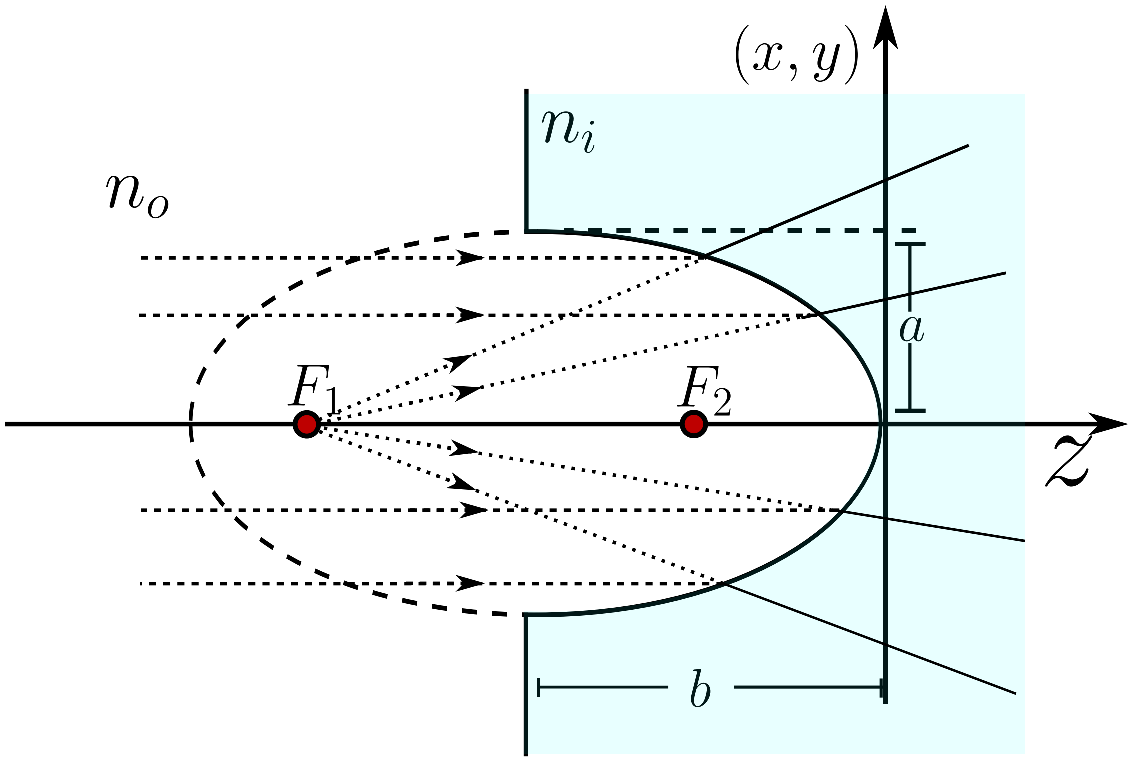


Figure 2.17: Ellipsoid diopter when $n_i > n_o$ and $z_i < 0$.

Having all these results on count, we can see in figure 2.17 that all the rays that becomes parallel to the z-axis and from infinity diverges as they would come from a point source from F_1.

2.5.2 Revolution Hyperboloid

In the case $n_i < n_o$ we have got that the equation 2.62 becomes an hyperboloid and, again, there are two possibilities, $z_i < 0, z_i > 0$, so the conical equation becomes

$$\left(\frac{n_i + no}{n_i z_i}\right)^2 \left[z - \frac{n_i z_i}{n_i + n_o}\right] - \frac{n_i + n_o}{z_i^2(n_o - n_i)}(x^2 + y^2) = 1. \tag{2.72}$$

If $z_i > 0$ we define again the following scalars

$$b^2 = \frac{n_i^2}{(n_i + n_o)^2} z_i^2,$$
$$a^2 = \frac{n_o - n_i}{(n_i + n_o)^2} z_i^2,$$

so the equation of the hyperboloid (see figure 2.18) becomes

$$\frac{1}{b^2}(z-b)^2 - \frac{1}{a^2}(x^2+y^2) = 1. \tag{2.73}$$

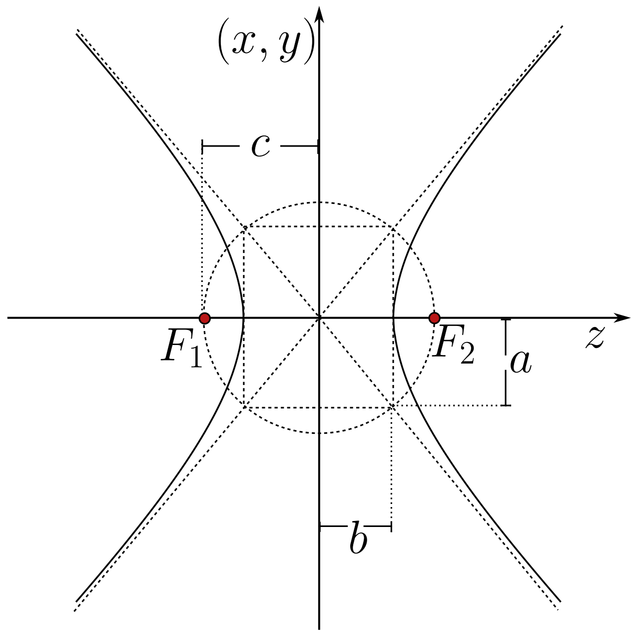


Figure 2.18: Geometrical representation of the two leaf hyperboloid with the parameters a, b and foci F_1 and F_2.

Again, to give a characterization of the conical surface we are going to define the segment $c = \overline{F_1 B}$ such that the eccentricity is given by

$$\epsilon = \frac{c}{b} = \frac{\sqrt{a^2+b^2}}{b} = \sqrt{1+\frac{a^2}{b^2}} = \frac{n_o}{n_i} > 1.$$

Thus the two Foci F_1 and F_2 are located at the distances

$$f_1 = b - c = b - \epsilon b = (1-\epsilon)b = \frac{n_i - n_o}{n_o + n_i} z_o, \tag{2.74}$$

$$f_2 = b + c = b + \epsilon b = (1+\epsilon)b = z_i. \tag{2.75}$$

In the figure 2.19 we can see the geometrical behavior of this first case, all the rays comes from the infinity and converges to a point which is located in the second foci F_2.

Now, for the second case, if $z_i < 0$ we define

$$b = -\frac{n_i}{n_i + n_o} z_o,$$

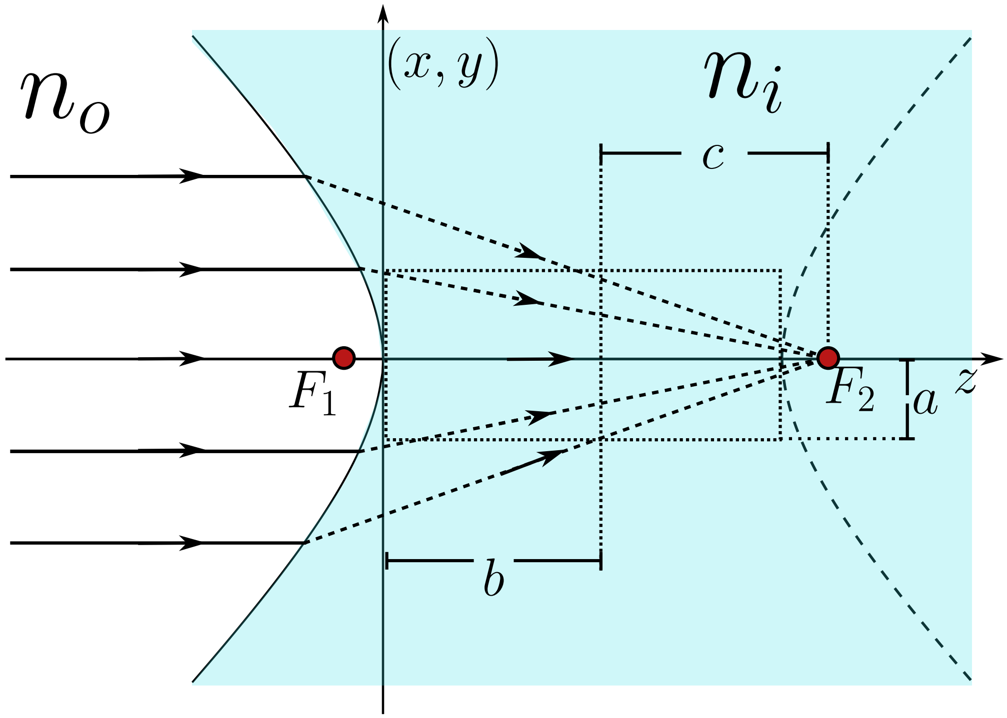


Figure 2.19: Geometrical representation of the first case of a two leaf hyperboloid.

in such way that $b > 0$ with this we write

$$\frac{1}{b^2}(z+b)^2 - \frac{1}{a^2}(x^2+y^2) = 1.$$

Again, we define the segment $c = \overline{F_1 B}$ such that the eccentricity is given by

$$\epsilon = \frac{c}{b} = \frac{\sqrt{a^2+b^2}}{b} = \sqrt{1+\frac{a^2}{b^2}} = \frac{n_o}{n_i} > 1.$$

As we can see in the figure 2.20 all the rays come from the infinity and interact with the hyperboloid diopter an the rays refracted in such geometrical way that they behave as they come from a object point source located in the foci of the first leaf at F_1.

2.6 Rigurously stigmatic surfaces by reflection

The reflective surfaces are the main components in the morden giant telescopes, for example, the James Webb telecopes is an array of many reflective surfaces which are electronically displaced such that the image formation gets the least aberration possible. Also, the reflective surfaces are better

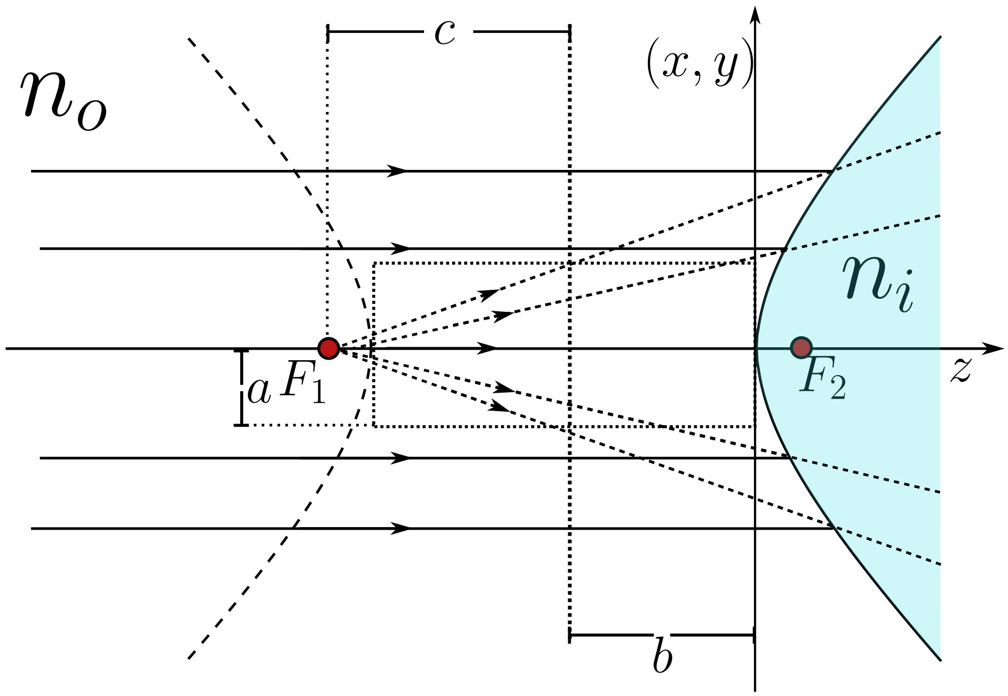


Figure 2.20: Geometrical representation of the first case of a two leaf hyperboloid.

than the refractive ones in certain cases, because the optical systems can be smaller and don't have chromatic aberrations.

Tradionally, the reflective surface are developed indepentdenly from the refractive surface. Under this formulations, the rigurously stigmatic surfaces by reflection can be obtained doing $n_i = -n_o$, where the Descartes ovals become a conic surface. To this case, the equation 2.42 becomes

$$\begin{aligned}&\left\{\left[(-n_o)^2 - n_o^2\right]\rho^2 - 2z\left[\left(n_o^2 z_o - (-n_o)^2 z_i\right]\right\}^2\\ &- 4(-n_o)n_o\left[(-n_o)z_i - n_o z_o\right]\left\{\left[n_o z_i - (-n_o)z_o\right]\rho^2 + 2z z_o z_i\left[(-n_o) - n_o\right]\right\} = 0,\end{aligned}$$

simplyfing we've got

$$\begin{aligned}&\left\{-2zn_o^2\left[z_o - z_i\right]\right\}^2 - 4n_o^3\left[z_i + z_o\right]\left\{n_o\left[z_i + z_o\right]\rho^2 - 4z z_o z_i n_o\right\} = 0,\\ &4z^2 n_o^4\left(z_o - z_i\right)^2 - 4n_o^4\left(z_o + z_i\right)^2\rho^2 + 16n_o^4 z z_o z_i\left(z_o + z_i\right) = 0,\end{aligned}$$

dividing all by $-16n_o^4 z_o z_i$ an having on count that $\rho^2 = x^2 + y^2 + z^2$ we've got

$$\begin{aligned}&\frac{(z_o + z_i)^2}{4z_o z_i}\left(x^2 + y^2\right) + \frac{(z_o - z_i)^2}{4z_o z_i}z^2 - \frac{(z_o + z_i)^2}{4z_o z_i}z^2 - z(z_o + z_i) = 0,\\ &\frac{(z_o + z_i)^2}{4z_o z_i}\left(x^2 + y^2\right) + z^2 - z(z_o + z_i) = 0,\end{aligned}$$

and completing the squares we've got

$$\left(z - \frac{z_o + z_i}{2}\right)^2 + \frac{(z_o + z_i)^2}{4 z_o z_i}\left(x^2 + y^2\right) = \left(\frac{z_o + z_i}{2}\right)^2, \tag{2.76}$$

which is precisely the equation of a conic of revolution around the z axis, exactly as it happened with the refractive surfaces, only that now the two indices have collapsed into a single reflective condition. The whole family of stigmatic mirrors is hidden inside 2.76, and the sign of the product $z_o z_i$ is what decides which conic we are looking at. When $z_o z_i > 0$ the coefficient multiplying $x^2 + y^2$ is positive and the surface closes onto itself as an ellipsoid, whereas when $z_o z_i < 0$ that coefficient turns negative and the surface opens as a hyperboloid. There is also a beautiful limit waiting in 2.76, because if we push the object to infinity, $z_o \to -\infty$, the conic degenerates into a paraboloid, which is the case that rules every telescope. We are going to walk through each of these situations, since each one corresponds to a real optical instrument.

2.6.1 The ellipsoidal mirror

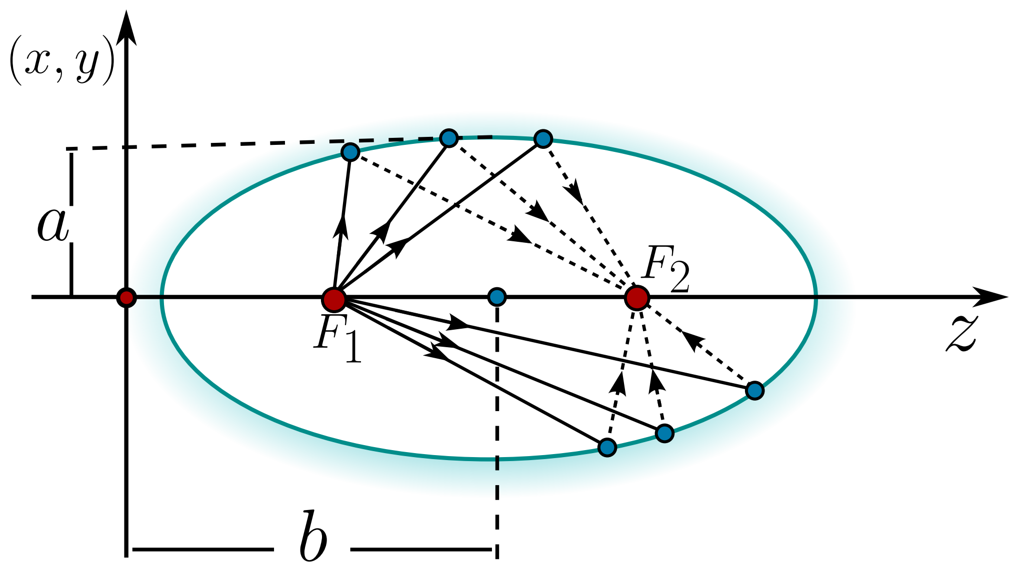


Figure 2.21: Ellipsoidal mirror. The object F_1 and the image F_2 coincide with the two foci of the ellipse, so every ray leaving one focus is reflected exactly onto the other.

Lets start with $z_o z_i > 0$, so that the object and the image sit on the same side of the vertex and the mirror is an ellipsoid. Dividing 2.76 by $\left(\frac{z_o+z_i}{2}\right)^2$ the equation takes the canonical shape

$$\frac{\left(z - \frac{z_o+z_i}{2}\right)^2}{\left(\frac{z_o+z_i}{2}\right)^2} + \frac{x^2 + y^2}{z_o z_i} = 1, \tag{2.77}$$

and if we baptise the two semi-axes as

$$b^2 = \frac{(z_o + z_i)^2}{4}, \qquad a^2 = z_o z_i, \tag{2.78}$$

we recognise a prolate ellipsoid centered at $z = b = \frac{z_o + z_i}{2}$, that is

$$\frac{1}{b^2}(z-b)^2 + \frac{1}{a^2}(x^2 + y^2) = 1. \tag{2.79}$$

That the ellipsoid comes out prolate, and not oblate, is not an accident, because

$$b^2 - a^2 = \frac{(z_o + z_i)^2}{4} - z_o z_i = \frac{(z_o - z_i)^2}{4} \geq 0,$$

so the semi-axis along the optical axis is always the largest one and the mirror is stretched along z. The distance from the center to each focus is then $c = \sqrt{b^2 - a^2} = \frac{|z_o - z_i|}{2}$, and the eccentricity becomes

$$\epsilon = \frac{c}{b} = \frac{\sqrt{b^2 - a^2}}{b} = \sqrt{1 - \frac{a^2}{b^2}} = \frac{|z_o - z_i|}{z_o + z_i} < 1, \tag{2.80}$$

which confirms once more that we are dealing with an ellipse. The really interesting thing is where the foci land, because measured from the vertex they sit at $z = b \pm c$, that is

$$b - c = \frac{z_o + z_i}{2} - \frac{|z_o - z_i|}{2}, \qquad b + c = \frac{z_o + z_i}{2} + \frac{|z_o - z_i|}{2}, \tag{2.81}$$

and no matter which of the two conjugate points is the larger one, these two numbers are nothing but z_o and z_i themselves. **The two foci of the ellipsoid are exactly the object point and the image point.**

This is the whole magic of the ellipsoidal mirror and the reason it is rigorously stigmatic. An ellipse has the geometric property that any ray leaving one focus is reflected straight into the other focus, because the sum of the two focal distances is constant along the whole curve, and a constant optical path is exactly the Fermat's principle at work. Placing a point source at one focus therefore produces a perfect point image at the other one, with no aberration at all. This is the principle behind the elliptical reflectors used to concentrate the light coming from one focus onto a detector sitting at the other, and it is the same property that makes the whispering galleries carry a voice from one focus of the room to the other.

When both z_o and z_i are negative the algebra is identical once we define $b = -\frac{z_o + z_i}{2} > 0$, so that the ellipsoid is now centered at $z = -b$ and its equation reads $\frac{1}{b^2}(z+b)^2 + \frac{1}{a^2}(x^2 + y^2) = 1$. The eccentricity is the same and the foci again fall on z_o and z_i, the only difference being that both conjugate points now live on the opposite side of the vertex.

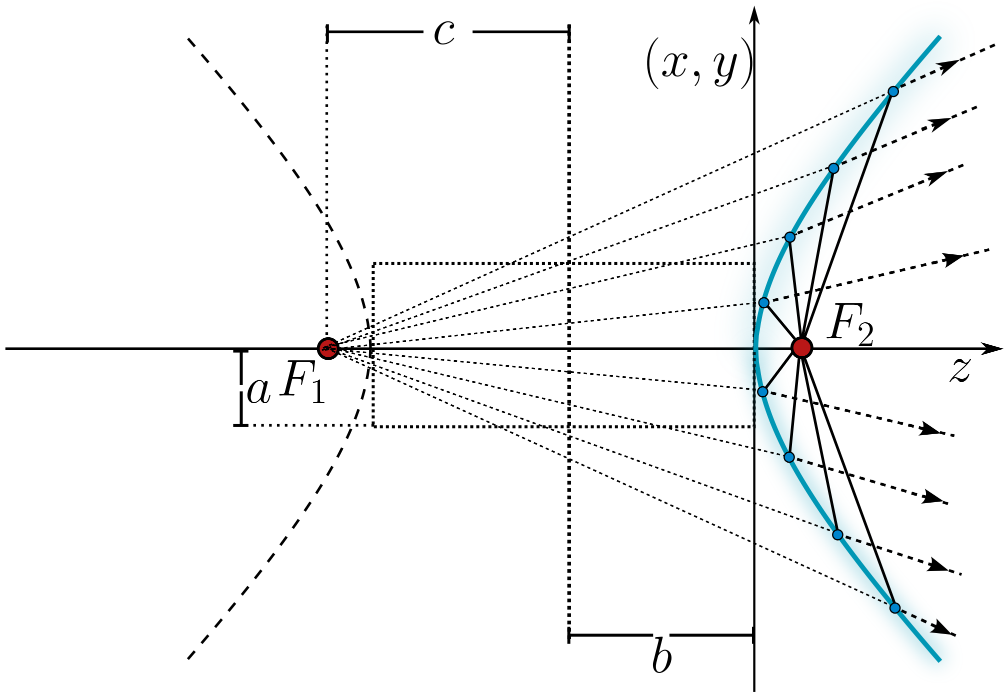


Figure 2.22: Hyperboloid mirror. The rays from the object F_1 diverge as they were emited by a virtual image locaed in F_2.

2.6.2 The hyperboloidal mirror

Now we let the product change sign, $z_o z_i < 0$, which happens when the object and the image sit on opposite sides of the mirror and one of them is virtual. The coefficient of $x^2 + y^2$ in 2.76 becomes negative, so writing $a^2 = -z_o z_i > 0$ while keeping $b^2 = \frac{(z_o+z_i)^2}{4}$ the canonical form is now

$$\frac{1}{b^2}(z-b)^2 - \frac{1}{a^2}(x^2+y^2) = 1, \tag{2.82}$$

a hyperboloid of two sheets around the optical axis. The focal distance is measured with a plus sign this time, $c = \sqrt{a^2+b^2}$, and a short computation gives again $c = \frac{|z_o - z_i|}{2}$, so the eccentricity is

$$\epsilon = \frac{c}{|b|} = \frac{\sqrt{a^2+b^2}}{|b|} = \sqrt{1+\frac{a^2}{b^2}} = \frac{|z_o - z_i|}{|z_o + z_i|} > 1, \tag{2.83}$$

which is greater than one exactly because $z_o z_i < 0$ forces $|z_o - z_i| > |z_o + z_i|$. And just as before, the foci $z = b \pm c$ collapse onto z_o and z_i, so once more the two conjugate points are the two foci of the conic.

The hyperboloidal mirror is therefore stigmatic between its two foci as well, but with a twist, because since the foci lie on opposite sheets one of the conjugate points is always virtual. This is precisely the behavior exploited by the convex secondary mirror of a Cassegrain telescope, where the light that is already converging toward the focus of the primary mirror is intercepted by a hyperboloidal secondary and redirected, without adding any aberration, to a more comfortable focus behind the primary. The stigmatic pair of the hyperboloid is what lets these compact folded telescopes keep a perfect image on the axis.

2.6.3 Special cases: the spherical and the plane mirror

Two very familiar mirrors are hiding inside 2.76 as degenerate cases. The first one appears when the object and the image coincide, $z_o = z_i = R$. The coefficient of $x^2 + y^2$ becomes $\frac{(2R)^2}{4R^2} = 1$ and the conic reduces to $z^2 - 2Rz + x^2 + y^2 = 0$, and completing the square by adding R^2 to both sides we get

$$(z - R)^2 + x^2 + y^2 = R^2, \tag{2.84}$$

which is simply a sphere of radius R centered at $z = R$. This tells us that a spherical mirror is rigorously stigmatic for one very special pair of points, both of them sitting together at the center of the sphere, since a source placed at the center sends every ray along a radius, the ray strikes the surface perpendicularly and comes straight back on itself, forming a perfect image right on top of the source. For any other position the sphere is only approximately stigmatic, and this is the origin of the spherical aberration that spoils cheap mirrors and that the conics above were designed to cure.

The second degenerate case is $z_o = -z_i$, for which the sum $z_o + z_i$ vanishes. Both the linear term and the whole right hand side of 2.76 die at once, and the equation collapses to $z^2 = 0$, that is, the plane $z = 0$. We recover the ordinary plane mirror, which is stigmatic for every point and always produces an image symmetric to the object at $z_i = -z_o$, which is nothing more than the everyday statement that a plane mirror puts your reflection as far behind the glass as you stand in front of it.

2.6.4 The parabolic mirror and objects at infinity

The most important mirror of all comes from letting the object recede to infinity, $z_o \to -\infty$, which is the situation of a telescope pointed at a star. Going back to 2.76 and expanding the square on the left, the terms $\frac{(z_o+z_i)^2}{4}$ appear on both sides and cancel,

$$z^2 - z(z_o + z_i) + \frac{(z_o + z_i)^2}{4} + \frac{(z_o + z_i)^2}{4z_o z_i}(x^2 + y^2) = \frac{(z_o + z_i)^2}{4},$$

leaving the expanded conic

$$z^2 - z(z_o + z_i) + \frac{(z_o + z_i)^2}{4z_o z_i}(x^2 + y^2) = 0.$$

Dividing everything by z_o and taking the limit $z_o \to -\infty$, the term z^2/z_o vanishes, the factor $\frac{z_o+z_i}{z_o} \to 1$ and $\frac{(z_o+z_i)^2}{4z_o^2 z_i} \to \frac{1}{4z_i}$, so we are left with the remarkably clean surface

$$x^2 + y^2 = 4z_i z, \tag{2.85}$$

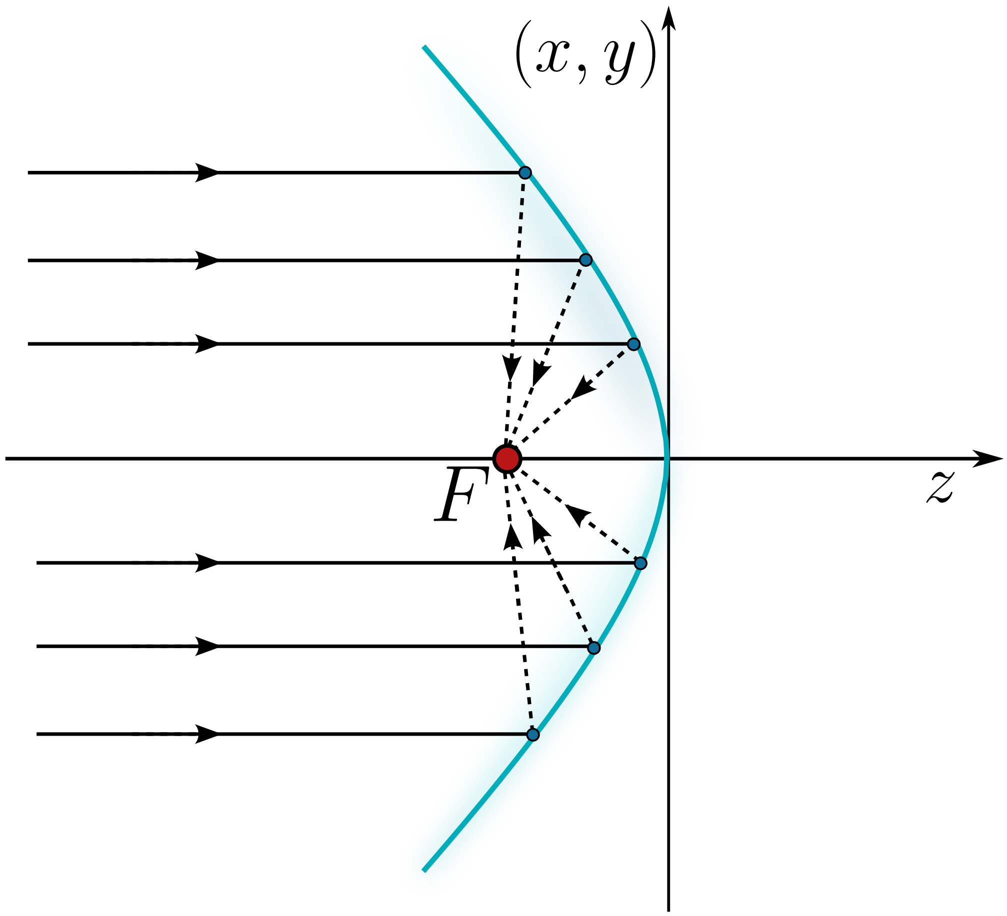


Figure 2.23: Parabolic mirror obtained as the limit $z_o \to -\infty$. The rays coming parallel to the axis from an object at infinity are all focused onto the single point $z = z_i$.

a paraboloid of revolution whose opening is fixed by the sign of z_i.

The paraboloid is the crown jewel of the reflective optics. A parabola has the defining property that every ray arriving parallel to its axis is reflected exactly through the focus, so a bundle of parallel rays coming from a star, which is an object at infinity, is gathered into a single perfect point at $z = z_i$, the focus of the mirror. There is no chromatic aberration, because the reflection does not depend on the wavelength, and there is no spherical aberration, because the surface is a true paraboloid instead of a sphere. This is why every serious reflecting telescope, from the first design of Newton to the segmented primary of the James Webb, uses a parabolic mirror as its light collecting surface, and it is also why the same shape reappears in the satellite dishes and the searchlights, where the ray tracing simply runs the other way and a source placed at the focus is turned into a parallel beam.

2.7 Explicit formulation of the Descartes ovoids

Everything we have built so far about rigorously stigmatic surfaces is contained in the quartic 2.42, and that equation has got one serious practical defect: it is *implicit*. It tells us whether a point (x, y, z) belongs to the ovoid, but it never hands us the surface itself. If we want to grind a lens, feed a profile to a numerically controlled machine, or trace a ray and ask where it meets the glass, what we need is a formula that returns the surface height directly, a function $z(\rho)$ rather than a condition $F(x, y, z) = 0$. The purpose of this section is to extract exactly that.

The route we shall follow is worth stating in advance, because it is a beautiful piece of strategy. Instead of attacking 2.42 head on, we notice that it is disguised: hidden inside that fourth degree expression there is a *quadratic* in z whose coefficients happen to depend on ρ. Once the disguise is removed, solving for z is nothing more than the quadratic formula every student already knows, and the entire family of Descartes ovoids collapses into a single closed expression.

2.7.1 The Descartes ovoid as a superconic

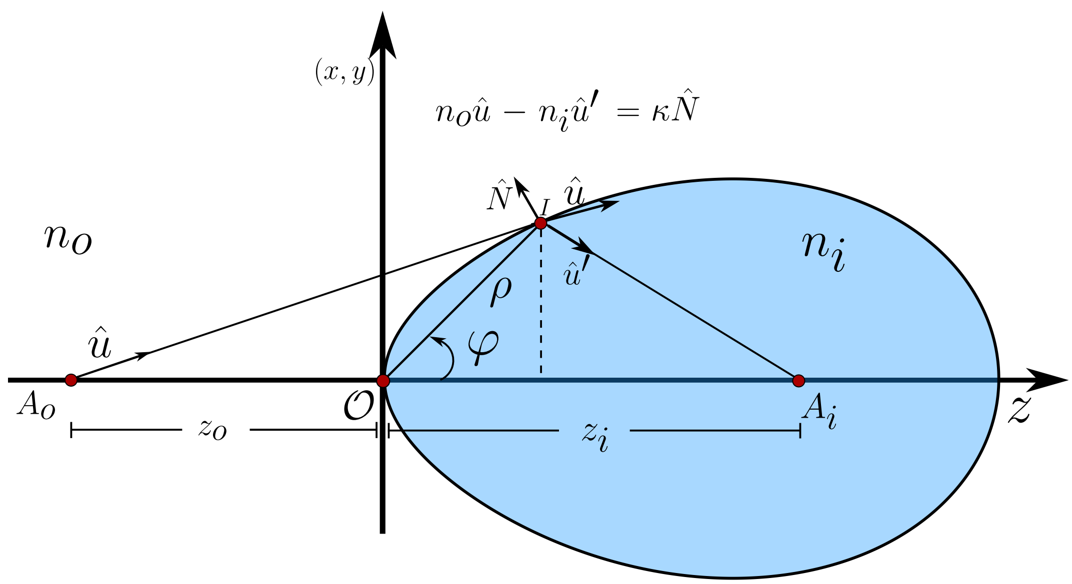


Figure 2.24: The rigorously stigmatic quartic surface described with the polar parameters ρ and φ on the meridional plane, with the object A_o at z_o and the image A_i at z_i. The inset recalls the vector Snell-Descartes law $n_o\hat{u} - n_i\hat{u}' = k\hat{N}$ obeyed at the point of incidence I.

We work on the meridional plane, the plane that contains the optical axis, and we describe a point I of the surface by the polar pair (ρ, φ) measured from the vertex $\mathcal{O}$, exactly as in figure 2.24. The radius ρ is the one already introduced in 2.40, that is $\rho^2 = x^2 + y^2 + z^2$, so that the transverse height of the point is $r^2 = \rho^2 - z^2$. Working with ρ instead of r is the single trick that keeps every expression below manageable, and the reason is easy to see: the optical path lengths that Fermat's principle compares are distances measured from a point, and ρ is precisely such a distance.

The family of surfaces we are about to meet has got a name of its own.

Definition 8 *(Superconic surface): A surface of revolution about the z axis is called a **superconic** if, on the meridional plane, it satisfies*

$$c_0 G z^2 - 2\left(1 + b_1\rho^2 + b_2\rho^4 + \cdots\right) z + \left(c_0 + c_1\rho^2 + c_2\rho^4 + \cdots\right)\rho^2 = 0, \tag{2.86}$$

where c_0 is the curvature of the surface at its vertex, called the paraxial curvature, and G is a dimensionless number called the conic constant. When every correction coefficient vanishes, $b_j = c_j = 0$ for $j \geq 1$, the surface reduces to the standard conic of revolution

$$c_0 G z^2 - 2z + c_0\rho^2 = 0. \tag{2.87}$$

The definition looks like an arbitrary generalisation, but it is not. Equation 2.87 is exactly the conic we have been meeting all through this chapter, the ellipsoid, the hyperboloid and the paraboloid of the previous sections, written in a single line. What 2.86 does is to let the two coefficients that a conic keeps rigidly constant become slowly varying functions of ρ, and the claim we are going to prove is that the Descartes ovoid is precisely such a surface, with only the first correction in each series switched on.

Let us take the ovoid 2.42 and divide it, term by term, by the quantity

$$\mathcal{D} \equiv 4n_i n_o z_i z_o \left(n_i - n_o\right)\left(n_i z_i - n_o z_o\right). \tag{2.88}$$

This looks unmotivated, and in the literature it usually appears without comment, so it is worth saying plainly where it comes from: $\mathcal{D}$ is simply the product of every factor that the four coefficients of the quartic have in common, so dividing by it is the fastest way to normalise the coefficient of z to exactly -2, which is the form 2.86 demands. Let us carry the division out coefficient by coefficient. Expanding the square in 2.42 we get

$$\begin{aligned}\left(n_i^2 - n_o^2\right)^2\rho^4 - 4z\left(n_i^2 - n_o^2\right)\left(n_i^2 z_i - n_o^2 z_o\right)\rho^2 + 4z^2\left(n_i^2 z_i - n_o^2 z_o\right)^2 \\ - 4n_i n_o\left(n_i z_i - n_o z_o\right)\left(n_o z_i - n_i z_o\right)\rho^2 - 8n_i n_o\left(n_i z_i - n_o z_o\right) z_i z_o\left(n_i - n_o\right) z = 0.\end{aligned}$$

Now we divide by $\mathcal{D}$ and look at one power of z at a time. The coefficient of z^2 becomes

$$\frac{4\left(n_i^2 z_i - n_o^2 z_o\right)^2}{\mathcal{D}} = \frac{\left(n_i^2 z_i - n_o^2 z_o\right)^2}{n_i n_o z_i z_o\left(n_i - n_o\right)\left(n_i z_i - n_o z_o\right)},$$

the part of the coefficient of z that carries no ρ becomes

$$\frac{-8n_i n_o\left(n_i z_i - n_o z_o\right) z_i z_o\left(n_i - n_o\right)}{4n_i n_o z_i z_o\left(n_i - n_o\right)\left(n_i z_i - n_o z_o\right)} = -2,$$

which is exactly the normalisation we were after, and the part of the coefficient of z that carries ρ^2 becomes

$$\frac{-4\left(n_i^2-n_o^2\right)\left(n_i^2 z_i-n_o^2 z_o\right)}{\mathcal{D}}=-\frac{\left(n_i+n_o\right)\left(n_i^2 z_i-n_o^2 z_o\right)}{n_i n_o z_i z_o\left(n_i z_i-n_o z_o\right)},$$

where the common factor n_i-n_o has cancelled between numerator and denominator. Finally the two purely radial terms give

$$\frac{-4 n_i n_o\left(n_i z_i-n_o z_o\right)\left(n_o z_i-n_i z_o\right)}{\mathcal{D}}=\frac{n_i z_o-n_o z_i}{z_i z_o\left(n_i-n_o\right)}, \qquad \frac{\left(n_i^2-n_o^2\right)^2}{\mathcal{D}}=\frac{\left(n_i-n_o\right)\left(n_i+n_o\right)^2}{4 n_i n_o z_i z_o\left(n_i z_i-n_o z_o\right)}.$$

Collecting the four results, the Descartes ovoid has become

$$f(z, \rho)=O G z^2-2\left(1+S \rho^2\right) z+\left(O+T \rho^2\right) \rho^2=0, \tag{2.89}$$

which is the superconic 2.86 with $c_0=O$ and with only the first coefficient of each series different from zero, $b_1=S$ and $c_1=T$. The four numbers that appeared along the way deserve a name.

Definition 9 *(Form parameters of a Descartes ovoid): The shape of the rigorously stigmatic surface that images the axial point z_o onto the axial point z_i across the interface between the media n_o and n_i is fixed by the four real numbers*

$$G=\frac{\left(n_i^2 z_i-n_o^2 z_o\right)^2}{n_i n_o\left(n_i z_i-n_o z_o\right)\left(n_i z_o-n_o z_i\right)}, \tag{2.90}$$

$$O=\frac{n_i z_o-n_o z_i}{z_i z_o\left(n_i-n_o\right)}, \tag{2.91}$$

$$T=\frac{\left(n_i-n_o\right)\left(n_i+n_o\right)^2}{4 n_i n_o z_i z_o\left(n_i z_i-n_o z_o\right)}, \tag{2.92}$$

$$S=\frac{\left(n_i+n_o\right)\left(n_i^2 z_i-n_o^2 z_o\right)}{2 n_i n_o z_i z_o\left(n_i z_i-n_o z_o\right)}, \tag{2.93}$$

called the form parameters and written together as $GOTS$. They are not independent: they always satisfy the constraint

$$S^2=G O T. \tag{2.94}$$

The constraint 2.94 is not an accident of notation, it is the algebraic fingerprint of stigmatism, and it is what separates a genuine Descartes ovoid from an arbitrary superconic. Let us verify it, since the calculation is short and the result will do a great deal of work for us in a moment. Squaring 2.93,

$$S^2=\frac{\left(n_i+n_o\right)^2\left(n_i^2 z_i-n_o^2 z_o\right)^2}{4 n_i^2 n_o^2 z_i^2 z_o^2\left(n_i z_i-n_o z_o\right)^2},$$

while multiplying 2.90, 2.91 and 2.92 together,

$$G O T=\frac{\left(n_i^2 z_i-n_o^2 z_o\right)^2}{n_i n_o\left(n_i z_i-n_o z_o\right)\left(n_i z_o-n_o z_i\right)} \cdot \frac{\left(n_i z_o-n_o z_i\right)}{z_i z_o\left(n_i-n_o\right)} \cdot \frac{\left(n_i-n_o\right)\left(n_i+n_o\right)^2}{4 n_i n_o z_i z_o\left(n_i z_i-n_o z_o\right)},$$

and the two coloured factors cancel in pairs, leaving

$$GOT = \frac{(n_i + n_o)^2 \left(n_i^2 z_i - n_o^2 z_o\right)^2}{4 n_i^2 n_o^2 z_i^2 z_o^2 \left(n_i z_i - n_o z_o\right)^2} = S^2,$$

which is 2.94.

Each of the four parameters carries a clear geometrical job. O is the paraxial curvature, that is, the curvature of the ovoid right at the vertex, and it is the only one of the four that survives in the paraxial world of the next section. G plays the part of the conic constant, and it is the parameter that decides which of the classical conics the ovoid most resembles: we shall sort the whole zoo by the value of G in the next section. The remaining two, S and T, are the genuinely new ingredients, the ones that make an ovoid an ovoid rather than a conic, and it is instructive to see when they die. Both 2.92 and 2.93 carry the product $z_i z_o$ in the denominator, so letting either the object or the image recede to infinity, $z_o \to \infty$ or $z_i \to \infty$, drives $S = T = 0$ and reduces 2.89 to the plain conic 2.87. This is a pleasing consistency check, because it is precisely the statement we proved geometrically earlier in this chapter: the conics of revolution are rigorously stigmatic exactly when one of their two conjugate points sits at infinity, and the ovoids are what one needs when both points are at a finite distance.

2.7.2 The explicit sag of the ovoid

With 2.89 in hand the hard part is over, because that expression is nothing more than a quadratic equation in z, with the coefficients

$$a = OG, \qquad b = -2\left(1 + S\rho^2\right), \qquad c = \left(O + T\rho^2\right)\rho^2.$$

The quadratic formula gives at once

$$\begin{aligned} z &= \frac{2\left(1 + S\rho^2\right) \pm \sqrt{4\left(1 + S\rho^2\right)^2 - 4\,OG\left(O + T\rho^2\right)\rho^2}}{2OG} \\ &= \frac{\left(1 + S\rho^2\right) \pm \sqrt{\left(1 + S\rho^2\right)^2 - OG\left(O + T\rho^2\right)\rho^2}}{OG}, \end{aligned}$$

and it pays to expand the discriminant carefully before going further,

$$\begin{aligned} \left(1 + S\rho^2\right)^2 - OG\left(O + T\rho^2\right)\rho^2 &= 1 + 2S\rho^2 + S^2\rho^4 - O^2 G\rho^2 - OGT\rho^4 \\ &= 1 + \left(2S - O^2 G\right)\rho^2 + \left(S^2 - GOT\right)\rho^4, \end{aligned}$$

so that the exact sag of the surface reads

$$z = \frac{\left(1 + S\rho^2\right) \pm \sqrt{1 + \left(2S - O^2 G\right)\rho^2 + \left(S^2 - GOT\right)\rho^4}}{OG}. \tag{2.95}$$

Two things now happen, and both are worth pausing on. The first is the choice of sign. A quadratic offers two roots, and here they correspond to the two sheets of the quartic, the near one and the far one; but only one of them is the surface we actually built, because our ovoid was constructed to pass through the vertex $\mathcal{O}$, so it must satisfy $z \to 0$ as $\rho \to 0$. Setting $\rho = 0$ in 2.95 gives $z = (1 \pm 1)/(OG)$, which is $2/(OG)$ for the plus sign and 0 for the minus sign. The physically relevant branch is therefore the one carrying the **minus** sign, and the plus sign describes the outer sheet of the quartic, the one drawn with dashed lines in figure 2.25.

The second is the constraint 2.94 finally earning its keep. The coefficient of ρ^4 under the radical is exactly $S^2 - GOT$, which vanishes identically for a Descartes ovoid. The quartic under the square root collapses to a quadratic, and 2.95 simplifies to

$$z = \frac{1}{OG}\left(1 + S\rho^2 - \sqrt{1 + (2S - O^2G)\,\rho^2}\right). \tag{2.96}$$

This is already explicit, but it is numerically clumsy, because for small ρ it is the difference of two nearly equal numbers, each close to one, and it hides the fact that z must vanish quadratically at the vertex. The cure is the standard one: rationalise, multiplying and dividing by the conjugate $1 + S\rho^2 + \sqrt{1 + (2S - O^2G)\rho^2}$. The numerator becomes

$$\begin{aligned}\left(1 + S\rho^2\right)^2 - \left[1 + \left(2S - O^2G\right)\rho^2\right] &= 1 + 2S\rho^2 + S^2\rho^4 - 1 - 2S\rho^2 + O^2G\rho^2 \\ &= S^2\rho^4 + O^2G\rho^2 = \rho^2\left(S^2\rho^2 + O^2G\right),\end{aligned}$$

and here the constraint acts a second time: replacing $S^2 = GOT$,

$$\rho^2\left(GOT\rho^2 + O^2G\right) = \rho^2\,OG\left(T\rho^2 + O\right),$$

so the factor OG cancels against the one already sitting in front of 2.96. We are left with the result we were hunting.

Definition 10 *(Explicit form of the Descartes ovoid): The rigorously stigmatic surface conjugating the axial points z_o and z_i across the interface $n_o\,|\,n_i$ is given, in closed form, by the sag*

$$z(\rho) = \frac{\left(O + T\rho^2\right)\rho^2}{1 + S\rho^2 + \sqrt{1 + (2S - O^2G)\,\rho^2}}, \tag{2.97}$$

together with the transverse coordinate

$$r(\rho) = \pm\sqrt{\rho^2 - z^2(\rho)}, \tag{2.98}$$

where G, O, T, S are the form parameters 2.90 to 2.93.

Equation 2.97 is the explicit formulation we set out to find, and it is worth appreciating how much it says. A single fraction, built from four numbers that are themselves simple combinations of n_o, n_i, z_o, z_i, contains every rigorously stigmatic refracting surface there is. Reading it near the vertex is instructive: for small ρ the square root tends to one and the denominator tends to two, so

$$z(\rho) \approx \frac{O}{2}\rho^2,$$

which is the parabola of curvature O, confirming once more that O is the paraxial curvature and showing that every ovoid, no matter how exotic, looks like the same simple paraboloid if we only stay close enough to the axis. This is the seed of the whole paraxial theory of the last section of this chapter.

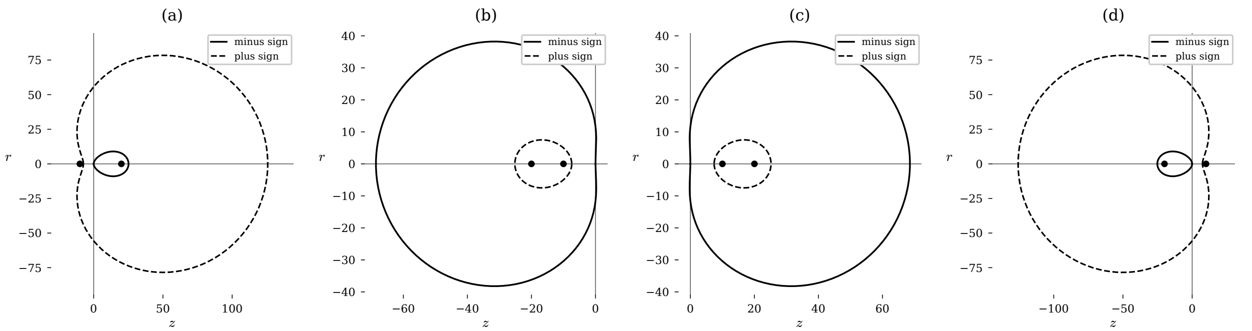


Figure 2.25: Meridional sections of Descartes ovoids for fixed indices $n_o = 1$ and $n_i = 1.7$ and varying conjugate positions. Solid curves are the physical branch, the minus sign of 2.95, which passes through the vertex; dashed curves are the plus sign, the outer sheet of the quartic. The four panels correspond to (z_o, z_i) equal to $(-10, 20)$, $(-10, -20)$, $(10, 20)$ and $(10, -20)$.

Figure 2.25 makes the two branches visible. The solid curve is the ovoid proper, and the dashed curve is its companion sheet; together they close the quartic. Changing the signs of z_o and z_i switches between real and virtual objects and images, and the surface responds by opening to the right or to the left, which is the geometrical way of saying that the same algebra serves a converging and a diverging element without any change of formula. Table 2.26 collects the degenerate cases, that is, the values that the form parameters take when the ovoid reduces to one of the classical conics we studied earlier in this chapter, and it can be read as a dictionary between the two languages.

2.7.3 The Schwarzschild equation as a degenerate case

The table above shows that whenever a conjugate point goes to infinity we get $T = S = 0$, and it is worth following that case explicitly, because the surface it produces is the one that dominates real optical design. Putting $T = S = 0$ in 2.97,

$$z(\rho) = \frac{O\rho^2}{1 + \sqrt{1 - O^2 G\rho^2}}, \tag{2.99}$$

an expression written, as everything in this section has been, in terms of the polar radius ρ. Optical designers, however, never use ρ: they use the transverse height r, the distance from the axis, because

Surface	n_i	z_o	z_i	G	O	T	S
Ellipsoidal refractive	$n_i > n_o$	∞	z_i	$-\left(\frac{n_o}{n_i}\right)^2$	$\frac{n_i}{z_i(n_i-n_o)}$	0	0
Hyperboloidal refractive	$n_i < n_o$	∞	z_i	$-\left(\frac{n_o}{n_i}\right)^2$	$\frac{n_i}{z_i(n_i-n_o)}$	0	0
Hyperboloidal refractive	$n_i > n_o$	z_o	∞	$-\left(\frac{n_i}{n_o}\right)^2$	$-\frac{n_o}{z_o(n_i-n_o)}$	0	0
Ellipsoidal refractive	$n_i < n_o$	z_o	∞	$-\left(\frac{n_i}{n_o}\right)^2$	$-\frac{n_o}{z_o(n_i-n_o)}$	0	0
Spherical refractive	n_i	z_o	z_o	$\frac{(n_i+n_o)^2}{n_i n_o}$	$\frac{1}{z_o}$	$\frac{(n_i+n_o)^2}{4n_i n_o z_o^3}$	$\frac{(n_o+n_i)^2}{2n_i n_o z_o^2}$
Ellipsoidal reflective	$-n_o$	$z_o > 0$	$z_i > 0$	$-\left(\frac{z_i-z_o}{z_i+z_o}\right)^2$	$\frac{z_i+z_o}{2z_i z_o}$	0	0
Ellipsoidal reflective	$-n_o$	$z_o < 0$	$z_i < 0$	$-\left(\frac{z_i-z_o}{z_i+z_o}\right)^2$	$\frac{z_i+z_o}{2z_i z_o}$	0	0
Hyperboloidal reflective	$-n_o$	$z_o > 0$	$z_i < 0$	$-\left(\frac{z_i-z_o}{z_i+z_o}\right)^2$	$\frac{z_i+z_o}{2z_i z_o}$	0	0
Hyperboloidal reflective	$-n_o$	$z_o < 0$	$z_i > 0$	$-\left(\frac{z_i-z_o}{z_i+z_o}\right)^2$	$\frac{z_i+z_o}{2z_i z_o}$	0	0
Paraboloidal reflective	$-n_o$	∞	z_i	-1	$\frac{1}{2z_i}$	0	0
Paraboloidal reflective	$-n_o$	z_o	∞	-1	$\frac{1}{2z_o}$	0	0

Figure 2.26: Degenerate cases of refractive and reflective surfaces according to the indices, the object position and the image position. For each combination the table lists the values taken by the form parameters, showing how the classical conics of revolution sit inside the general ovoid as the particular cases $T = S = 0$.

that is what an aperture stop actually limits and what a lens grinder actually measures. So we must change variables using $\rho^2 = z^2 + r^2$. The source we are following states the outcome in one line, so let us do the work in full, since the manipulation is a nice example of how a disguised equation can be untangled.

Start by clearing the denominator in 2.99 and inserting $\rho^2 = z^2 + r^2$,

$$z\left(1+\sqrt{1-O^2G\left(z^2+r^2\right)}\right) = O\left(z^2+r^2\right),$$

then isolate the radical and square both sides,

$$\begin{aligned} z\sqrt{1-O^2G\left(z^2+r^2\right)} &= O\left(z^2+r^2\right) - z, \\ z^2 - O^2Gz^2\left(z^2+r^2\right) &= O^2\left(z^2+r^2\right)^2 - 2Oz\left(z^2+r^2\right) + z^2. \end{aligned}$$

The lone z^2 cancels on both sides, and every surviving term carries a common factor $z^2 + r^2$, which is nonzero away from the vertex, so we may divide by it,

$$-O^2Gz^2 = O\left(z^2+r^2\right) - 2z.$$

Dividing once more by O and gathering everything on one side we obtain a genuine quadratic in z,

$$O\left(G+1\right)z^2-2z+Or^2=0, \tag{2.100}$$

whose roots are

$$z=\frac{1\pm\sqrt{1-O^2\left(G+1\right)r^2}}{O\left(G+1\right)}.$$

Once again only the minus sign passes through the vertex, and once again the expression is best rationalised, the numerator becoming $1-\left[1-O^2(G+1)r^2\right]=O^2(G+1)r^2$, so that

$$z(r)=\frac{Or^2}{1+\sqrt{1-\left(G+1\right)O^2r^2}}. \tag{2.101}$$

Equation 2.101 is the celebrated **Schwarzschild equation** for conic surfaces of revolution, the single most used formula in the whole of optical design, and it is written in every lens catalogue in the world. In the standard notation the paraxial curvature is called c_0 and the conic constant is called K, so that comparing with 2.101 we can read off the dictionary $O=c_0$ and $G=K$. What we have shown is therefore a genuine generalisation: **the explicit ovoid 2.97 is the Schwarzschild equation extended to the case where both conjugate points lie at a finite distance**, with G generalising the conic constant, O remaining the vertex curvature, and the extra pair S, T measuring exactly how far the surface has departed from being a plain conic.

2.8 Special Descartes ovoids

The conic constant G defined in 2.90 sorts the ovoids into families in the same way that the eccentricity sorts the conics, and since the numerator of 2.90 is a perfect square, the sign of G is decided entirely by the sign of the product in its denominator. Let us go through the five cases, keeping in mind that in every one of them the sag is still given by the same formula 2.97, only with the corresponding restriction on the parameters.

2.8.1 Spherovoids, $G=0$

Since the numerator of 2.90 is squared, G can vanish only if that numerator vanishes,

$$\left(n_i^2z_i-n_o^2z_o\right)^2=0 \qquad\Longleftrightarrow\qquad n_i^2z_i=n_o^2z_o. \tag{2.102}$$

The same factor $n_i^2z_i-n_o^2z_o$ sits in the numerator of S in 2.93, so the condition kills two parameters at once, $G=S=0$, and the surface is described by O and T alone. The sag 2.97 then loses its square root entirely, since $\sqrt{1+0}=1$ and the denominator is just 2,

$$z(\rho)=\frac{1}{2}\left(O+T\rho^2\right)\rho^2. \tag{2.103}$$

This is the simplest ovoid there is, a pure quartic with no radical at all. If in addition $T = 0$ we are left with $z = \frac{1}{2}O\rho^2$, and recalling $\rho^2 = z^2 + r^2$ this reads $2z = O(z^2 + r^2)$, that is $z^2 + r^2 - 2z/O = 0$, which upon completing the square becomes $(z - 1/O)^2 + r^2 = 1/O^2$: a sphere of radius $1/O$ centred on the axis at $z = 1/O$. So T is the parameter that measures the departure of a spherovoid from a true sphere, and this is the precise sense in which T labels the family of ovoids.

2.8.2 Parovoids, $G = -1$

Setting $G = -1$ in 2.90 gives the condition

$$\left(n_i^2 z_i - n_o^2 z_o\right)^2 = -n_i n_o \left(n_i z_i - n_o z_o\right)\left(n_i z_o - n_o z_i\right), \tag{2.104}$$

and the sag follows from 2.97 by replacing $-O^2G$ with $+O^2$,

$$z(\rho) = \frac{\left(O + T\rho^2\right)\rho^2}{1 + S\rho^2 + \sqrt{1 + (2S + O^2)\,\rho^2}}. \tag{2.105}$$

The value $G = -1$ is special for a reason that 2.100 makes transparent: it is exactly the value at which the quadratic term of the Schwarzschild quadratic vanishes, so that the equation degenerates from a conic with two vertices into a parabola with one. The parovoid is therefore the ovoid analogue of the paraboloid, which is why it carries that name.

2.8.3 Hyperovoids, $G < -1$

The condition is the inequality version of 2.104,

$$\left(n_i^2 z_i - n_o^2 z_o\right)^2 < -n_i n_o \left(n_i z_i - n_o z_o\right)\left(n_i z_o - n_o z_i\right), \tag{2.106}$$

with the sag still given by the general expression

$$z(\rho) = \frac{\left(O + T\rho^2\right)\rho^2}{1 + S\rho^2 + \sqrt{1 + (2S - O^2G)\,\rho^2}}. \tag{2.107}$$

These are the ovoids that generalise the hyperboloid of revolution, the surface we found earlier in this chapter when we asked for rigorous stigmatism with one point at infinity and the light going from a denser to a rarer medium.

2.8.4 Prolate ellipsovoids, $G > 0$

Because the numerator of 2.90 is a square, G is positive precisely when its denominator is positive, and since $n_i n_o > 0$ always, the condition reduces to

$$\left(n_i z_i - n_o z_o\right)\left(n_i z_o - n_o z_i\right) > 0, \tag{2.108}$$

with the sag again given by 2.97. These surfaces generalise the prolate ellipsoid, the egg shaped one, elongated along the optical axis.

2.8.5 Oblate ellipsovoids, $0 > G > -1$

The remaining window is the one the source we are following leaves empty, so let us fill it in. Two conditions must hold simultaneously. First $G < 0$, which as above requires the denominator of 2.90 to be negative,

$$(n_i z_i - n_o z_o)(n_i z_o - n_o z_i) < 0. \tag{2.109}$$

Second $G > -1$, which we obtain by multiplying that inequality through by the denominator; since the denominator is now negative the inequality reverses, and we get

$$\left(n_i^2 z_i - n_o^2 z_o\right)^2 < -n_i n_o (n_i z_i - n_o z_o)(n_i z_o - n_o z_i). \tag{2.110}$$

Comparing 2.110 with the hyperovoid condition 2.106 shows something worth noticing: the two inequalities are formally identical, and what separates an oblate ellipsovoid from a hyperovoid is not that inequality at all but the accompanying sign condition 2.109, which fixes on which side of -1 the ratio falls. The sag is once more 2.97, and the surfaces generalise the oblate ellipsoid, the one flattened along the optical axis, the shape of a lentil rather than of an egg.

2.9 Descartes ovoids and aspheric surfaces

The rigorously stigmatic surfaces we have built are perfect, but they are perfect for exactly one pair of points. Move the object anywhere else and the perfection evaporates, which is a severe limitation for any instrument that must look at an extended scene rather than at a single star. The practical way around this is to give up on exactness and to ask instead for a surface that is very nearly stigmatic over a useful range, and the standard tool for the job is the aspheric surface.

The starting point is the Schwarzschild equation 2.101, which we now write in the conventional notation of the optical designer,

$$z = \frac{c_0' r^2}{1 + \sqrt{1 - (1 + K')\, c_0'^2 r^2}}, \tag{2.111}$$

where c_0' is the vertex curvature and $K' = -e^2$ is the Schwarzschild constant, written in terms of the eccentricity e of the conic. The idea is then simply to deform this conic by adding higher order terms in r, giving the surface enough free parameters to bend the emerging wavefront back into shape.

Definition 11 *(Aspheric surface): An aspheric surface of revolution is a conic of revolution corrected by a series of even powers of the transverse height,*

$$z = \frac{c_0' r^2}{1 + \sqrt{1 - (1 + K')\, c_0'^2 r^2}} + \sum_{j \geq 2} A_{2j} r^{2j}, \tag{2.112}$$

where the numbers A_{2j} are called the asphericity coefficients. The conics of revolution are the degenerate case $A_{2j} = 0$ for every j.

Only even powers appear, and the reason is pure symmetry rather than convenience: the surface is a surface of revolution about the axis, so it cannot distinguish r from $-r$, and any odd power would break that symmetry. It is worth being clear about what 2.112 does and does not achieve. An aspheric surface is an *open* surface, built as a power series rather than as the closed algebraic object 2.97, and therefore it is not equivalent to a Descartes ovoid and does not possess rigorous stigmatism at all, except in the trivial case where every A_{2j} vanishes and we are back to a conic. What it does possess is flexibility. Each coefficient A_{2j} can be tuned to cancel a particular order of aberration in the emerging wavefront, so that in place of one perfect pair of points one obtains a whole region of the field that is imaged almost perfectly. The trade is a familiar one in physics: exactness at a single point is exchanged for uniform, controlled imperfection over a useful domain, and it is this trade that makes modern camera lenses and telescope correctors possible.

2.10 Approximate stigmatism in the spherical diopter

Rigorous stigmatism and the ovoids that realise it are beautiful, but we have just admitted their limitation: they work for one pair of conjugate points and no others. Real instruments must cope with objects wherever they happen to be, so we now change our question. Instead of asking which surface is exactly stigmatic, we fix the surface to be the easiest one to manufacture, the sphere, and ask how well it images and how its imaging fails. This is the study of **approximate stigmatism**, and it is where practical optics begins.

The choice of the sphere is not arbitrary. A sphere is the only surface that a grinding process produces spontaneously: rub two glass blanks together with abrasive between them, with random relative motion, and the only shape invariant under all those rotations is a sphere. Historically this is why spherical surfaces dominated optics for three centuries, and it is why the distortions peculiar to the sphere, above all spherical aberration, became the central problem of instrument making.

Consider then the arrangement of figure 2.27: a spherical cap $\mathcal{D}$ of radius R with its centre of curvature at C, its vertex V at the origin, an axial object point O at z_o and an axial candidate image point O' at z_i. A ray leaves O, strikes the surface at the point I located by the angle φ measured at C, and continues to O'. We need the two path lengths, and both follow from the law of cosines applied to the triangles OCI and $O'CI$. Writing the position of I as $C + R(-\cos\varphi, \sin\varphi)$ and the position of O as $C + (z_o - R, 0)$, the squared distance is

$$l_o^2 = (R\cos\varphi + z_o - R)^2 + R^2\sin^2\varphi = R^2 + (z_o - R)^2 + 2R\,(z_o - R)\cos\varphi,$$

where the cross term survives and the two squared sines and cosines have combined into R^2. The same computation on the image side gives the companion expression, so that

$$l_o = \left[R^2 + (z_o - R)^2 + 2R\,(z_o - R)\cos\varphi\right]^{1/2}, \tag{2.113}$$

$$l_i = \left[R^2 + (z_i - R)^2 + 2R\,(z_i - R)\cos\varphi\right]^{1/2}. \tag{2.114}$$

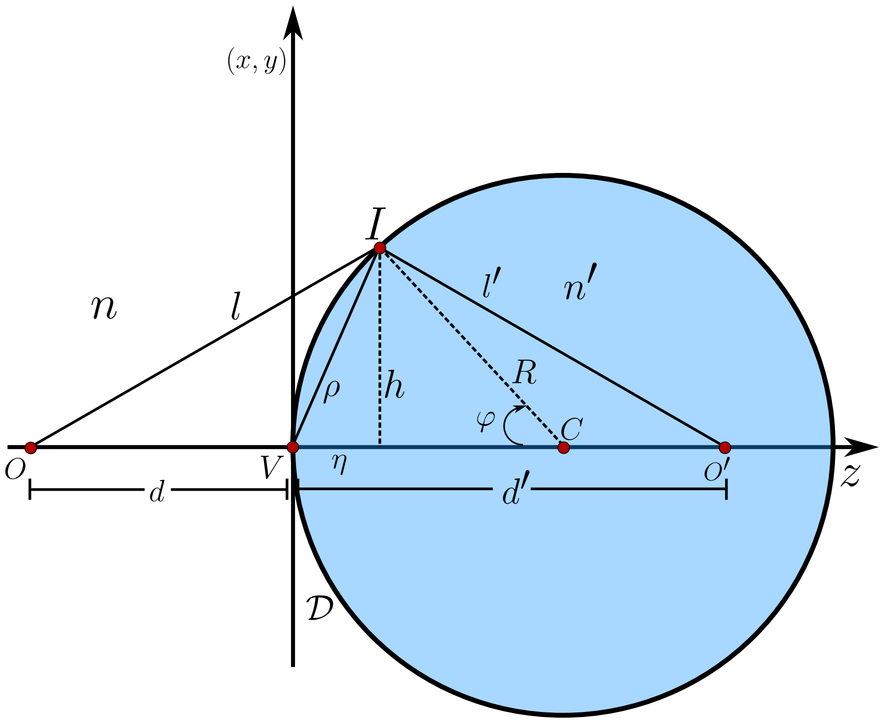


Figure 2.27: Image formation by a spherical diopter $\mathcal{D}$ of radius R and centre C, separating the media n and n', which in our notation are n_o and n_i. The ray leaves the axial object O at z_o, meets the surface at I, and reaches O' at z_i; the paths are $l = l_o$ and $l' = l_i$, the sagitta is η, the transverse height is h, the chord from the vertex is ρ and φ is the angle subtended at the centre of curvature.

The optical path from O to O' through I is $L = n_o l_o + n_i l_i$, and Fermat's principle demands that it be stationary with respect to the only freedom the ray has got, namely the choice of the point I, that is, the angle φ. Imposing $\mathrm{d}L/\mathrm{d}\varphi = 0$ and differentiating 2.113 and 2.114 implicitly, $2l_o\,\mathrm{d}l_o/\mathrm{d}\varphi = -2R(z_o - R)\sin\varphi$, we obtain

$$n_o \frac{R\,(z_o - R)}{l_o}\sin\varphi + n_i \frac{R\,(z_i - R)}{l_i}\sin\varphi = 0. \tag{2.115}$$

For any ray that is not travelling along the axis we have $\sin\varphi \neq 0$, so the common factor $R\sin\varphi$ may be cancelled, leaving $n_o(z_o - R)/l_o + n_i(z_i - R)/l_i = 0$. Splitting each numerator,

$$\frac{n_o z_o}{l_o} - \frac{n_o R}{l_o} + \frac{n_i z_i}{l_i} - \frac{n_i R}{l_i} = 0,$$

and moving the two terms carrying R to the other side gives the exact conjugation relation of the spherical diopter,

$$\frac{n_i}{l_i} + \frac{n_o}{l_o} = \frac{1}{R}\left[n_i \frac{z_i}{l_i} + n_o \frac{z_o}{l_o}\right]. \tag{2.116}$$

Equation 2.116 is exact, and it is also a disappointment, because l_o and l_i still depend on φ through 2.113 and 2.114. In other words, the position z_i that satisfies it is different for every ray, which is precisely the statement that the sphere is *not* rigorously stigmatic. Notice carefully what we have and have not demanded: we required only that Fermat's principle hold for the individual ray through I, that is, that small variations $\Delta\varphi$ produce negligible variations in L, and not that $L(O, O')$ be the same constant for all rays. Consequently O and O' are not the Young points of the previous sections, no true image is formed, and from here onwards we must speak of approximate stigmatism and, in place of aplanatism, of isoplanatism.

2.10.1 Gaussian optics: the paraxial approximation

The whole difficulty lives in the φ dependence of the path lengths, so the simplest possible cure is to allow only those rays for which φ is small. Restricting the beam to a narrow cylinder about the axis and keeping only the leading term of every expansion, $\sin\varphi \approx \varphi$ and $\cos\varphi \approx 1$, the two path lengths collapse to the axial distances themselves, $l_o \approx -z_o$ and $l_i \approx z_i$, the minus sign appearing because a real object sits at negative z_o while l_o is a positive length. Substituting into 2.116,

$$\frac{n_i}{z_i} + \frac{n_o}{-z_o} = \frac{1}{R}\left[n_i\frac{z_i}{z_i} + n_o\frac{z_o}{-z_o}\right] = \frac{1}{R}\left[n_i - n_o\right],$$

which is the central formula of elementary optics.

$$\boxed{\frac{n_i}{z_i} - \frac{n_o}{z_o} = \frac{n_i - n_o}{R}} \tag{2.117}$$

Every trace of φ has disappeared. That is the whole point: within this approximation the image position no longer depends on which ray we followed, so all paraxial rays from O do meet at the same O', and an approximate image exists. The pair O, O' satisfying 2.117 are called conjugate points, and O' is the paraxial image of O. The right hand side depends only on the surface and the two media, never on the object, so it deserves a name of its own.

Definition 12 *(Vergence): The vergence of a spherical diopter of radius R separating the media n_o and n_i is*

$$\mathcal{V} = \frac{n_i - n_o}{R}, \tag{2.118}$$

a quantity that measures the power of the surface to bend a wavefront, and which is measured in dioptres when R is expressed in metres.

Two particular conjugates are important enough to be named. Sending the object to infinity, $z_o \to \infty$, the term n_o/z_o dies and 2.117 gives the position at which incoming parallel light is gathered, while sending the image to infinity gives the object position from which light emerges parallel. These are the image and object focal distances,

$$f_o = \frac{n_o R}{n_o - n_i}, \tag{2.119}$$

$$f_i = \frac{n_i R}{n_i - n_o}. \tag{2.120}$$

Their ratio is worth a remark that elementary treatments often skip: $f_i/f_o = -n_i/n_o$, so the two focal lengths of a single refracting surface are **not** equal, and they become equal in magnitude only when the surface is bounded by the same medium on both sides. The familiar symmetric picture of a thin lens in air is therefore a special case, not the general rule.

In these conditions the optical path is approximately independent of the point I, which is what approximate stigmatism means: an image is formed, it carries distortions, but those distortions are small and, more importantly, they do not change as the object is displaced in a transverse plane. That invariance is itself a property worth naming.

Definition 13 *(Isoplanatism): An optical system that is approximately stigmatic, and whose distortions stay invariant under a transverse translation of the object, is called **isoplanatic**.*

Isoplanatism is the working substitute for aplanatism, and it is far more than a consolation prize. If the defect a system introduces is the same everywhere in the field, then the image is the object convolved with one fixed blur, the system behaves as a linear filter with a single impulse response, and the entire machinery of Fourier optics becomes available to describe and even to undo it. A system whose defects varied from point to point would admit no such description. We should also note, following the same reasoning, that the approximations made here are not special to the sphere: the identical expansion can be carried out for hyperboloidal and ellipsoidal diopters, and for reflecting surfaces as well.

2.10.2 The thin lens

A single refracting surface is rarely useful on its own, because the image it forms lies inside the glass. The simplest practical element is therefore the **thin lens**, a piece of transparent material of index n_L bounded by two spherical surfaces of radii R_1 and R_2, whose thickness is small enough that we may treat both vertices as sitting at the same point on the axis. The source we are following presents this object only through photographs, so let us supply the calculation, which is a direct and very satisfying application of 2.117.

The method is to apply the conjugation formula twice and to let the image of the first surface act as the object of the second. Let the lens sit in a medium of index n, let the object be at z_o, and call z_1 the position of the intermediate image formed by the first surface alone. Applying 2.117 to that surface, where light passes from n into n_L,

$$\frac{n_L}{z_1} - \frac{n}{z_o} = \frac{n_L - n}{R_1}, \tag{2.121}$$

and applying it to the second surface, where light passes from n_L back into n and where the object is the point z_1 just found,

$$\frac{n}{z_i} - \frac{n_L}{z_1} = \frac{n - n_L}{R_2}. \tag{2.122}$$

The thinness of the lens is what allows us to use the same coordinate z_1 in both equations, since the two vertices coincide. Adding 2.121 and 2.122, the intermediate position cancels identically,

$$\frac{n}{z_i} - \frac{n}{z_o} = \frac{n_L - n}{R_1} - \frac{n_L - n}{R_2} = (n_L - n)\left(\frac{1}{R_1} - \frac{1}{R_2}\right),$$

and dividing through by n we arrive at the conjugation formula of the thin lens,

$$\frac{1}{z_i} - \frac{1}{z_o} = \left(\frac{n_L}{n} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \equiv \frac{1}{f}, \tag{2.123}$$

the right hand side being the **lensmaker's equation** for the focal length f. The cancellation of z_1 is the whole content of the calculation and deserves emphasis: it says that a thin lens has got a single focal length that depends only on the glass and on the two radii, and not at all on where the object happens to be. Setting $z_o \to \infty$ gives $z_i = f$, so f is indeed the distance at which parallel light is brought to a focus, which is exactly what a lens shows happening to sunlight.

Notice also that 2.123 is unchanged if the lens is turned around, since reversing the lens swaps R_1 with R_2 and flips both of their signs, leaving $1/R_1 - 1/R_2$ untouched. A thin lens therefore has got the same focal length whichever way it faces, a fact that is far from obvious and that is emphatically false once the thickness is restored or once aberrations are taken into account, as the next subsection will show.

2.10.3 Spherical aberration

The paraxial world is comfortable but it is a fiction, since every real instrument has got an aperture of finite size and therefore admits rays that leave the axis by a non negligible amount. To see what those rays do we must go back and keep the terms that the paraxial approximation threw away. This is where spherical aberration is born.

We begin with a purely geometrical relation visible in figure 2.27. The point I lies at a height h above the axis and at a distance η from the vertex measured along it, the quantity called the **sagitta**. Since C is at a distance R from the vertex, projecting CI on the axis gives immediately

$$\cos\varphi = \frac{R - \eta}{R} = 1 - \frac{\eta}{R}. \tag{2.124}$$

On the other hand ρ, the chord from the vertex to I, satisfies $\rho^2 = \eta^2 + h^2$, while the right triangle built on C gives $R^2 = (R - \eta)^2 + h^2$. Expanding the latter, $R^2 = R^2 - 2R\eta + \eta^2 + h^2$, the squared radii cancel and we are left with $2R\eta = \eta^2 + h^2 = \rho^2$, that is

$$\eta = \frac{\rho^2}{2R} \qquad \Longrightarrow \qquad \frac{\eta}{R} = \frac{\rho^2}{2R^2} = 1 - \cos\varphi. \tag{2.125}$$

This little identity is the bridge between the angular description and the aperture description of a ray, and it is what will let us express everything in terms of the single quantity ρ, the distance from the axis at which the ray strikes the surface.

Now rewrite the exact path length 2.113. Expanding the square,

$$\begin{aligned} l_o^2 &= R^2 + z_o^2 - 2Rz_o + R^2 + 2Rz_o\cos\varphi - 2R^2\cos\varphi \\ &= z_o^2 + 2R^2\left(1-\cos\varphi\right) - 2Rz_o\left(1-\cos\varphi\right), \end{aligned}$$

and doing the same on the image side we obtain the exact pair

$$l_o = \left[z_o^2 + 2R^2\left(1-\cos\varphi\right) - 2Rz_o\left(1-\cos\varphi\right)\right]^{1/2} = -z_o\left[1+\left(\frac{2R^2}{z_o^2}-\frac{2R}{z_o}\right)\left(1-\cos\varphi\right)\right]^{1/2}, \tag{2.126}$$

$$l_i = \left[z_i^2 + 2R^2\left(1-\cos\varphi\right) - 2Rz_i\left(1-\cos\varphi\right)\right]^{1/2} = z_i\left[1+\left(\frac{2R^2}{z_i^2}-\frac{2R}{z_i}\right)\left(1-\cos\varphi\right)\right]^{1/2}. \tag{2.127}$$

Substituting $1-\cos\varphi = \rho^2/2R^2$ from 2.125 into the bracket,

$$\left(\frac{2R^2}{z_o^2}-\frac{2R}{z_o}\right)\frac{\rho^2}{2R^2} = \frac{\rho^2}{z_o^2} - \frac{\rho^2}{Rz_o} = \frac{\rho^2}{z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right),$$

and therefore

$$l_o = -z_o\left[1+\frac{\rho^2}{z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right)\right]^{1/2}, \tag{2.128}$$

$$l_i = z_i\left[1+\frac{\rho^2}{z_i}\left(\frac{1}{z_i}-\frac{1}{R}\right)\right]^{1/2}. \tag{2.129}$$

These are still exact. Everything that follows is controlled approximation.

The conjugation relation 2.116 needs the reciprocals, so we expand them binomially. For $\rho \ll z_o$, $\rho \ll z_i$ and $\rho \ll R$ the corrections are small and $(1+x)^{-1/2} \approx 1 - x/2$ gives

$$\frac{1}{l_o} = \frac{-1}{z_o}\left[1+\frac{\rho^2}{z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right)\right]^{-1/2} \approx \frac{-1}{z_o}\left[1-\frac{\rho^2}{2z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right)\right], \tag{2.130}$$

$$\frac{1}{l_i} = \frac{1}{z_i}\left[1+\frac{\rho^2}{z_i}\left(\frac{1}{z_i}-\frac{1}{R}\right)\right]^{-1/2} \approx \frac{1}{z_i}\left[1-\frac{\rho^2}{2z_i}\left(\frac{1}{z_i}-\frac{1}{R}\right)\right]. \tag{2.131}$$

Note that we have kept one term more than Gaussian optics did: setting $\rho = 0$ here returns exactly the paraxial substitution $l_o \to -z_o$, $l_i \to z_i$ of the previous subsection, so the ρ^2 terms are precisely the first correction beyond Gauss. Inserting 2.130 and 2.131 into 2.116 and noting that on the right hand side the factors z_i and z_o cancel against the leading $1/z_i$ and $-1/z_o$,

$$\begin{aligned} &\frac{n_i}{z_i}\left[1-\frac{\rho^2}{2z_i}\left(\frac{1}{z_i}-\frac{1}{R}\right)\right] - \frac{n_o}{z_o}\left[1-\frac{\rho^2}{2z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right)\right] \\ &\qquad = \frac{1}{R}\left(n_i\left[1-\frac{\rho^2}{2z_i}\left(\frac{1}{z_i}-\frac{1}{R}\right)\right] - n_o\left[1-\frac{\rho^2}{2z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right)\right]\right). \end{aligned} \tag{2.132}$$

Expanding both sides and moving the paraxial part to the right,

$$\frac{n_i}{z_i}-\frac{n_o}{z_o}=\frac{n_i-n_o}{R}-\frac{1}{R}\frac{n_i\rho^2}{2z_i}\left(\frac{1}{z_i}-\frac{1}{R}\right)+\frac{1}{R}\frac{n_o\rho^2}{2z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right)+\frac{n_i}{z_i}\frac{\rho^2}{2z_i}\left(\frac{1}{z_i}-\frac{1}{R}\right)-\frac{n_o}{z_o}\frac{\rho^2}{2z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right), \tag{2.133}$$

and now the four correction terms pair off beautifully. The two carrying n_i share the factor $n_i\rho^2(1/z_i-1/R)/(2z_i)$ and differ only by $1/z_i-1/R$, and the same happens on the object side, so each pair produces a perfect square,

$$\frac{n_i}{z_i}-\frac{n_o}{z_o}=\frac{n_i-n_o}{R}+\rho^2\left[\frac{n_i}{2z_i}\left(\frac{1}{z_i}-\frac{1}{R}\right)^2-\frac{n_o}{2z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right)^2\right]. \tag{2.134}$$

Equation 2.134 is the conjugation formula of the spherical diopter corrected to first order beyond Gauss, and the bracket multiplying ρ^2 is **spherical aberration**. The physical reading is immediate and is worth stating plainly: because the correction is proportional to ρ^2, rays that strike the surface further from the axis obey a different conjugation relation and therefore cross the axis at a different place. There is no single image point, only a continuous smear of crossing points, whose envelope is the surface known as the caustic. Rearranging 2.134 so that everything belonging to the image side sits on the left and everything belonging to the object side on the right,

$$n_i\left[\frac{1}{z_i}-\frac{1}{R}-\rho^2\frac{1}{2z_i}\left(\frac{1}{z_i}-\frac{1}{R}\right)^2\right]=n_o\left[\frac{1}{z_o}-\frac{1}{R}-\rho^2\frac{1}{2z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right)^2\right], \tag{2.135}$$

which exhibits a quantity conserved across the surface.

Definition 14 *(Marginal invariant): The quantity*

$$\mathcal{M}=n\left[\frac{1}{z}-\frac{1}{R}-\rho^2\frac{1}{2z}\left(\frac{1}{z}-\frac{1}{R}\right)^2\right] \tag{2.136}$$

takes the same value on both sides of a spherical diopter. For paraxial rays, $\rho\to 0$, it reduces to the Abbe invariant

$$\mathcal{A}=n\left[\frac{1}{z}-\frac{1}{R}\right]. \tag{2.137}$$

In practice one wants the aberration expressed through the object distance alone, since that is the quantity an experimenter controls. For moderate apertures the correction term is already small, so we commit only a second order error by evaluating it with the paraxial relation 2.117. That relation gives $1/z_i=(1/n_i)\,(n_o/z_o+(n_i-n_o)/R)$, whence

$$\frac{1}{z_i}-\frac{1}{R}=\frac{n_o}{n_i}\frac{1}{z_o}+\frac{n_i-n_o}{n_iR}-\frac{1}{R}=\frac{n_o}{n_i}\frac{1}{z_o}-\frac{n_o}{n_iR}=\frac{n_o}{n_i}\left(\frac{1}{z_o}-\frac{1}{R}\right),$$

a compact and very useful relation in its own right. Substituting it into the bracket of 2.134 and simplifying, the aberration term becomes

$$\frac{n_i}{2z_i}\left(\frac{1}{z_i}-\frac{1}{R}\right)^2-\frac{n_o}{2z_o}\left(\frac{1}{z_o}-\frac{1}{R}\right)^2\approx\frac{1}{2}\left(\frac{n_o}{n_i}\right)^2\left(\frac{1}{z_o}-\frac{1}{R}\right)^2\left(\frac{n_i-n_o}{n_o}\right)\left(\frac{n_o}{R}-\frac{n_i+n_o}{z_o}\right), \quad (2.138)$$

so that the conjugation formula reads

$$\frac{n_i}{z_i}-\frac{n_o}{z_o}=\frac{n_i-n_o}{R}+\rho^2\frac{1}{2}\left(\frac{n_o}{n_i}\right)^2\left(\frac{1}{z_o}-\frac{1}{R}\right)^2\left(\frac{n_i-n_o}{n_o}\right)\left(\frac{n_o}{R}-\frac{n_i+n_o}{z_o}\right) \quad (2.139)$$

$$=\mathcal{V}+\Delta\mathcal{V}_+. \quad (2.140)$$

The identical manipulation carried out on the image side instead gives

$$\frac{n_i}{z_i}-\frac{n_o}{z_o}=\frac{n_i-n_o}{R}-\rho^2\frac{1}{2}\left(\frac{n_i}{n_o}\right)^2\left(\frac{1}{z_i}-\frac{1}{R}\right)^2\left(\frac{n_o-n_i}{n_i}\right)\left(\frac{n_i}{R}-\frac{n_i+n_o}{z_i}\right) \quad (2.141)$$

$$=\mathcal{V}-\Delta\mathcal{V}_-, \quad (2.142)$$

where $\mathcal{V}$ is the vergence 2.118 and $\pm\Delta\mathcal{V}_\pm$ is the change in vergence produced by the marginal rays. It is convenient to absorb these corrections into an effective radius by defining

$$\frac{1}{R_+}=\rho^2\frac{1}{2}\left(\frac{n_o}{n_i}\right)^2\left(\frac{1}{z_o}-\frac{1}{R}\right)^2\left(\frac{1}{n_o}\right)\left(\frac{n_o}{R}-\frac{n_i+n_o}{z_o}\right), \quad (2.143)$$

$$\frac{1}{R_-}=\rho^2\frac{1}{2}\left(\frac{n_i}{n_o}\right)^2\left(\frac{1}{z_i}-\frac{1}{R}\right)^2\left(\frac{1}{n_i}\right)\left(\frac{n_i}{R}-\frac{n_i+n_o}{z_i}\right), \quad (2.144)$$

so that $\Delta\mathcal{V}_+=(n_i-n_o)/R_+$ and $\Delta\mathcal{V}_-=(n_o-n_i)/R_-$, and both 2.139 and 2.141 take the single compact form

$$\frac{n_i}{z_i}-\frac{n_o}{z_o}=(n_i-n_o)\left(\frac{1}{R}\pm\frac{1}{R_\pm}\right). \quad (2.145)$$

The interpretation could not be neater: **including the marginal rays amounts to nothing more than correcting the curvature of the surface**, replacing $1/R$ by $1/R\pm1/R_\pm$. A marginal ray behaves as though it had struck a sphere of a slightly different radius from the one the paraxial ray sees, and since the correction carries the factor ρ^2, each annulus of the aperture acts as if it belonged to a slightly different lens.

The focal length for marginal rays

Since the aberration term depends on the height ρ at which the ray crosses the surface, different rays are deviated by different amounts and the system does not possess one focal length but a continuum of them, one for each zone of the aperture. The most important of these is the focus of the outermost rays for an object at infinity, the marginal image focus f_M, which we obtain from 2.139 by isolating n_i/z_i, dividing by n_i and letting $z_o \to \infty$,

$$\begin{aligned}\frac{1}{f_M} &= \lim_{z_o\to\infty}\left(\frac{n_o}{n_i z_o} + \frac{n_i - n_o}{n_i R} + \frac{\rho^2}{n_i}\frac{1}{2}\left(\frac{n_o}{n_i}\right)^2\left(\frac{1}{z_o} - \frac{1}{R}\right)^2\left(\frac{n_i - n_o}{n_o}\right)\left(\frac{n_o}{R} - \frac{n_i + n_o}{z_o}\right)\right) \\ &= \frac{n_i - n_o}{n_i R}\left(1 + \frac{\rho^2}{2R^2}\left(\frac{n_o}{n_i}\right)^2\right). \end{aligned} \tag{2.146}$$

The three limits are elementary: the first term vanishes, the squared bracket tends to $1/R^2$ and the last bracket tends to n_o/R, and collecting the surviving factors gives exactly the coefficient displayed.[1] Using $\rho^2/2R^2 = 1 - \cos\varphi$ from 2.125, the same result reads, in terms of the angle subtended at the centre of curvature,

$$\frac{1}{f_M} = \frac{n_i - n_o}{n_i R}\left[1 + \left(\frac{n_o}{n_i}\right)^2 - \left(\frac{n_o}{n_i}\right)^2\cos\varphi\right]. \tag{2.147}$$

Both expressions say the same physical thing, and it is the thing every student has seen without recognising it. Since $1/f_M$ is *larger* than the paraxial value $(n_i - n_o)/n_i R$, the marginal focal length is **shorter** than the paraxial one: rays entering near the rim of a converging surface are bent too strongly and cross the axis closer to the glass than rays entering near the axis. The difference between the two crossing points, $f_{\text{paraxial}} - f_M$, is called the longitudinal spherical aberration, and 2.146 tells us that it grows as ρ^2, that is, as the square of the aperture radius.

This single fact explains a great deal of everyday optical practice. It is why stopping down a camera lens sharpens the image, since closing the diaphragm cuts off precisely the outer annuli that carry the largest error. It is why a bright point source seen through a cheap lens is surrounded by a soft halo rather than being a clean point, the halo being the cross section of the caustic. And it is why the ovoids and conics of the earlier sections mattered at all: those surfaces were constructed to make this term vanish identically, which is the precise sense in which a Descartes ovoid is the cure for the disease that 2.146 diagnoses. Finally, note that the correction carries the factor $(n_o/n_i)^2$, so a surface bounded by media of similar index aberrates less; splitting the bending of a ray across several weakly refracting surfaces instead of one strong surface is the oldest and still the most effective trick in the design of well corrected objectives.

[1] The source followed here prints this result, and the one that follows it, with the opposite sign in front of the correction. That sign is a slip: an exact Snell trace through a spherical surface confirms 2.146 as written, and, as argued in the text, the opposite sign would push the marginal focus *away* from the vertex, contradicting the observed behaviour of a simple lens. We have also corrected an evident misprint, $(n_o/n_o)^2$ for $(n_o/n_i)^2$, in 2.147.

Chapter 3

Electromagnetic Optics and Polarisation

Light is the slender thread that ties electricity and magnetism into a single weave.

"We can scarcely avoid the inference that light consists in the transverse undulations of the same medium which is the cause of electric and magnetic phenomena."

– James Clerk Maxwell

3.1 A historical context of electromagnetic optics and polarisation

The idea that light is an electromagnetic phenomenon - and that its transverse vibration carries a hidden orientation we call polarisation - is one of the great unifications of nineteenth-century physics. Before light could be understood as a travelling disturbance of electric and magnetic fields, it had to shed two older skins: Newton's corpuscles and the purely mechanical aether of the early wave theorists. The story of polarisation, in particular, is the story of how a seemingly minor curiosity - light reflected from a window behaving differently from ordinary light - forced physicists to accept that the optical vibration is transverse, and ultimately vectorial.

The first quantitative encounter with polarisation came in 1808, when Étienne-Louis Malus, looking through a calcite crystal at sunlight reflected from the windows of the Luxembourg Palace in Paris, noticed that the intensity of the transmitted beam changed as he rotated the crystal. From these observations he extracted the law that bears his name, $I = I_0 \cos^2 \theta$, relating the transmitted intensity to the angle between the analyser and the plane of polarisation Malus (1809). Malus had discovered that reflection could endow light with the same directional property that double refraction revealed, without invoking any crystal at all. A few years later, in 1815, David Brewster established the law that the polarisation upon reflection is complete at a particular angle, the Brewster angle, for which the reflected and refracted rays are perpendicular Brewster (1815).

These facts were deeply puzzling within the longitudinal wave picture inherited from sound. The resolution came from Augustin-Jean Fresnel and François Arago, who between 1816 and 1819 showed experimentally that two beams polarised at right angles do not interfere. Fresnel drew the inevitable, audacious conclusion: the optical vibration must be *transverse* to the direction of propagation. From this single hypothesis he derived the laws governing the amplitudes of reflected and refracted light - the Fresnel equations - which remain in use today. Transversality made polarisation a natural attribute of light: a wave oscillating in a plane perpendicular to its motion can do so along infinitely many directions, and the choice of that direction *is* its state of polarisation.

A decisive step toward a quantitative description of partially polarised light was taken by George Gabriel Stokes in 1852. Recognising that realistic beams are never perfectly polarised, Stokes introduced four measurable quantities - today called the Stokes parameters - that completely characterise the state of polarisation, including the degree of polarisation, of any beam, coherent or not Stokes (1852). Decades later, Henri Poincaré gave these parameters a beautiful geometric home: every state of polarisation corresponds to a point on the surface of a sphere, the Poincaré sphere, on which the poles represent circular polarisation and the equator linear polarisation. This geometric picture is the one that reappears, in the scalar reduction of diffraction theory, when one decomposes the field onto two opposite states of the sphere.

The grand synthesis arrived with James Clerk Maxwell. In his 1865 memoir *A Dynamical Theory of the Electromagnetic Field* Maxwell (1865), Maxwell showed that the equations governing electricity and magnetism admit wave solutions propagating at a speed numerically equal to the measured speed of light, and concluded that "light is an electromagnetic disturbance." Polarisation was thereby identified with the direction of the electric field vector $\vec{E}$, and the whole edifice of physical optics became a chapter of electromagnetism. The experimental confirmation came in 1887, when Heinrich

Hertz generated and detected electromagnetic waves in the laboratory, verifying that they reflect, refract and polarise exactly as light does Hertz (1893).

In the twentieth century the description of polarisation was recast in the language of linear algebra. R. Clark Jones, in a celebrated series of papers beginning in 1941, introduced the calculus of two-component complex vectors and 2×2 matrices that now bears his name, ideally suited to fully polarised, coherent light Jones (1941). For partially polarised or depolarising systems, the complementary 4×4 Mueller calculus acting on the Stokes vector provides the appropriate tool. The modern treatment of all these matters - transversality, the Fresnel equations, the StokesPoincaré formalism and the Jones and Mueller calculi - is presented with classic completeness in Born and Wolf's *Principles of Optics* Born and Wolf (1999), the reference against which later treatments are still measured.

From Malus's rotating crystal to the polarisation-encoded qubits of quantum information, the orientation of the optical field has proven to be far more than an ornament: it is a genuine degree of freedom of light, vectorial in nature, and the bridge between the scalar optics of rays and the full electromagnetic description developed in the chapters that follow.

3.2 The vector behavior of the electromagnetic radiation

Until this part of the lectures, we've studied the light as a ray, however, a better model for the light becomes with the 4 Maxwell's equations, which are:

$$\boldsymbol{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}, \qquad \textit{Gauss Law for the electric field} \tag{3.1}$$

$$\boldsymbol{\nabla} \cdot \vec{B} = 0, \qquad \textit{Gauss Law for the magnetic field} \tag{3.2}$$

$$\boldsymbol{\nabla} \times \vec{E} = -\frac{\partial}{\partial t}\vec{B}, \qquad \textit{Faraday-Lenz law} \tag{3.3}$$

$$\boldsymbol{\nabla} \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial}{\partial t}\vec{E} + \mu_0 \vec{J}, \qquad \text{Ampére-Maxwell Law..} \tag{3.4}$$

The equations from 3.1 to 3.4 have got such elegant form thanks to the vector notation given by the physicist Oliver Heaviside and thanks to some of the most useful theorems of the mathematical analysis, the divergence (Gauss) and curl (Stokes) theorems, which allow a smoth transition from the integral form of the Maxwell equations to the differential ones.

The Gauss Law for the electric field 3.1 is really easy to read when we know the geometric interpretatio4n of the divergence operator $\nabla \cdot$ which quantifies how much the electric field $\vec{E}$ goes into (also gets out) from a point, but the right hand side of the equation 3.1 tells us that the presence of any *charge density* ρ produces sources and sinks of electric field (positive charge densities produce sources and negative charge densities produce sinks of electric field).

The Gauss Law for the magnetic field 3.2 tells us that **alwyas the sources of magnetic fields are compensated by sinks of magnetic fields**, also it is interpreted as **there are no magnetic fields monopoles**.

The equation 3.3 is known as the Faraday-Lenz law, the Faraday law tells us that every circulation of electric field produces a variation of magnetic field and every time variation of magnetic field produces circulations of electric field, however there is something importart, because if we take the equation 3.3 as $\nabla \times \vec{E} = \partial_t \vec{B}$ without the minus sign there wont be energy conservation, so the minus sign is introduced by Lenz and this equation can describe the beahavior of eddy currents.

The fourth equation 3.4 describes the circulations of magnetic fields, due to the Gauss Law 3.2 there wont be any polar magnetic field, it is only possible to have axial-like magnetic fields so, this circulations can only be created by two ways, from the presence of density currents $\vec{J}$ or by time variation of the electric fields, Maxwell added this term such that the continuity equation

$$\frac{\partial}{\partial t}\rho + \boldsymbol{\nabla} \cdot \vec{J} = 0, \tag{3.5}$$

is satisfied.

Now,because we're interested in the study of the electromagnetic radiation *light* we must look for a wave equation, to do this, we're going to use the following mathematical identity:

$$\boldsymbol{\nabla} \times \left(\boldsymbol{\nabla} \times \vec{B}\right) = \boldsymbol{\nabla}\left(\boldsymbol{\nabla} \cdot \vec{A}\right) - \nabla^2 \vec{A}. \tag{3.6}$$

From the Faraday-Lenz law 3.3 we've got applying the curl to both sides

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla} \times \vec{E}\right) = -\boldsymbol{\nabla} \times \left(\frac{\partial}{\partial t}\vec{B}\right),$$

now, befores we're going further we're going to suppose that the electromagnetic fields are smooth vector functions, or at least of class $\mathcal{C}^2 := \left\{f : D^2 f \in \mathcal{C}\right\}$ (this means all the second derivaties are continuos), therefore because these are continuous we can use the identity 3.6 and interchange the curl with the time derivative on $\vec{B}$, thus

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla} \cdot \vec{E}\right) - \nabla^2 \vec{E} = -\frac{\partial}{\partial t}\left(\boldsymbol{\nabla} \times \vec{B}\right),$$

to solve the left hand side we are going to use the Gauss law of the electric field 3.1 and for the right hand side we're going to use the Ampére-Maxwell law 3.4 such that we've got

$$\boldsymbol{\nabla}\left(\frac{\rho}{\epsilon_0}\right) - \nabla^2 \vec{E} = -\frac{\partial}{\partial t}\left(\mu_0 \epsilon_0 \frac{\partial}{\partial t}\vec{E} + \mu_0 \vec{J}\right),$$

$$\nabla^2 \vec{E} - \frac{1}{C^2}\frac{\partial^2}{\partial t^2}\vec{E} = \frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho + \mu_0 \frac{\partial}{\partial t}\vec{J}, \tag{3.7}$$

where $C^2 = \frac{1}{\epsilon_0 \mu_0}$.

The 3.7 is the formal wave equation for the electric field. In the left hand side we find the traditional well known wave equation but on the right hand side we can see some physical information about how can those waves be generated. We can see that any gradient of electric charge densities produces electric waves and that time variations of the current density source them too, with a plus sign, because the derivation used the minus sign of the Lenz law twice: once when $\partial_t \vec{B}$ was replaced by $-\boldsymbol{\nabla} \times \vec{E}$ in going from 3.4 to the line above, and a second time when the whole equation was rearranged to bring $\nabla^2 \vec{E} - \frac{1}{C^2}\partial_t^2 \vec{E}$ to the left, and the two minus signs cancel each other. Now, we've shown that there are electric waves so we're going to proof that there are also magnetic waves.

From the Ampére-Maxwell law 3.4 we've got applying the curl on both sides

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla} \times \vec{B}\right) = \boldsymbol{\nabla} \times \left(\mu_0 \epsilon_0 \frac{\partial}{\partial t}\vec{E} + \mu_0 \vec{J}\right),$$

now, under our assumptions and also $\rho, \vec{J}$ are at leat $\mathcal{C}^2$, then we can apply the identity 3.6 and interchange the time derivatide with the curl on $\vec{E}$, thus

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla} \cdot \vec{B}\right) - \nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial}{\partial t}\left(\boldsymbol{\nabla} \times E\right) + \mu_0 \boldsymbol{\nabla} \times \vec{J}.$$

On the left side we can use the Gauss law for the magnetic field 3.2, and on the right hand side we can use the Faraday-Lenz law thus

$$-\nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial}{\partial t}\left(-\frac{\partial}{\partial t}\vec{B}\right) + \mu_0 \boldsymbol{\nabla} \times \vec{J},$$

$$\nabla^2 \vec{B} - \frac{1}{C^2}\frac{\partial^2}{\partial t^2}\vec{B} = -\mu_0 \boldsymbol{\nabla} \times \vec{J}. \tag{3.8}$$

The equation 3.8 is the wave equation for the magnetic field, on the left hand side we can recognize the traditional wave terms and on the right hand side we can see the sources of magnetic waves, because there are no magnetic monopoles there is not term for gradients of such charges, only the circulations of density currents produces this magnetic waves.

Befores going any further we must stop and think about something important, if the light is both electric and magnetic radiation, why the equations 3.7 and 3.8 are only coupled by the density current, or even more, in the vacuum, both equation are independent, so if we think on photons we can question ourselves, ***Where are the photons? Are the photons on the electric or magnetic field*** We are so used to learn they're on both, but how?

3.3 Maxwell equations are the Schrödinger equation for photons

The sentence *photons are in the electromagnetic radiation* is easy to beliave but hard to proof from a classical point of view. We could be tempted by the devil and add both equation 3.7 and 3.8 and get something like

$$\nabla^2\left(\vec{E} + \vec{B}\right) - \frac{1}{C^2}\frac{\partial^2}{\partial t^2}\left(\vec{E} + \vec{B}\right) = \frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho + \mu_0\left(\frac{\partial}{\partial t} - \boldsymbol{\nabla}\times\right)\vec{J}, \tag{3.9}$$

which is correct, completely correct **in a world where does not exist physics and analysis**.

From an analysis point of view, the electric field $\vec{E}$ and the magnetic field $\vec{B}$ have got completely different natures, the electric field is a polar vector function, and the magnetic field is an axial vector function, so both fields live on differente vector spaces and the addition $\vec{E}+\vec{B}$ does not have any sense.

To solve this problem, there is an interesing vector field called the ***Riemann-Silberstein field*** which is a correct mathematical (and physical as we we'll see) superposition of the electric and magnetic fields given by

$$\vec{F}=\vec{E}+iC\vec{B}. \tag{3.10}$$

That imaginary unit i may be seen innocent, however adding this complex topology to $\vec{E},\vec{B}$ solves the superposition problem. It is called for Riemann because he was the first one who studied (alongside Gauss) the electromagnetism using complex numbers, and for Riemann because he used the vector field 3.10 to rewrite the Einstein's field equations in a quaternions formulation. This may be seen as an innocent superpostion but we're going to see that this will allow us, from a classical point of view, to arrive to the correct *photon wave function* and showing that the Maxwell equations acting on 3.10 (which behaves as the photon field) are also quantum equations.

Lets rewrite the Maxwell equations 3.1-3.4 in terms of the photon field $\vec{F}$. First, just multiply the Gauss law for the magnetic field 3.2 by iC and add to the electric Gauss law 3.1 and we'll have

$$\boldsymbol{\nabla}\cdot\left(\vec{E}+iC\vec{B}\right)=\frac{\rho}{\epsilon_0}. \tag{3.11}$$

And, also multiply the Ampére-Maxwell law 3.4 by iC and add it to the Faraday-Lenz law 3.3 we will have

$$\boldsymbol{\nabla}\times\left(\vec{E}+iC\vec{B}\right)=\frac{i}{C}\frac{\partial}{\partial t}\left(\vec{E}+iC\vec{B}\right)+iC\mu_0\vec{J}. \tag{3.12}$$

Joning both equations and using the definitions of $\vec{F}$ we'll wave that the four Maxwell's equations becomes two

$$\boldsymbol{\nabla}\cdot\vec{F}=\frac{\rho}{\epsilon_0}, \qquad \text{Gauss law for the photon field} \tag{3.13}$$

$$\boldsymbol{\nabla}\times\vec{F}=\frac{i}{C}\frac{\partial}{\partial t}\vec{F}+iC\mu_0\vec{J}, \qquad \text{Faraday-Lenz-Ampére-Maxwell laws for the photon field.} \tag{3.14}$$

The physical interpretaion of the equations 3.13 and 3.14 becomes clear. The Gauss law 3.13 tells us that the presence of charge densities produces sources and sinks of photon fields (electromagnetic fields), and the second equation 3.14 says that time variatons of the photon field produces circulations of photon fields and also, the mere presence of density currents produces also circulations of this photon fields. Now it is becoming clearer the relation between where are the photons and how can it be seen in the elctromagnetic field at classical levels.

Now, we're going to proof that the set of equations 3.13 and 3.14 contains two wave equation. One traditional wave equation for $\vec{F}$ and an Schrödigner wave equation for $\vec{F}$, showing this beauty behavior both classical and quantum of the Maxwells equation. Lets apply the curl of the second law 3.14 to both sides as follows

$$\boldsymbol{\nabla}\times\left(\boldsymbol{\nabla}\times\vec{F}\right)=\frac{i}{C}\boldsymbol{\nabla}\times\left(\frac{\partial}{\partial t}\vec{F}\right)+iC\mu_0\boldsymbol{\nabla}\times\vec{J},$$

applying the identity 3.6 on the left han side and permutting th time derivative and the curl in the right hand side as

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\vec{F}\right)-\nabla^2\vec{F}=\frac{i}{C}\frac{\partial}{\partial t}\left(\boldsymbol{\nabla}\times\vec{F}\right)+iC\mu_0\boldsymbol{\nabla}\times\vec{J},$$

using the Gauss law 3.13 and the FLAM law 3.14 we'll have

$$\frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho-\nabla^2\vec{F}=\frac{i}{C}\frac{\partial}{\partial t}\left(\frac{i}{C}\frac{\partial}{\partial t}\vec{F}+iC\mu_0\vec{J}\right)+iC\mu_0\boldsymbol{\nabla}\times\vec{J},$$

carrying out the time derivative on the right hand side, and using $\frac{i}{C}\cdot\frac{i}{C}=-\frac{1}{C^2}$ and $\frac{i}{C}\cdot iC\mu_0=-\mu_0$, we've got

$$\frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho-\nabla^2\vec{F}=-\frac{1}{C^2}\frac{\partial^2}{\partial t^2}\vec{F}-\mu_0\frac{\partial}{\partial t}\vec{J}+iC\mu_0\boldsymbol{\nabla}\times\vec{J},$$

and moving $\nabla^2\vec{F}$ and $\frac{1}{C^2}\partial_t^2\vec{F}$ to the same side, that is multiplying the whole equation by -1 and swapping it around, we arrive at the *classical wave equation for the photon field*

$$\nabla^2\vec{F}-\frac{1}{C^2}\frac{\partial^2}{\partial t^2}\vec{F}=\frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho+\mu_0\left(\frac{\partial}{\partial t}-iC\boldsymbol{\nabla}\times\right)\vec{J}. \tag{3.15}$$

As a check, notice that $\nabla^2-\frac{1}{C^2}\partial_t^2$ is a linear operator and $\vec{F}=\vec{E}+iC\vec{B}$ is a linear combination, so 3.15 must equal 3.7 plus iC times 3.8, and indeed $\left[\frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho+\mu_0\partial_t\vec{J}\right]+iC\left[-\mu_0\boldsymbol{\nabla}\times\vec{J}\right]=\frac{1}{\epsilon_0}\boldsymbol{\nabla}\rho+\mu_0\left(\partial_t\vec{J}-iC\boldsymbol{\nabla}\times\vec{J}\right)$, exactly the right hand side above.

We can see that looks very similar to the sinister equation 3.9 but that was a really forced one insted of our beauty 3.15 which was calculated from the Maxwell's equations, and we see the origin of this electromagnetic or photon fields, these waves are originated as from charge density gradientes, and time variations and circulations of the density current, now the behavior of the photons, from a classical perspective can be seen in the presence of charges and currents.

Now, to find out the quantum wave equation from the classical equation 3.14 we are going to use, the following algebraic identity:

$$\vec{A}\times\vec{B}=-i\left(\vec{A}\cdot\vec{S}\right)\vec{B},\qquad \vec{S}=S_x\hat{i}+S_y\hat{j}+S_z\hat{k} \tag{3.16}$$

$$S_x=\begin{bmatrix}0&0&0\\0&0&-i\\0&i&0\end{bmatrix},\ S_y=\begin{bmatrix}0&0&i\\0&0&0\\-i&0&0\end{bmatrix},\ S_z=\begin{bmatrix}0&-i&0\\i&0&0\\0&0&0\end{bmatrix}.$$

The matrices S_x, S_y, S_z are called teh photon spin 1 matrices, and $\vec{A}, \vec{B}$ are arbitrary vectors. Uusing the identity 3.16 on 3.14 we'll have

$$\begin{aligned}
\boldsymbol{\nabla} \times \vec{F} &= -i\left(\boldsymbol{\nabla} \cdot \vec{S}\right)\vec{F}, \quad \text{using the product rule} \\
\boldsymbol{\nabla} \times \vec{F} &= -i\left(\vec{S} \cdot \boldsymbol{\nabla}\right)\vec{F}, \\
\frac{i}{C}\frac{\partial}{\partial t}\vec{F} &= -i\left(\vec{S} \cdot \boldsymbol{\nabla}\right)\vec{F} - iC\mu_0\vec{J},
\end{aligned}$$

multipliying both sides by $C\hbar$ we'll have

$$i\hbar\frac{\partial}{\partial t}\vec{F} = C\left(\vec{S} \cdot \frac{\hbar}{i}\boldsymbol{\nabla}\right)\vec{F} - i\hbar C^2\mu_0\vec{J}, \tag{3.17}$$

which is the Schrödinger-like equation for the photon field $\vec{F}$ with the perturbation of the density currents $\vec{J}$ if we're on the vacuum, we'll have the well known as the ***Birula's photon wave function*** Białynicki-Birula (1994)

$$i\hbar\frac{\partial}{\partial t}\vec{F} = C\left(\vec{S} \cdot \frac{\hbar}{i}\boldsymbol{\nabla}\right)\vec{F}. \tag{3.18}$$

Therefore the Maxwells equation by themselves are classical and quantum, contains the wave functions that describes the photon behavior and, as we we'll see further in this lectures, we can carry on this result to general relativity using a tensor that I named **Riemann-Silberstein tenstor** which even condensates the four equations to just one.

3.4 Properties of the electromagnetic waves

From now on we're going to focus on the electric wave equation 3.7 and not the complete electromagnetic wave equation 3.15 for three reasons:

1. For now, we're going to focus on the radiation propagation on the vacuum, so the electric field intensity $||\vec{E}||$ is a lot bigger than the magnetic field intensity $||\vec{B}||$ with a relation approximadelety like $||\vec{E}|| \approx C||\vec{B}||$.

2. The interaction between light and matter is really hard to study when we consider the electric polarization and even more when we consider all the magnetic properties and effects. This topics are going to be study in the lectures on quantum optics.

3. This is still a lecture on fundamentals, so we're going to start from the basic notions and getting richer models while we advance.

3.4.1 Principal solutions of the electric wave equation

Due to the last reasons, we're going to to focus on the study of the electric wave equation

$$\nabla^2\vec{E}(\vec{r},t) = \frac{1}{C^2}\frac{\partial^2\vec{E}(\vec{r},t)}{\partial t^2}. \tag{3.19}$$

We're going to solver this equation in a physical way (latter on this lectures we're going to solve it in a more sophisticaed way). We know from a physical point of view that $\vec{E}$ should be a propagation function with a constant velcity $\vec{v}$ (because there is no matter so the momentum is conserved), therefore we could suppose the electric wave have got the following expression

$$\vec{E}(\vec{r},t) = \vec{u}(\vec{k}\cdot\vec{r}\pm\omega t, t), \tag{3.20}$$

where we have got that $\omega^2/k^2 = C^2$. The two most popular solutions in the basics literature are the plane wave equation

$$\vec{E}(\vec{r},t) = E(\vec{r},t)e^{i\left(\vec{k}\cdot r-\omega t+\phi\right)}\hat{e}(\vec{r},t), \tag{3.21}$$

where $E(\vec{r},t)$ is the amplitude, $\vec{k}$ is the waver number, a kind of spatial frequency, ω is the angular frequency, ϕ is the proper fase, an $\hat{e}$ is the unit vector that describes the vector behavior of the electric wave, also we will identify $\hat{e}$ as the *polarisation*.

The other one, very popular is the spherical wave

$$\vec{E}(\vec{r},t) = \frac{E(t)}{||\vec{r}||}e^{i(kr-\omega t+\phi)}\hat{e}. \tag{3.22}$$

Both expressions are usually written down as if they had fallen from the sky, but each of them is a genuine solution of the wave equation 3.19 and can be obtained directly from it. Let us do it carefully, first for the plane wave and then for the spherical one.

In the vacuum the wave equation 3.19 acts on every Cartesian component of $\vec{E}$ in exactly the same way and without mixing them, so for the moment we can forget the vector character, write $\vec{E}(\vec{r},t) = \psi(\vec{r},t)\hat{e}$ with $\hat{e}$ a fixed unit vector, and solve the scalar equation

$$\nabla^2\psi = \frac{1}{C^2}\frac{\partial^2\psi}{\partial t^2}. \tag{3.23}$$

A natural way to attack it is to separate the space and time dependence, proposing $\psi(\vec{r},t) = R(\vec{r})T(t)$. Substituting into 3.23 and dividing by RT we get

$$\frac{\nabla^2 R}{R} = \frac{1}{C^2}\frac{\ddot{T}}{T}.$$

The left hand side depends only on the position and the right hand side only on the time, so the only way for them to be equal for every $\vec{r}$ and every t is that both are the same constant. We call that constant $-k^2$, where the minus sign is chosen so that the time part stays bounded and oscillatory, as a wave must be. The time equation then becomes

$$\ddot{T} + C^2 k^2 T = 0,$$

and defining $\omega = Ck$, which is precisely the relation $\omega^2/k^2 = C^2$ we anticipated, its solution is $T(t) = e^{\mp i\omega t}$. The spatial part becomes the Helmholtz equation

$$\nabla^2 R + k^2 R = 0. \tag{3.24}$$

Now we ask for the simplest possible geometry of the wavefronts. If we want the surfaces of constant phase to be planes, the field must depend on the position only through a single linear combination of the coordinates, that is, through $\vec{k}\cdot\vec{r}$ for some fixed vector $\vec{k}$. This suggests

$$R(\vec{r}) = e^{i\vec{k}\cdot\vec{r}},$$

and since $\nabla^2 e^{i\vec{k}\cdot\vec{r}} = -\left|\left|\vec{k}\right|\right|^2 e^{i\vec{k}\cdot\vec{r}}$, the Helmholtz equation 3.24 is satisfied as long as $\left|\left|\vec{k}\right|\right| = k$. The vector $\vec{k}$ that was born as a mere separation constant turns out to point along the direction of propagation and to have a length fixed by the frequency. Putting the space and time parts back together, restoring the amplitude E_0 and a constant phase ϕ, and recovering the polarisation vector, we arrive at

$$\vec{E}(\vec{r},t) = E_0\, e^{i\left(\vec{k}\cdot\vec{r}-\omega t+\phi\right)}\hat{e},$$

which is exactly the plane wave 3.21. If instead of the constant E_0 and the fixed $\hat{e}$ we allow a slowly varying amplitude $E(\vec{r},t)$ and polarisation $\hat{e}(\vec{r},t)$, we recover the modulated waves that we will use later, whose wavefronts are still approximately planes over regions small compared with the modulation.

There is still one condition we have not used, and it is the one that makes the wave truly electromagnetic. In the vacuum the Gauss law 3.1 reads $\boldsymbol{\nabla}\cdot\vec{E} = 0$, and evaluating it on our solution

$$\boldsymbol{\nabla}\cdot\left(E_0\, e^{i\left(\vec{k}\cdot\vec{r}-\omega t+\phi\right)}\hat{e}\right) = i\,E_0\left(\vec{k}\cdot\hat{e}\right)e^{i\left(\vec{k}\cdot\vec{r}-\omega t+\phi\right)} = 0,$$

which can only hold if $\vec{k}\cdot\hat{e} = 0$. The polarisation is forced to be perpendicular to the direction of propagation, and we recover, now as a theorem instead of a hypothesis, the transversality that Fresnel had to postulate a century and a half earlier.

The plane wave describes a field whose wavefronts are infinite parallel planes, an idealisation that is excellent far from the source but says nothing about how the radiation spreads out from a small emitter. For that situation the natural symmetry is not translational but spherical, so we now look for a field that depends on the position only through the distance $r = ||\vec{r}||$ to the source. Working again with the scalar amplitude $\psi(r,t)$, we need the Laplacian of a function that depends on r alone, which in spherical coordinates is

$$\nabla^2\psi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right).$$

This expression hides a very useful identity. If we expand it and compare it with the second derivative of the product $r\psi$, we find

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) = \frac{1}{r}\frac{\partial^2}{\partial r^2}(r\psi)\,, \tag{3.25}$$

which turns the awkward radial Laplacian into an ordinary second derivative acting on $r\psi$. Using 3.25, the wave equation 3.23 for a spherically symmetric field becomes

$$\frac{1}{r}\frac{\partial^2}{\partial r^2}(r\psi) = \frac{1}{C^2}\frac{\partial^2\psi}{\partial t^2}.$$

Multiplying both sides by r and noticing that r does not depend on the time, so it can go inside the time derivative on the right, we get

$$\frac{\partial^2}{\partial r^2}(r\psi) = \frac{1}{C^2}\frac{\partial^2}{\partial t^2}(r\psi)\,.$$

If we now baptise the combination $u(r,t) = r\,\psi(r,t)$, this is nothing but the one dimensional wave equation

$$\frac{\partial^2 u}{\partial r^2} = \frac{1}{C^2}\frac{\partial^2 u}{\partial t^2}, \tag{3.26}$$

whose general solution, by d'Alembert, is the sum of a disturbance travelling outwards and one travelling inwards,

$$u(r,t) = f(r - Ct) + g(r + Ct).$$

The term $g(r + Ct)$ represents a wave collapsing towards the source, so for radiation escaping from an emitter we keep only the outgoing part $u(r,t) = f(r - Ct)$. Undoing the change $u = r\psi$ we finally obtain

$$\psi(r,t) = \frac{f(r - Ct)}{r},$$

and the factor $1/r$, which we had to guess before, now appears on its own as a direct consequence of the geometry. Choosing a monochromatic profile $f(r - Ct) = E_0\,e^{ik(r-Ct)} = E_0\,e^{i(kr-\omega t)}$, again with $\omega = Ck$, and restoring the phase and the polarisation we reach the spherical wave 3.22,

$$\vec{E}(\vec{r},t) = \frac{E_0}{r}\,e^{i(kr-\omega t+\phi)}\hat{e}.$$

It is worth pausing on the $1/r$. The amplitude of the field falls off as the inverse of the distance, so its square, which measures the intensity, falls off as $1/r^2$. Since the area of a spherical wavefront grows exactly as r^2, the total power crossing every sphere centred on the source is the same, and energy is conserved as the wave expands. This is why the field goes as $1/r$ and not as $1/r^2$: the inverse square law belongs to the intensity, not to the field itself.

The plane wave and the spherical wave look like very different objects, and yet they are intimately related, because far enough from the source the spherical wave is indistinguishable from a plane wave. To see it, place ourselves at a large distance r_0 from the emitter along a fixed direction $\hat{n} = \vec{r}_0/r_0$, and look at the field only inside a small region around that point, writing $\vec{r} = \vec{r}_0 + \vec{\delta}$ with $\left\|\vec{\delta}\right\| \ll r_0$. The distance to the source is then

$$r = \left\|\vec{r}_0 + \vec{\delta}\right\| = r_0 \sqrt{1 + \frac{2\,\hat{n}\cdot\vec{\delta}}{r_0} + \frac{\left\|\vec{\delta}\right\|^2}{r_0^2}} \approx r_0 + \hat{n}\cdot\vec{\delta},$$

where we kept only the first order in $\vec{\delta}/r_0$. The phase of the spherical wave becomes, up to the constant kr_0,

$$kr \approx kr_0 + k\,\hat{n}\cdot\vec{\delta} = kr_0 + \vec{k}\cdot\vec{\delta}, \qquad \vec{k} = k\,\hat{n},$$

so it depends on the position through the single linear combination $\vec{k}\cdot\vec{\delta}$, which is exactly the signature of a plane wave travelling along $\hat{n}$. The amplitude, on the other hand, is $E_0/r \approx E_0/r_0$, which over the small region hardly changes and can be treated as the constant E_0/r_0. Putting both together, in the neighbourhood of the point the field reads

$$\vec{E}(\vec{r},t) \approx \frac{E_0}{r_0}\, e^{i\left(kr_0 + \vec{k}\cdot\vec{\delta} - \omega t + \phi\right)}\hat{e},$$

which is precisely a plane wave 3.21 of amplitude E_0/r_0 and wavevector $\vec{k} = k\hat{n}$. In physical terms, a small patch of a sphere of enormous radius is almost flat, and the observer far from the source sees the curved wavefronts as if they were planes. This is the regime in which the plane wave, despite carrying infinite energy across an infinite front, becomes a legitimate and extremely convenient approximation, and it is the reason why so much of optics can be done with plane waves even though every real source radiates spherically.

3.4.2 Beams with a particular phase and structured light

The plane wave and the spherical wave are the two solutions that every textbook writes down first, and for a good reason, since they are the simplest geometries the wave equation admits. It would be a serious mistake, however, to believe that they are the only solutions, or even the most important ones. Both are idealisations that no laboratory ever produces exactly, because the plane wave has an infinite transverse extent and carries an infinite power, and the pure spherical wave demands a perfectly point like source radiating in every direction at once. The light we actually generate and use, above all the light of a laser, comes in the form of *beams*, that is, fields well confined around a propagation axis that carry a finite power. What distinguishes one beam from another, and what grants each one its remarkable properties, is not so much its amplitude as the way its *phase* is organised across the wavefront. This subsection is a short tour through the most relevant of these structured beams.

Before writing any of them it helps to have a common equation from which they all descend. Almost every beam of interest travels mostly along one axis, which we take to be z, with only a gentle spreading in the transverse plane, and this is what we call the *paraxial regime*. We therefore look for solutions of the Helmholtz equation 3.24 of the form

$$\vec{E}(\vec{r},t) = u(x,y,z)\, e^{i(kz-\omega t)}\hat{e},$$

where the exponential carries the fast oscillation along the axis and u is a slowly varying envelope that sculpts the beam. Substituting this into the Helmholtz equation and cancelling the common factor we obtain

$$\nabla_\perp^2 u + \frac{\partial^2 u}{\partial z^2} + 2ik\frac{\partial u}{\partial z} = 0, \qquad \nabla_\perp^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2},$$

and because the envelope varies slowly over a wavelength we may neglect $\partial_z^2 u$ against $2k\,\partial_z u$, arriving at the *paraxial wave equation*

$$i\frac{\partial u}{\partial z} = -\frac{1}{2k}\nabla_\perp^2 u. \tag{3.27}$$

It is impossible not to notice that 3.27 is formally identical to the Schrödinger equation of a free particle in two dimensions, with the propagation coordinate z playing the role of the time and $1/k$ playing the role of $\hbar/m$. This is not a coincidence but another face of the deep link between Maxwell and Schrödinger we met earlier in this chapter, and it tells us that the whole catalogue of structured beams is, in a precise mathematical sense, the catalogue of the wave packets of a free particle.

The simplest and most important solution of 3.27 is the one whose transverse profile is a Gaussian. Writing $\rho^2 = x^2 + y^2$ for the transverse radius, its electric field is

$$\vec{E} = E_0\,\frac{w_0}{w(z)}\exp\left(-\frac{\rho^2}{w(z)^2}\right)\exp\left(i\left[kz - \omega t + \frac{k\rho^2}{2R(z)} - \zeta(z)\right]\right)\hat{e}, \tag{3.28}$$

where the three functions that govern the beam are the width, the radius of curvature and the Gouy phase,

$$w(z) = w_0\sqrt{1 + \left(\frac{z}{z_R}\right)^2}, \qquad R(z) = z\left[1 + \left(\frac{z_R}{z}\right)^2\right], \qquad \zeta(z) = \arctan\frac{z}{z_R}, \tag{3.29}$$

and $z_R = \frac{kw_0^2}{2} = \frac{\pi w_0^2}{\lambda}$ is the *Rayleigh range*, the distance over which the beam stays approximately collimated, while w_0 is the *waist*, the minimal radius the beam reaches at $z = 0$. Reading the phase of 3.28 term by term makes the physics appear. The piece kz is the ordinary plane wave advance along the axis. The quadratic piece $k\rho^2/2R(z)$ is exactly the phase that a spherical wave of radius $R(z)$ carries near the axis, so it tells us that the wavefronts are neither planes nor full spheres but gently curved caps, flat at the waist where $R \to \infty$, most strongly curved around the Rayleigh range, and flattening again far away where $R(z) \approx z$ and the beam finally looks spherical. The last piece, the *Gouy phase* $\zeta(z)$, is an extra amount of phase, close to π in total, that the beam accumulates as it crosses the focus, an anomaly with no counterpart in the plane wave and measurable by interferometry. The amplitude factor $w_0/w(z)$ in front of the Gaussian keeps the power constant while the beam widens, since as $w(z)$ grows the peak drops so that the integrated intensity is conserved. The beam spreads with a divergence half angle $\theta = \lambda/\pi w_0$, and this reciprocal relation between waist and divergence is nothing but the diffraction limit and, once more, an uncertainty relation between the transverse position and the transverse momentum of the field. The Gaussian beam is the fundamental transverse mode of a laser resonator, the famous TEM_{00}, and it is the most confined beam that diffraction permits, which is why it is the default description of any well behaved laser and the starting point for the design of almost every optical system that works with coherent light.

The Gaussian is only the lowest member of a whole family. Since 3.27 is linear, it admits complete sets of higher order modes that share the same Gaussian envelope but decorate it with nodal structure. In Cartesian coordinates these are the *Hermite-Gauss* modes, indexed by two integers and built from Hermite polynomials, which display a rectangular grid of bright lobes. In cylindrical coordinates the natural family is that of the *Laguerre-Gauss* modes, indexed by a radial number p and an azimuthal number ℓ, whose field is

$$\begin{aligned}\vec{E}_{p\ell} = E_0\, \frac{w_0}{w(z)} \left(\frac{\rho\sqrt{2}}{w(z)}\right)^{|\ell|} L_p^{|\ell|}\left(\frac{2\rho^2}{w(z)^2}\right) \exp\left(-\frac{\rho^2}{w(z)^2}\right) e^{i\ell\varphi} \\ \times \exp\left(i\left[kz - \omega t + \frac{k\rho^2}{2R(z)} - (2p + |\ell| + 1)\,\zeta(z)\right]\right)\hat{e},\end{aligned} \tag{3.30}$$

where $L_p^{|\ell|}$ are the associated Laguerre polynomials and φ is the azimuthal angle. Two features of 3.30 deserve attention. The Gouy phase now grows in proportion to the mode order $2p + |\ell| + 1$, so different modes advance their phase at different rates, a fact that governs how a superposition of them rotates and reshapes as it propagates. Far more striking is the azimuthal factor $e^{i\ell\varphi}$, which winds the phase through ℓ complete turns around the axis and forces the field to vanish exactly on the axis, giving these beams their characteristic doughnut profile with a dark core. The wavefront is then no longer a stack of caps but a set of intertwined helices, an *optical vortex* of topological charge ℓ. This helical phase is not a mere decoration, because a field carrying $e^{i\ell\varphi}$ transports orbital angular momentum, and each of its photons carries exactly $\ell\hbar$ of it, a quantity entirely separate from the spin angular momentum $\pm\hbar$ associated with circular polarisation. For this reason the Laguerre-Gauss beams have become one of the central tools of modern optics. They are used in the *optical tweezers* to trap microscopic particles and set them spinning, in super resolution microscopy where the dark core of a doughnut beam switches off the fluorescence everywhere except at the very centre, and in classical and quantum communications alike, where the unbounded integer ℓ offers a new and in principle limitless alphabet that lets a single photon encode much more than a single bit.

All the beams written so far spread as they travel, because they are paraxial and diffraction is unavoidable for any field of finite width. It is natural to ask whether the exact Helmholtz equation admits a beam that does not spread at all, and it does. If we demand a field whose transverse intensity profile is literally independent of z, separation of variables in cylindrical coordinates leads to the *Bessel beam*

$$\vec{E} = E_0\, J_\ell(k_\perp \rho)\, e^{i\ell\varphi}\, e^{i(k_z z - \omega t)}\hat{e}, \qquad k_\perp^2 + k_z^2 = k^2, \tag{3.31}$$

where J_ℓ is the Bessel function of order ℓ. Its transverse pattern, a bright central spot when $\ell = 0$ or a vortex ring when $\ell \neq 0$, surrounded by concentric rings, is exactly the same at every plane z, so the beam is said to be *non-diffracting*. The picture behind 3.31 is illuminating, since it is the coherent superposition of plane waves whose wavevectors all lie on a cone of fixed half angle $\arctan(k_\perp/k_z)$, and the interference of that cone regenerates the same profile plane after plane. This conical construction also explains the beam's most surprising property, its *self-healing*, because if an obstacle blocks the central spot the outer rays that build the cone continue past it and reconstruct the pattern a short distance downstream. The ideal Bessel beam pays for these gifts with an infinite power, since its rings extend forever and each one carries a comparable share of energy, so in practice it is produced with an axicon or a ring aperture and remains non-diffracting only over a finite range. Its long, thin and self-repairing focus makes it invaluable for light sheet microscopy, for precise optical alignment and for material processing whenever a large depth of focus is required.

There is a last beam whose behaviour borders on the paradoxical. Looking again at the paraxial equation 3.27, now in a single transverse dimension, one finds a solution written in terms of the Airy function,

$$u(x, z=0) = \mathrm{Ai}\left(\frac{x}{x_0}\right) e^{\,ax/x_0}, \qquad a > 0, \tag{3.32}$$

where x_0 sets the transverse scale and the exponential aperture e^{ax/x_0} is introduced to make the total energy finite. As it propagates, the *Airy beam* shows two properties that seem to contradict everything the plane wave taught us. It is non-diffracting, keeping its intensity profile over long distances, and it is *self-accelerating*, since its main lobe does not travel in a straight line but follows a parabolic trajectory $x \propto z^2$, so the beam bends sideways even though no force and no index gradient act on it. Nothing is being violated here, because the centre of mass of the whole field does travel straight, and it is only the intensity pattern that curves, in the same way that the crest of a water wave may move differently from the water itself. Like the Bessel beam it is also self-healing, and these curved, resilient trajectories have been used to route light around obstacles, to guide electric discharges along curved plasma channels, to sweep microscopic particles along a chosen path and, once again, in advanced microscopy.

Seen together, this family teaches a lesson that the plane and the spherical waves, taken alone, keep hidden, namely that what organises a beam and endows it with a useful behaviour is the shape of its phase across the wavefront. A phase linear in the transverse coordinate, $e^{i\vec{k}_\perp\cdot\vec{\rho}}$, simply tilts the beam. A phase quadratic in ρ, $e^{ik\rho^2/2R}$, focuses or defocuses it and is exactly what a thin lens imprints. A phase linear in the azimuth, $e^{i\ell\varphi}$, gives it a vortex and orbital angular momentum. A conical phase produces the non-diffracting Bessel beam, and a cubic phase produces the self bending Airy beam. Modern devices such as the spatial light modulators and the metasurfaces let us paint almost any of these phase profiles onto a wavefront at will, and this ability to engineer the phase is the whole discipline known today as *structured light*. The plane wave and the spherical wave remain useful as building blocks and as limiting cases, but from here on the reader should keep in mind that they are only two convenient points in a vastly richer landscape of solutions.

3.4.3 From 3D wavefronts to 2D wavefronts

As we've seen the electromagnetic radiation has got many presentations, however, there is a very strong formulation where the light is studied in $2D$, in fact, the following chapter treats the electromagnetic radiation as a $2D$ field.

From the mathematical point of view there is a whole theory about arbitrary surfaces or complex geometries, here, we're going to use the properties of *smooth manifolds*[1] This is thanks to the nature

[1] A smooth manifold if a set which can be completely covered by subsets and there are smooth functions from the manifold to $\mathbb{R}^n$.

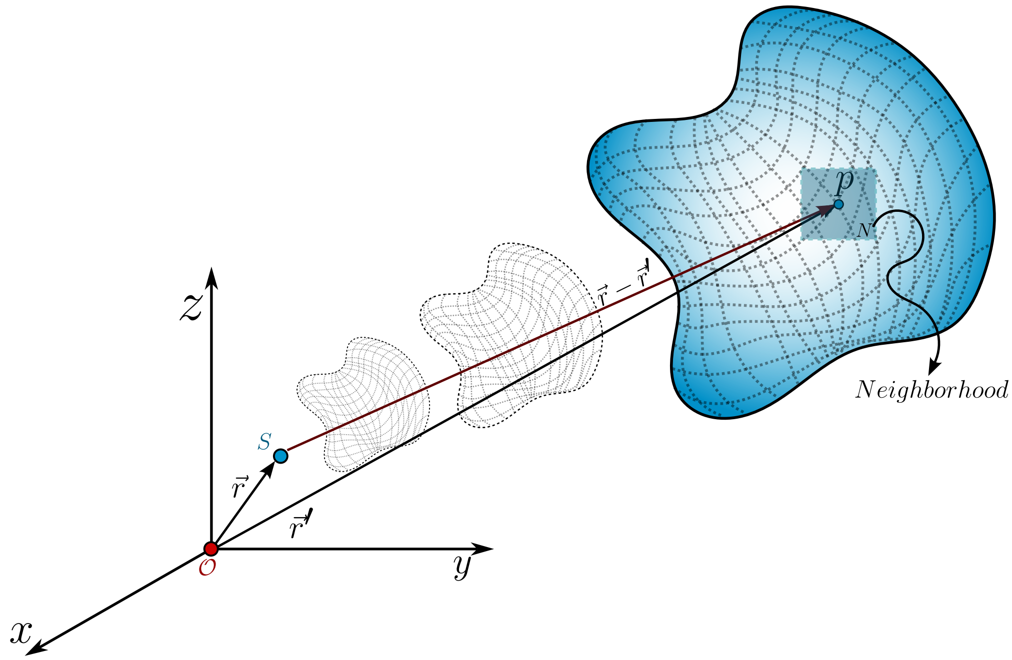


Figure 3.1: A source $\mathcal{S}$ produces an arbitrary kind of wave which propagates in the space-time, then, for any given time and position we can take an arbitrary point p and, due to the smooth manifold nature of this wave, there exists a neighborhood N of p where the wavefront is plane.

of the physical solution of the wave equations, so, given any arbitrary wavefront as shown 3.1 we can choose any point a set and observer on a neighborhood N, and because the wavefront behaves as an smooth manifold then this neighborhood is $\mathbb{R}^2$.

What we are doing is thinkg on this surface like the earth, because we're son tiny we see the earth as plane but it is not. The same phenomena is here, we step on and observer across the point p on the wavefront and for us there is a neighborhood N where it behaves exactly as $\mathbb{R}^2$ ad that is going to allow us to start the study of the polarisation.

3.5 Light's polarisation

In the electromagnetic theory the word *polarisation* is used as a tendency of the electric dipoles to orientate in a given microscopic volume. However, this word also is used to describe the **electric field orientation** in its propagation. Those two definitions then seem completely indepent, however, as we are going to show in the microscopic theory of radiation these two are completely related.

The polarisation of light is one of the most usefull properties of light, there are many applications on real life which uses this property, for example, the cancer cells stole the glucose from the healthy ones, so there is a higher sugar concentration in cancer cells than the healthy ones, so if we irradiate a sample with linearly polarized light and study the transmited radiation, we're going to see that the light which went across the cancer cells changed their polarization, rotating a given agle (which depends on the concentration of glucose) so we can detected the presence of cancer cells using the polarisation.

3.5.1 Monochromatic and deterministic formulation: Jones spinors and the polarization ellipse

As we've said, we're going to focus on the electric field behavior in a $2D$ formulation, so the electric field is going to be a vector field $\vec{E} \in \mathcal{C}^2\left(\mathbb{C}^2\right)$ which means it is *a complex vector function which their second derivatives are continuous.* Its general form is given by

$$\vec{E}(\vec{r},t;\omega) = \begin{bmatrix} \mathcal{E}_1 e^{i\left(\vec{k}\cdot\vec{r}-\omega t+\phi_1\right)} \\ \mathcal{E}_2 e^{i\left(\vec{k}\cdot\vec{r}-\omega t+\phi_2\right)} \end{bmatrix}, \tag{3.33}$$

where we're considering only one frequency ω so this describes the electric field of a monocrhomatic field and as we can see, this vector field is a plane wave.

We can factor the propagation phase $\exp\left[i\left(\vec{k}\cdot\vec{r}-\omega t\right)\right]$ in both components ad we will have the following vector field

$$\vec{e} = \begin{bmatrix} \mathcal{E}_1 e^{i\phi_1} \\ \mathcal{E}_2 e^{i\phi_2} \end{bmatrix}. \tag{3.34}$$

The vector field 3.34 is known as the *Jones vector* but, in fact, their true nature ist vectorial, the field 3.34 is a **spinor**[2] a we're going to see this spinor behavior in this chapter.

The Jones spinor 3.34 is a vector that describes how the electric field oscillates, in general, under our assumptions[3] is an ellipse, well known as the polarization ellipse. To get the ellipse from 3.34 we're going to define the following electric field componentes, given by

$$\frac{E_1}{\mathcal{E}_1} e^{-i\phi_2} = \frac{1}{2} e^{i\triangle\phi}, \tag{3.35}$$

$$\frac{E_2}{\mathcal{E}_2} e^{-i\phi_1} = \frac{1}{2} e^{-i\triangle\phi}, \tag{3.36}$$

$$\triangle\phi = \phi_1 - \phi_2.$$

Now, we're going to calculate the magnitude of the difference between 3.35 and 3.36, and because these are complex number, it is given by

$$\left(\frac{E_1}{\mathcal{E}_1} e^{-i\phi_2} - \frac{E_2}{\mathcal{E}_2} e^{-i\phi_1}\right)\left(\frac{E_1}{\mathcal{E}_1} e^{-i\phi_2} - \frac{E_2}{\mathcal{E}_2} e^{-i\phi_1}\right)^* = \frac{1}{4}\left(e^{i\triangle\phi} - e^{-i\triangle\phi}\right)\left(e^{i\triangle\phi} - e^{-i\triangle\phi}\right)^*,$$

[2] We can see a spinor a $1/2$ tensor. From a geometric point of view, we say that a vector $\vec{v}$ is, in fact, and spinor if it needs two complete rotation of 360° to arrive its original position

[3] deterministic, $2D$, monocrhomatic, space-time independent, propagation on vacuum

where $*$ represents the complex conjugate. In the right hand side we can see tha the difference between those two exponentials are the sine of the phase difference $\triangle$ thus

$$\frac{1}{4}\left(e^{i\triangle\phi}-e^{-i\triangle\phi}\right)\left(e^{i\triangle\phi}-e^{-i\triangle\phi}\right)^{*}=\frac{1}{4}\left(2i\sin\triangle\phi\right)\left(-2i\sin\triangle\phi\right)=\sin^{2}\triangle\phi. \tag{3.37}$$

In the left hand side we've got the following

$$\left(\frac{E_1}{\mathcal{E}_1}e^{-i\phi_2}-\frac{E_2}{\mathcal{E}_2}e^{-i\phi_1}\right)\left(\frac{E_1}{\mathcal{E}_1}e^{-i\phi_2}-\frac{E_2}{\mathcal{E}_2}e^{-i\phi_1}\right)^{*}=$$
$$\frac{|E_1|^2}{\mathcal{E}_1^2}-\frac{E_1E_2^*}{\mathcal{E}_1\mathcal{E}_2}e^{i\triangle\phi}-\frac{E_2E_1^*}{\mathcal{E}_2\mathcal{E}_1}e^{-i\triangle\phi}+\frac{|E_2|^2}{\mathcal{E}_2^2},$$

the crossed terms in blue corresponds to the same quantity but one is the complex conjugate of the other, so if we factor the minus sign we'll have the real part of that complex number, so, the left hand side is given by

$$\frac{|E_1|^2}{\mathcal{E}_1^2}+\frac{|E_2|^2}{\mathcal{E}_2^2}-2\frac{Re\left\{E_1E_2^*\right\}}{\mathcal{E}_1\mathcal{E}_2}\cos\triangle\phi. \tag{3.38}$$

Combining the LHS 3.38 and the RHS 3.37 we will have the *polarisation ellipse*

$$\frac{|E_1|^2}{\mathcal{E}_1^2}+\frac{|E_2|^2}{\mathcal{E}_2^2}-2\frac{Re\left\{E_1E_2^*\right\}}{\mathcal{E}_1\mathcal{E}_2}\cos\triangle\phi=\sin^2\triangle\phi. \tag{3.39}$$

Then we've shown the electric field lives on an geometric space on $\mathcal{C}^2$ which is a real rotated elllipse, before going further we should notice something.

The Jones spinor 3.34 is not an unique thing of the electric field, also describes the magnetic field under these assumptions but also **describes any complex field with 2 degrees of freedom**, for example the gravitational waves have got a polarisation ellipse, $2D$ phonons have polarisation ellipse, and so on.

From a mathematical point of view the origin of the ellipse is obvious, the Jones spinor 3.34 corresponds to two unit complex circles which rotate asincronically, so the general figure is an ellipse not a circle, but, there is a further a more fundamental question, and it is **why does the nature choose an elllipse as the fundamental description of the orientation of the electric field** The short answer is: $No, the nature does not choose that$, the long answer is going to be shown in the theory of coherence chapter.

How to build the polarisation ellipse: Phasors

Given any Jones spinor 3.34 how can we construct the polarisation ellipse? To answer this we're going to use phasors which is a formal word for a polar form of a complex number.

Because we've got a $2D$ vector we've got two components, two polar complex numbers $\mathcal{E}_m e^{i\phi_m}$ with $m = 1, 2$ so we can describe these two phasors as two circunferences as is shown in 3.2. The

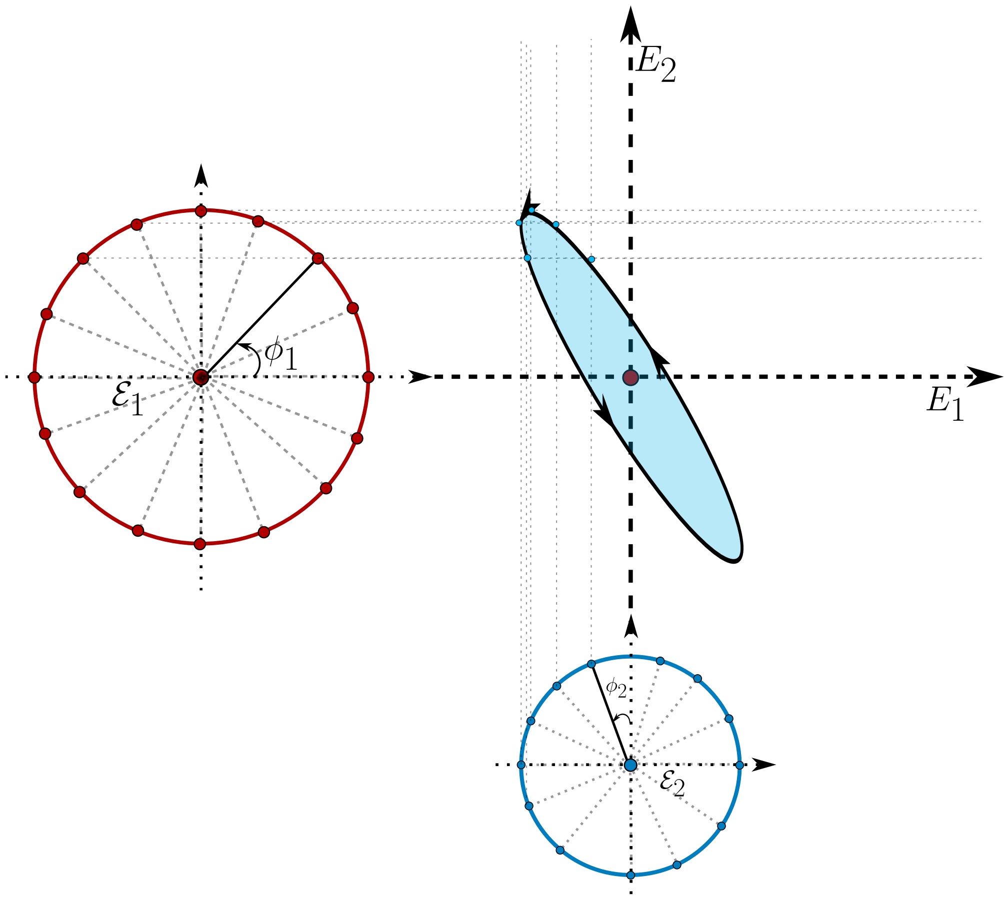


Figure 3.2: The polarisation ellipse can be reconstructred taking all the possible rotations of the two phasors (which corresponds to the compoenentes of the Jones spinor).

x component or $1-st$ component is a circunference with radius $\mathcal{E}_1$ and the initial point on this circunference is located at an angle ϕ_1 respect to the $x-axis$ (shown as the red circunference in the figure 3.2). The second phasor, corresponding to the y component of the jones spinor is another circunference but the radius is $\mathcal{E}_2$ and the initial point is located at an angle of ϕ_2 but respect to the $y-axis$.

To construct the polarisation ellipse, we start with the two initial points an draw two straight lines, for the horizontal phasor (the red one) the lines are horizontal, and for the vertical phasor (the blue one) the lines are vertical. In the intersection of the two rects we draw a point, and we desplace both initial points by the same angle, lets call it $\delta\phi$ until we arrive at the initial point. The set of all intersection points draw the polarisation ellipse, which their semi-axes are going to be $\mathcal{E}_1, \mathcal{E}_2$ and the orientation is going to depends on $\triangle\phi$ if $\triangle\phi = \frac{\pi}{2} + n\pi$ with $n \in \mathbb{Z}$ the ellipse isn't going to be orientated also $\triangle\phi$ conditionates if the electric field is going to rotate clockwise or counterclockwise.

How to build the polarisation ellipse: Ellipse angles

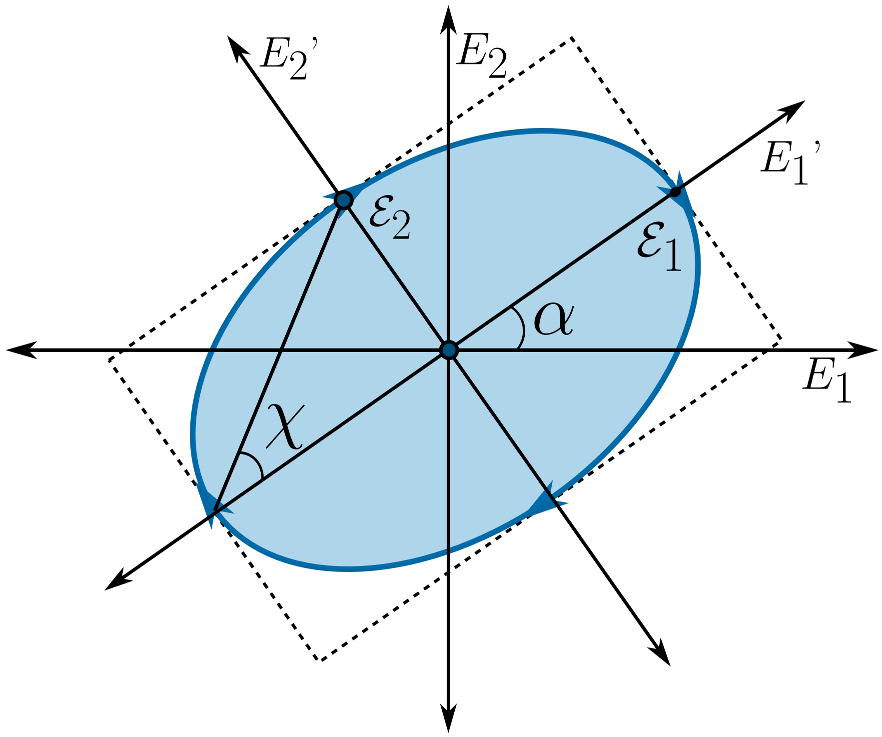


Figure 3.3: The general form of the polarisation ellipse can be described with only two angles; the orientation angle α and the elipticity angle χ.

It is not very comfortable to work with phasor and even we can't see the ellipse inmediately with the Jones spinor, so we would like a more visual or efficient way to interpretate the polarisation. First we must notice that the electric field amplitudes only change the size of the ellipse, but the relative phases ϕ_1, ϕ_2 are the ones who determine the form of the ellipse, so we can work out with a Jones spinor as an unit vector, this means $\vec{e} \rightarrow \hat{e}$ such that $\hat{e} \cdot \hat{e} = 1$.

The general polarisation ellipse can be seen in 3.3 which can be fully described using just two angles α and χ.

Definition 15 *(Characteristic polarisation angles): We define the two characteristic polarisation angles as the orientation angle $-\pi \leq \alpha \leq \pi$ which describes how orientated is the ellipse respect to the horizontal axis, and the elipticity angle $-\pi/2 \leq \chi \leq \pi/2$ which describes completily the form of the ellipse.*

How can we translate the Jones spinor with characteristic angles? To do this, lets remind that without any loss of generality we can say the Jones spinor is unitary ($\hat{e} \cdot \hat{e} = 1$) and lets

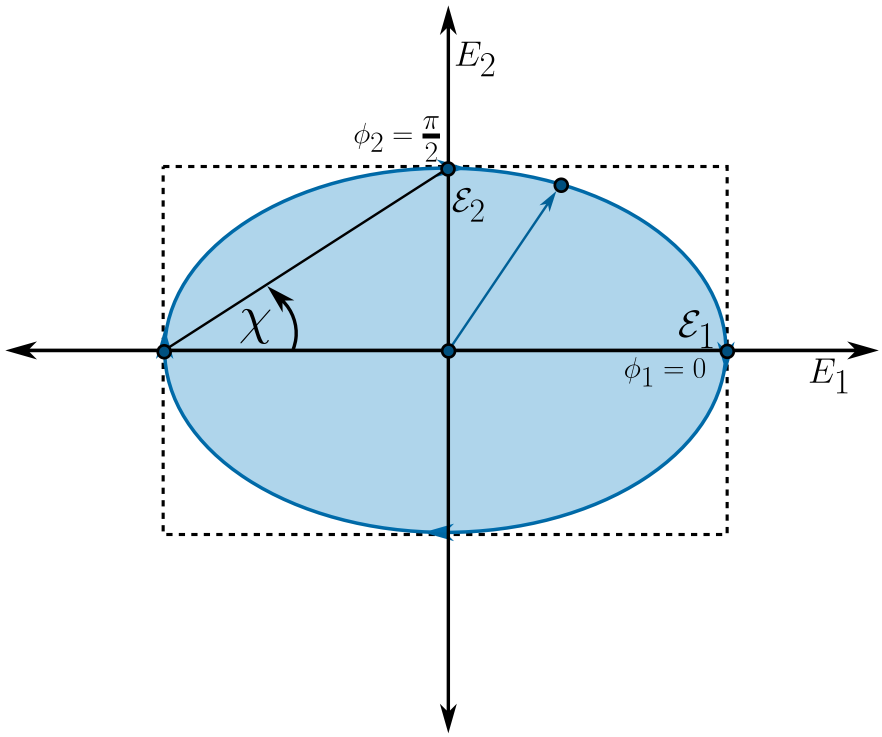


Figure 3.4: The polarisation ellipse is completely unrotated due to the relative phases between the componentes of the electric field.

consider the following phases $\phi_1 = 0$ and $\phi_2 = \pi/2$ and we'll have a polarization elllipse as in shown in 3.4.

As we can see, the polarization vector can be describe using a polar parametrization with the angle χ, so the Jones spinor becomes

$$\hat{e}(\chi) = \begin{bmatrix} \cos\chi \\ i\sin\chi \end{bmatrix}, \tag{3.40}$$

the i term comes from the phase $\phi_2 = \pi/2$ where $\exp(\pi/2) = i$. Now, to construct the full polarization states shown in 3.3 we only have to rotate the Jones spinor 3.40 an angle α, so, the polarisation states is given by

$$\hat{e}(\alpha,\chi) = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} \begin{bmatrix} \cos\chi \\ i\sin\chi \end{bmatrix},$$

$$\hat{e}(\alpha,\chi) = \begin{bmatrix} \cos\alpha\cos\chi - i\sin\alpha\sin\chi \\ \sin\alpha\cos\chi + i\cos\alpha\sin\chi \end{bmatrix}$$

Definition 16 *(Polarisation convention): From now on in this lectures whenever we're working with polarisation states in terms of the angles* $(\alpha.\chi)$ *instead of using* $\hat{e}(\alpha,\chi)$ *we're going to use the ket* *notation, therefore* $\hat{e}(\alpha, \chi) \longmapsto |\alpha; \chi\rangle$ *where we use the semicolon* ; *instead of the comma* , *because the angles are two parameters and does not represente a two level ket* ($|\alpha, \chi\rangle$) *(however polarisation is a two level system).*

Having on count this convention, we've got that the general description of a polarisation state in terms of the characteristic angles is given by

$$|\alpha; \chi\rangle = \begin{bmatrix} \cos\alpha\cos\chi - i\sin\alpha\sin\chi \\ \sin\alpha\cos\chi + i\cos\alpha\sin\chi \end{bmatrix}. \tag{3.41}$$

Because the polarisation is a $2D$ vector in this formulation it can be written as the superposition of, at leat, two linearly independent polarisation states. There is a convention, whenever we talk about polarisation we always write it as the superposition of two ortognal polarisation states, so, how can we build this orthogonal states? We have got two answers, using the orientation angle or the elipticity angle. Using the orientation angle is the most obvious and inmediate way, because we only have to put the two ellipses orthogonal, this means taking the state $|\alpha; \chi\rangle$ and rotate it like $|\alpha + \frac{\pi}{2}; \chi\rangle$.

To think about orthogonal states using elipticity we only have got to think about that given any two orthogonal polarisation states $|0; \chi\rangle$, $|0; \chi'\rangle$ the following two states $|\alpha; \chi\rangle$ and $|0; \chi'\rangle$ are also orthogonal because the rotation transformation preservers the relative angle between vectors. Having this on count, lets suppose that $\chi' = \chi + \theta$ and by orthogonality lets calculate the value of the angle θ that guarantees the orthogonality between polarisation states. Lets calculate the dot product

$$\begin{aligned}
&\langle 0; \chi | 0; \chi + \theta\rangle = [\cos\chi, -i\sin\chi] \begin{bmatrix} cos(\chi+\theta) \\ i\sin(\chi+\theta) \end{bmatrix} = 0, \\
&\cos\chi\cos(\chi+\theta) + \sin\chi\sin(\chi+\theta) = 0, \\
&\cos(\chi + \theta - \chi) = 0, \\
&\cos\theta = 0 \longrightarrow \theta = (2n-1)\frac{\pi}{2} \quad n \in \mathbb{Z},
\end{aligned}$$

where we have used the cosine of the sum of two angles, so for the cosine of θ to vanish it must ve an odd multiple of $pi/2$, without any loss of generality lets take $n = 1$, so $\theta = \pi/2$ and the orthogonal state can be obtained just taking a displacemente of $\pi/2$ on the elipticity, then any polarization state can be writeen as

$$|\alpha; \chi\rangle = c_1 |\alpha'; \chi'\rangle + c_2 \left|\alpha'; \chi' + \frac{\pi}{2}\right\rangle. \tag{3.42}$$

The canonical polarisation states: horizontal, vertical, diagonal and circular

Equation 3.41 already contains the whole catalogue of polarisation states, so it is worth stopping here and finding explicitly the handful of states that we are going to meet again and again in the laboratory: the two linear states aligned with the coordinate axes, the two diagonal linear states, and the two circular states. For every one of them we want three things: the amplitudes and phases

$\mathcal{E}_1, \mathcal{E}_2, \phi_1, \phi_2$ of the original spinor 3.34, normalised so that $\mathcal{E}_1^2 + \mathcal{E}_2^2 = 1$, and the pair of characteristic angles (α, χ) that reproduce that spinor in 3.41.

Since $|\alpha; \chi\rangle$ has got a real first term in its combination and a purely imaginary second term whenever we compare it componentwise with a target spinor $\vec{e} = [e_1, e_2]^\mathsf{T}$, it is convenient to split 3.41 into its real and imaginary parts once and for all,

$$\cos\alpha\cos\chi = \mathrm{Re}\{e_1\}, \quad -\sin\alpha\sin\chi = \mathrm{Im}\{e_1\}, \quad \sin\alpha\cos\chi = \mathrm{Re}\{e_2\}, \quad \cos\alpha\sin\chi = \mathrm{Im}\{e_2\}. \tag{3.43}$$

Every state below is obtained by feeding a particular $\vec{e}$ into the system 3.43 and solving for α, χ inside their allowed ranges $-\pi \le \alpha \le \pi$ and $-\pi/2 \le \chi \le \pi/2$.

Horizontal linear polarisation. The field only oscillates along the x axis, so the second component of the spinor must vanish, $\mathcal{E}_2 = 0, \mathcal{E}_1 = 1$. A common phase multiplying the whole spinor has got no physical meaning, since it is only a redefinition of the origin of time, so we are free to set $\phi_1 = 0$; the phase ϕ_2 plays no role at all because it multiplies a null amplitude. The normalised spinor is

$$\vec{e}_H = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \phi_1 = 0, \quad \mathcal{E}_2 = 0. \tag{3.44}$$

Plugging $e_1 = 1, e_2 = 0$ into 3.43 gives $\cos\alpha\cos\chi = 1$ together with $\sin\alpha\sin\chi = \sin\alpha\cos\chi = \cos\alpha\sin\chi = 0$, and because $\cos\alpha\cos\chi = 1$ is the maximum value that a product of two cosines can reach, it forces $\cos\alpha = \cos\chi = 1$, that is

$$\boxed{\alpha = 0, \qquad \chi = 0} \qquad \Longrightarrow \qquad |0; 0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \tag{3.45}$$

Vertical linear polarisation. By the same argument, now the field only oscillates along y, so $\mathcal{E}_1 = 0, \mathcal{E}_2 = 1$ and we set $\phi_2 = 0$ (again, ϕ_1 is irrelevant),

$$\vec{e}_V = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \phi_2 = 0, \quad \mathcal{E}_1 = 0. \tag{3.46}$$

Feeding $e_1 = 0, e_2 = 1$ into 3.43 gives $\sin\alpha\cos\chi = 1$, which forces $\sin\alpha = \cos\chi = 1$, so

$$\boxed{\alpha = \frac{\pi}{2}, \qquad \chi = 0} \qquad \Longrightarrow \qquad |\pi/2; 0\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{3.47}$$

Diagonal linear polarisation, at $\pi/4$ from the horizontal. Now both components carry the same amplitude, $\mathcal{E}_1 = \mathcal{E}_2 = 1/\sqrt{2}$, and, this being the new ingredient, they also share the same phase, $\phi_1 = \phi_2 = 0$, so that $\triangle\phi = \phi_1 - \phi_2 = 0$ and

$$\vec{e}_D = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \phi_1 = \phi_2 = 0. \tag{3.48}$$

Here 3.43 reads $\cos\alpha\cos\chi = \sin\alpha\cos\chi = 1/\sqrt{2}$ and $\sin\alpha\sin\chi = \cos\alpha\sin\chi = 0$. If $\sin\chi$ were different from zero the last two equations would force $\sin\alpha = \cos\alpha = 0$ at once, which is impossible, so we must have $\chi = 0$; the first two equations then reduce to $\cos\alpha = \sin\alpha = 1/\sqrt{2}$, giving

$$\boxed{\alpha = \frac{\pi}{4}, \qquad \chi = 0}. \tag{3.49}$$

Antidiagonal linear polarisation, at $-\pi/4$ from the horizontal. The amplitudes are the same as in the diagonal case, but now the two components oscillate in opposition, $\phi_1 = 0, \phi_2 = \pi$, so that $e^{i\phi_2} = -1$, $\triangle\phi = -\pi$, and

$$\vec{e}_A = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}, \qquad \phi_1 = 0, \quad \phi_2 = \pi. \tag{3.50}$$

The same reasoning used above forces again $\chi = 0$, and now $\cos\alpha = 1/\sqrt{2}, \sin\alpha = -1/\sqrt{2}$, so

$$\boxed{\alpha = -\frac{\pi}{4}, \qquad \chi = 0}. \tag{3.51}$$

It is worth pausing on the four linear states 3.45, 3.47, 3.49 and 3.51: all of them share $\chi = 0$, and this is not a coincidence, since the ellipticity angle measures how far the ellipse has departed from a straight segment, and every linear state is nothing but a degenerate ellipse of zero ellipticity. The orientation angle α is the only one that changes from state to state, and it does exactly what its name promises, it tells us the tilt of that segment with respect to the x axis.

Circular polarisations. We now ask for the opposite extreme, the states in which the two orthogonal oscillations have got the same amplitude, $\mathcal{E}_1 = \mathcal{E}_2 = 1/\sqrt{2}$, but are shifted by a quarter of a cycle rather than being in phase or in opposition, $\phi_1 = 0, \phi_2 = \mp\pi/2$, so that $e^{i\phi_2} = \pm i$, $\triangle\phi = \pm\pi/2$, and

$$\vec{e}_\pm = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\\pm i\end{bmatrix}, \qquad \phi_1 = 0, \quad \phi_2 = \mp\frac{\pi}{2}. \tag{3.52}$$

Plugging $e_1 = 1/\sqrt{2}, e_2 = \pm i/\sqrt{2}$ into 3.43, the real part of the second line, $\sin\alpha\cos\chi = 0$, together with the imaginary part of the first line, $\sin\alpha\sin\chi = 0$, can only be satisfied inside the allowed ranges by $\alpha = 0$ (the case $\cos\chi = 0$ is discarded because it would force the left hand side of $\cos\alpha\cos\chi = 1/\sqrt{2}$ to vanish). With $\alpha = 0$ the remaining equations collapse to $\cos\chi = 1/\sqrt{2}$ and $\sin\chi = \pm 1/\sqrt{2}$, so

$$\boxed{\alpha = 0, \qquad \chi = \pm\frac{\pi}{4}} \qquad \Longrightarrow \qquad |0; \pm\pi/4\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\\pm i\end{bmatrix}. \tag{3.53}$$

Notice that the orientation angle α is completely undetermined by the shape of the state, and indeed it must be so, a circle has got no preferred direction to be tilted towards, so setting $\alpha = 0$ is only a matter of convenience. What really distinguishes the two circular states is the sign of χ, and this sign is directly inherited from the sign of $\triangle\phi = \phi_1 - \phi_2$, exactly as anticipated when we built the ellipse from the two phasors: a positive $\triangle\phi$ produces $\chi < 0$ and a negative $\triangle\phi$ produces $\chi > 0$. To see the rotation itself, we restore the time dependence at a fixed point in space and take the real part of the spinor 3.52,

$$E_1(t) = \frac{1}{\sqrt{2}}\cos(\omega t), \qquad E_2(t) = \frac{1}{\sqrt{2}}\cos\left(\omega t \mp \frac{\pi}{2}\right) = \pm\frac{1}{\sqrt{2}}\sin(\omega t),$$

so that at $\omega t = 0$ the vector (E_1, E_2) points along $+x$, and an instant later it has acquired a small positive component along $+y$ for the upper sign and along $-y$ for the lower sign. In other words, the state with $\chi = +\pi/4$ traces a counterclockwise circle and the state with $\chi = -\pi/4$ traces a clockwise circle in the x-y plane as time goes on. Whether counterclockwise is called left or right circular polarisation depends on whether the observer is defined to look along the propagation direction or against it, a convention we postpone fixing until the Poincaré sphere is introduced; for the moment we simply keep the two states labelled by the unambiguous quantity $\chi = \pm\pi/4$, the largest ellipticity angle allowed, exactly as $\chi = 0$ was the smallest.

Together, the six states 3.45, 3.47, 3.49, 3.51 and 3.53 already reveal the whole geometric role of the pair (α, χ): α only rotates a state that has got some elongation, while χ interpolates continuously between a straight segment at $\chi = 0$ and a circle at $|\chi| = \pi/4$, with its sign fixing the sense in which the tip of the electric field turns; every other value of χ in between will describe, as we should expect by now, a genuine ellipse of intermediate eccentricity.

3.5.2 Time evolution of the polarisations: Stokes parameters and the Poincaré sphere

The mathematics developed so far describes a beam of light whose polarisation is perfectly defined at every instant, the situation formalised by the Jones spinor 3.34. Real light is seldom so cooperative: sunlight, the light of an incandescent lamp, or even laser light after it has bounced once off a rough surface, is generally a superposition of many independent wavetrains whose relative phase drifts randomly and far too fast to be tracked. Nothing in the deterministic polarisation ellipse built in the previous subsection can describe such a beam, and it was precisely this gap that George Gabriel Stokes set out to close in 1852, in a paper read before the Cambridge Philosophical Society and modestly titled *On the composition and resolution of streams of polarized light from different sources* (Stokes, 1852). Stokes, who a few years later became Lucasian Professor of Mathematics at Cambridge (the same chair later held by Dirac and, more recently, by Hawking), and whose name today is more readily attached to the Navier-Stokes equations of fluid motion, to Stokes' theorem of vector calculus and to the Stokes shift of fluorescence, proposed four real quantities, built from simple combinations of intensities measured behind a polariser and a quarter wave plate held at a handful of fixed orientations, that could characterise the polarisation of any beam whatsoever, fully polarised, fully unpolarised, or anything in between. Unlike the Jones spinor 3.34, which needs a single coherent phase to be written down at all, Stokes' four numbers are honest time averages, and they remain perfectly well defined even when no such phase exists.

It is a curious fact that this construction, so evidently useful and so close in spirit to the density matrix that quantum mechanics would introduce some seventy years later for exactly the same reason (to describe mixtures that a single wavefunction cannot capture), attracted almost no attention for the rest of the nineteenth century. Stokes himself never returned to it: he went on to serve as secretary and later as president of the Royal Society, absorbed by the administrative life such offices demanded, and his scientific attention drifted to fluorescence, to the wave theory of light in the aether, and above all to hydrodynamics. Nor did the wider community pick it up. Optics in the second half of the nineteenth century was still occupied with problems that were, by construction, coherent and deterministic, interference, diffraction, the propagation of well defined wavefronts, and the statistical treatment of an ensemble of independent wavetrains had no natural home in that programme. Even the form in which Stokes left his result, four numbers without an accompanying algebra of matrices to act on them, made it look more like a table of measurements than like a theory. His four coefficients, not yet known by the name *Stokes parameters* that later authors would give them, waited in the pages of the *Transactions of the Cambridge Philosophical Society* for the better part of a century.

The first person to give this problem a genuinely new geometric life, although without building explicitly on Stokes' algebra, was Henri Poincaré. Poincaré (1854 to 1912) is often remembered as the last of the great universalist mathematicians, equally at home in celestial mechanics, where his study of the three body problem contained, without his fully realising it, the first seeds of chaos theory, in topology, where the conjecture that bears his name would only be settled a century later, and in the foundations of mathematical physics, where his work on the Lorentz group anticipated several of the kinematic ingredients of special relativity. In 1892, in the second volume of his lecture course *Théorie Mathématique de la Lumière*, Poincaré proposed representing every state of polarisation as a single point on the surface of a sphere (Poincaré, 1892). He built this sphere directly from the orientation and ellipticity angles of the polarisation ellipse, the very angles α and χ of the ket 3.41, so that the poles of the sphere fall on the two circular states, its equator is entirely covered by the linear states, exactly as anticipated in the historical overview at the beginning of this chapter, and every other point corresponds to one particular ellipse. It was only later, once the sphere and Stokes' four numbers were laid side by side, that it became clear the coordinates Poincaré had been plotting are, up to normalisation by the total intensity, precisely the three last Stokes parameters: two men, working four decades apart with entirely different tools, one through operational measurements of intensity and the other through the geometry of spherical trigonometry, had arrived at the same object from opposite directions. The sphere has carried Poincaré's name ever since, even though, as we shall see, its natural coordinates are Stokes'.

The parameters finally found the audience they deserved when a new generation of physicists needed to follow partially polarised light through processes that destroy coherence outright, above all multiple scattering in planetary atmospheres, in colloidal suspensions, and in the interstellar medium, a regime where the Jones calculus (Jones, 1941), powerful as it is, simply cannot be used, since it presupposes a single coherent amplitude that scattering does not leave behind. Soleillet, in 1929, and Francis Perrin, in a 1942 study of light scattered by opalescent media (Perrin, 1942), independently rebuilt essentially the same four component, matrix based description that the problem demanded, at first without fully recognising it as Stokes' own. The decisive act of recognition, and the paper usually credited with rescuing Stokes' parameters from a century of obscurity, restoring his original notation and giving the whole subject the name by which it is still taught, was published by Walter McMaster in the *American Journal of Physics* in 1954 (McMaster, 1954). Subrahmanyan Chandrasekhar's monumental treatise on radiative transfer, appearing in the same decade, adopted the Stokes vector as the natural object to propagate through a scattering atmosphere and cemented its place in astrophysics (Chandrasekhar, 1950). From ellipsometry and remote sensing to the polarisation maps of the cosmic microwave background and to modern quantum optics, where the Stokes parameters reappear as the expectation values of the photon's Pauli operators, the formalism that Stokes wrote down in 1852, that the nineteenth century forgot, and that Poincaré unknowingly rediscovered forty years later in the language of spheres, is today one of the most used tools in the whole of optics.

Let us now build these four parameters explicitly, starting from nothing more than the polarisation ellipse 3.39 itself, and see why Stokes was forced to define them as time averages rather than simply reading them off the field.

The ellipse 3.39 is a fixed curve, but the physical field is not a curve, it is a point running around that curve once every optical period. To say this with equations, the real components E_1, E_2 that satisfy 3.39 at every instant are themselves functions of time,

$$E_1(t) = \mathcal{E}_1 \cos(\omega t - \phi_1), \qquad E_2(t) = \mathcal{E}_2 \cos(\omega t - \phi_2), \tag{3.54}$$

the real part of the electric field 3.34 at a fixed point in space (its constant spatial phase $\vec{k} \cdot \vec{r}$ has simply been absorbed into ϕ_1, ϕ_2). Therefore the expression for the polarisation ellipse is given by

$$\frac{|E_1(t)|^2}{\mathcal{E}_1^2} + \frac{|E_2(t)|^2}{\mathcal{E}_2^2} - 2\frac{Re\left\{E_1(t)E_2^*(t)\right\}}{\mathcal{E}_1\mathcal{E}_2} \cos \triangle\phi(t) = \sin^2 \triangle\phi(t) \tag{3.55}$$

One can check directly that eliminating t between the two equations 3.54, exactly as was done with the phasors 3.35 and 3.36, returns 3.39; the ellipse is nothing but the trace, frozen in space, left behind by the point $(E_1(t), E_2(t))$ as it oscillates.

A single period of visible light lasts a few femtoseconds, $T = \lambda/C \sim 2 \times 10^{-15}$ s for green light, while even the fastest photodetector available in a laboratory needs picoseconds to respond, and the human eye needs tens of milliseconds. In either case an enormous number of optical cycles, somewhere between 10^6 and 10^{13} of them depending on the instrument, go by before any detector can register a single reading; the point $(E_1(t), E_2(t))$ has swept around the ellipse an enormous number of times before any instrument manages to read anything at all. No detector, in other words, can ever measure the instantaneous field $E(t)$ itself, only some average of it taken over a great many periods. This is not a limitation that better technology could one day remove, it is what intensity means: the energy flux carried by a wave is proportional to the square of the field, so what a detector physically integrates is always $\left\langle E^2 \right\rangle$, never E. This is exactly the operation Stokes performed, building his four parameters out of quadratic, time averaged combinations of the oscillating field rather than out of the ellipse itself, and it is why they, unlike the Jones spinor, remain meaningful even when no single coherent phase exists to write down. Over one period $T = 2\pi/\omega$, define the time average of any function $f(t)$ as $\langle f \rangle \equiv \frac{1}{T}\int_0^T f(t)\, dt$. The first two averages we need are the mean square amplitudes,

$$\langle E_1^2 \rangle = \frac{\mathcal{E}_1^2}{T}\int_0^T \cos^2(\omega t - \phi_1)\, dt = \frac{\mathcal{E}_1^2}{2T}\int_0^T \left[1 + \cos(2\omega t - 2\phi_1)\right] dt = \frac{\mathcal{E}_1^2}{2T}\left[T + 0\right] = \frac{\mathcal{E}_1^2}{2},$$

where we used $\cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$ and the fact that $\cos(2\omega t - 2\phi_1)$ completes exactly two full oscillations as t runs from 0 to T, so its integral over that stretch vanishes; the identical computation gives $\langle E_2^2\rangle = \mathcal{E}_2^2/2$.

The remaining piece of information lives in how E_1 and E_2 oscillate together, and for that we need the cross average

$$\begin{aligned}\langle E_1E_2\rangle &= \frac{\mathcal{E}_1\mathcal{E}_2}{T}\int_0^T \cos(\omega t-\phi_1)\cos(\omega t-\phi_2)\,dt\\ &= \frac{\mathcal{E}_1\mathcal{E}_2}{2T}\int_0^T [\cos(\phi_1-\phi_2)+\cos(2\omega t-\phi_1-\phi_2)]\,dt = \frac{\mathcal{E}_1\mathcal{E}_2}{2}\cos\triangle\phi,\end{aligned}$$

where we used $\cos A\cos B = \frac{1}{2}[\cos(A-B)+\cos(A+B)]$, the phase difference $\triangle\phi = \phi_1-\phi_2$ already defined near 3.35, and, once again, that the piece oscillating at 2ω averages to zero over a full period.

This single average, however, only ever gives us $\cos\triangle\phi$, and a beam with $\triangle\phi$ and one with $-\triangle\phi$ (opposite handedness, exactly as when we built the circular states 3.53) would be indistinguishable from $\langle E_1E_2\rangle$ alone. To tell them apart we need a second, independent combination, and the natural one to try is E_2 sampled a quarter of a period earlier,

$$\tilde{E}_2(t) \equiv E_2\left(t-\frac{T}{4}\right) = \mathcal{E}_2\cos\left(\omega t-\frac{\pi}{2}-\phi_2\right) = \mathcal{E}_2\sin(\omega t-\phi_2),$$

using $\cos(x-\pi/2)=\sin x$ and $\omega T/4 = \pi/2$; physically, this quarter period delay is exactly what a birefringent plate cut to a quarter of a wavelength does to whichever component crosses it, so $\tilde{E}_2$ is not a mere mathematical trick but a genuinely measurable field. Its cross average with E_1 is

$$\begin{aligned}\langle E_1\tilde{E}_2\rangle &= \frac{\mathcal{E}_1\mathcal{E}_2}{T}\int_0^T \cos(\omega t-\phi_1)\sin(\omega t-\phi_2)\,dt\\ &= \frac{\mathcal{E}_1\mathcal{E}_2}{2T}\int_0^T [\sin(2\omega t-\phi_1-\phi_2)+\sin\triangle\phi]\,dt = \frac{\mathcal{E}_1\mathcal{E}_2}{2}\sin\triangle\phi,\end{aligned}$$

where now we used $\cos A\sin B = \frac{1}{2}[\sin(A+B)-\sin(A-B)]$ with $A-B = \phi_2-\phi_1 = -\triangle\phi$, together with the vanishing, as always, of the term oscillating at 2ω.

Having these time averages the time-dependent polarisation ellipse 3.55 becomes

$$\frac{\mathcal{E}_1^2}{2\mathcal{E}_1^2}+\frac{\mathcal{E}_2^2}{2\mathcal{E}_2^2}-\frac{2\mathcal{E}_1\mathcal{E}_2}{2\mathcal{E}_1\mathcal{E}_2}\cos^2\triangle\phi = \sin^2\triangle\phi,$$

multiplying both side by $4\mathcal{E}_1^2\mathcal{E}_2^2$ we'll have

$$2\mathcal{E}_1^2\mathcal{E}_2^2+2\mathcal{E}_1^2\mathcal{E}_2^2-(2\mathcal{E}_1\mathcal{E}_2\cos\triangle\phi)^2 = (2\mathcal{E}_1\mathcal{E}_2\sin\triangle\phi)^2\,,$$

by mathematical convenience we're going to add zero as $0 = \mathcal{E}_1^4+\mathcal{E}_2^4-\mathcal{E}_1^4-\mathcal{E}_2^4$ as follows

$$\mathcal{E}_1^4 + \mathcal{E}_2^4 + 2\mathcal{E}_1^2\mathcal{E}_2^2 - \mathcal{E}_1^4 - \mathcal{E}_2^4 + 2\mathcal{E}_1^2\mathcal{E}_2^2 - (2\mathcal{E}_1\mathcal{E}_2\cos\triangle\phi)^2 = (2\mathcal{E}_1\mathcal{E}_2\sin\triangle\phi)^2\,,$$
$$\left(\mathcal{E}_1^2+\mathcal{E}_2^2\right)^2 - \left(\mathcal{E}_1^2-\mathcal{E}_2^2\right)^2 - (2\mathcal{E}_1\mathcal{E}_2\cos\triangle\phi)^2 = (2\mathcal{E}_1\mathcal{E}_2\sin\triangle\phi)^2$$

We finally have everything we need. Stokes defined his four parameters, up to the overall factor of two needed to cancel the one half that every average above carries, as

$$S_0 \equiv 2\langle E_1^2\rangle + 2\langle E_2^2\rangle = \mathcal{E}_1^2 + \mathcal{E}_2^2, \tag{3.56}$$
$$S_1 \equiv 2\langle E_1^2\rangle - 2\langle E_2^2\rangle = \mathcal{E}_1^2 - \mathcal{E}_2^2, \tag{3.57}$$
$$S_2 \equiv 4\langle E_1E_2\rangle = 2\mathcal{E}_1\mathcal{E}_2\cos\triangle\phi, \tag{3.58}$$
$$S_3 \equiv 4\langle E_1\tilde{E}_2\rangle = 2\mathcal{E}_1\mathcal{E}_2\sin\triangle\phi. \tag{3.59}$$

Four numbers built out of only two amplitudes and one phase difference cannot be independent, and indeed squaring and adding 3.57 to 3.59 gives

$$\begin{aligned} S_1^2+S_2^2+S_3^2 &= \left(\mathcal{E}_1^2-\mathcal{E}_2^2\right)^2 + 4\mathcal{E}_1^2\mathcal{E}_2^2\cos^2\triangle\phi + 4\mathcal{E}_1^2\mathcal{E}_2^2\sin^2\triangle\phi \\ &= \mathcal{E}_1^4 - 2\mathcal{E}_1^2\mathcal{E}_2^2 + \mathcal{E}_2^4 + 4\mathcal{E}_1^2\mathcal{E}_2^2\left(\cos^2\triangle\phi+\sin^2\triangle\phi\right) \\ &= \mathcal{E}_1^4 + 2\mathcal{E}_1^2\mathcal{E}_2^2 + \mathcal{E}_2^4 = \left(\mathcal{E}_1^2+\mathcal{E}_2^2\right)^2, \end{aligned}$$

where the whole phase difference $\triangle\phi$ has disappeared through the single identity $\cos^2\triangle\phi+\sin^2\triangle\phi = 1$, leaving, by 3.56,

$$S_1^2+S_2^2+S_3^2 = S_0^2. \tag{3.60}$$

This is the equality Stokes was really after: of the four numbers describing a fully polarised beam, only three are independent, exactly as only three are needed to fix a point on a sphere of given radius, a hint of the Poincaré sphere we will make precise shortly.

Every one of these four combinations is quadratic in the field, and this is precisely what lets us rewrite them, exactly, in terms of the complex components $e_1 \equiv \mathcal{E}_1 e^{i\phi_1}$ and $e_2 \equiv \mathcal{E}_2 e^{i\phi_2}$ of the spinor 3.34, without a single cosine or sine left in sight. Since $e_1e_1^* = \mathcal{E}_1^2$, $e_2e_2^* = \mathcal{E}_2^2$, and $e_1e_2^* = \mathcal{E}_1\mathcal{E}_2e^{i(\phi_1-\phi_2)} = \mathcal{E}_1\mathcal{E}_2\left(\cos\triangle\phi + i\sin\triangle\phi\right)$, comparing real and imaginary parts term by term with 3.58 and 3.59 shows that

$$\mathcal{E}_1^2 \pm \mathcal{E}_2^2 = e_1e_1^* \pm e_2e_2^*, \qquad 2\mathcal{E}_1\mathcal{E}_2\cos\triangle\phi = e_1e_2^* + e_2e_1^*, \qquad 2\mathcal{E}_1\mathcal{E}_2\sin\triangle\phi = i\left(e_1^*e_2 - e_1e_2^*\right),$$

the last equality following from $e_1^*e_2 - e_1e_2^* = \mathcal{E}_1\mathcal{E}_2\left(e^{-i\triangle\phi} - e^{i\triangle\phi}\right) = -2i\,\mathcal{E}_1\mathcal{E}_2\sin\triangle\phi$, so multiplying through by i removes the leftover $i^2 = -1$ and leaves a real number, exactly as a physical intensity must be.

Definition 17 *(Stokes parameters): Given the Jones spinor $\vec{e} = [e_1, e_2]^\mathsf{T}$ of a fully polarised,* *monochromatic beam 3.34, its Stokes parameters are the four real numbers*

$$S_0 = e_1 e_1^* + e_2 e_2^*, \tag{3.61}$$
$$S_1 = e_1 e_1^* - e_2 e_2^*, \tag{3.62}$$
$$S_2 = e_1 e_2^* + e_2 e_1^*, \tag{3.63}$$
$$S_3 = i\left(e_1^* e_2 - e_1 e_2^*\right). \tag{3.64}$$

Each of the four carries its own, immediate physical meaning. $S_0 = \mathcal{E}_1^2 + \mathcal{E}_2^2$ is simply the total intensity of the beam, the same quantity we normalised to one when we wrote the Jones ket 3.41 as a unit vector. $S_1 = \mathcal{E}_1^2 - \mathcal{E}_2^2$ measures how much the beam prefers the horizontal axis over the vertical one, positive for a beam closer to $|0;0\rangle$ and negative for one closer to $|\pi/2;0\rangle$. $S_2 = 2\mathcal{E}_1\mathcal{E}_2 \cos\triangle\phi$ measures the same kind of preference, only rotated by forty five degrees, between the two diagonal states 3.49 and 3.51. And $S_3 = 2\mathcal{E}_1\mathcal{E}_2 \sin\triangle\phi$ measures the preference between the two senses of circulation 3.53.

Together, (S_1, S_2, S_3) are nothing but three Cartesian coordinates. Writing the components of the general ket 3.41 as $e_1 = \cos\alpha\cos\chi - i\sin\alpha\sin\chi$ and $e_2 = \sin\alpha\cos\chi + i\cos\alpha\sin\chi$, for which $S_0 = 1$ since 3.41 is already a unit vector, and substituting into 3.62 to 3.64, a short calculation using $\cos^2\theta - \sin^2\theta = \cos 2\theta$ and $2\sin\theta\cos\theta = \sin 2\theta$ on each term gives

$$S_1 = S_0 \cos 2\alpha \cos 2\chi, \qquad S_2 = S_0 \sin 2\alpha \cos 2\chi, \qquad S_3 = -S_0 \sin 2\chi, \tag{3.65}$$

so that $(S_1, S_2, S_3)/S_0$ is, quite literally, a point on a sphere of unit radius, parametrised by twice the orientation angle and twice the ellipticity angle: this is exactly the Poincaré (see the figure 3.5) sphere of the historical discussion above, and 3.65 is the dictionary between its Cartesian coordinates and the angles α, χ used throughout this chapter.

As a consistency check, squaring and adding the three equalities 3.65 reproduces the identity 3.60 once more, $S_1^2 + S_2^2 + S_3^2 = S_0^2 \cos^2 2\chi \left(\cos^2 2\alpha + \sin^2 2\alpha\right) + S_0^2 \sin^2 2\chi = S_0^2 \left(\cos^2 2\chi + \sin^2 2\chi\right) = S_0^2$, this time because every fully polarised beam corresponds to a single, definite pair of angles α, χ, that is, to a single point sitting exactly on the surface of the Poincaré sphere.

Real light, as we already remarked in the historical discussion, is rarely so cooperative: sunlight, or light that has scattered even once, is better described as a rapidly and randomly fluctuating superposition of many such states, with $\alpha(t)$ and $\chi(t)$ themselves wandering on a timescale far shorter than any detector can follow, so that what actually gets measured is not the Stokes vector of one ellipse but a further time average over many different ones. Averaging a family of unit vectors that point in different directions on the sphere never gives a vector as long as the individual ones were, so for such partially polarised light the equality 3.60 is generally replaced by the inequality $S_1^2 + S_2^2 + S_3^2 < S_0^2$, with the extreme case $S_1 = S_2 = S_3 = 0$ describing completely unpolarised light, a beam with a perfectly well defined intensity but no preferred point on the Poincaré sphere at all. We will not develop this idea further here, since it belongs properly to the theory of coherence taken up in 6, but it is worth keeping in mind already that the four numbers built in this subsection say everything there is to say about polarisation, deterministic or not, while the single ket 3.41 only ever describes the former.

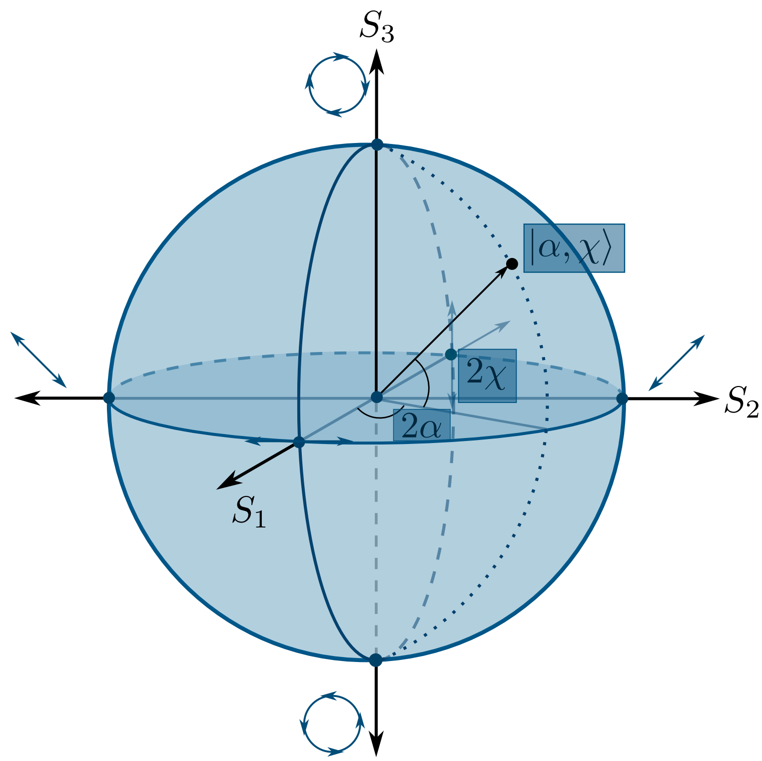


Figure 3.5: In the Poincaré any polarisation state is a point on the surface on the sphere, howerver if the state is partially polarized it would be a pointe inside the poincaré sphere whose distance from the center is $\sqrt{S_1^2 + S_2^2 + S_3^2}$.

In the poincaré sphere 3.5 and the Stokes parameters 3.65 we see an interesting relation, given any state $|\alpha;\chi\rangle$ on the Poincaré sphere the characteristic angles aren't the same, these two are the double, so a full roation on the Jones spinor becomes two full rotations of the Stokes vector $\vec{S} = [S_1, S_2, S_3]^T$. In fact, we've proven in 3.42 that two orthogonal polarisation states are separated an angle of $\pi/2$ but, on the Poincaré sphere this angle is the double, so two orthogonal states are separated π from the other, this means **any two orthogonal polarization states are diametrally opposite on the Poincaré sphere**.

Why the Stokes vector is not a vector

It is customary, and we shall keep the custom, to arrange the four numbers 3.61 to 3.64 into a column and to call the result the *Stokes vector*. The name is convenient and completely standard, but taken literally it is wrong, and the reason is not a pedantic one, since it is the very same doubling of angles that makes the Poincaré sphere work in the first place.

In physics an object does not earn the name of vector by being a list of numbers. It earns it by transforming in a definite way under the symmetry group of the space it is supposed to live in, so that a rotation of the laboratory axes by an angle θ takes its components v_i into $R_{ij}(\theta)v_j$. The triple (S_1, S_2, S_3) fails this test at the first attempt. Turn the laboratory axes by θ about the propagation direction; the orientation of every ellipse is now measured from the new x axis, so $\alpha \to \alpha - \theta$, and 3.65 gives at once

$$S_1' = S_1 \cos 2\theta + S_2 \sin 2\theta, \qquad S_2' = -S_1 \sin 2\theta + S_2 \cos 2\theta, \qquad S_3' = S_3. \tag{3.66}$$

The pair (S_1, S_2) does rotate, but by twice the angle through which the laboratory was turned. The cleanest way to see how fatal this is, is to turn the laboratory by half a turn, $\theta = \pi$. Every polarisation ellipse is centrally symmetric, so it is carried exactly onto itself, no state of light has changed, and every measurable quantity must come back unchanged; and indeed 3.66 returns $S_i' = S_i$, because $2\theta = 2\pi$. A genuine vector of the transverse plane would have come back as $-\vec{v}$ instead. So (S_1, S_2, S_3) is not a vector of the physical space in which the experiment sits, and there is no direction one could point at in the laboratory and call the direction of S_3.

The second failure is algebraic rather than geometric. A vector lives in a vector space, and a vector space is closed under multiplication by any real number, in particular by -1. But $-\vec{S}$ has got $S_0 < 0$, a beam of negative intensity, which is not a state of anything, so no physical Stokes vector possesses an additive inverse. What the physical Stokes vectors form is not a vector space but a convex cone, the region carved out by $S_0 \geq 0$ together with the inequality $S_0^2 \geq S_1^2 + S_2^2 + S_3^2$ discussed above. Even the addition that this cone does admit is not the one a physicist would naively expect: adding two Stokes vectors describes the superposition of two mutually *incoherent* beams, which was precisely Stokes' purpose, and it is emphatically not the coherent superposition of two fields, for which one must add the Jones spinors first and only afterwards form the four averages. The Stokes parameters are quadratic in the field, and quadratic quantities never add the way the field itself does.

If the four numbers are not the components of a vector, then components of what are they? Of an operator. It is enough to collect the very same products that appear in 3.61 to 3.64 into a single array.

Definition 18 *(Polarisation matrix): The polarisation matrix, also called the coherence matrix, of a beam with Jones spinor $\vec{e} = [e_1, e_2]^\mathsf{T}$ is the Hermitian, positive semidefinite matrix*

$$\Phi \equiv \left\langle \vec{e}\,\vec{e}^{\,\dagger} \right\rangle = \begin{bmatrix} \langle e_1 e_1^* \rangle & \langle e_1 e_2^* \rangle \\ \langle e_2 e_1^* \rangle & \langle e_2 e_2^* \rangle \end{bmatrix}, \tag{3.67}$$

the averages being taken over the response time of the detector, exactly as in the construction of the Stokes parameters themselves.

Take now the four Hermitian matrices

$$\sigma_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \sigma_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad \sigma_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \sigma_3 = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}, \tag{3.68}$$

the identity together with the three Pauli matrices, ordered and signed to match the conventions we have been using. Any Hermitian 2×2 matrix can be expanded in this set, and carrying out the four traces term by term against 3.67 reproduces exactly the definitions 3.61 to 3.64,

$$S_\mu = \operatorname{tr}\left(\Phi\sigma_\mu\right), \qquad \Phi = \frac{1}{2}\sum_{\mu=0}^{3} S_\mu\sigma_\mu. \tag{3.69}$$

The S_μ are therefore the coordinates of a Hermitian *operator* in the basis $\{\sigma_\mu\}$, which is a tensorial object living in an internal, abstract space, and not a vector of the laboratory. This reading also explains, in one line, the inequality that partially polarised light forced upon us, because writing 3.69 out gives

$$\Phi = \frac{1}{2}\begin{bmatrix} S_0 + S_1 & S_2 + iS_3 \\ S_2 - iS_3 & S_0 - S_1 \end{bmatrix}, \qquad \det\Phi = \frac{1}{4}\left(S_0^2 - S_1^2 - S_2^2 - S_3^2\right) \geq 0,$$

the determinant being non negative precisely because Φ is positive semidefinite, that is, because no measurement of intensity may ever return a negative number. That single physical demand is the entire content of $S_1^2 + S_2^2 + S_3^2 \leq S_0^2$, with equality, a vanishing determinant and a rank one matrix, describing fully polarised light.

There is, finally, one honest sense in which the word vector can be rescued. The combination $S_0^2 - S_1^2 - S_2^2 - S_3^2$ that has just appeared is exactly a Minkowski interval, with S_0 in the role of the time component and (S_1, S_2, S_3) in that of the spatial one, and the physical states fill the forward light cone of an abstract four dimensional space: fully polarised light sits on the cone itself, where the interval vanishes, partially polarised light strictly inside it, and completely unpolarised light on the time axis, where the three spatial components vanish together. The real matrices that describe the action of optical elements on this four tuple, the Mueller matrices to which we will come later, then turn out to belong to a group closely related to the Lorentz group, an observation already made by McMaster when he rescued the formalism (McMaster, 1954). The Stokes four tuple is a four vector of that abstract space, never a three vector of the laboratory, and keeping the traditional name is harmless as long as we remember which space it is a vector of.

3.5.3 The bridge between Jones and Stokes: the stereographic projection

We now own two different languages for one and the same physical object. The Jones formalism describes a polarisation state with two complex numbers 3.34, and the Stokes formalism describes it with four real ones, 3.61 to 3.64, which for fully polarised light are constrained by 3.60 to live on the surface of the Poincaré sphere 3.5. Two languages that speak about the same thing must be translatable into each other, and the translation here is not a mere change of symbols: it is a piece of geometry, and a very old and very beautiful one, the stereographic projection. What follows is a deliberately phenomenological tour of that bridge, close in spirit to the way Azzam and Bashara present it in their treatise on polarised light (Azzam and Bashara, 1977), where the reader will also find all the fine print we happily skip here.

What the stereographic projection is

Stereographic projection is the oldest recipe known for drawing a sphere on a flat sheet of paper. Take a sphere resting on a plane, single out the point N of the sphere diametrically opposite to the point of contact, and call it the centre of projection. To find where a point P of the sphere lands on the paper, draw the straight line joining N to P and follow it until it pierces the plane; the piercing point Q is the image of P. That is the whole construction, and figure 3.6 shows it at a glance.

Definition 19 *(Stereographic projection): Let S^2 be a sphere resting on a plane $\mathbb{R}^2$, and let $N \in S^2$ be the point diametrically opposite to the point of contact. The stereographic projection from N assigns to every point $P \in S^2$, with $P \neq N$, the unique point $Q \in \mathbb{R}^2$ at which the straight line through N and P meets the plane. The map is one to one, it sends circles drawn on the sphere to circles or straight lines drawn on the plane, and it preserves the angle at which any two curves cross. The centre N alone has no image, so the plane is completed with a single extra point, the point at infinity, declared to be the image of N.*

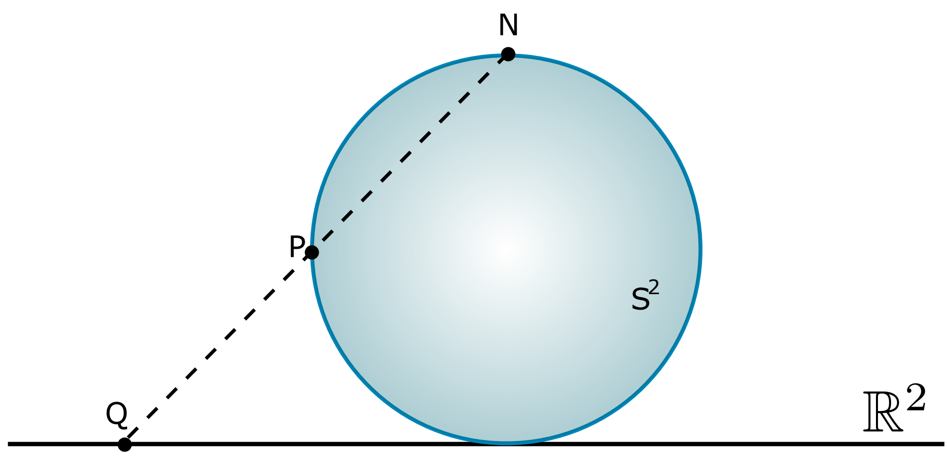


Figure 3.6: The stereographic projection. A ray leaving the centre of projection N and passing through a point P of the sphere S^2 meets the plane $\mathbb{R}^2$ at a single point Q. Points near the contact point fall near the origin, and points near N fly off towards infinity.

The last clause of the definition is the one that carries all the physics. The plane by itself is not enough to hold the whole sphere, since one point of the sphere always escapes; but as soon as we agree that all the ways of running off to infinity on the plane are one and the same destination, the accounting becomes exact and the sphere and the completed plane become the same object seen twice. This completed plane, $\mathbb{C} \cup \{\infty\}$, is what mathematicians call the extended complex plane, and the sphere carrying it is what they call the Riemann sphere. The construction is far older than that name: Hipparchus and Ptolemy already used it to draw star charts, and the medieval astrolabe is nothing but a stereographic map of the celestial sphere engraved on a brass disc, which works precisely because the projection turns circles into circles and keeps angles honest.

Why a polarisation state is a single complex number

Nothing in the previous paragraph mentions light, so we must now ask what a sphere and a plane have to do with polarisation. The answer starts with a simple count. The Jones spinor 3.34 carries two complex numbers, that is, four real ones, the two amplitudes $\mathcal{E}_1, \mathcal{E}_2$ and the two phases ϕ_1, ϕ_2; but two of those four say nothing whatsoever about polarisation. The overall size of the spinor is the intensity of the beam, and dimming the light does not change the shape of its ellipse; the overall phase is only a choice of the origin of time, and starting the clock a little earlier does not change that shape either. Multiplying the whole spinor by any complex number therefore leaves the polarisation state untouched, and what survives such a multiplication is only the *ratio* of the two components.

Definition 20 *(Complex polarisation ratio): Given the Jones spinor $\vec{e} = [e_1, e_2]^\mathsf{T}$ of a fully polarised beam 3.34, its complex polarisation ratio is the number*

$$w \equiv \frac{e_2}{e_1} = \frac{\mathcal{E}_2}{\mathcal{E}_1}\, e^{i(\phi_2-\phi_1)} = \frac{\mathcal{E}_2}{\mathcal{E}_1}\, e^{-i\triangle\phi} \;\in\; \mathbb{C}\cup\{\infty\}, \tag{3.70}$$

with the value $w = \infty$ reserved for the case $e_1 = 0$. Two Jones spinors describe the same polarisation state if and only if they share the same w.

Both pieces of w are quantities we have already been using throughout this chapter, only now packed into one symbol. Its modulus $|w| = \mathcal{E}_2/\mathcal{E}_1$ is the ratio of the two amplitudes, which decides how elongated the ellipse is and towards which axis it leans, and its argument $\arg w = -\triangle\phi$ is the phase lag between the two oscillations, which decides how open the ellipse is and in which sense the field turns. Feeding the six canonical states of the previous subsection into 3.70 gives, immediately,

$$w_H = 0, \qquad w_V = \infty, \qquad w_D = 1, \qquad w_A = -1, \qquad w_\pm = \pm i, \tag{3.71}$$

read off from the spinors 3.44, 3.46, 3.48, 3.50 and 3.52. Notice that the list already forces the point at infinity upon us: vertical polarisation has got no finite ratio at all, since its first component vanishes, and it is a perfectly ordinary state of light that no honest map is allowed to leave out. So the natural home of polarisation is not the plane $\mathbb{C}$ but the completed plane $\mathbb{C}\cup\{\infty\}$, and by the previous subsubsection that home is a sphere.

The Poincaré sphere is that sphere

The sphere we have just been handed is not a new one. Rest the Poincaré sphere on the complex plane of w, so that it touches the plane at the origin, and project from the point N on top, as in figure 3.7. Every reading of the resulting picture is a statement about light.

The point of contact carries $w = 0$, and by 3.71 that is horizontal polarisation; the centre of projection is sent to infinity, and that is vertical polarisation. The two states that our coordinate axes single out therefore sit at the two poles of the picture, diametrically opposite each other, exactly as they should, since they are orthogonal states and we already knew that orthogonal states are

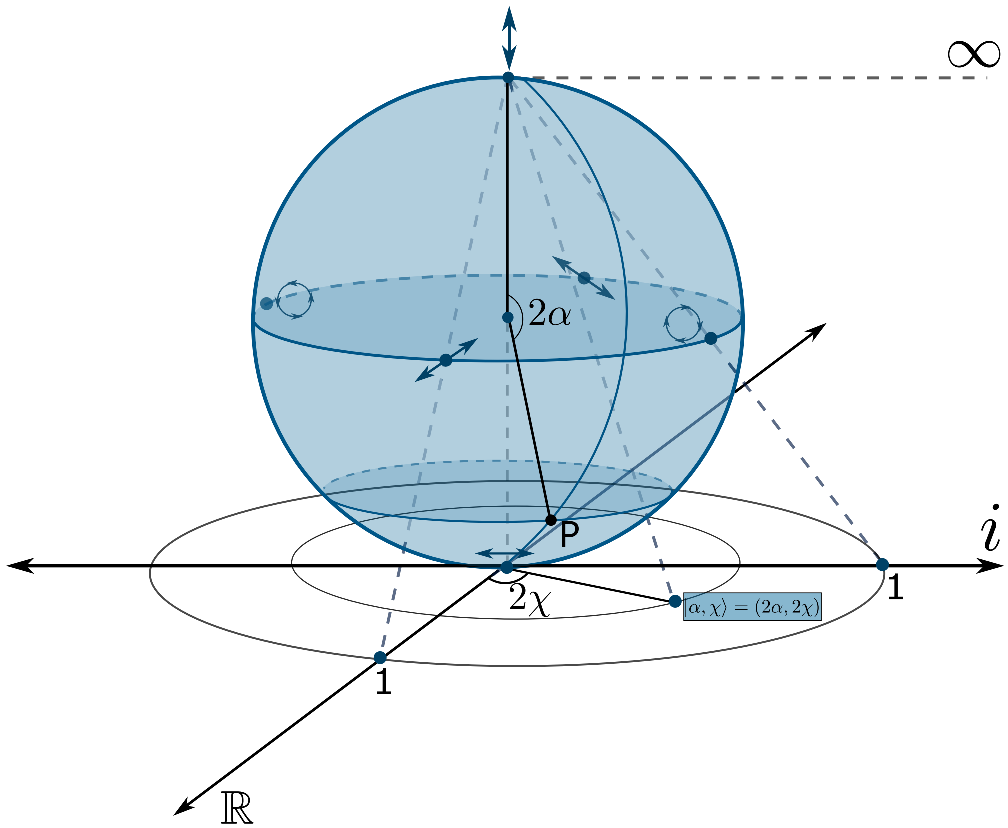


Figure 3.7: The Poincaré sphere resting on the complex plane of the ratio w. Horizontal polarisation sits at the point of contact, $w = 0$, vertical polarisation sits at the centre of projection and is sent to the point at infinity, and the equator, where the two amplitudes are equal, is mapped onto the unit circle $|w| = 1$. The angles 2α and 2χ that locate a state P on the sphere are twice the orientation and ellipticity angles of its ellipse.

antipodal on the Poincaré sphere. Halfway between them lies the equator, the set of states for which the two amplitudes are equal, and it is mapped onto the unit circle $|w| = 1$. On that circle sit the diagonal state at $w = 1$, the antidiagonal at $w = -1$, and the two circular states at $w = \pm i$, each pair again antipodal, again orthogonal. Moving radially outwards from the origin means pouring amplitude into the second component and turning a horizontal state gradually into a vertical one; going around at fixed radius means sliding the phase difference $\triangle\phi$ and opening or closing the ellipse without touching its amplitudes.

The antipodal rule survives the trip down to the plane in a form worth recording. The state orthogonal to $\vec{e} = [e_1, e_2]^\mathsf{T}$ is $[-e_2^*, e_1^*]^\mathsf{T}$, whose ratio is $e_1^*/(-e_2^*)$, so on the plane the antipodal map reads

$$w \longmapsto -\frac{1}{w^*}, \tag{3.72}$$

which indeed exchanges 0 with ∞, 1 with -1, and $+i$ with $-i$, that is, horizontal with vertical, diagonal with antidiagonal, and each circular state with the other one. A statement that on the sphere was a plain geometric fact, going to the other side, becomes on the plane a short algebraic rule.

The dictionary between the two formalisms is now explicit. Dividing the Stokes parameters 3.61 to 3.64 by S_0 and writing them in terms of the ratio 3.70 gives

$$\frac{S_1}{S_0} = \frac{1-|w|^2}{1+|w|^2}, \qquad \frac{S_2}{S_0} = \frac{2\,\mathrm{Re}\{w\}}{1+|w|^2}, \qquad \frac{S_3}{S_0} = \frac{-2\,\mathrm{Im}\{w\}}{1+|w|^2}, \tag{3.73}$$

and these three expressions are, letter for letter, the classical formulae of the inverse stereographic projection, the ones that lift a point of the plane back up to the sphere. The sign carried by S_3 is only the mirror convention of the drawing, and says which of the two circular states we agree to place on the upper half of the imaginary axis. Nothing else in 3.73 is a matter of convention: the Jones ratio and the Stokes vector are the same information, once written on the plane and once written on the sphere.

A flat atlas of every polarisation state

Because the projection loses nothing, we may throw the sphere away and keep only the map. Figure 3.8 does exactly that: it is the complex plane of w with the polarisation ellipse of each state drawn at the point that represents it, a complete atlas of polarised light on a single sheet.

Three features of the atlas deserve to be read out loud. First, the whole real axis is filled with linear states: a real w means $\triangle\phi = 0$ or $\triangle\phi = \pi$, the two components oscillate in step or in opposition, and the ellipse collapses to a segment, whose tilt sweeps through every possible orientation as w runs from one end of the axis to the other. Second, the imaginary axis carries the states whose ellipse is aligned with the coordinate axes, growing from the flat horizontal segment at the origin, through wider and wider ellipses, to a perfect circle at $w = \pm i$, and then closing again into the vertical segment far away. Third, and most striking, the edges of the atlas in every direction, $+\infty$, $-\infty$, $+i\infty$ and $-i\infty$ alike, all carry the same vertical polarisation, because on the sphere all those escape routes converge on the single point N. The plane looks infinite, but it is not: it closes up on itself, and the object it really depicts is compact, bounded and without edges, a sphere.

This is the sense in which the Jones and the Stokes formalisms are one theory. A polarisation state is a point; call it w and you are doing Jones calculus on the plane, call it $(S_1, S_2, S_3)/S_0$ and you are doing Stokes calculus on the sphere, and 3.73 carries you back and forth at will. The choice between them is a choice of convenience, and each has got its own: the sphere makes orthogonality and the doubling of angles visible at a glance, while the plane reduces a state to a single number that can simply be divided or multiplied. The next subsection exploits precisely this, because once a state is a single complex number, what an optical element does to light becomes a transformation of the plane onto itself, and the geometry of the sphere tells us in advance which transformations are allowed.

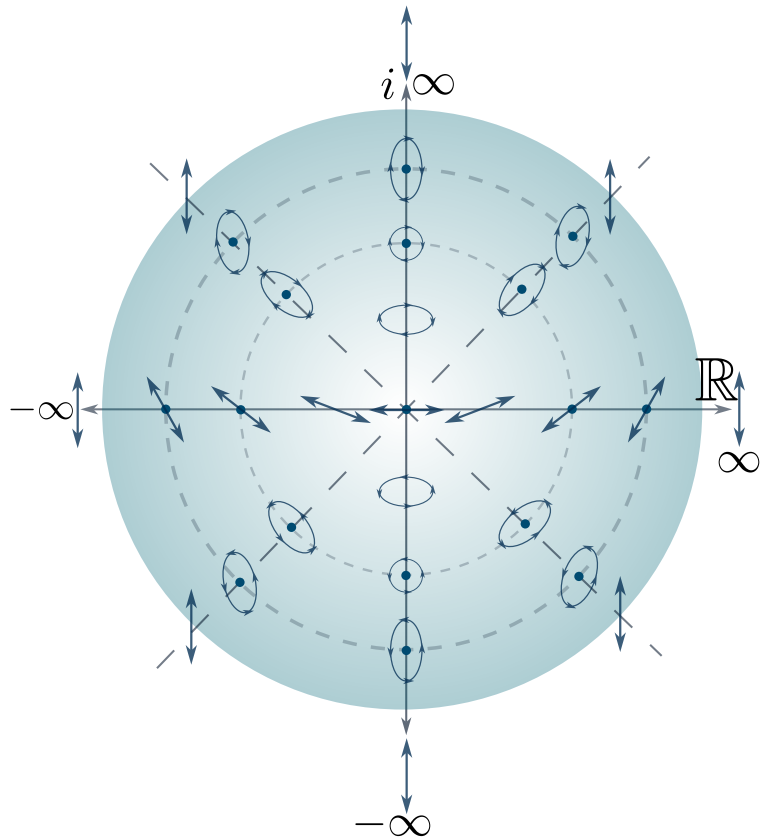


Figure 3.8: The flattened Poincaré sphere. Each point of the complex plane of w carries the ellipse of the state it represents. The origin is horizontal polarisation, the real axis carries every linear state, the imaginary axis carries the ellipses aligned with the coordinate axes together with the two circular states, and every direction of escape to infinity leads to one and the same state, the vertical one.

3.5.4 Transformations of Polarisation: Jones matrices

Every polarisation state built so far is a fixed vector in $\mathbb{C}^2$, but light does not stay the same forever: it crosses lenses, reflects off mirrors, is dimmed by tinted glass, or travels through crystals whose very atomic arrangement singles out a preferred direction in space. Every one of these processes, as long as it acts linearly and does not depend on where or when the light arrives (that is, as long as it is a genuine, homogeneous optical element and not, say, a nonlinear medium or a time varying modulator), can only do one thing to the Jones spinor 3.34: multiply it by a fixed matrix.

Definition 21 *(Jones matrix): A linear, homogeneous optical element acting on a fully polarised, monochromatic beam is represented by a matrix $J \in \mathbb{C}^{2\times 2}$, its Jones matrix, such that the outgoing spinor 3.34 is related to the incoming one by*

$$\vec{e}\,' = J\,\vec{e}.$$

Almost every optical element used in a real laboratory falls into one of three families, according to what kind of matrix J it is: retarders, rotators and dichroics. We build all three from the same simple idea, that of writing J in the basis of its own eigenstates, and we let the physics of each device tell us what those eigenvalues should be.

Retarders

Physically, a retarder, also called a wave plate, is cut from a birefringent crystal, quartz, calcite or mica, whose internal atomic arrangement is not the same along every direction, so that the refractive index it presents to light depends on how that light is polarised. Every birefringent plate has, built into it, two mutually orthogonal privileged directions, its fast axis and its slow axis, and light polarised exactly along one of them propagates through unchanged in form, picking up only an ordinary propagation phase, but at a different rate along the fast axis, index n_f, than along the slow one, index n_s. After a thickness d, the component travelling along the slow axis has fallen behind the one along the fast axis by a phase

$$\delta = \frac{2\pi d}{\lambda}\,(n_s - n_f)\,, \tag{3.74}$$

called the retardance, while any other input, being a superposition of the two axes, comes out with its ellipse reshaped, because the two halves of the superposition have rotated relative to each other in the complex plane.

Definition 22 *(Retarder): Let $|\psi_1\rangle = |\alpha;\chi\rangle$ and $|\psi_2\rangle = |\alpha+\pi/2;-\chi\rangle$ be a pair of orthogonal Jones kets 3.41*[4]*. A retarder of eigenstates $|\psi_1\rangle, |\psi_2\rangle$ and retardance δ is the Jones matrix*

$$W(\alpha,\chi;\delta) = e^{i\delta/2}\,|\psi_1\rangle\langle\psi_1| + e^{-i\delta/2}\,|\psi_2\rangle\langle\psi_2|\,. \tag{3.75}$$

Equation 3.75 is exactly the spectral decomposition of W: $|\psi_1\rangle, |\psi_2\rangle$ are, by construction, its eigenvectors, with eigenvalues $e^{\pm i\delta/2}$ of unit modulus, so W is unitary and preserves the total intensity $\vec{e}^{\dagger}\vec{e}$ of any beam it acts on, exactly as an element that only dephases, and never absorbs, should.

The case $\chi = 0$ is by far the most common, since it is what an ordinary birefringent plate, cut with its fast axis in the plane of the crystal, produces; we call it a linear retarder, because both eigenstates $|\alpha;0\rangle = [\cos\alpha, \sin\alpha]^{\mathsf{T}}$ and $|\alpha+\pi/2;0\rangle = [-\sin\alpha, \cos\alpha]^{\mathsf{T}}$ are linearly polarised. Substituting them into 3.75 and writing $e^{\pm i\delta/2} = \cos(\delta/2) \pm i\sin(\delta/2)$, the four entries work out to

$$\begin{aligned}
W_{11} &= \cos^2\alpha\, e^{i\delta/2} + \sin^2\alpha\, e^{-i\delta/2} = \cos\frac{\delta}{2} + i\sin\frac{\delta}{2}\cos 2\alpha,\\
W_{22} &= \sin^2\alpha\, e^{i\delta/2} + \cos^2\alpha\, e^{-i\delta/2} = \cos\frac{\delta}{2} - i\sin\frac{\delta}{2}\cos 2\alpha,\\
W_{12} = W_{21} &= \sin\alpha\cos\alpha\left(e^{i\delta/2} - e^{-i\delta/2}\right) = i\sin\frac{\delta}{2}\sin 2\alpha,
\end{aligned}$$

[4]Writing out the components of $|\alpha;\chi\rangle$ and $|\alpha+\pi/2;-\chi\rangle$ and multiplying term by term shows that $\langle\psi_1|\psi_2\rangle = 0$ identically, for every α and every χ, so this pair is always orthogonal.

so that the linear retarder of fast axis at α and retardance δ is

$$W(\alpha, 0; \delta) = \begin{bmatrix} \cos\frac{\delta}{2} + i\sin\frac{\delta}{2}\cos 2\alpha & i\sin\frac{\delta}{2}\sin 2\alpha \\ i\sin\frac{\delta}{2}\sin 2\alpha & \cos\frac{\delta}{2} - i\sin\frac{\delta}{2}\cos 2\alpha \end{bmatrix}. \tag{3.76}$$

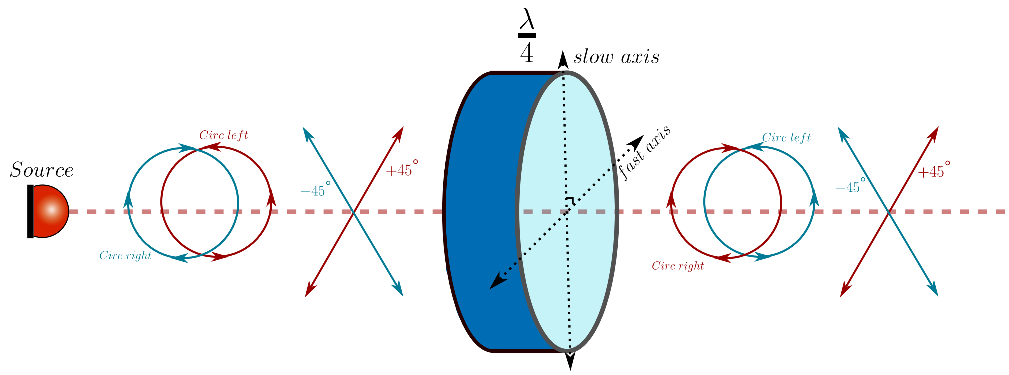


Figure 3.9: A quarter wave plate retarder is a retarder with $\delta = \pi/2$ and when it has got its fast axis oriented with the horizontal one it transforms the polarization in a very specific way. The $\pm 45°$ diagonal states transform into circular polarization states, and the circular polarization states transform into $\pm 45°$ diagonal states.

Two particular values of δ are so useful that they have earned their own names. When $\delta = \pi/2$, a quarter of a full cycle, we speak of a quarter wave plate (see figure 3.9); taking the fast axis horizontal, $\alpha = 0$, equation 3.76 becomes, up to the irrelevant global phase $e^{i\pi/4}$,

$$W_{\lambda/4} = \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix}, \tag{3.77}$$

whose action on the diagonal state 3.49 is immediate to check: $|\pi/4; 0\rangle = \frac{1}{\sqrt{2}}[1, 1]^{\mathsf{T}}$ is sent to $\frac{1}{\sqrt{2}}[1, -i]^{\mathsf{T}} = |0; -\pi/4\rangle$, one of the circular states 3.53. This is exactly why a quarter wave plate is the standard tool for turning linear polarisation into circular, and back.

When instead $\delta = \pi$, half a full cycle, we get the half wave plate,

$$W(\alpha, 0; \pi) = i \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{bmatrix}, \tag{3.78}$$

and, up to the global phase i, this matrix is a reflection: acting on a linear state it sends $|\theta; 0\rangle$ to $|2\alpha - \theta; 0\rangle$, reflecting its orientation about the fast axis, so a half wave plate rotates any linear input by twice the angle between the input and the fast axis, and it is the tool of choice whenever an arbitrary rotation of a linear polarisation is needed. Acting instead on a circular state, a short calculation with 3.78 shows that $|0; \pi/4\rangle$ is sent to $|0; -\pi/4\rangle$ for every value of α, so a half wave plate always reverses the handedness of a circular input, whatever its own orientation happens to be.

Nothing prevents us from choosing genuinely elliptical eigenstates, $0 < |\chi| < \pi/4$: the resulting elliptical retarder is a real, if less common, device, built from a birefringent material whose axes have additionally been twisted by a small optical activity, so that its privileged states are ellipses rather than straight lines, and it is worth constructing its Jones matrix properly, both because it is the fully general case of 3.75 and because doing so will let us obtain the rotator, further on, in a single line.

Write the first eigenstate $|\psi_1\rangle = |\alpha;\chi\rangle$ of 3.41 as $|\psi_1\rangle = [a, b]^\mathsf{T}$ with

$$a = \cos\alpha\cos\chi - i\sin\alpha\sin\chi, \qquad b = \sin\alpha\cos\chi + i\cos\alpha\sin\chi,$$

so that $|a|^2 + |b|^2 = 1$, since $|\psi_1\rangle$ is a unit vector. Comparing component by component with $|\psi_2\rangle = |\alpha + \pi/2; -\chi\rangle$ shows the compact relation $|\psi_2\rangle = [-b^*, a^*]^\mathsf{T}$, so that the matrix built from the two orthonormal eigenstates as columns,

$$U(\alpha,\chi) = \begin{bmatrix} |\psi_1\rangle & |\psi_2\rangle \end{bmatrix} = \begin{bmatrix} a & -b^* \\ b & a^* \end{bmatrix}, \tag{3.79}$$

is unitary, $U^\dagger U = UU^\dagger = I$, precisely because $|\psi_1\rangle, |\psi_2\rangle$ are orthonormal. Equation 3.79 is nothing but the change of basis matrix from the eigenbasis $\{|\psi_1\rangle, |\psi_2\rangle\}$, in which the retarder is diagonal by definition, to the standard basis in which we write the Jones spinor 3.34. Writing $\Lambda(\delta) = \operatorname{diag}\left(e^{i\delta/2}, e^{-i\delta/2}\right)$ for the retarder expressed in its own eigenbasis, this change of basis gives a second, completely explicit way of writing 3.75,

$$W(\alpha,\chi;\delta) = U(\alpha,\chi)\,\Lambda(\delta)\,U(\alpha,\chi)^\dagger, \tag{3.80}$$

and carrying out the product explicitly,

$$U\Lambda U^\dagger = \begin{bmatrix} a & -b^* \\ b & a^* \end{bmatrix} \begin{bmatrix} e^{i\delta/2} & 0 \\ 0 & e^{-i\delta/2} \end{bmatrix} \begin{bmatrix} a^* & b^* \\ -b & a \end{bmatrix} = \begin{bmatrix} |a|^2 e^{i\delta/2} + |b|^2 e^{-i\delta/2} & ab^*\left(e^{i\delta/2} - e^{-i\delta/2}\right) \\ a^* b\left(e^{i\delta/2} - e^{-i\delta/2}\right) & |b|^2 e^{i\delta/2} + |a|^2 e^{-i\delta/2} \end{bmatrix}.$$

Squaring and multiplying the two expressions for a, b above gives $|a|^2 - |b|^2 = \cos 2\alpha\cos 2\chi$, $|a|^2 + |b|^2 = 1$ and $ab^* = \frac{1}{2}\sin 2\alpha\cos 2\chi - \frac{i}{2}\sin 2\chi$, exactly the same three combinations already met while building the linear retarder; using them, together with $e^{\pm i\delta/2} = \cos(\delta/2) \pm i\sin(\delta/2)$, the elliptical retarder takes the closed form

$$W(\alpha,\chi;\delta) = \begin{bmatrix} \cos\dfrac{\delta}{2} + i\sin\dfrac{\delta}{2}\cos 2\alpha\cos 2\chi & \sin\dfrac{\delta}{2}\sin 2\chi + i\sin\dfrac{\delta}{2}\sin 2\alpha\cos 2\chi \\ -\sin\dfrac{\delta}{2}\sin 2\chi + i\sin\dfrac{\delta}{2}\sin 2\alpha\cos 2\chi & \cos\dfrac{\delta}{2} - i\sin\dfrac{\delta}{2}\cos 2\alpha\cos 2\chi \end{bmatrix}. \tag{3.81}$$

Setting $\chi = 0$ in 3.81 gives $\cos 2\chi = 1, \sin 2\chi = 0$ and collapses it exactly back to the linear retarder 3.76, as it must: the fully general elliptical case was, after all, only ever hiding the extra angle χ inside the change of basis matrix 3.79.

Rotators

Let us push the general elliptical retarder 3.81 to the opposite extreme from the linear case and choose circular eigenstates, $\alpha = 0, \chi = \pi/4$, that is $|\psi_1\rangle = |0; \pi/4\rangle$ and $|\psi_2\rangle = |0; -\pi/4\rangle$ of 3.53. Since $\cos 2\chi = \cos(\pi/2) = 0$ and $\sin 2\chi = \sin(\pi/2) = 1$, every term in 3.81 carrying a factor of $\cos 2\chi$ vanishes identically and we are left with

$$W(0, \pi/4; \delta) = \begin{bmatrix} \cos\dfrac{\delta}{2} & \sin\dfrac{\delta}{2} \\ -\sin\dfrac{\delta}{2} & \cos\dfrac{\delta}{2} \end{bmatrix}, \tag{3.82}$$

and the right hand side is exactly the rotation matrix already used in 3.41 to build every ellipse out of the unrotated form 3.40. A retarder of circular eigenstates, in other words, is nothing but a rotation of the whole Jones plane: it sends $|\alpha; \chi\rangle$ to $|\alpha - \delta/2; \chi\rangle$ for every α and every χ, leaving the ellipticity untouched and shifting only the orientation.

Definition 23 *(Rotator): A rotator by an angle θ is the Jones matrix*

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \tag{3.83}$$

equivalently the circular retarder 3.82 of retardance $\delta = -2\theta$, so that $R(\theta)\,|\alpha; \chi\rangle = |\alpha + \theta; \chi\rangle$ for every α, χ.

Physically, this is exactly what happens inside an optically active medium: a solution of chiral sugar molecules, or a slab of quartz cut along its optic axis, where it is not the two linear directions that see different refractive indices, but the two circular ones. The phenomenon is called circular birefringence, and a beam of any polarisation crossing a thickness d of such a medium has its ellipse rotated, rigidly and without distortion, by an angle proportional to d; the same net effect, a pure rotation with no change of ellipticity, is produced by a Faraday rotator, a piece of glass or crystal placed in a strong magnetic field along the beam, in which it is the magnetic field rather than molecular chirality that splits the two circular eigenstates. Notice that $R(\theta)$, unlike a generic retarder, is a real matrix: rotators are the only non-trivial polarisation transformations whose Jones matrix has got no imaginary part at all, a reflection of the fact that neither optical activity nor the Faraday effect privileges any particular linear direction in space, only a sense of circulation.

Dichroics and polarisers

Every element considered so far is unitary: it reshapes the polarisation ellipse but never changes the total intensity. A dichroic element breaks this on purpose. Physically, it is again built from an anisotropic material, but now one whose two privileged directions are absorbed differently rather than merely retarded differently, because the microscopic charges responsible for the absorption, free electrons in a metal wire, iodine chains in a stretched polymer, the delocalised electrons of an elongated organic molecule, can oscillate much more easily along one direction than along the perpendicular one. Mathematically we only have to repeat the construction 3.75 once more, replacing the two phase factors of unit modulus by two real, non-negative amplitude transmittances.

Definition 24 *(Dichroic): A dichroic element of eigenstates $|\psi_1\rangle, |\psi_2\rangle$ (orthogonal, as in the retarder) and amplitude transmittances $t_1, t_2 \in \mathbb{R}$, $t_1, t_2 \geq 0$, is the Jones matrix*

$$D = t_1 |\psi_1\rangle \langle\psi_1| + t_2 |\psi_2\rangle \langle\psi_2| . \tag{3.84}$$

Because t_1, t_2 are real, D in 3.84 is Hermitian rather than unitary, unless $t_1 = t_2$, the trivial case in which nothing happens to the polarisation, only to the overall brightness: light along $|\psi_1\rangle$ comes out multiplied by t_1 and light along $|\psi_2\rangle$ by t_2, and any state in between comes out reshaped towards whichever eigenstate survives best.

Taking $\chi = 0$ once again gives the linear dichroic, and repeating the same short calculation that led to 3.76, now with t_1, t_2 in place of $e^{\pm i\delta/2}$,

$$D(\alpha, 0; t_1, t_2) = \begin{bmatrix} t_1 \cos^2\alpha + t_2 \sin^2\alpha & (t_1 - t_2)\sin\alpha\cos\alpha \\ (t_1 - t_2)\sin\alpha\cos\alpha & t_1 \sin^2\alpha + t_2\cos^2\alpha \end{bmatrix}. \tag{3.85}$$

The most important member of this family is obtained by pushing the contrast between the two eigenstates to its limit, $t_1 = 1$ and $t_2 = 0$: the surviving axis is transmitted perfectly and the other is extinguished completely. Equation 3.85 then collapses to

$$P(\alpha) = \begin{bmatrix} \cos^2\alpha & \sin\alpha\cos\alpha \\ \sin\alpha\cos\alpha & \sin^2\alpha \end{bmatrix} = |\alpha; 0\rangle \langle\alpha; 0| , \tag{3.86}$$

which we recognise, on the right, as the orthogonal projector onto the linear state $|\alpha; 0\rangle$.

Definition 25 *(Ideal linear polariser): An ideal linear polariser of transmission axis at angle α is the limiting linear dichroic 3.85 with $t_1 = 1, t_2 = 0$, that is, the orthogonal projector $P(\alpha) = |\alpha; 0\rangle \langle\alpha; 0|$ of 3.86.*

This is the deepest sense in which a polariser is a simple device: it does not really transform a polarisation state, it projects it, discarding every component along $|\alpha + \pi/2; 0\rangle$ and keeping only the shadow that the incoming ellipse casts onto its own transmission axis. Every partial linear dichroic 3.85 with $t_2 \neq 0$ is only an imperfect version of this same projection, letting some of the unwanted component leak through.

Circular dichroism is built the same way, with $|\psi_1\rangle = |0; \pi/4\rangle$ and $|\psi_2\rangle = |0; -\pi/4\rangle$ in 3.84. Specialising a, b from the change of basis matrix 3.79 to $\alpha = 0, \chi = \pi/4$ gives $a = 1/\sqrt{2}$, $b = i/\sqrt{2}$, so

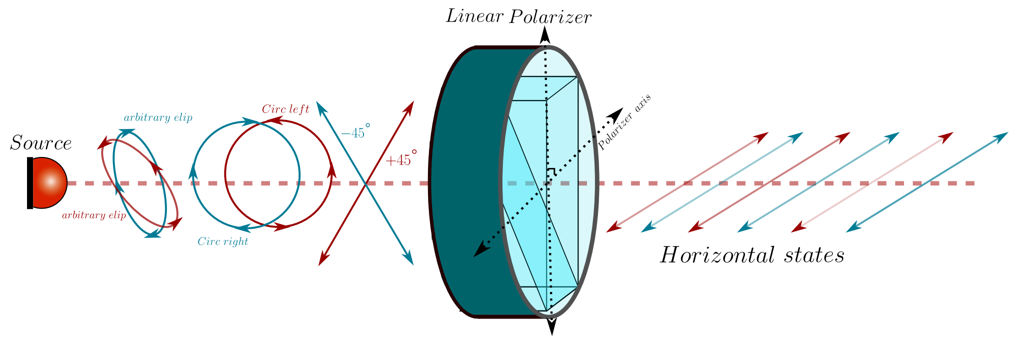


Figure 3.10: A linear polariser with its polariser axis oriented parallel to the horizontal axis proyects all the incident states to horizontal ones, however the transmited intensity is reduced by the relative angle between the incident state $|\alpha;\chi\rangle$ and the polariser axis α_{pol} in this case $\alpha_{pol} = 0°$.

that $|a|^2 = |b|^2 = 1/2$ and $ab^* = -i/2$; repeating, with these values, the same matrix multiplication that led to 3.81, but with the real transmittances t_1, t_2 in place of the phases $e^{\pm i\delta/2}$,

$$D_{\circlearrowleft}(t_1,t_2) = \begin{bmatrix} \dfrac{t_1+t_2}{2} & -i\dfrac{t_1-t_2}{2} \\ i\dfrac{t_1-t_2}{2} & \dfrac{t_1+t_2}{2} \end{bmatrix}, \tag{3.87}$$

describing a material that absorbs its two circular eigenstates differently. Circular dichroism is a much smaller effect than linear dichroism in most materials, but it is an extremely sensitive probe of molecular chirality, proteins, sugars and DNA are all optically active and mildly circularly dichroic, which is why circular dichroism spectroscopy is a standard tool in structural biology and chemistry.

Stepping back from the algebra, what actually makes a material dichroic is always some built in anisotropy at the microscopic level: an elongated molecule, a stretched polymer chain, or a crystal lattice, along which charges are freer to respond to the oscillating electric field of light in one direction than in the perpendicular one, so the material absorbs, rather than merely slows down, one polarisation component more strongly than the other. A polariser, more generally, is simply any device, dichroic or not, designed to output light of a well controlled polarisation state from an input that may not have one at all; dichroism is only one of the physical mechanisms that nature and engineering have found to build one.

Three real polarisers illustrate how different those mechanisms can be. Tourmaline, the mineral in which dichroism was first noticed, is naturally dichroic, but the polariser found today in almost every pair of sunglasses, camera filter and liquid crystal display is Edwin Land's Polaroid sheet, a film of polyvinyl alcohol stretched in one direction to align its long polymer chains and then doped with iodine, whose mobile electrons run freely along the chains and absorb the field component parallel to them, the linear dichroic 3.85 realised in a plastic film. A wire grid polariser applies the same physical idea to a metal instead of a dye: an array of parallel conducting wires, spaced much closer together than the wavelength of the light, lets the free electrons in the wires oscillate and reflect the field component parallel to them, while the perpendicular component, unable to drive a current across the gaps between wires, passes through almost unaffected; because it works by reflection rather than absorption, it survives far higher optical powers than a dyed polymer sheet, and it is the polariser of choice in the infrared and in microwave optics. Finally, the Glan-Thompson prism takes a completely different route, using no absorption at all: two prisms of calcite, a strongly birefringent crystal, are cut and cemented together so that, at the internal interface, the ordinary ray strikes beyond its critical angle and is totally internally reflected out of the beam, while the extraordinary ray, seeing a different refractive index, is transmitted straight through. Being lossless for the transmitted beam and free of any absorbing dye, this kind of polariser can handle the intense, focused light of a laser without being damaged, which the simple dichroic sheet cannot.

3.5.5 Transformations of Polarisation: Mueller matrices

The Jones calculus of the previous subsection is exact, elegant and, for a large class of problems, entirely sufficient. It is also, in a very precise sense, too narrow. A Jones matrix acts on the spinor 3.34, and that spinor exists only when the beam has got a single, well defined phase relation between its two components, that is, only when the light is fully polarised. The moment we point our instrument at sunlight, at a cloud, at a sheet of paper, at a biological tissue or at any surface rough on the scale of a wavelength, the object we must transform is no longer a spinor but the four Stokes parameters 3.61 to 3.64, and no 2×2 complex matrix will do the job.

There is a second and deeper failure, and it is worth stating before we build anything. A Jones matrix can never *depolarise*. Whatever it does to a fully polarised beam, the outgoing beam is again fully polarised, so a formalism built on Jones matrices alone cannot describe the single most common thing that real matter does to light. We shall prove this in a moment, and the proof will also tell us exactly what has to be added.

The formalism that repairs both defects was assembled by Hans Mueller at the Massachusetts Institute of Technology in the early nineteen forties, building on the earlier scattering work of Soleillet and of Perrin (Perrin, 1942), and it was the same wave of activity that rescued the Stokes parameters from the obscurity described at the beginning of this section. Its later systematisation owes a great deal to McMaster (McMaster, 1954) and to Chandrasekhar's treatment of radiative transfer (Chandrasekhar, 1950), where polarised light must be propagated through a scattering atmosphere and where Jones spinors are simply not available.

From the Jones matrix to the coherence matrix

The bridge between the two calculi is the polarisation matrix 3.67, and building it is a one line calculation whose consequences occupy the rest of this subsection. Let a linear, homogeneous element with Jones matrix J act on the beam, so that $\vec{e}\,' = J\vec{e}$. The outgoing polarisation matrix is, by definition,

$$\Phi' = \left\langle \vec{e}\,'\vec{e}\,'^{\dagger} \right\rangle = \left\langle J\vec{e}(J\vec{e})^{\dagger} \right\rangle = \left\langle J\,\vec{e}\,\vec{e}^{\,\dagger} J^{\dagger} \right\rangle = J \left\langle \vec{e}\,\vec{e}^{\,\dagger} \right\rangle J^{\dagger} = J\,\Phi\,J^{\dagger}. \tag{3.88}$$

Every step is elementary, but the third one is the whole point and deserves to be said out loud: the matrix J could be taken outside the time average because the *element* is deterministic even when the *light* is not. The waveplate does not flicker; only the field does. This is precisely why 3.88 remains valid for partially polarised and even for completely unpolarised light, where the spinor $\vec{e}$ has got no meaning at all and only the averages survive.

Notice also the shape of the transformation. The field transforms linearly, $\vec{e} \to J\vec{e}$, but the polarisation matrix transforms by a **congruence**, $\Phi \to J\Phi J^\dagger$, with J appearing twice. Since the Stokes parameters are themselves quadratic in the field, this is exactly what we should have expected, and it is the algebraic reason why the Mueller matrix we are about to build is quadratic in the Jones matrix.

The Mueller matrix

Because the Stokes parameters are linear coordinates of Φ through 3.69, and because 3.88 is linear in Φ, the map induced on the four Stokes parameters must be linear as well.

Definition 26 *(Mueller matrix): A linear, homogeneous optical element acting on a beam of arbitrary degree of polarisation is represented by a real 4×4 matrix M, its Mueller matrix, such that the outgoing Stokes vector is related to the incoming one by*

$$S'_\mu = \sum_{\nu=0}^{3} M_{\mu\nu} S_\nu, \qquad \textit{that is} \qquad \vec{S}\,' = M\vec{S}. \tag{3.89}$$

When the element also possesses a Jones matrix, the two descriptions must agree, and forcing them to agree hands us the Mueller matrix explicitly. Insert 3.88 into the first of 3.69 and then expand the incoming Φ with the second,

$$S'_\mu = \mathrm{tr}\left(\Phi'\sigma_\mu\right) = \mathrm{tr}\left(J\Phi J^\dagger\sigma_\mu\right) = \mathrm{tr}\left(J\left[\frac{1}{2}\sum_{\nu=0}^{3} S_\nu\sigma_\nu\right]J^\dagger\sigma_\mu\right) = \sum_{\nu=0}^{3}\left[\frac{1}{2}\mathrm{tr}\left(\sigma_\mu J\sigma_\nu J^\dagger\right)\right]S_\nu,$$

where the trace has been taken through the finite sum and the cyclic property $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ has been used to bring σ_μ to the front. Comparing with 3.89 we can read off the coefficients at once.

Definition 27 *(Mueller matrix of a Jones element): An element whose action on fully polarised light is given by the Jones matrix J has got the Mueller matrix*

$$\boxed{M_{\mu\nu} = \frac{1}{2}\,tr\left(\sigma_\mu\, J\, \sigma_\nu\, J^\dagger\right)} \tag{3.90}$$

with σ_μ the four matrices 3.68. A Mueller matrix that can be written in this form is called ***non depolarising****, or a* ***Mueller-Jones*** *matrix.*

Formula 3.90 is the dictionary we were after, and three of its properties are worth extracting immediately.

Taking the complex conjugate of 3.90 and using $\overline{\operatorname{tr} A} = \operatorname{tr} A^\dagger$ together with the hermiticity $\sigma_\mu^\dagger = \sigma_\mu$,

$$\overline{M_{\mu\nu}} = \frac{1}{2}\operatorname{tr}\left(\left[\sigma_\mu J \sigma_\nu J^\dagger\right]^\dagger\right) = \frac{1}{2}\operatorname{tr}\left(J\sigma_\nu J^\dagger \sigma_\mu\right) = M_{\mu\nu},$$

the last step being once more the cyclic property. So M is a real matrix, as it must be, since it maps four measurable intensities into four measurable intensities.

Replacing J by $e^{i\theta}J$ leaves 3.90 untouched, because the phase from J cancels against the conjugate phase from $J^\dagger$. The map $J \mapsto M$ is therefore not injective: all Jones matrices differing by an overall phase collapse onto the same Mueller matrix. This is not a defect but an improvement, since that phase was never measurable in the first place, and we shall see in a moment that it is the same two to one correspondence that produces the doubling of angles on the Poincaré sphere.

If light crosses first the element J_1 and then the element J_2, the combined Jones matrix is $J_2 J_1$, and substituting it into 3.90 while inserting the resolution $\frac{1}{2}\sum_\lambda \sigma_\lambda \operatorname{tr}(\sigma_\lambda\,\cdot\,)$ between the two factors shows that

$$M\left(J_2 J_1\right) = M\left(J_2\right) M\left(J_1\right), \tag{3.91}$$

so that 3.90 is a group homomorphism and an optical bench is described, exactly as in the Jones calculus, by the ordered product of the matrices of its elements.

Retarders are rotations of the Poincaré sphere

Let us now compute a Mueller matrix explicitly, and let us choose the simplest retarder, the linear one with its fast axis along x, whose Jones matrix is 3.76 with $\alpha = 0$, that is $W = \operatorname{diag}\left(e^{i\delta/2}, e^{-i\delta/2}\right)$. We need the four objects $W\sigma_\nu W^\dagger$. The first two are immediate, since W is unitary and diagonal and therefore commutes with σ_0 and σ_1,

$$W\sigma_0 W^\dagger = WW^\dagger = \sigma_0, \qquad W\sigma_1 W^\dagger = \sigma_1.$$

The interesting ones are the other two. Multiplying out,

$$W\sigma_2 W^\dagger = \begin{bmatrix} 0 & e^{i\delta} \\ e^{-i\delta} & 0 \end{bmatrix} = \cos\delta \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + \sin\delta \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix} = \cos\delta\,\sigma_2 + \sin\delta\,\sigma_3,$$

where we simply split $e^{\pm i\delta} = \cos\delta \pm i\sin\delta$ and recognised the two Pauli matrices, and in exactly the same way

$$W\sigma_3 W^\dagger = \begin{bmatrix} 0 & ie^{i\delta} \\ -ie^{-i\delta} & 0 \end{bmatrix} = -\sin\delta\,\sigma_2 + \cos\delta\,\sigma_3.$$

Since $\frac{1}{2}\operatorname{tr}(\sigma_\mu\sigma_\nu) = \delta_{\mu\nu}$, the coefficients just found are literally the columns of M, and we obtain

$$M_W(\delta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\delta & -\sin\delta \\ 0 & 0 & \sin\delta & \cos\delta \end{bmatrix}. \tag{3.92}$$

Read this matrix slowly, because it is the central geometrical fact of the whole subsection. The first row and the first column are those of the identity, so S_0 is untouched: a retarder absorbs nothing, exactly as its unitarity promised. The remaining 3×3 block leaves S_1 alone and rotates the pair (S_2, S_3) by the angle δ. In other words, on the Poincaré sphere the element performs a **rigid rotation by the retardance δ about the S_1 axis**, and the S_1 axis is precisely the diameter joining the two eigenstates of this retarder, the horizontal and vertical states 3.45 and 3.47. The eigenstates are the fixed points of the motion, as fixed points of a rotation must be, and every other state is carried around them on a circle of constant latitude.

Nothing in this conclusion depends on our having chosen $\alpha = 0$. Repeating the computation with the general elliptical retarder 3.81 is a longer but entirely mechanical exercise, and it yields the following statement, which we record as the geometrical heart of the Mueller calculus.

Theorem 1 *(Retarders as rotations): The Mueller matrix of a retarder of retardance δ whose eigenstates are $|\alpha;\chi\rangle$ and its orthogonal partner is*

$$M_W = \begin{bmatrix} 1 & \vec{0}^{\mathsf{T}} \\ \vec{0} & R\left(\hat{n}, \delta\right) \end{bmatrix}, \tag{3.93}$$

where $R(\hat{n}, \delta)$ is the ordinary 3×3 rotation by the angle δ about the unit vector

$$\hat{n} = \left(\cos 2\alpha \cos 2\chi,\ \sin 2\alpha \cos 2\chi,\ -\sin 2\chi\right), \tag{3.94}$$

which by 3.65 is exactly the point of the Poincaré sphere representing the eigenstate $|\alpha;\chi\rangle$.

Every waveplate on an optical bench is therefore a rotation of the Poincaré sphere (see figure 3.11), and the two numbers that specify it have got transparent meanings: the orientation of its axis fixes *about which diameter* the sphere turns, and its retardance fixes *by how much*. Three familiar devices become obvious.

A **quarter wave plate**, $\delta = \pi/2$, turns the sphere by a quarter of a full turn. Starting from a linear state on the equator and choosing the axis at $\pi/4$ from it, a quarter turn lands exactly on a pole, which is the statement that a quarter wave plate converts linear light into circular light, as we found by direct matrix multiplication in the previous subsection.

A **half wave plate**, $\delta = \pi$, turns the sphere by half a full turn about an equatorial axis. A rotation by π about the axis $\hat{n}$ sends any vector $\vec{v}$ to $2(\hat{n} \cdot \vec{v})\hat{n} - \vec{v}$, so a linear state at azimuth 2θ on the equator, acted on by a plate whose axis sits at azimuth 2α, is carried to azimuth $4\alpha - 2\theta$. Dividing by two, the physical polarisation direction goes from θ to $2\alpha - \theta$: **a half wave plate reflects the plane of polarisation about its own fast axis**. Applying the same rule to the poles, $\vec{v} = (0, 0, \pm 1)$ with $\hat{n}$ equatorial gives $\vec{v} \to -\vec{v}$, so the two circular states are exchanged and the handedness of any beam is reversed.

A **rotator** 3.83 is, as we proved, a circular retarder, so its axis 3.94 is the polar axis $\hat{n} = (0, 0, -1)$ and it spins the sphere about the line joining the two circular states. A rotator that turns the

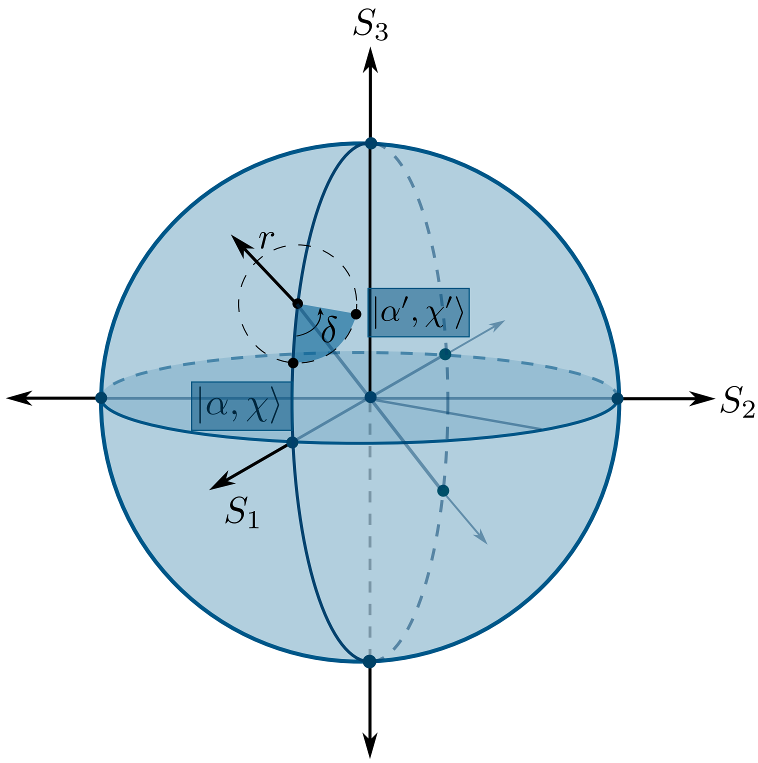


Figure 3.11: An initial state $|\alpha;\chi\rangle$ under the effect of a rotated retarder with retar δ and eigenstate r is rotated an angle of δ arounde the r-axis to a final state $|\alpha'.\chi'\rangle$.

physical polarisation by an angle θ has got retardance $\delta = 2\theta$, so it rotates the Poincaré sphere by 2θ. This is the doubling of angles once more, and we can now see its true origin: the map 3.90 restricted to unitary Jones matrices is the classical two to one homomorphism from $SU(2)$ onto the rotation group $SO(3)$. Turning a waveplate through a full physical revolution, $\theta : 0 \to \pi$, returns the Jones matrix not to $+I$ but to $-I$, while the Mueller matrix has already completed a full 2π rotation and returned to the identity. The polarisation spinor is a spinor in exactly the sense that an electron wavefunction is, and the Poincaré sphere is its Bloch sphere.

Polarisers, dichroics and the Malus law

Not every element is unitary, and the non unitary ones move the sphere in a completely different way. Take first the ideal linear polariser 3.86, whose Jones matrix is the projector $J = |\psi\rangle\langle\psi|$ onto the transmitted state. Feeding it into 3.90 and writing $\hat{n}$ for the Stokes vector of that state, the result is the strikingly simple

$$M_P = \frac{1}{2}\begin{bmatrix} 1 & \hat{n}^{\mathsf{T}} \\ \hat{n} & \hat{n}\,\hat{n}^{\mathsf{T}} \end{bmatrix}. \tag{3.95}$$

This matrix has got rank one in a strong sense: whatever comes in, the outgoing Stokes vector is proportional to $(1, \hat{n})$, that is, the entire Poincaré sphere is crushed onto the single point $\hat{n}$. The map is catastrophically non invertible, which is the geometric expression of the obvious physical fact that a polariser destroys information and cannot be undone.

The Malus law now drops out with no effort at all. Send in linear light at an angle θ, whose Stokes vector is $\vec{S} = (1, \cos 2\theta, \sin 2\theta, 0)$, through a polariser whose axis is at α, so that $\hat{n} = (\cos 2\alpha, \sin 2\alpha, 0)$. The transmitted intensity is the zeroth component of 3.95 acting on $\vec{S}$,

$$S_0' = \frac{1}{2}\left(1 + \hat{n} \cdot \vec{S}_{123}\right) = \frac{1}{2}\left[1 + \cos 2\alpha \cos 2\theta + \sin 2\alpha \sin 2\theta\right] = \frac{1}{2}\left[1 + \cos\left(2\theta - 2\alpha\right)\right] = \cos^2\left(\theta - \alpha\right),$$

which is the Malus law, obtained here as a statement about the angle between two vectors on the Poincaré sphere rather than as a statement about a projection in the laboratory. Feeding unpolarised light $\vec{S} = (1, 0, 0, 0)$ into the same formula gives $S_0' = 1/2$ together with a fully polarised output, which is the equally familiar fact that a polariser transmits exactly half of natural light.

The general linear dichroic 3.85, with real transmittances t_1 and t_2 along its two eigenstates, interpolates between the retarder and the polariser and is more interesting still. Taking $J = \mathrm{diag}(t_1, t_2)$ in 3.90 and computing the four congruences as before,

$$M_D = \begin{bmatrix} a & b & 0 & 0 \\ b & a & 0 & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & c \end{bmatrix}, \qquad a = \frac{t_1^2 + t_2^2}{2}, \quad b = \frac{t_1^2 - t_2^2}{2}, \quad c = t_1 t_2, \tag{3.96}$$

and a one line computation shows that these three numbers are not independent,

$$a^2 - b^2 = \frac{\left(t_1^2 + t_2^2\right)^2 - \left(t_1^2 - t_2^2\right)^2}{4} = t_1^2 t_2^2 = c^2.$$

The relation $a^2 - b^2 = c^2$ is the signature of a hyperbolic rotation. Writing $a = c\cosh\zeta$ and $b = c\sinh\zeta$, which the constraint permits, 3.96 becomes

$$M_D = t_1 t_2 \begin{bmatrix} \cosh\zeta & \sinh\zeta & 0 & 0 \\ \sinh\zeta & \cosh\zeta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \tanh\zeta = \frac{t_1^2 - t_2^2}{t_1^2 + t_2^2} \equiv D, \tag{3.97}$$

an overall attenuation multiplying a **Lorentz boost** along the S_1 axis, with rapidity ζ and with the quantity D, called the diattenuation, playing the part of the velocity β.

This is the promise made at the end of the discussion of why the Stokes vector is not a vector, now delivered. There we found that the physical states fill the forward light cone of an abstract Minkowski space, with the interval $S_0^2 - S_1^2 - S_2^2 - S_3^2$ measuring the departure from full polarisation. We can now name the transformations: **retarders are the rotations of that Minkowski space and dichroics are its boosts**, and together they generate the proper orthochronous Lorentz group.

The geometry is faithful in every detail. A boost along $\hat{n}$ drags points of the sphere toward $\hat{n}$, crowding them ever closer as the rapidity grows, and this is exactly what a partial polariser does: it does not turn a state, it pulls it toward its own transmission axis. The ideal polariser 3.95 is the limit $t_2 \to 0$, that is $D \to 1$ and $\zeta \to \infty$, the infinite boost that carries every point to the same destination, which is why the sphere collapses.

What no Jones matrix can do: depolarisation

We can now prove the claim made at the very beginning. Suppose the incoming light is fully polarised, so that by the discussion of the Stokes identity its polarisation matrix has $\det \Phi = 0$, equivalently $\Phi = \vec{e}\,\vec{e}^{\dagger}$ has rank one. Then by 3.88 and the multiplicative property of the determinant,

$$\det \Phi' = \det\left(J \Phi J^{\dagger}\right) = \left|\det J\right|^{2} \det \Phi = 0, \tag{3.98}$$

so the outgoing light is fully polarised as well. **A Jones element maps pure states to pure states and can never reduce the degree of polarisation.** The Poincaré sphere may be rotated, boosted or crushed, but a point on its surface never moves into its interior.

Yet the interior is where most of the light in the universe lives. The resolution is that a depolarising sample is not one Jones matrix but a *statistical ensemble* of them, fluctuating faster than the detector can follow or varying from point to point across the illuminated spot: the rough surface presents a different microfacet to each part of the beam, the tissue offers a different scattering history to each photon, the turbulent atmosphere a different birefringence at each instant. Averaging 3.88 over that ensemble and using the linearity of 3.90 in the pair $(J, J^{\dagger})$,

$$M = \langle M\left(J\right)\rangle = \sum_{k} p_k\, M\left(J_k\right), \qquad p_k \geq 0, \quad \sum_{k} p_k = 1, \tag{3.99}$$

a convex combination of non depolarising matrices. Convex sums of rotations are not rotations, and this is where the degree of polarisation is lost, in exactly the way that averaging unit vectors pointing in different directions was found to shorten the Stokes vector. The archetype is the ideal depolariser

$$M_{\Delta} = \operatorname{diag}\left(1, d_1, d_2, d_3\right), \qquad |d_j| \leq 1, \tag{3.100}$$

which shrinks the sphere into an ellipsoid with semiaxes d_1, d_2, d_3 centred on the origin, and which for $d_1 = d_2 = d_3 = 0$ collapses everything onto the centre, that is, onto completely unpolarised light of unchanged intensity.

A parameter count makes the size of the gap plain. A general Mueller matrix carries 16 real numbers. A Jones matrix carries 8, of which one is the unmeasurable global phase, so the non depolarising Mueller matrices form a family of only 7 parameters sitting inside a 16 dimensional space. The overwhelming majority of physically admissible Mueller matrices therefore have got no Jones counterpart whatsoever, and this is the precise sense in which the Mueller calculus is not a reformulation of the Jones calculus but a genuine enlargement of it.

However, we'll see in the chapter of they of coherence that our work in the GOTS can use the Jones matrices to explain with great precission the depolarisation phenomena.

Why this matters

It is worth closing with the reason this formalism is used far outside the optics laboratory. A Mueller matrix is 16 real numbers, and every one of them is measurable: illuminate the sample with four known input states, analyse each output with four known projections, and the resulting 16 intensities invert to give M. The technique is called Mueller polarimetry, and what it delivers is a complete linear optical fingerprint of the sample, containing its birefringence, its dichroism and, crucially, its depolarising power, all of them separable from one another by decomposing M into a product of a retarder, a dichroic and a depolariser.

The depolarising part is often the most informative, precisely because it is the part Jones could not describe. Healthy and cancerous tissue differ markedly in how strongly they depolarise, since the difference in the arrangement of collagen fibres changes the statistics of the ensemble in 3.99, and polarimetric imaging is being developed as a label free diagnostic on exactly this basis. In remote sensing the depolarisation of scattered sunlight distinguishes spherical droplets from irregular ice crystals in a cloud, which is why the Stokes formalism entered atmospheric physics through the radiative transfer of Chandrasekhar (Chandrasekhar, 1950). In semiconductor metrology, ellipsometry measures the Mueller matrix of a wafer to determine film thicknesses to sub nanometre precision (Azzam and Bashara, 1977). And in astronomy the polarisation of starlight, scrambled by interstellar dust, maps the galactic magnetic field.

There is finally a structural lesson that reaches well beyond optics. The polarisation matrix Φ is a 2×2 Hermitian, positive semidefinite matrix of unit trace once normalised, which is precisely the definition of the density matrix of a two level quantum system; the Poincaré sphere is its Bloch sphere; a Jones matrix is a unitary evolution; and a general Mueller matrix is the classical counterpart of a quantum channel, a completely positive map, with 3.99 playing the role of the ensemble decomposition of that channel. Depolarisation is decoherence. Everything we have built in this subsection, the rotations, the boosts, the convex mixing that destroys purity, reappears unchanged in the theory of open quantum systems, and a bench with a laser and two waveplates is one of the cheapest and most transparent laboratories in which those ideas can be watched happening.

3.5.6 The Jones theorems

Jones did not merely propose a convenient notation. In the series of papers that runs through the *Journal of the Optical Society of America* between 1941 and 1956, and above all in the second of them, written jointly with Henry Hurwitz (Hurwitz and Jones, 1941), he proved a handful of structural theorems that answer a question no amount of matrix multiplication answers by itself: given that an optical bench may contain an arbitrary number of waveplates, polarisers and rotators arranged in any order, **what is the most general thing such a bench can possibly do to a beam of light, and how few elements are needed to do it?**

The answers are remarkably restrictive, and that is exactly what makes them useful. They tell the instrument designer that certain constructions are impossible, that certain others are redundant, and that any device whatsoever, however baroque, is equivalent to a short standard chain. Before stating them we need to fix what equivalence means and to classify the primitive elements by the kind of matrix they possess.

Definition 28 *(Optical equivalence): Two optical systems are* ***optically equivalent*** *if their Jones matrices differ at most by a global phase factor, that is, if $J_2 = e^{i\gamma} J_1$ for some real γ. Equivalently, by 3.90, they possess the same Mueller matrix, and therefore no measurement of any kind performed on the outgoing light can distinguish them.*

The three primitive devices built in the previous subsections fall into sharply distinct algebraic classes, and the whole of what follows is a consequence of that classification. A retarder 3.75 is **unitary** and, once the irrelevant global phase is removed, **unimodular**, $\det W = 1$; the reader may check this directly on the linear retarder 3.76, whose determinant is $\cos^2 \frac{\delta}{2} + \sin^2 \frac{\delta}{2} \left(\cos^2 2\alpha + \sin^2 2\alpha\right) = 1$. A rotator 3.83 is **real orthogonal** with unit determinant, hence a particular case of a retarder, as we already proved. A linear dichroic 3.85, and its limiting case the ideal polariser 3.86, is **real symmetric and positive semidefinite**, while a general elliptical dichroic 3.84 is **Hermitian and positive semidefinite**. Finally, a cascade is the ordered product of the matrices of its elements. Every theorem below is a statement about what such products can look like.

Theorem I: every system has an eigenpolarisation

Theorem 2 *(Existence of eigenpolarisations): Every optical system with Jones matrix J possesses at least one polarisation state that it leaves unchanged in form, altering it at most in amplitude and phase. Generically there are exactly two such states, and they are mutually orthogonal if and only if J is a normal matrix, $JJ^\dagger = J^\dagger J$.*

Proof. The eigenvalue equation $J\vec{e} = \lambda\vec{e}$ has a non trivial solution precisely when $\det(J - \lambda I) = 0$. Writing $J = \begin{bmatrix} j_{11} & j_{12} \\ j_{21} & j_{22} \end{bmatrix}$, this determinant is the quadratic

$$\lambda^2 - (j_{11} + j_{22})\,\lambda + (j_{11}j_{22} - j_{12}j_{21}) = \lambda^2 - (\operatorname{tr} J)\,\lambda + \det J = 0, \tag{3.101}$$

a polynomial of degree two with complex coefficients. By the fundamental theorem of algebra it possesses at least one root $\lambda_1 \in \mathbb{C}$, and for that root the matrix $J - \lambda_1 I$ is singular, so its kernel contains a non zero vector $\vec{e}_1$. That vector is a Jones spinor 3.34 and therefore a genuine polarisation state, and by construction $J\vec{e}_1 = \lambda_1\vec{e}_1$: the outgoing spinor is the incoming one multiplied by a single complex number, whose modulus is an amplitude change and whose argument is a phase change, neither of which alters the shape or the orientation of the ellipse. When the discriminant of 3.101 does not vanish the two roots are distinct and the two eigenvectors are linearly independent, giving two eigenpolarisations.

For the second part, recall the spectral theorem: a complex square matrix admits an *orthonormal* basis of eigenvectors if and only if it is normal. Since orthogonality of the two eigenspinors in the

Hermitian product is precisely orthogonality of the two polarisation states, the equivalence follows at once. ■

The theorem has got more content than it first appears. All three primitive elements are normal: a retarder is unitary and a dichroic is Hermitian, and both classes satisfy $JJ^\dagger = J^\dagger J$ trivially. Their eigenstates are therefore always orthogonal, which is why we were able to build every one of them in the previous subsections from a pair of orthogonal states $|\psi_1\rangle, |\psi_2\rangle$ using a spectral decomposition. **A cascade, however, need not be normal.** Take a linear retarder of retardance $\delta = 1.2$ with its axis at $\alpha = 0.4$, followed by a linear dichroic with transmittances $t_1 = 0.9$ and $t_2 = 0.3$ along the coordinate axes. The commutator $[J, J^\dagger]$ of the product does not vanish, and the two eigenpolarisations turn out to satisfy $|\langle e_1|e_2\rangle| \approx 0.46$, which is very far from zero.

Elements whose eigenpolarisations are orthogonal are called **homogeneous**, and the others **inhomogeneous**. The lesson is worth stating plainly, because it is a common source of confusion: a waveplate followed by a polariser is not, in general, describable as some other waveplate followed by some other polariser with the same axes, and the reason is that the composite object has stopped being normal. The eigenstates it does possess are no longer a pair of orthogonal ellipses but a skew pair, and this is one of the ways in which real optical systems misbehave.

Theorem II: retarders and rotators

Theorem 3 *(First equivalence theorem): An optical system containing any number of retarders and rotators, in any order, is optically equivalent to a system containing only two elements: one linear retarder followed by one rotator.*

Proof. Every retarder and every rotator has got a unitary, unimodular Jones matrix, that is, an element of the group $SU(2)$. Since $SU(2)$ is a group it is closed under multiplication, so the whole cascade, whatever its length and order, is again a single element $U \in SU(2)$. The theorem therefore reduces to a purely algebraic claim: every $U \in SU(2)$ can be written as $U = R(\theta)\, W(\alpha, 0; \delta)$ for suitable θ, α, δ.

The cleanest way to see this is to move to the Poincaré sphere, where we already know what these objects do. By the theorem on retarders as rotations, the linear retarder $W(\alpha, 0; \delta)$ acts on the sphere as a rotation by δ about the equatorial axis 3.94 sitting at azimuth 2α, since $\chi = 0$ places $\hat{n}$ on the equator; and the rotator $R(\theta)$, being a circular retarder, acts as a rotation about the polar axis. So the claim becomes: *every rotation of the sphere is a rotation about an equatorial axis followed by a rotation about the polar axis.*

This is the Euler angle decomposition in disguise. Write the general rotation in the ZYZ convention as $\mathcal{R} = Z(\phi)\, Y(\vartheta)\, Z(\psi)$, which is possible for any element of $SO(3)$. Now insert the identity $Z(\psi)Z(-\psi)$ and regroup,

$$\mathcal{R} = Z(\phi)Y(\vartheta)Z(\psi) = \underbrace{Z\,(\phi+\psi)}_{\text{polar}}\ \underbrace{[Z(-\psi)\,Y(\vartheta)\,Z(\psi)]}_{\text{equatorial}},$$

and observe that the bracket is a rotation by the angle ϑ about the axis $Z(-\psi)\hat{y}$, which lies in the equatorial plane because $\hat{y}$ does and Z preserves that plane. The first factor is a rotation about the polar axis. Since ϕ, ϑ, ψ range over all of $SO(3)$, so do the three parameters θ, α, δ of our two element chain. Finally, the Mueller map 3.90 restricted to $SU(2)$ is onto $SO(3)$, and its only ambiguity is the sign of U, which is a global phase and hence optically invisible. The two systems are therefore optically equivalent. ■

The physical reading is the one that matters. **No matter how many waveplates you stack, at whatever angles, the result is a single elliptical retarder.** It can absorb nothing, because unitarity is preserved under multiplication, and it can depolarise nothing, by 3.98. Everything a pile of birefringent crystal can do is to rotate the Poincaré sphere once, and three numbers suffice to say how. This is why the elliptical retarder we constructed by a change of basis is not an academic curiosity but a piece of hardware: any (α, χ, δ) you like can be realised with one ordinary waveplate and one rotator, and in the laboratory the combination of two quarter wave plates and a half wave plate is the standard universal polarisation controller built on exactly this theorem.

Theorem III: partial polarisers and rotators

Theorem 4 *(Second equivalence theorem): An optical system containing any number of linear partial polarisers and rotators, in any order, is optically equivalent to a system containing only two elements: one linear partial polariser followed by one rotator.*

Proof. A linear partial polariser 3.85 has a *real symmetric* matrix, and a rotator 3.83 has a *real orthogonal* one. Products of real matrices are real, so the entire cascade is described by some real matrix A, non singular as long as no element is an ideal polariser. Furthermore $\det A > 0$, because every factor has positive determinant, namely $t_1 t_2 > 0$ for a partial polariser and $+1$ for a rotator.

We now invoke the real polar decomposition, and since it is the engine of both this theorem and the next, let us prove it rather than quote it. Consider the matrix AA^T. It is symmetric, since $\left(AA^\mathsf{T}\right)^\mathsf{T} = AA^\mathsf{T}$, and it is positive definite, because for any non zero real vector $\vec{v}$ we have $\vec{v}^\mathsf{T} AA^\mathsf{T} \vec{v} = \left\|A^\mathsf{T}\vec{v}\right\|^2 > 0$, the strict inequality holding because A^T is invertible. A real symmetric positive definite matrix can be diagonalised by an orthogonal matrix, $AA^\mathsf{T} = V \operatorname{diag}(\mu_1, \mu_2)\, V^\mathsf{T}$ with $\mu_1, \mu_2 > 0$, so we may define

$$S \equiv V \operatorname{diag}\left(\sqrt{\mu_1}, \sqrt{\mu_2}\right) V^\mathsf{T}, \tag{3.102}$$

which is again real, symmetric, positive definite, invertible, and satisfies $S^2 = AA^\mathsf{T}$. Setting $R \equiv S^{-1}A$ we verify directly that

$$RR^\mathsf{T} = S^{-1}A\, A^\mathsf{T} S^{-1} = S^{-1}S^2 S^{-1} = I,$$

where we used that S^{-1} is symmetric. So R is real orthogonal, and $\det R = \det A / \det S > 0$ forces $\det R = +1$, making R a proper rotation, that is, a rotator $R(\theta)$. We have obtained $A = SR$ with S

a real symmetric positive definite matrix, which is exactly a linear partial polariser with some axis and some pair of transmittances, followed by a rotator. ■

Note where each hypothesis was used, because the theorem is sharp. Reality of the matrices is what forced the unitary factor to be a mere rotation instead of a general retarder; had we allowed elliptical partial polarisers, the argument would have produced a retarder in its place and the statement would fail. Physically this says something clean: **a stack of absorbing but non birefringent elements can never manufacture birefringence**, it can only attenuate anisotropically and turn the plane of polarisation. Dichroism and birefringence do not generate each other.

Theorem IV: the general equivalence theorem

Theorem 5 *(Third equivalence theorem): Every non singular Jones matrix J factorises uniquely as*

$$J = H\,U, \tag{3.103}$$

with H Hermitian positive definite and U unitary. Consequently any optical system whatsoever, containing any number of retarders, partial polarisers and rotators in any order, is optically equivalent to one elliptical partial polariser followed by one elliptical retarder, and therefore, expanding each of these through Theorem II, to a chain of four primitive elements: two retarders, one partial polariser and one rotator.

Proof. The construction copies the real case with daggers in place of transposes. Consider $JJ^\dagger$. It is Hermitian, since $\left(JJ^\dagger\right)^\dagger = JJ^\dagger$, and positive definite, because for any non zero $\vec{v} \in \mathbb{C}^2$,

$$\vec{v}^\dagger JJ^\dagger \vec{v} = \left(J^\dagger \vec{v}\right)^\dagger \left(J^\dagger \vec{v}\right) = \left\|J^\dagger \vec{v}\right\|^2 > 0,$$

strictly, because $J^\dagger$ is invertible and so cannot annihilate $\vec{v}$. By the spectral theorem a Hermitian matrix is diagonalisable by a unitary one, $JJ^\dagger = V\,\mathrm{diag}\,(\mu_1, \mu_2)\,V^\dagger$ with $\mu_1, \mu_2 > 0$ real, and we define its positive square root

$$H \equiv V\,\mathrm{diag}\,(\sqrt{\mu_1}, \sqrt{\mu_2})\,V^\dagger, \tag{3.104}$$

which is Hermitian, positive definite, invertible and satisfies $H^2 = JJ^\dagger$. Put $U \equiv H^{-1}J$. Then, using $H^\dagger = H$ and hence $\left(H^{-1}\right)^\dagger = H^{-1}$,

$$UU^\dagger = H^{-1}J\,J^\dagger H^{-1} = H^{-1}H^2H^{-1} = I,$$

so U is unitary and $J = HU$ as claimed. For uniqueness, suppose $J = H'U'$ with H' Hermitian positive definite and U' unitary. Then $JJ^\dagger = H'U'U'^\dagger H' = H'^2$, and since a positive definite Hermitian matrix has got exactly one positive definite Hermitian square root, $H' = H$, whence $U' = H^{-1}J = U$. ■

The dictionary back to hardware is immediate. The factor H, being Hermitian and positive definite, is by 3.84 an elliptical partial polariser, with its two eigenstates and its two real transmittances

$\sqrt{\mu_1}, \sqrt{\mu_2}$. The factor U, being unitary, is by 3.75 an elliptical retarder, which Theorem II lets us build from one linear retarder and one rotator. And H itself, an elliptical partial polariser, is a linear partial polariser conjugated by a retarder that carries the linear eigenstates onto the elliptical ones. Counting the free parameters confirms that nothing has been lost or invented: a general $J \in GL(2, \mathbb{C})$ has got eight real parameters, matched exactly by four for H plus four for U in 3.103, and equally by the hardware chain of two retarders at two parameters each, one partial polariser at three, and one rotator at one, totalling eight.

This is the deepest of the three theorems and it deserves a sentence of its own. **Every linear, non depolarising optical system in existence, however many elements it contains, does exactly two things and in this order: it attenuates the two components of the field unequally, and then it retards one relative to the other.** Diattenuation followed by birefringence, and nothing else is possible. In the language of the previous subsection the statement is even more striking, because a Hermitian positive H maps to a Lorentz boost and a unitary U to a rotation: 3.103 is nothing but the familiar decomposition of a Lorentz transformation into a boost times a rotation, transplanted to the Poincaré sphere. The result is not a historical curiosity either. Its descendant at the level of Mueller matrices, in which a measured M is factorised into a depolariser, a retarder and a diattenuator, is the standard tool by which a modern polarimeter turns sixteen raw numbers into three physically interpretable objects.

Theorem V: reversibility

The last theorem concerns what happens when the light is sent back through the system, and it is the one with the most conspicuous practical consequence.

Theorem 6 *(Reversibility): If a reciprocal optical system has Jones matrix J for propagation in one direction, then for propagation in the opposite direction its Jones matrix is the transpose J^{T}, the transverse axes being referred to the direction of travel.*

Proof. Reversing the beam does two things: it traverses the elements in the opposite order, and it traverses each individual element backwards. The first is automatic in the calculus, since if the forward system is $J = J_n \cdots J_2 J_1$ then the reversed one is $\tilde{J}_1 \tilde{J}_2 \cdots \tilde{J}_n$, where $\tilde{J}_k$ denotes the kth element traversed backwards. The identity $(J_n \cdots J_1)^{\mathsf{T}} = J_1^{\mathsf{T}} \cdots J_n^{\mathsf{T}}$ reverses the order automatically, so the theorem holds for the whole system as soon as it holds for each element, $\tilde{J}_k = J_k^{\mathsf{T}}$. It therefore suffices to check the primitives.

A linear retarder 3.76 is a **symmetric** matrix, its two off diagonal entries being equal to $i \sin \frac{\delta}{2} \sin 2\alpha$, so $W^{\mathsf{T}} = W$: a waveplate looks the same from either side, which is exactly what a slab of birefringent crystal must do, since its fast and slow axes are properties of the crystal and not of the direction of travel. A linear dichroic 3.85 is likewise symmetric, and for the same reason. A rotator 3.83 is orthogonal, so $R(\theta)^{\mathsf{T}} = R(\theta)^{-1} = R(-\theta)$: run backwards, an optically active medium rotates the plane of polarisation by the opposite angle. This is the correct physics, because in a sugar solution the sense of rotation is tied to the handedness of the molecules and to the direction of propagation together, so reversing the beam reverses the apparent sense. Every primitive obeys the rule, and therefore so does every system built from them. ■

The consequences are best seen in a double pass, the situation of a beam that goes through a system, reflects off a mirror and returns. Ignoring the mirror itself, the round trip matrix is $J^{\mathsf{T}}J$. For a quarter wave plate with its axis along x this gives, since W is symmetric,

$$W\left(0,0;\tfrac{\pi}{2}\right)^{\mathsf{T}} W\left(0,0;\tfrac{\pi}{2}\right) = W\left(0,0;\tfrac{\pi}{2}\right)^2 = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} = W\left(0,0;\pi\right), \tag{3.105}$$

precisely a half wave plate: **double passing a retarder doubles its retardance**, which is the everyday trick by which a quarter wave plate placed in front of a mirror rotates linear polarisation by ninety degrees and lets a polarising beam splitter separate the outgoing beam from the returning one. For a rotator, by contrast,

$$R(\theta)^{\mathsf{T}} R(\theta) = R(-\theta)R(\theta) = I, \tag{3.106}$$

so **double passing an optically active medium undoes the rotation exactly**. Send polarised light through a tube of sugar solution, reflect it and send it back, and it emerges with its original azimuth.

And here is the payoff. A Faraday rotator, mentioned when we introduced the rotators, looks identical to an optically active medium on a single pass, yet its sense of rotation is fixed by the external magnetic field alone and not by the direction of propagation. On the return pass it therefore contributes $R(\theta)$ again rather than $R(-\theta)$, so the round trip gives $R(\theta)R(\theta) = R(2\theta)$ instead of the identity. The Faraday rotator **violates the reversibility theorem**, and it is allowed to do so precisely because a magnetic field breaks the time reversal symmetry on which reciprocity rests. This single fact is the entire working principle of the optical isolator, the device that lets light out of a laser but not back into it, without which no high power laser or fibre optic communication link would be stable. A theorem about transposes, proved in three lines, is what tells us that such a device cannot be built from waveplates and sugar and must be built from magnetised glass.

What the theorems say together

Standing back, the five results form a single picture. Theorem I says every system has a state it preserves, but warns that cascades of well behaved elements need not be well behaved themselves. Theorems II and III say that the two families of devices are closed and do not generate each other: birefringence stacks into birefringence and dichroism into dichroism. Theorem IV says that when the two families are mixed, the outcome is still astonishingly restricted, one diattenuator and one retarder and nothing more, eight real numbers in total. Theorem V says that this whole structure is blind to the direction of travel unless a magnetic field is present to break the symmetry.

The unifying statement is group theoretic, and it is worth making explicit because it will reappear whenever linear systems are studied. The non depolarising optical elements form the group $GL(2,\mathbb{C})$, of which the retarders form the unitary subgroup and the dichroics the positive Hermitian cone, and

3.103 is the statement that the group is the product of the two. Everything a bench can do to fully polarised light is an element of that group, and everything it can do to partially polarised light is, by the ensemble argument 3.99, a convex mixture of such elements. The Jones theorems are, in the end, the complete answer to the question of what polarisation optics is capable of.

3.5.7 The elliptic general Malus law

3.5.8 The law of elliptic birrefringents

Chapter 4

Interference and Interferometers

When two waves meet, light may add to light and yield darkness; in that paradox lies the whole of physical optics.

"When two undulations, from different origins, coincide either perfectly or very nearly in direction, their joint effect is a combination of the motions belonging to each."
– Thomas Young

4.1 A historical context of interference and interferometers

Interference is the most direct fingerprint of the wave nature of light: when two beams overlap, their amplitudes - not their intensities - add, and the result can be brighter than either beam alone or, astonishingly, complete darkness. From this simple superposition principle grew the interferometer, an instrument that turns the invisible phase of a light wave into a visible pattern of fringes, and which became one of the most precise measuring devices ever built.

Although Robert Hooke and Isaac Newton both studied the coloured fringes produced by thin films - the phenomenon still known as Newton's rings, carefully described in Newton's *Opticks* of 1704 Newton (1704) - neither interpreted them as evidence of waves. Newton's immense authority, and his corpuscular theory, held the field for a century. The decisive break came in 18011804 with Thomas Young, whose double-slit experiment showed that light passing through two narrow apertures produces a pattern of bright and dark bands that can only be explained if the two beams alternately reinforce and cancel one another Young (1804). Young introduced the very word “interference” and, with it, the principle of superposition that underlies all of wave optics.

Young's qualitative insight was forged into a quantitative theory by Augustin-Jean Fresnel, who combined the principle of interference with Huygens's secondary wavelets to compute diffraction and interference patterns in full detail, and whose recognition that light is a transverse wave (developed with Arago) tied interference firmly to polarisation: only components sharing a polarisation can interfere. By the middle of the nineteenth century the wave theory was secure, and interference had become a tool. Hippolyte Fizeau and Léon Foucault used interferometric methods to compare the speed of light in different media, and Fizeau exploited the visibility of fringes to infer stellar properties - an idea far ahead of its time.

The instrument that defines the modern era is the Michelson interferometer. Albert A. Michelson, seeking ever more precise measurements, built in the 1880s a device that splits a beam in two, sends the halves along perpendicular arms, and recombines them so that any difference in optical path appears as a shift of the fringes. With it, Michelson and Edward Morley performed in 1887 their famous experiment to detect the motion of the Earth through the luminiferous aether; the null result Michelson and Morley (1887) became one of the experimental pillars on which Einstein's special relativity would later rest. Michelson also turned his interferometer into a metrological standard, measuring the length of the standard metre in wavelengths of light, work for which he received the Nobel Prize in 1907.

Almost simultaneously, Charles Fabry and Alfred Perot introduced in 1899 an interferometer of a different kind, based on multiple reflections between two highly reflecting parallel surfaces Fabry and Perot (1899). The FabryPerot interferometer produces extremely sharp transmission fringes and offers very high spectral resolving power; it is the direct ancestor of the optical resonators and cavities that define the modes of every laser. Other configurations followed - the MachZehnder interferometer, with its separated paths ideal for probing transparent samples, and the Sagnac interferometer, sensitive to rotation and the heart of modern optical gyroscopes.

The visibility of the fringes in any of these instruments is not merely an experimental nuisance: it is a direct measure of the coherence of the light, and it was precisely the analysis of fringe visibility that motivated the statistical theory of coherence developed in a later chapter. In the twentieth century, interferometry reached almost unimaginable sensitivity: the Laser Interferometer Gravitational-Wave Observatory (LIGO), a pair of kilometre-scale Michelson interferometers, detected in 2015 the passage of a gravitational wave by measuring a change in arm length thousands of times smaller than a proton Abbott et al. (2016). From Young's two slits to the detection of colliding black holes, the interferometer has remained the same idea - read the phase of a wave by letting it interfere with itself - refined to extraordinary perfection.

4.2 How does interference occur?

Lets consider two sources of electromagnetic field, under the conditions we've mentioned on this lectueres, we're going only to focus on the electric component, therefore, lets suppose we've got the following two sources of electric field

$$\vec{E}_1\left(\vec{r},t\right) = \vec{E}_1(\vec{r})e^{-i\omega_1 t}, \tag{4.1}$$

$$\vec{E}_2\left(\vec{r},t\right) = \vec{E}_2(\vec{r})e^{-i\omega_2 t}, \tag{4.2}$$

where the complex field amplitudes are given by

$$\vec{E}_1(\vec{r}) = E_1(\vec{r})e^{i\varphi_1(\vec{r})}\hat{e}_1, \tag{4.3}$$

$$\vec{E}_2(\vec{r}) = E_2(\vec{r})e^{i\varphi_2(\vec{r})}\hat{e}_2. \tag{4.4}$$

Theses complex field amplitudes are almost general, here the polarization vectores $\hat{e}_1, \hat{e}_2$ are'nt space-time dependent, which, in general, is what it's observed. The phases $\varphi_1(\vec{r}), \varphi_1(\vec{r})$ represent an arbitrary wavefront behavior,also $E_1(\vec{r}), E_2(\vec{r})$ represent arbitrary amplitudes. Thus the expressions above are general experessions for arbitrary sources of radiation.

The interference appears in the superposition of two or more electric fields, therefore, to study the interference of these two field in a space-time point $(\vec{r},t)$ we're going to add these two as follows

$$\vec{E}(\vec{r},t) = \vec{E}_1(\vec{r},t) + \vec{E}_2(\vec{r},t). \tag{4.5}$$

As we've explained before, measuring the electric field is really hardo to do, so what we study and see are the irradiencies or radiations intensity, also the frequencies $\{\omega_1, \omega_2\}$ are too fast time oscillations for the human eye and most of the detectors, so we only can study their time averages. Having all this on count, lets calculate the average of the irradiances

$$\left\langle \left|\left|\vec{E}(\vec{r},t)\right|\right|^2 \right\rangle = \left\langle \left|\left|\vec{E}_1(\vec{r},t)\right|\right|^2 \right\rangle + \left\langle \left|\left|\vec{E}_2(\vec{r},t)\right|\right|^2 \right\rangle + 2\left\langle Re\left\{\vec{E}_1^*(\vec{r},t)\cdot\vec{E}_2(\vec{r},t)\right\}\right\rangle,$$

$$\left\langle \left|\left|\vec{E}(\vec{r},t)\right|\right|^2 \right\rangle = \left|\left|\vec{E}_1(\vec{r})\right|\right|^2 + \left|\left|\vec{E}_2(\vec{r})\right|\right|^2 + 2Re\left\{\vec{E}_1^*(\vec{r})\cdot\vec{E}_2(\vec{r})\left\langle e^{-i(\omega_1-\omega_2)t}\right\rangle\right\}.$$

The first two terms corresponds to **the total field intensity** and the third terms is called **the interference term**because it contains the interaction of both fields.

From a mathematical point of view, to make the interaction non zero, both frequencies must be the same $\omega_1 = \omega_2$, this, like most books say, should imply that the interference only ocurrs with two sources with the same frequency or on identical photons but, this is not completely correct. The frequency-difference exponential $\exp\left(-i\left[\omega_1 - \omega_2\right]t\right)$ is, in general, not zero, so **there are a lot ultrafast oscillations and interferences** what we see, and what we detected using measuring instruments is the time average, and in average, these are zero.

Before going further, lets define the field intensisties as

$$\left|\left|\vec{E}_1(\vec{r})\right|\right|^2 = I_1(\vec{r}), \tag{4.6}$$

$$\left|\left|\vec{E}_2(\vec{r})\right|\right|^2 = I_2(\vec{r}). \tag{4.7}$$

The interference terms is then given by

$$\vec{E}_1^*(\vec{r}) \cdot \vec{E}_2(\vec{r}) = E_1(\vec{r})E_2(\vec{r})e^{i[\phi_2(\vec{r})-\phi_1(\vec{r})]}\hat{e}_1^* \cdot \hat{e}_2, \tag{4.8}$$

then the field intensity is

$$I = I_1 + I_2 + 2Re\left\{\left(\hat{e}_1^* \cdot \hat{e}_2\right)e^{i\triangle\phi(\vec{r})}\right\}\underbrace{E_1(\vec{r})E_2(\vec{r})}_{(I_1 I_2)^{1/2}}. \tag{4.9}$$

The functional $\triangle\phi = \phi_2(\vec{r}) - \phi_1(\vec{r})$ is known as the phase-difference of the waves. The intereference term is zero if the **polarisations are orthogonal** this means $\hat{e}_1^* \cdot \hat{e}_2 = 0$. Without any loss of generality an only for mathematical convenience[1] we're going to consider both sources have got the same polarisation, this means $\hat{e}_1 = \hat{e}_2 \to \hat{e}_1^* \cdot \hat{e}_1 = 1$. Thus, the total intensity becomes

$$I = I_1 + I_2 + 2\left(I_1 I_2\right)^{1/2}\cos\triangle\phi = \left(I_1 + I_2\right)\left(1 + 2\frac{\left(I_1 I_2\right)^{1/2}}{I_1 + I_2}\cos\triangle\phi\right). \tag{4.10}$$

The equation 4.10 is the our most general expression for the interference. Lets notice that the intereference terms is composed in two parts: the first one is a relation between the two intensities I_1, I_2 and the second one only depends of the phase-difference of the two sources by a cosine, then, **the real interference becomes in the phases not in the intensities**, the first terms only **modulates the visibility** of the interference, and we then can call it $\mathcal{V}$ and write it as

$$\mathcal{V} = 2\frac{\left(I_1 I_2\right)^{1/2}}{I_1 + I_2} = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}, \tag{4.11}$$

$$I_{min} = \left(I_1^{1/2} - I_2^{1/2}\right)^2, \quad I_{max} = \left(I_1^{1/2} + I_2^{1/2}\right)^2, \tag{4.12}$$

this term $\mathcal{V}$ is called the **visibility factor** which module the constrast on the interference fringes.

[1] In fact, the polarisation here is only an contrast modulator, however, when we study the spatial coherence, the polarisation is even more important and conditionates the behavior of the interference.

4.2.1 Interference of plane waves

The expression 4.10 is, as we said, the general interference expression, so lets apply it to two plane wave sources. For the plane wave we've got that the complex amplitudes are given by

$$E_1(\vec{r},t) = E_1 e^{i\left(\vec{r}\cdot\vec{k}+\varphi_1\right)}, \qquad E_2(\vec{r},t) = E_2 e^{i\left(\vec{r}\cdot\vec{k}+\varphi_2\right)}, \tag{4.13}$$

where, we are going to suppose, just for to make the maths a bit easier but the methodology is the same, that both amplitudes are the same $E_1 = E_2$.

Because, we are considering that the both waves have got the same frequency $\omega_1 = omega_2$ this implies that both have the same wave number amplitude, this means $||\vec{k}_1|| = ||\vec{k}_2|| = k$. Then, we will have that the phase-difference term is given by

$$\triangle\phi = \left(\vec{k}_2 - \vec{k}_1\right)\cdot\vec{r} + \varphi_2 - \varphi_1. \tag{4.14}$$

The interference 4.10 having on count that $I_1 = I_2 = I_0$ is then

$$I = (I_0 + I_0)\left(1 + 2\frac{(I_0 I_0)^{1/2}}{I_0 + I_0}\cos\triangle\phi\right) = 2I_0\left[1 + \cos\left(\vec{K}\cdot r + \triangle\varphi\right)\right], \tag{4.15}$$

where $\vec{K} = \vec{k}_2 - \vec{k}_1$ is the interferences vector. To understand bettwer the intereference, we're going to do a bit of geometry (see figure 4.1).

Because we're interested on the interference plane, we will take to $\vec{r} = X\hat{x}$ on the interference plane, where $\hat{x} = \vec{K}/||\vec{K}||$, then the spatial dependce gets the form

$$\vec{K}\cdot\vec{r} = KX = 2\kappa X\sin\left(\frac{\alpha}{2}\right), \tag{4.16}$$

and because K is a wave number vector on the interference plane, it can be written in terms of its wave-lenght,which orresponds to the interfringe distance Λ as

$$K = \frac{2\pi}{\Lambda}, \tag{4.17}$$

then

$$\vec{K}\cdot\vec{r} = \frac{2\pi}{\Lambda}X. \tag{4.18}$$

To calculate the relation between the interfringe distance Λ an the known parameters of the wave, we're going to equate the equations 4.16 and 4.18 therefore

$$\frac{2\pi}{\Lambda}X = \frac{2\pi}{\lambda}X\sin\left(\frac{\alpha}{2}\right), \tag{4.19}$$

$$\Lambda = \frac{\lambda}{2\sin\left(\frac{\alpha}{2}\right)}, \tag{4.20}$$

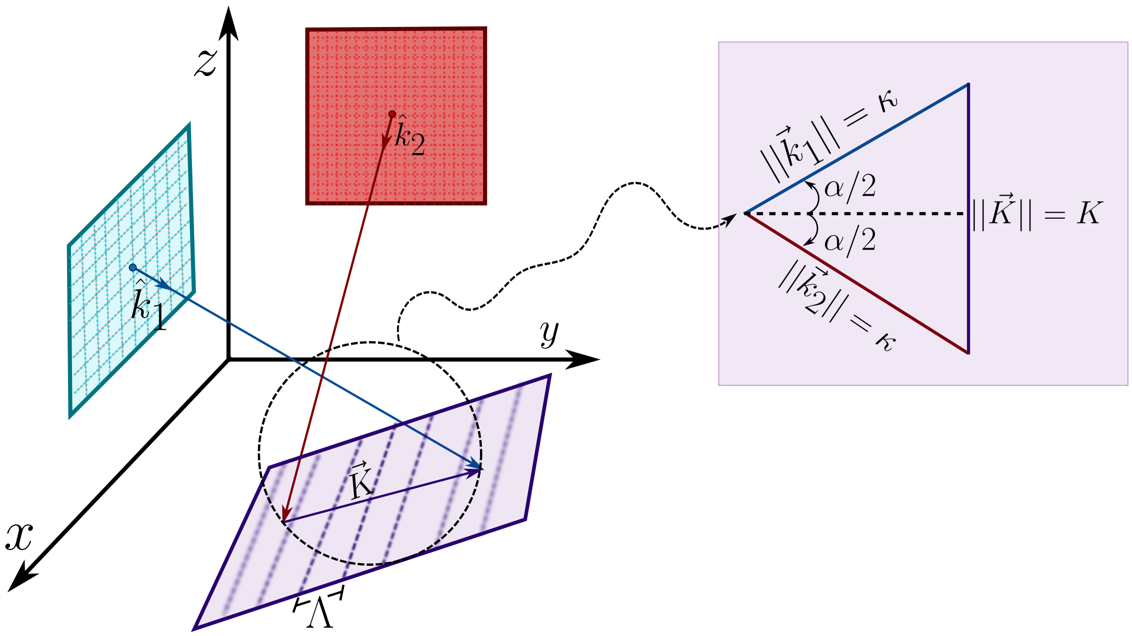


Figure 4.1: Two plane waves with unitary number vector $\hat{k}_1, \hat{k}_2$ propagate and interferece producing a pattern of interference characterized by a vector $\vec{K}$ and an interfringe distance Λ.

where we used that $\kappa = \frac{2\pi}{\lambda}$. With this term, we finally got that the total intensity in the interference of two plane wave is given by

$$I = 2I_0 \left[1 + \cos\left(\frac{2\pi}{\Lambda} X + \phi \right) \right]. \tag{4.21}$$

We can see that the interfringe distance Λ behaves in a very logique way. In $\alpha \to 0$ then $\Lambda \to \infty$ this means, when the two wave number vector $\hat{k}_1, \hat{k}_2$ gets each time more parallel, the interfringe distance tends to infinity, this means, there won't be any interference pattern.

In the other way, if $\alpha \to \pi$ then $\Lambda \to min$ this means, when the two wave number vectors tend to be 180° separeted the interfrence distance is the minimum, being clrear for us to see the interference pattern.

4.2.2 Interference of spherical waves

In a similar way to the last section about plane waves, we're going to consider two radiation sources which propagte as spherical waves, so the electric field waves are given by

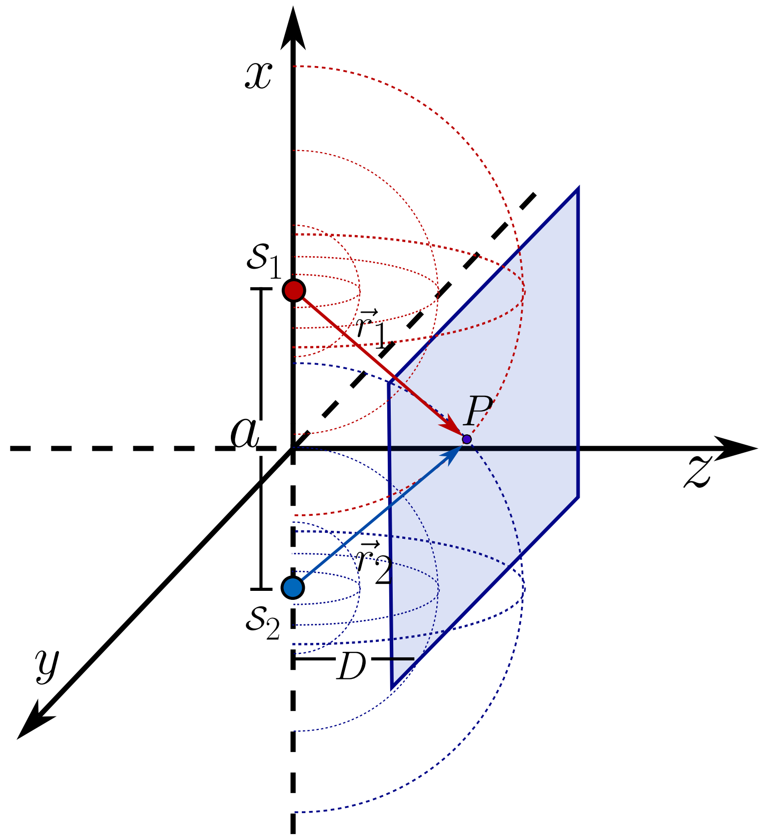


Figure 4.2: Two plane sources of spherical waves interfere in a point P located on the interference plane. The two sources are separated a distance a an the distance from both surface to the interference plane is D.

$$E_1(\vec{r}) = \frac{E_1}{r_1} e^{i(kr_1+\varphi_1)}, \tag{4.22}$$

$$E_2(\vec{r}) = \frac{E_2}{r_2} e^{i(kr_2+\varphi_2)}. \tag{4.23}$$

And, again, we're going to consider, without any loss of generality just for aesthetical purposes, that both have got the sample amplitude $E_1 = E_2 = E_0$. Also, lets suppose a point P on the interference plane and that the two point sources are separated a distance a $[m]$ and both sources are separated a dsitance D $[m]$ as is shown in the figure 4.2.

We have got that $r_1 = \overline{\mathcal{S}_1 P}$ and $r_2 = \overline{\mathcal{S}_2 P}$, then the fase difference is given by

$$\triangle\phi(r) = \kappa\,(r_2 - r_1) + \triangle\varphi. \tag{4.24}$$

And we are going to consider that the field amplitudes, in a neighborhood of P change very little with r_1 and r_2, and then

$$\frac{E_0}{r_1} \approx \frac{E_0}{r_2} = E, \tag{4.25}$$

the intensity is then given by

$$I = 2I_0 \left\{1 + \cos\left[\kappa\left(r_2 - r_1\right) + \triangle\varphi\right]\right\}, \tag{4.26}$$

before going further, see that the **surfaces of equal intensity $\mathbf{r_2} - \mathbf{r_1} = cte$ are hyperboloids with focii in $\mathcal{S}_1, \mathcal{S}_2$.**

In a plane parallel to $\overline{\mathcal{S}_1\mathcal{S}_2}$ at a distance $z = D$ two hyperbolids are produced. For $r_1 \approx r_2$ and $D \gg a$ this hiperboloids are practically planes. Therefore

$$r_1 = \left[D^2 + \left(x - \frac{a}{2}\right)^2 + y^2\right]^{1/2} = D\left[1 + \frac{x^2}{D^2} + \frac{y^2}{D^2} + \frac{a^2}{4D^2} - \frac{ax}{D^2}\right]^{1/2}, \tag{4.27}$$

applying taylor series to first order of $\sqrt{1+x} \approx 1 + x/2$ we've got

$$r_1 \approx D\left[1 + \frac{x^2 + y^2}{2D^2} + \frac{a^2}{2D^2} - \frac{ax}{2D^2}\right], \tag{4.28}$$

and in the same way for r_2

$$r_2 \approx D\left[1 + \frac{x^2 + y^2}{2D^2} + \frac{a^2}{2D^2} + \frac{ax}{2D^2}\right]. \tag{4.29}$$

Substracting 4.29 from 4.27 we then we will have that the phase difference is then

$$\triangle\phi = \kappa\left(r_2 - r_1\right) + \triangle\varphi \approx \frac{\kappa a X}{D} + \triangle\varphi, \tag{4.30}$$

then the total intensity in the interference is given by

$$I = 2I_0\left[1 + \cos\left(\kappa\frac{aX}{D} + \varphi_2 - \varphi_1\right)\right], \tag{4.31}$$

where again

$$KX = \frac{\kappa a X}{D} = \frac{2\pi}{\lambda}\frac{aX}{D},$$

but $K = 2\pi/\Lambda$, being again K the magnitude of the resultante wave number vector on the plane interference, and Λ the interfringe distance. Thus the relation between Λ an the known paramaters of both waves is

$$\frac{2\pi}{\Lambda} = \frac{2\pi a X}{D} \longrightarrow \Lambda = \frac{\lambda D}{a}.$$

With this result, we can see that when the two sources get closer $a \to 0$ the interfringe $\Lambda \to \infty$ this means, the interference is going to disappear when the two point sources get together.

For example, lets consider two monocrhomatic plane waves, which have got complex amplitudes

$$\vec{E}_1 = \mathcal{E} e^{i\varphi_1} \hat{e}_1, \qquad \vec{E}_2 = \mathcal{E} e^{i\varphi_2} \hat{e}_2, \tag{4.32}$$

where $\hat{e}_1, \hat{e}_2$ are thei polarisation vectors and their wave number vectors are given by

$$\vec{k}_1 = [k_{1x}, k_{1y}, k_{1z}]^T \qquad \vec{k}_2 = [k_{2x}, k_{2y}, k_{2z}]^T . \tag{4.33}$$

Now, lets answer the following questions:

1. What relation should be between $\vec{k}_1, \vec{k}_2$ to realize the interference?
2. What's the expression for the visibility?
3. What's the plane where the interference fringes have the least distance? What is the orientation of the plane and the interference fringes distances on that plane?

Lets answer each part following the methodology we have already built. The observable quantity is never the instantaneous field but the time average of the irradiance, which for our two waves reads

$$\left\langle \left|\left|\vec{E}\right|\right|^2 \right\rangle = I_1 + I_2 + 2\,\mathrm{Re}\left\{ \vec{E}_1^* \cdot \vec{E}_2 \left\langle e^{-i(\omega_1-\omega_2)t} \right\rangle \right\} . \tag{4.34}$$

The average $\left\langle e^{-i(\omega_1-\omega_2)t} \right\rangle$ vanishes for any detector whose integration time T satisfies $|\omega_1 - \omega_2| T \gg 1$, so the interference term survives only when $\omega_1 = \omega_2 = \omega$. This first condition is **temporal, not geometric**, and the statement grants it by declaring both waves monochromatic and of the same nature. Its immediate translation to the space of wave number vectors, in a homogeneous and isotropic medium of index n, is

$$\left|\left|\vec{k}_1\right|\right| = \left|\left|\vec{k}_2\right|\right| = \kappa = \frac{n\omega}{c} = \frac{2\pi}{\lambda}. \tag{4.35}$$

This equality of the magnitudes is the only relation that the interference truly requires, and it does not constrain the directions at all. It is worth saying this out loud, because the wording of the question tempts us to answer that the vectors must be parallel, and it is exactly the opposite. If $\vec{k}_1 = \vec{k}_2$ the interference term does exist but no longer depends on the position, the illumination becomes uniform and there are no fringes. For an observable pattern we need the interference vector

$$\vec{K} \equiv \vec{k}_2 - \vec{k}_1 \neq \vec{0}, \tag{4.36}$$

whose magnitude follows at once from the equality of the moduli 4.35,

$$\left|\left|\vec{K}\right|\right|^2 = 2\kappa^2(1 - \cos\alpha) = 4\kappa^2 \sin^2\frac{\alpha}{2} \quad \Longrightarrow \quad K = 2\kappa \sin\frac{\alpha}{2}, \tag{4.37}$$

with α the angle between $\vec{k}_1$ and $\vec{k}_2$. A third condition, this time genuinely geometric, comes from the interference term carrying the factor $\hat{e}_1^* \cdot \hat{e}_2$, which vanishes for orthogonal polarisations. If we want the maximum contrast we need $\hat{e}_1 = \hat{e}_2 = \hat{e}$, and since both waves are transverse, so that $\hat{e} \perp \vec{k}_1$ and $\hat{e} \perp \vec{k}_2$, this common direction is fixed up to a sign,

$$\hat{e} = \frac{\vec{k}_1 \times \vec{k}_2}{\left\| \vec{k}_1 \times \vec{k}_2 \right\|}. \tag{4.38}$$

In words, two non collinear plane waves reach unit visibility only when they are polarised perpendicular to the plane that contains $\vec{k}_1$ and $\vec{k}_2$, which is the quantitative form of the Arago and Fresnel statement we met before.

For the second part, with $\omega_1 = \omega_2$ the interference term survives the average and the total irradiance is

$$I(\vec{r}) = I_1 + I_2 + 2\sqrt{I_1 I_2}\; |\hat{e}_1^* \cdot \hat{e}_2| \cos\left[\triangle\phi(\vec{r}) + \delta\right], \qquad \triangle\phi(\vec{r}) = \vec{K} \cdot \vec{r} + \varphi_2 - \varphi_1, \tag{4.39}$$

where $I_j = |\mathcal{E}_j|^2$ and $\delta = \arg(\hat{e}_1^* \cdot \hat{e}_2)$. As the cosine sweeps the whole interval $[-1, 1]$ when $\vec{r}$ runs over the space, the extremes of the irradiance are read directly,

$$I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2}\,|\hat{e}_1^* \cdot \hat{e}_2|\,, \qquad I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2}\,|\hat{e}_1^* \cdot \hat{e}_2|\,, \tag{4.40}$$

and the Michelson visibility, the same one we use as the contrast modulator, follows without any approximation

$$\mathcal{V} = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2}\,|\hat{e}_1^* \cdot \hat{e}_2|\,. \tag{4.41}$$

With it the irradiance takes the canonical form

$$I(\vec{r}) = (I_1 + I_2)\left[1 + \mathcal{V}\cos\triangle\phi(\vec{r})\right], \tag{4.42}$$

which cleanly separates the three roles, since $I_1 + I_2$ fixes the mean level, $\mathcal{V}$ fixes the contrast and $\triangle\phi$ carries all the physics of the interference. Notice that $\mathcal{V} = 1$ demands both equal intensities and parallel polarisations, but also that $\mathcal{V}$ is remarkably robust against an imbalance of the intensities, since a ratio as large as $I_1/I_2 = 2$ still leaves $\mathcal{V} = 0.943$. We will make use of this insensitivity in the next example.

For the third part, the surfaces of constant phase, and therefore the fringes seen as objects living in the three dimensional space, are the planes $\vec{K} \cdot \vec{r} = \text{cte}$, perpendicular to $\vec{K}$ and separated by

$$\Lambda_0 = \frac{2\pi}{K} = \frac{\lambda}{2\sin(\alpha/2)}. \tag{4.43}$$

What the question asks is on which observation plane P the cut of this family produces the closest fringes. Let $\hat{n}$ be the normal of P and decompose $\vec{K}$ into a part along $\hat{n}$ and a part $\vec{K}_\parallel$ lying inside the plane. Only $\vec{K}_\parallel$ modulates the field over P, so the observed fringes are straight lines perpendicular to $\vec{K}_\parallel$ with a separation

$$\Lambda(\hat{n}) = \frac{2\pi}{\left|\left|\vec{K}_\parallel\right|\right|} = \frac{\Lambda_0}{\sin\gamma}, \qquad \gamma = \angle(\vec{K}, \hat{n}). \tag{4.44}$$

Since $\Lambda(\hat{n}) \geq \Lambda_0$ for every orientation, with equality only when $\sin\gamma = 1$, the fringes are closest exactly when the observation plane contains the direction of $\vec{K}$,

$$P: \quad \hat{n}\cdot\vec{K} = 0, \qquad \Lambda_{\min} = \frac{2\pi}{\left|\left|\vec{k}_2 - \vec{k}_1\right|\right|} = \frac{\lambda}{2\sin(\alpha/2)}, \tag{4.45}$$

while any other tilt dilates the fringes by the factor $1/\sin\gamma$. This is a whole one parameter family of planes rather than a single one, and two of its members deserve a name. The first is the plane spanned by $\vec{k}_1$ and $\vec{k}_2$, whose normal is $\hat{n} = (\vec{k}_1 \times \vec{k}_2)/||\vec{k}_1 \times \vec{k}_2||$, and it contains $\vec{K}$ because $\vec{K}$ is a linear combination of $\vec{k}_1$ and $\vec{k}_2$, being moreover the same plane to which the polarisation had to be perpendicular in the first part. The second is the plane perpendicular to the bisector of the two beams, which is the one used as a screen in the laboratory, and it belongs to the optimal family thanks to the identity

$$\vec{K}\cdot(\vec{k}_1 + \vec{k}_2) = \left|\left|\vec{k}_2\right|\right|^2 - \left|\left|\vec{k}_1\right|\right|^2 = 0. \tag{4.46}$$

So the interference vector is always perpendicular to the bisector, the natural screen is already the optimal one and there is no need to tilt it. On it the fringes are straight lines parallel to $\vec{k}_1 + \vec{k}_2$ with the separation $\lambda/[2\sin(\alpha/2)]$.

4.3 Interferometers

4.3.1 Young interferometer

For most of the eighteenth century the nature of light seemed a settled matter. The authority of Newton, whose *Opticks* Newton (1704) accounted for reflection, refraction and the colours of thin films in terms of tiny corpuscles, had pushed into the background the wave picture that Huygens had proposed in his *Traité de la Lumière* Huygens (1690). A corpuscle travels in a straight line and carries momentum, so the rectilinear propagation of light and the sharpness of shadows looked like immediate consequences of the theory, and the enormous prestige of Newton did the rest. The wave idea survived only as a minority opinion, lacking the one thing that would have made it irresistible, a phenomenon that particles simply cannot imitate.

That phenomenon was interference, and it was Thomas Young who brought it to light. In his Bakerian lecture of 1801, published as *On the Theory of Light and Colours* Young (1802), Young stated what he called the principle of the superposition of undulations, the plain but revolutionary idea that where two portions of light meet their disturbances add, so that two bright beams can combine to give darkness. A few years later, in a second Bakerian lecture Young (1804), he turned the principle into an experiment, letting light pass through a small opening and then splitting the emerging beam so that its two halves overlapped on a screen, where they painted a regular pattern of bright and dark bands. In his *Course of Lectures on Natural Philosophy* Young (1807) the arrangement acquired the form we still teach, two closely spaced apertures illuminated by a common source, and from the spacing of the fringes Young was able to do something no one had done before, to measure the wavelength of light and to find it astonishingly small, a fraction of a thousandth of a millimetre.

The importance of this result is difficult to overstate. The dark bands were the direct proof that light added to light can produce less light, something no theory of independent particles can explain, and they gave the wave theory the empirical victory it had lacked for a century. The reception, however, was hostile, since to attack Newton in the England of 1804 was close to heresy, and Young's papers were ridiculed in the reviews of the day. Vindication came from the continent a decade later, when Fresnel and Arago placed the wave theory on a rigorous mathematical footing and extended it to diffraction and polarisation. The experiment itself never left the stage. It is the archetype of every wavefront division interferometer, the simplest device in which the phase difference between two coherent beams is turned into a visible intensity pattern, and its complete modern treatment belongs to any serious optics text Born and Wolf (1999). More than that, when the same experiment is repeated with electrons, atoms or single photons sent one at a time, the fringes still build up, and the two slit arrangement becomes the cleanest illustration of quantum superposition, the phenomenon that Feynman called the only mystery of quantum mechanics Feynman et al. (1965). The precision descendants of Young's screen reach today the kilometre sized interferometers that detect gravitational waves Abbott et al. (2016).

To see the whole machinery of the previous section at work on this instrument, lets analyse the Young interferometer quantitatively, with its two slits illuminated by a spherical wave coming from a point source, arranged as in the figure 4.3. We place the two slits $\mathcal{S}_1$ and $\mathcal{S}_2$ symmetrically at $x = \pm d/2$ on the plane $z = 0$, separated by a distance d, and the observation screen at $z = D$. The point source $\mathcal{S}$ sits at $(x, z) = (-a, -b)$, so that b is its distance to the slit plane and a its transverse displacement from the symmetry axis, with $a = 0$ meaning the source on the axis, and we measure the coordinate x on the screen from that axis.

The methodology of the chapter applies in two chained stages, and this is really the only structural idea of the problem. The first stage goes from the source to the slits and decides two things, namely with what amplitude and with what phase the field reaches each slit. The second stage goes from the slits to the screen and is always the interference of two secondary spherical waves, exactly the one we treated in the spherical wave section, with the slits acting as the sources $\mathcal{S}_1, \mathcal{S}_2$ separated by d and the screen at a distance D. Writing the field that leaves the slit j as $\mathcal{E}_j e^{i\psi_j}$ and the path from that slit to the point P on the screen as s_j, the total relative phase at P is

$$\triangle\phi(x) = \kappa(s_2 - s_1) + (\psi_2 - \psi_1), \tag{4.47}$$

and the irradiance obeys the canonical expression 4.42 with the visibility 4.41 evaluated on the amplitudes that reach each slit.

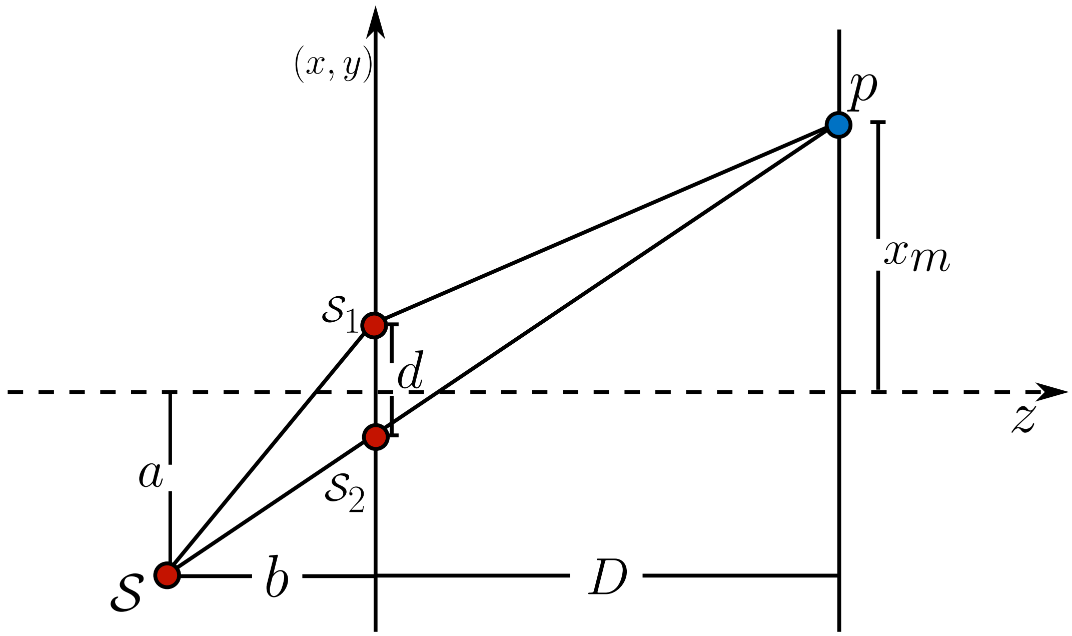


Figure 4.3: Young interferometer illuminated by a spherical wave from a point source $\mathcal{S}$ placed off axis. The source to slit stage sets the amplitude and the phase reaching each slit $\mathcal{S}_1, \mathcal{S}_2$, and the slit to screen stage is the interference of two secondary spherical waves.

The second stage is common to any illumination, so we dispatch it once. With $s_j = [D^2 + (x \mp d/2)^2]^{1/2}$ and the same Taylor expansion of the chapter,

$$s_2 - s_1 \approx \frac{(x + d/2)^2 - (x - d/2)^2}{2D} = \frac{xd}{D}, \tag{4.48}$$

valid in the paraxial regime $D \gg d, |x|$, from which

$$\kappa(s_2 - s_1) \approx \frac{2\pi}{\lambda}\frac{xd}{D} = \frac{2\pi}{\Lambda}x, \qquad \Lambda = \frac{\lambda D}{d}. \tag{4.49}$$

This is precisely the interfringe of the spherical wave section with $a \to d$. Everything that distinguishes one illumination from another sits, then, in the term $\psi_2 - \psi_1$.

For the spherical illumination of this example the source at $(-a, -b)$ emits the wave $E = (E_0/r)e^{i\kappa r}$ of the chapter, and now the two slits are no longer equivalent, because their distances to the source

$$r_1 = \left[b^2 + \left(a + \frac{d}{2}\right)^2\right]^{1/2}, \qquad r_2 = \left[b^2 + \left(a - \frac{d}{2}\right)^2\right]^{1/2} \tag{4.50}$$

differ as soon as $a \neq 0$, and this affects both the phase and the amplitude. In phase, expanding as in the chapter with $b \gg d, a$,

$$r_1 - r_2 \approx \frac{(a + d/2)^2 - (a - d/2)^2}{2b} = \frac{ad}{b} \quad \Longrightarrow \quad \psi_2 - \psi_1 = \kappa(r_2 - r_1) = -\frac{\kappa a d}{b}. \tag{4.51}$$

In amplitude, the slit j receives E_0/r_j, so $I_j \propto 1/r_j^2$ and the visibility 4.41 becomes

$$\mathcal{V} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2} = \frac{2/(r_1 r_2)}{1/r_1^2 + 1/r_2^2} = \frac{2 r_1 r_2}{r_1^2 + r_2^2}. \tag{4.52}$$

Putting the two stages together in 4.47, the total phase is

$$\triangle\phi(x) = \frac{\kappa d}{D} x - \frac{\kappa a d}{b} = \frac{2\pi d}{\lambda}\left(\frac{x}{D} - \frac{a}{b}\right), \tag{4.53}$$

and the irradiance on the screen reads

$$I(x) = (I_1 + I_2)\left\{1 + \mathcal{V}\cos\left[\frac{2\pi d}{\lambda}\left(\frac{x}{D} - \frac{a}{b}\right)\right]\right\}, \qquad \mathcal{V} = \frac{2 r_1 r_2}{r_1^2 + r_2^2}. \tag{4.54}$$

The qualitative difference with a plane illumination, for which $\mathcal{V} = 1$ always, is not in the position of the fringes but in the contrast, because an off axis spherical source gives $\mathcal{V} < 1$.

The first part asks for the value of a that reduces the contrast by 10% with respect to $a = 0$. At $a = 0$ we have $r_1 = r_2$ and $\mathcal{V} = 1$, and we ask for $\mathcal{V}(a) = 0.9$. Writing $t = r_1/r_2 \geq 1$, the condition $2t/(1 + t^2) = 9/10$ is the quadratic $9t^2 - 20t + 9 = 0$, whose larger root is

$$t = \frac{10 + \sqrt{19}}{9} = 1.5954 \quad \Longrightarrow \quad \frac{r_1^2}{r_2^2} = 2.5454. \tag{4.55}$$

A contrast drop of only 10% already requires one slit to be some 60% farther from the source than the other, which shows how insensitive the visibility is to an amplitude imbalance, exactly as we anticipated in the second part of the first example. Substituting the distances 4.50,

$$\frac{b^2 + (a + d/2)^2}{b^2 + (a - d/2)^2} = t^2 \quad \Longleftrightarrow \quad a^2 - \frac{10}{\sqrt{19}}\, d\, a + b^2 + \frac{d^2}{4} = 0, \tag{4.56}$$

a quadratic in a whose formal solution is

$$a = \frac{5d}{\sqrt{19}} \pm \sqrt{\frac{81 d^2}{76} - b^2}. \tag{4.57}$$

Here the physics jumps out of the discriminant, since real solutions exist only if $b \leq \frac{9d}{2\sqrt{19}} = 1.0324\, d$. In a real Young interferometer one always has $b \gg d$, typically $d \sim 10^{-4}$ m and $b \sim 10^{-1}$ m, so the discriminant is negative and no displacement of a point source can degrade the contrast by 10%. The paraxial estimate confirms it, since with $t = 1 + \eta$ and $\eta \approx (r_1 - r_2)/b \approx ad/b^2$ the visibility is $\mathcal{V} \approx 1 - \eta^2/2 = 1 - \frac{1}{2}(ad/b^2)^2$, so $\mathcal{V} = 0.9$ would need $ad/b^2 = \sqrt{0.2} \approx 0.447$, that is $a \approx 0.45\, b^2/d$, a gigantic displacement that destroys the paraxial approximation long before it touches the contrast. The physical conclusion is that a strictly point source always produces fringes of essentially unit visibility, wherever it is placed, because moving it translates the pattern but does not erase it. The version of the question that does have a solution is the one where a is the width of an incoherent thermal source instead of the displacement of a point source, which belongs to the study of spatial coherence and gives a visibility $\mathcal{V}(a) = |\mathrm{sinc}(\pi a d/\lambda b)|$ that falls to its first zero at $a = \lambda b/d$. The lesson to keep is the contrast between the two readings, since displacing a point source acts on the phase and slides the fringes, while broadening an incoherent source acts on the coherence and washes them out.

The second part asks for the position x_m of the m-th maximum. The maxima of the irradiance 4.54 are the zeros modulo 2π of the phase 4.53,

$$\frac{2\pi d}{\lambda}\left(\frac{x_m}{D} - \frac{a}{b}\right) = 2\pi m, \qquad m \in \mathbb{Z}, \tag{4.58}$$

from where

$$x_m = m\frac{\lambda D}{d} + \frac{aD}{b} = m\Lambda + \frac{aD}{b}. \tag{4.59}$$

The pattern keeps its full periodicity $\Lambda = \lambda D/d$ and is shifted as a rigid block by aD/b toward the side opposite to the displacement of the source, consistent with the figure, where the source lies below the axis while x_m is measured above. Geometrically, the zero order maximum sits where the total path difference vanishes, that is on the straight line that joins the source with the midpoint of the slits and continues to the screen, a line that leaves $(-a, -b)$, passes through the origin and reaches the screen at $x = aD/b$.

The third part asks for a condition between a and b so that the origin is a minimum. A minimum at $x = 0$ requires the total phase there to be an odd multiple of π. Evaluating 4.53 at $x = 0$,

$$\triangle\phi(0) = -\frac{2\pi d}{\lambda}\frac{a}{b} = -(2m+1)\pi, \qquad m = 0, 1, 2, \ldots \tag{4.60}$$

which is the same as demanding that the path difference introduced by the source to slits stage be an odd number of half wavelengths, $ad/b = (2m+1)\lambda/2$, that is

$$\frac{a}{b} = \frac{(2m+1)\lambda}{2d}, \qquad m = 0, 1, 2, \ldots \tag{4.61}$$

The condition depends on a and b only through their ratio, as it must, since the only thing a point source communicates to the system is the direction from which it illuminates it. The smallest case $m = 0$ gives $a/b = \lambda/2d$, which for typical numbers, $\lambda = 633\,\mathrm{nm}$ and $d = 0.2\,\mathrm{mm}$, is $a/b \approx 1.6 \times 10^{-3}$, so a displacement of about 1.6 mm for every metre of b is already enough to turn the central maximum into a minimum.

4.3.2 Fresnel biprism

The double mirror that Fresnel used to prove the wave nature of light Fresnel (1819) has one practical drawback, namely that two separate flat mirrors have to be tilted and held at an almost invisible angle between them, which makes the alignment tedious and unstable. Fresnel's own answer to this difficulty was to replace the two mirrors by a single piece of glass ground into two thin prisms joined base to base, with the shared edge running along the optical axis. This is the *Fresnel biprism*, and it produces exactly the same effect, two coherent virtual images of a single point source, but out of one rigid element instead of two separately adjustable ones.

Before setting up the biprism itself, it is worth pausing on the principle that makes this whole family of instruments work, since it is the same principle that will let us handle every wavefront division interferometer to come. When a wave leaves a real point source and passes through an optical element, be it a lens, a mirror or a prism, its wavefront is in general reshaped into something far too complicated to describe by hand. There is, however, a special situation in which nothing of interest is lost, namely when the element is *stigmatic* for the pair of points involved, in the sense already introduced in the previous chapter (see figure 2.12), a spherical wavefront leaving the object point is transformed into another spherical wavefront, indistinguishable in its geometry from the one that a genuine point source sitting at the image point would radiate. Whenever this holds, Fermat's principle guarantees that every ray reaching the image has travelled the same optical path, so the phase downstream of the element is exactly the phase of a fictitious spherical wave centred on that image, whether the image is real or virtual. This is the *method of images*: rather than propagating the true, complicated wavefront through the optical element, we replace the element by the point sources it images and let the machinery of two point source interference, already built in the previous section, take care of everything past that plane.

This single observation is what turns an ordinary mirror or prism into an interferometer. If one primary source is imaged twice, once through each of two branches of the same optical element, the observer downstream no longer sees one wave but two, radiated by two virtual sources $\mathcal{S}_1$ and $\mathcal{S}_2$. These two sources are coherent by construction rather than by assumption, since both wavefronts are literally two halves of the same original wave, carrying the same frequency and a phase relationship fixed once and for all by the geometry of the element; the condition $\omega_1 = \omega_2$ that opened this chapter is here automatically satisfied and never has to be imposed by hand, unlike the case of two genuinely independent sources. Once the positions of $\mathcal{S}_1$ and $\mathcal{S}_2$ are known, the problem stops being about mirrors or prisms altogether and becomes, exactly, the interference of two spherical waves solved earlier in this chapter, with the separation a and the source to screen distance D read directly off the geometry of the particular device. In this sense every wavefront division interferometer, the double mirror, the biprism, and the others we will meet, differs from the next only in how it manufactures its pair of virtual sources; the fringes that follow are always the same fringes.

The method is only as trustworthy as the stigmatism of the imaging step, and that has to be checked separately for every device, since it is never automatic. A plane mirror is rigorously and unconditionally stigmatic for every object point whatsoever, as we saw when it appeared as the degenerate conic $z_o = -z_i$ of the reflective surfaces 2.76, which is why Fresnel's original double mirror needs no further justification beyond the law of reflection. A prism is a different matter, because among refracting interfaces only the very special Cartesian ovoids are rigorously stigmatic, and an ordinary prism face is not one of them; it can only be stigmatic approximately, for a narrow paraxial bundle of rays clustered near a fixed angle of incidence. Showing that this approximate stigmatism holds well enough for a thin prism, and pinning down exactly how well, is therefore the first task in developing the biprism, and it is where we begin.

Let a point source $\mathcal{S}$ sit at a distance z_1 from the biprism, and let the screen of observation stand a further distance z_2 beyond it, so that $D = z_1 + z_2$ is the total distance from the source to the screen (see figure 4.4). Each half of the biprism is a thin prism of apex angle α and refractive index n, and the whole instrument is symmetric about the optical axis. The method of images consists of replacing the biprism entirely by the two virtual sources it creates, which turns the problem back into the two point source interference we already solved.

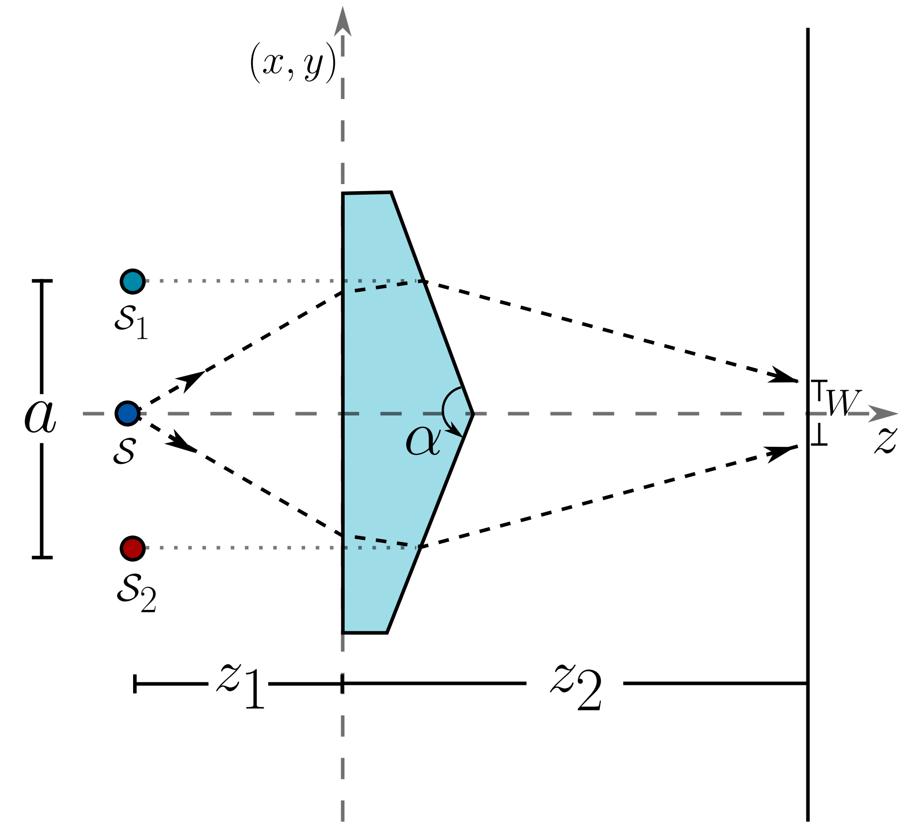


Figure 4.4: Method of images applied to the Fresnel biprism. Each half of the biprism produces a virtual point image of the source, $\mathcal{S}_1$ from the upper half and $\mathcal{S}_2$ from the lower half, separated by a and located in the same transverse plane as $\mathcal{S}$. Beyond the biprism the two beams overlap only inside the shaded strip of width W on the screen.

A thin prism deviates a ray by an angle that, to first order in the angle of incidence and near the condition of minimum deviation, does not depend on exactly where the ray crosses the prism nor, to that same order, on its precise angle of incidence,

$$\delta = (n-1)\alpha. \tag{4.62}$$

This independence is the whole reason the method of images works here, because if the deviation changed from ray to ray the upper half of the biprism would not produce a point image but an aberrated smear. It is worth checking how good 4.62 really is, since later we will be pushed toward angles that are not so small. Comparing it with the exact deviation at symmetric incidence, $\delta_{\text{exact}} = 2\arcsin[n\sin(\alpha/2)] - \alpha$, for $n = 1.5$ the relative error is only 0.04% at $\alpha = 0.05$ rad, 0.16% at $\alpha = 0.10$ rad, 0.35% at $\alpha = 0.15$ rad and 0.63% at $\alpha = 0.20$ rad, so the thin prism approximation survives well into the range we will need, though it stops being free.

Consider a ray leaving $\mathcal{S}$ with paraxial slope m and crossing the upper half of the biprism. It strikes the prism plane at a height $y_p = z_1 m$ and leaves with the new slope $m - \delta$, deviated toward the axis. Extending this outgoing ray backward by the same distance z_1 to the plane of the source, its apparent point of origin is at a height

$$y_v = y_p - z_1(m - \delta) = z_1 m - z_1 m + z_1\delta = z_1\delta, \tag{4.63}$$

which does not depend on m. Every ray that the upper half deviates, whatever its original slope, appears to come from the same point, so the upper half of the biprism produces a rigorously stigmatic virtual image $\mathcal{S}_1$ at a height $z_1\delta$ above the axis, in the very same transverse plane as the source. By the mirror symmetry of the instrument the lower half produces $\mathcal{S}_2$ at $-z_1\delta$. The biprism has disappeared from the problem, replaced by two mutually coherent point sources separated by

$$a = 2z_1\delta = 2z_1(n-1)\alpha, \tag{4.64}$$

sitting at a distance $D = z_1 + z_2$ from the screen. From here on the problem is precisely the interference of two spherical waves that we already solved, so the irradiance on the screen, with y the transverse coordinate, takes the canonical form

$$I(y) = (I_1 + I_2)\left[1 + \mathcal{V}\cos\left(\kappa\frac{a}{D}y\right)\right], \tag{4.65}$$

with a fringe spacing given directly by the two point source result

$$\Lambda = \frac{\lambda D}{a} = \frac{\lambda(z_1 + z_2)}{2z_1(n-1)\alpha}. \tag{4.66}$$

It helps to read 4.66 in the form that makes the underlying physics transparent, writing $\Lambda = \lambda/\beta$ with

$$\beta = \frac{a}{D} = 2(n-1)\alpha\,\frac{z_1}{z_1 + z_2}, \tag{4.67}$$

the local angle at which the two beams cross on the screen. This is exactly the same quantity that appeared in the interference of two plane waves, where $\Lambda = \lambda/[2\sin(\beta/2)]$, and the two expressions differ by only 0.12% at $\beta = 0.1$ rad, so identifying $\Lambda \approx \lambda/\beta$ here is entirely legitimate. What 4.67 makes plain is that

$$\beta \leq 2(n-1)\alpha, \tag{4.68}$$

with equality only in the limits $z_2 \to 0$, the screen pressed against the biprism, or $z_1 \to \infty$, a collimated beam. The crossing angle can never exceed the full deviation the biprism is able to imprint on a beam, and the divergence of the spherical wave leaving a finite z_1 can only make it smaller. A biprism illuminated by a point source is, in this sense, always a weaker interferometer than the same biprism illuminated by a plane wave.

As a concrete example, take a biprism of index $n = 1.5$ illuminated by light of wavelength $\lambda = 632\,\text{nm}$, and ask for the geometry that produces a fringe spacing $\Lambda = 10\lambda$. Setting $\Lambda = 10\lambda$ in 4.66 means $\beta = 1/10$, and since $n = 1.5$ gives $2(n-1) = 1$ exactly, the condition collapses to something remarkably clean,

$$\alpha \frac{z_1}{z_1 + z_2} = \frac{1}{10}\,\text{rad} \quad \Longleftrightarrow \quad \alpha = \frac{1}{10}\left(1 + \frac{z_2}{z_1}\right)\text{rad}. \tag{4.69}$$

This is not a single value of α but a whole one parameter family, since fixing the ratio z_2/z_1 fixes α and vice versa, and every member of the family shares the same image separation $a = D/10$, a tenth of the total source to screen distance, which is a strikingly large separation for two virtual images meant to interfere. There is a bound hiding in 4.69 that is easy to miss, because $z_2/z_1 \geq 0$ forces

$$\alpha \geq \frac{1}{10}\,\text{rad} = 5.73^\circ, \tag{4.70}$$

with the minimum reached only in the collimated limit. A prism angle of nearly six degrees is not what one usually pictures under the instruction to keep α small, and this is worth stating openly rather than hiding it, because the two requirements pull in opposite directions. A fringe spacing of $10\lambda = 6.32\,\mu\text{m}$ is an extraordinarily fine spacing to demand, comparable to the period of a diffraction grating, and producing it requires the two beams to cross at $\beta = 0.1\,\text{rad}$, a full tenth of a radian, while a biprism can only ever supply a crossing angle of $2(n-1)\alpha$. The solution exists and is entirely consistent, but it sits at the edge of the thin prism regime rather than deep inside it, and the error check performed above shows that the approximation 4.62 still holds at $\alpha = 0.1\,\text{rad}$, but by a margin of only 0.16% rather than by a comfortable one. It is also worth noticing that λ never appears in the condition on α alone, which makes sense once we remember that Λ was specified in units of λ and the pure geometry of the biprism carries no memory of colour.

A condition on α is not yet the whole answer, because it says nothing about whether a useful pattern is actually observable. The two beams emerging from the biprism only overlap inside a finite strip of the screen, since the ray that grazes the shared edge from above is deviated downward by δ and reaches the screen at a height $-z_2\delta$, while the ray grazing the edge from below is deviated upward and reaches $+z_2\delta$. Outside this interval only one of the two beams arrives and there is no interference at all, so the field of overlap has a width

$$W = 2z_2\delta = 2z_2(n-1)\alpha, \tag{4.71}$$

and the number of fringes that fit inside it follows at once,

$$N = \frac{W}{\Lambda} = \frac{z_2(n-1)\alpha}{5\lambda}. \tag{4.72}$$

Evaluated on the design condition 4.69, with $2(n-1) = 1$ for $n = 1.5$, this becomes $N = z_2(z_1+z_2)/(100\,z_1\lambda)$, and it is instructive to see it on real numbers.

z_1	z_2	α	a	N
1 m	1 cm	$0.101\,\text{rad} = 5.79^\circ$	10.1 cm	160
0.5 m	5 mm	$0.101\,\text{rad} = 5.79^\circ$	5.05 cm	80
∞ (collimated)	1 cm	$0.100\,\text{rad} = 5.73^\circ$	–	158

With the screen a centimetre from the biprism and the source a metre away, the design delivers 160 fringes of $6.32\,\mu$m inside a field of overlap only about a millimetre wide, so resolving them in practice needs a microscope objective or a detector with sub micron pixels, a condition that deserves to be stated explicitly rather than left implicit in the numbers. The visibility over this whole field stays extremely close to unity, since on the axis of symmetry the two virtual sources are exactly equidistant from the observation point and $\mathcal{V} = 1$ exactly, and even away from the axis the two distances remain equal up to second order corrections, so the same robustness of the contrast against small imbalances that we established for the Young interferometer is at work here, only now it holds by the exact symmetry of the construction rather than by an accident of the numbers.

It remains to see what happens when the point source is displaced off the optical axis, since a real source is never perfectly centred. Repeat the same construction with the source at $\mathcal{S} = (-z_1, s)$, a small transverse offset s away from the axis. A ray leaving $\mathcal{S}$ with slope m strikes the upper half of the biprism at height $y_p = s + z_1 m$ and leaves with slope $m - \delta$, so its virtual origin is now at

$$y_v = s + z_1 m - z_1(m - \delta) = s + z_1\delta, \tag{4.73}$$

once again independent of m, so the images remain perfectly stigmatic under decentring. The two virtual sources simply follow the displacement,

$$\mathcal{S}_1 = s + z_1\delta, \qquad \mathcal{S}_2 = s - z_1\delta, \tag{4.74}$$

and three consequences follow from 4.74 at once.

The fringe spacing does not change. The separation is still $a = 2z_1\delta$ and the distance to the screen is still $D = z_1 + z_2$, both untouched by s, so $\Lambda = \lambda D/a$ is invariant to first order in the displacement. It is invariant to second order as well, and for a reason worth stating rather than taking for granted, since the deviation of a prism is stationary at the condition of minimum deviation, $d\delta/di = 0$ there, so tilting the incident beam slightly changes δ only at order $(\Delta i)^2$. The biprism is therefore intrinsically robust against a decentred source, more so than the Young interferometer with its fixed slits, where no such stationarity protects the spacing.

The whole fringe system translates rigidly by exactly s. Both virtual images move together with the source, so the plane of zero path difference, the perpendicular bisector of $\mathcal{S}_1\mathcal{S}_2$, is simply the plane $y = s$, and the zero order fringe sits there regardless of z_2. There is no geometric amplification of the kind we found for the Young interferometer with fixed slits, where the shift on the screen was sD/b rather than s itself, and the difference is that there the source and its two images move independently of the fixed slits, while here the source and both of its images move together as a rigid unit.

The field of overlap translates too, but in the opposite direction and by a different amount, $-z_2 s/z_1$. The ray grazing the shared edge from above now leaves $\mathcal{S}$ with slope $-s/z_1$ and acquires the extra deviation $\mp\delta$, reaching the screen at $y = -z_2 s/z_1 \mp z_2\delta$, so the window keeps its width $W = 2z_2\delta$ but is now centred at $-z_2 s/z_1$. Since the fringes and the window move in opposite directions, the central fringe eventually walks out of the observable field once

$$|s| > \frac{z_1 z_2 \delta}{z_1 + z_2} = \frac{z_2\beta}{2}. \tag{4.75}$$

With the numbers of the table above, $z_1 = 1\,\mathrm{m}$, $z_2 = 1\,\mathrm{cm}$ and $\beta = 0.1$, this bound is $|s| < 0.5\,\mathrm{mm}$, a generous tolerance for the alignment of the source, yet a displacement of only $s = 6.3\,\mu\mathrm{m}$ is already enough to slide the whole pattern by one complete fringe. The Fresnel biprism is therefore, at the same time, an interferometer remarkably tolerant of a coarse misalignment of the source, since the fringe spacing survives it exactly, and an extremely sensitive detector of a fine transverse displacement, since the fringe position tracks that displacement one to one, a duality that turns out to be useful whenever the instrument is repurposed as a position sensor rather than as a simple demonstration of interference.

4.3.3 Michelson and Morley interferometer

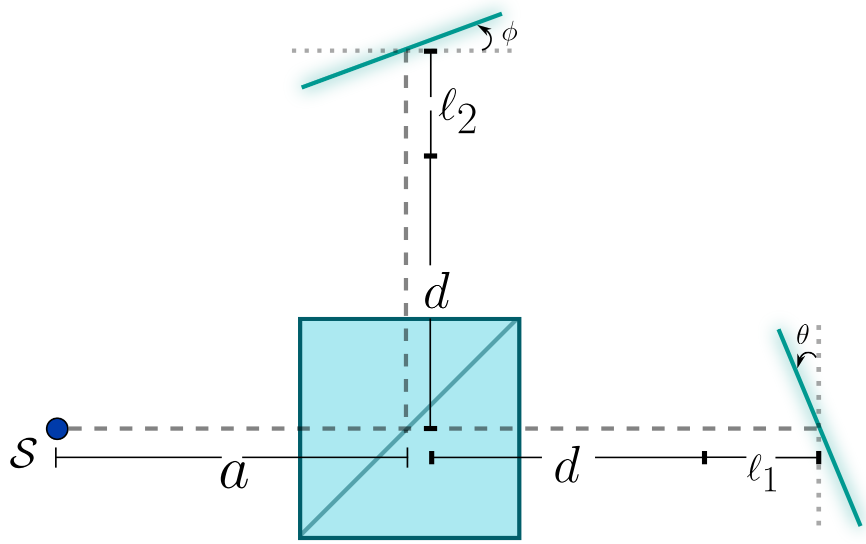


Figure 4.5: General configuration of the Michelson interferometer. A source placed a distance a from the beamsplitting cube illuminates it, and the beam is divided along two perpendicular arms. In arm 1 the light travels a total distance $d + \ell_1$ before striking mirror 1, tilted an angle θ away from the orientation that would send the beam straight back along itself; in arm 2 it travels a total distance $d + \ell_2$ before striking mirror 2, tilted an angle ϕ in the same sense. The two reflected beams return to the beamsplitter and recombine there, ready to interfere.

The configuration sketched above did not spring fully formed from the 1887 experiment already described at the opening of this chapter. Michelson built and tested a first, considerably simpler version of it in 1881, while on leave from the United States Naval Academy and working in Hermann von Helmholtz's laboratory in Berlin Michelson (1881). That prototype was mounted directly on a stone pier in the basement of the Astrophysical Observatory in Potsdam, in an attempt to damp out the ordinary tremors of the building, and it already embodied the essential idea sketched in figure 4.5, a beam split and sent along two perpendicular paths before being brought back together. Its single, short arm and its residual sensitivity to vibration, however, produced a fringe shift far too small and unsteady to draw any firm conclusion, and Alfred Potier and Lord Rayleigh soon pointed out besides that Michelson's original analysis had understated the expected shift by a factor of two, an error traceable to treating the motion of the Earth as though it affected both arms in the same way. Correcting the theory raised the predicted shift, but what the measurement truly needed was a much longer optical path and a mounting immune to vibration, and supplying both is exactly what occupied Michelson once he moved to the Case School of Applied Science and joined forces with Edward Morley.

The apparatus that resulted rested on a massive sandstone slab floated on a pool of mercury, so that it could be rotated smoothly about a vertical axis without flexing, and each arm was folded by a set of auxiliary mirrors to stretch its effective length to roughly eleven metres, some ten times longer than in the Potsdam attempt, which is what finally brought the sensitivity of the instrument within reach of the aether drift the theory predicted, only for none to appear Michelson and Morley (1887). The same principle, two beams recombined after independent and adjustable paths, proved useful well beyond that null result. Four decades later Michelson, together with Francis Pease, mounted a twenty foot version of essentially the same interferometer across the aperture of the 100 inch Hooker telescope at Mount Wilson, and used the visibility of the fringes it produced, precisely the visibility introduced earlier in this chapter, to measure for the first time the angular diameter of a star other than the Sun, Betelgeuse, from nothing more than the separation between the two outer mirrors at which the fringes first washed out Michelson and Pease (1921). The same architecture went on to become the working principle of Fourier transform spectroscopy, where a single continuously scanned mirror encodes an entire spectrum into one interferogram, and later still of optical coherence tomography, which uses exactly this idea to image tissue a few micrometres beneath the surface.

Chapter 5

Theory of Diffraction

Where geometry promised sharp shadows, light insisted on painting fringes.

"Diffraction is any deviation of light rays from rectilinear paths which cannot be interpreted as reflection or refraction."
– Arnold Sommerfeld

5.1 A historical context of diffraction theory: from shadows to the vector world

Diffraction - that phenomenon in which light bends around obstacles and paints patterns of light and shadow where geometry promised only sharp silhouettes - is one of the richest stories in physics. Its evolution spans from Renaissance observations to the vector equations that today model nanotechnologies, and the tale shows how humankind slowly learned to tame the undulations of light.

The story begins around 1660, when the Italian Jesuit Francesco Maria Grimaldi observed something disconcerting: when sunlight was passed through a small hole into a dark room, the edge of the shadow cast by an object was not sharp; instead, coloured fringes and illuminated zones appeared where the ray geometry of Euclid predicted darkness. In his posthumous book *Physico-Mathesis de Lumine* (1665) Grimaldi (1665), Grimaldi coined the term *diffractio* (from the Latin *dis-* + *fractus*, "to break into pieces"), suggesting that light was not merely reflected or refracted but also "broke" when skirting obstacles. Nevertheless, in an age dominated by Newton's corpuscular theory, these ideas remained in the penumbra.

The first theory capable of qualitatively explaining diffraction arrived in 1690 with Christiaan Huygens. In his *Traité de la Lumière* Huygens (1690), he proposed that every point of a wavefront acts as a source of new secondary wavelets (the **Huygens principle**), and that the envelope of these wavelets generates the propagating front. Although he did not explicitly incorporate interference - which limited the predictive power of his model - his construction already suggested that light could deviate when passing near edges.

The true leap occurred in the nineteenth century. Thomas Young, with his double-slit experiment (1801) Young (1804), demonstrated that light interferes with itself; but it was Augustin-Jean Fresnel who, between 1815 and 1820, fused the Huygens principle with the principle of interference, creating the **HuygensFresnel principle**. In his *Mémoire sur la diffraction de la lumière* Fresnel (1819), Fresnel cast diffraction as an integral superposition of secondary wavelets with relative phases. His expression for the intensity at a point P,

$$I(P) \propto \left| \int_{\text{aperture}} \frac{Ae^{i(kr-\omega t)}}{r} K(\theta)\, dS \right|^2, \tag{5.1}$$

where $K(\theta)$ is the *obliquity factor*, accounted quantitatively for the observed diffraction patterns. The climactic moment came in 1818: Siméon Denis Poisson, using Fresnel's theory, predicted that a circular disc should project a bright spot at the very centre of its shadow (the **PoissonArago spot**). When Dominique Arago verified it experimentally, the wave theory triumphed over Newtonian corpuscularism.

Powerful as it was, Fresnel's theory remained heuristic. In 1883 Gustav Kirchhoff gave it mathematical rigour through the wave equation and Green's theorem. In his memoir *Zur Theorie der Lichtstrahlen* Kirchhoff (1883), he derived the **Kirchhoff integral formula**

$$U(P) = \frac{1}{4\pi} \iint_{\Sigma} \left(U \frac{\partial}{\partial n} \left(\frac{e^{iks}}{s} \right) - \frac{e^{iks}}{s} \frac{\partial U}{\partial n} \right) d\Sigma, \tag{5.2}$$

where U is the scalar field (assuming monochromatic light and uniform polarisation), s the distance from the source point to P, and Σ the integration surface. This formulation - although it assumes mutually inconsistent boundary conditions, as pointed out by Henri Poincaré - became the cornerstone of the **scalar theory of diffraction**, valid whenever polarisation effects are negligible.

Kirchhoff's inconsistencies were resolved in 1896 by Arnold Sommerfeld Sommerfeld (1896). By eliminating the need to impose Dirichlet and Neumann conditions simultaneously, he derived rigorous solutions for the diffraction by a plane screen:

$$U(P) = \frac{1}{i\lambda} \iint_{\text{aperture}} U_0 \frac{e^{ikr}}{r} \cos\theta \, dS \quad \text{(RayleighSommerfeld diffraction)}. \tag{5.3}$$

This approach, later detailed in his *Vorlesungen über Theoretische Physik* Sommerfeld (1954), allowed diffraction patterns to be computed without unphysical infinities, extending the theory to arbitrary apertures.

The scalar theory, however, ignored the vector nature of light. In 1908 Gustav Mie, studying the diffraction (scattering) of light by spheres - today called **Mie scattering** - solved Maxwell's equations directly for the electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$ Mie (1908). Shortly afterwards, Peter Debye, in his 1909 thesis, introduced inhomogeneous vector waves to describe the focusing of beams, laying the foundations of the **vector theory of diffraction**.

The modern synthesis was consolidated by Max Born and Emil Wolf in *Principles of Optics* (1959) Born and Wolf (1999), where it was shown that the scalar approximation fails for systems with high numerical aperture, anisotropic or chiral media, and structures smaller than the wavelength. The vector theory, built directly on Maxwell's curl equations

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \tag{5.4}$$

allows one to model polarisation effects, surface plasmons and metasurfaces.

The invention of the laser in 1960 boosted practical applications. Joseph Goodman, in *Introduction to Fourier Optics* (1968) Goodman (2005), reinterpreted diffraction as a **linear, space-invariant system**, in which the far-field pattern is the Fourier transform of the aperture. This viewpoint enabled the design of holograms, diffractive lenses and computational aberration correction. Today the field enjoys a golden age: sub-wavelength **metasurfaces** that shape wavefronts through vector diffraction Yu et al. (2011), electron and neutron diffraction used to map atomic structures, and diffraction in nonlinear media probed with femtosecond lasers. From Grimaldi's fringes to engineered wavefronts, diffraction theory has continuously challenged and refined our understanding of light. In every diffracted corner we still hear an echo of Fresnel's question: how far can light bend before it breaks our certainties?

5.2 Scalar theory of diffraction

Electromagnetic waves are inherently vectorial: at each point $\vec{r}$ the electric field may exhibit a different state of polarisation. This property is expressed mathematically as

$$\vec{E}(\vec{r},t) = E(\vec{r},t)\,\hat{\vec{e}}(\vec{r},t), \tag{5.5}$$

where $E(\vec{r},t)$ denotes the amplitude of the field and $\hat{\vec{e}}(\vec{r},t)$ represents the polarisation vector, which besides varying in space may evolve in time in a stochastic manner.

The associated wave equation then takes the form

$$\nabla^2\left[E(\vec{r},t)\hat{\vec{e}}(\vec{r},t)\right] - \mu\varepsilon\frac{\partial^2}{\partial t^2}\left[E(\vec{r},t)\hat{\vec{e}}(\vec{r},t)\right] = 0, \tag{5.6}$$

whose direct solution presents notable complexities. To tackle this problem one resorts to simplifying strategies, such as decomposing the field into scalar components. These scalar solutions may afterwards be combined to reconstruct the full vector behaviour.

The temporal variation of amplitude, phase and polarisation is analysed rigorously through coherence theory (a topic explored in depth in Chapter 6). Nevertheless, under certain approximations the temporal dependence of the amplitude is attributed solely to the intrinsic oscillations of the field, while the polarisation is regarded as static. The electric field then simplifies to

$$\vec{E}(\vec{r},t) = E(\vec{r},t)\,\hat{\vec{e}}(\vec{r}), \tag{5.7}$$

allowing its expansion in an orthogonal polarisation basis,

$$\vec{E}(\vec{r},t) = E_1(\vec{r},t)\hat{\vec{e}}_1 + E_2(\vec{r},t)\hat{\vec{e}}_2. \tag{5.8}$$

Here $\hat{\vec{e}}_1$ and $\hat{\vec{e}}_2$ correspond to opposite polarisation states on the Poincaré sphere. Each scalar component E_i individually satisfies the wave equation

$$\nabla^2 E_i - \mu\varepsilon\frac{\partial^2 E_i}{\partial t^2} = 0 \quad (i = 1, 2), \tag{5.9}$$

thereby decoupling the vector problem into manageable scalar equations.

In situations where one component is negligible (for instance $E_2 \to 0$), the field reduces to

$$\vec{E}(\vec{r},t) = E(\vec{r},t)\,\hat{\vec{e}}, \tag{5.10}$$

defining a purely scalar case. This formalism underlines how the general vector description can emerge from the controlled superposition of scalar solutions.

5.3 The scalar wave equation (Helmholtz equation)

Let $\Psi(\mathbf{r}, t)$ be an arbitrary component of the electromagnetic field. This quantity satisfies the classical wave equation

$$\nabla^2 \Psi(\mathbf{r}, t) - \mu\varepsilon \frac{\partial^2 \Psi(\mathbf{r}, t)}{\partial t^2} = 0, \tag{5.11}$$

where Ψ represents a polychromatic field. To decompose the field temporally we apply the Fourier transform

$$\Psi(\mathbf{r}, t) = \int_{\mathbb{R}^+} U(\mathbf{r}, \nu)\, \exp(-2\pi \mathrm{i} \nu t)\, d\nu. \tag{5.12}$$

Substituting (5.12) into (5.11), the equation is re-expressed as

$$\int_{\mathbb{R}^+} \left[\nabla^2 U(\mathbf{r}, \nu) + \frac{\omega^2}{v^2} U(\mathbf{r}, \nu) \right] \exp(-2\pi \mathrm{i} \nu t)\, d\nu = 0, \tag{5.13}$$

which leads directly to the Helmholtz equation for each spectral component:

$$(\nabla^2 + k^2) U(\mathbf{r}, \nu) = 0. \tag{5.14}$$

Here $k = 2\pi/\lambda$ denotes the wavenumber, and $\lambda = c/(n\nu)$ is the wavelength in a medium of refractive index n. This formulation describes mathematically how each monochromatic component of the field (characterised by its complex amplitude U) obeys an equation independent of the temporal frequency.

5.4 The Kirchhoff scalar boundary-value problem

The scalar theory of diffraction, based on solutions of the wave equation under boundary conditions, describes the propagation of fields in paraxial approximations or with smooth phase modulations. Despite its scalar formulation, the formalism - developed by Kirchhoff by means of Green's functions - is adaptable to vector fields.

The original inconsistency of Kirchhoff, pointed out by Poincaré, lies in the over-determination of the boundary conditions (prescribing both the field and its normal derivative on a plane), which violates the uniqueness theorems of differential equations. Sommerfeld resolved this by redefining the Green's function so as to eliminate the redundancy, guaranteeing mathematical consistency.

We therefore start from the physical entity under study - the electromagnetic field - and as a first step it is appropriate to recall its wave equation

$$\nabla^2 \vec{E}(\vec{r},t) - \frac{1}{v^2}\frac{\partial^2 \vec{E}(\vec{r},t)}{\partial t^2} = 0, \tag{5.15}$$

which is of a vector nature. We may restrict our phenomenon to the study of the propagation of its components, grouping the vector character into the polarisation vector $\hat{e}(\vec{r},t)$. In order to avoid possible spacetime correlations between the evolution of the components and the polarisation of the field, the dependence of the polarisation vector on these variables is dropped, so that

$$\left(\nabla^2 E(\vec{r},t) - \frac{1}{v^2}\frac{\partial^2 E(\vec{r},t)}{\partial t^2}\right)\hat{e} \quad \longrightarrow \quad (\nabla^2 + k^2)E(\vec{r},t) = 0, \tag{5.16}$$

thus obtaining the Helmholtz equation for the scalar electric field.

5.4.1 Green's theorem

Green's theorem, a direct corollary of Gauss's theorem, establishes fundamental relations between volume and surface integrals for scalar fields. Let $f(\vec{r})$ and $g(\vec{r})$ be scalar functions with continuous second derivatives in a domain V. We start from the differential identity

$$\nabla \cdot [f(\vec{r})\nabla g(\vec{r})] = f(\vec{r})\nabla^2 g(\vec{r}) + \nabla f(\vec{r}) \cdot \nabla g(\vec{r}). \tag{5.17}$$

Subtracting the analogous expression with f and g interchanged, we obtain

$$\nabla \cdot [f\nabla g - g\nabla f] = f\nabla^2 g - g\nabla^2 f. \tag{5.18}$$

Integrating this relation over a volume V and applying the Gauss divergence theorem,

$$\int_V \nabla \cdot [f(\vec{r})\nabla g(\vec{r}) - g(\vec{r})\nabla f(\vec{r})]\, dV = \int_V \left(f\nabla^2 g - g\nabla^2 f\right) dV, \tag{5.19}$$

which allows the volume integral on the right-hand side to be converted into a surface integral over ∂V:

$$\oint_{\Sigma(V)} [f(\vec{r})\nabla g(\vec{r}) - g(\vec{r})\nabla f(\vec{r})] \cdot d\vec{\sigma} = \int_V \left(f\nabla^2 g - g\nabla^2 f\right) dV, \tag{5.20}$$

where:

- $d\vec{\sigma} = \hat{n}\, d\sigma$ is the differential surface element oriented along the outward normal $\hat{n}$ to the surface $\Sigma(V)$.
- $\frac{\partial}{\partial n} = \hat{n} \cdot \nabla$ denotes the directional derivative along $\hat{n}$.

- The operator ∇ acts exclusively on the spatial coordinates $\vec{r}$.

This identity allows the solution of inhomogeneous differential equations (e.g. the Helmholtz equation) to be expressed in terms of integrals over the sources and the boundary conditions. The presence of $\hat{n}$ emphasises that the surface contribution depends critically on the geometry of the volume V.

5.4.2 The HelmholtzKirchhoff integral theorem

The HelmholtzKirchhoff integral theorem adapts Green's formalism to the propagation of electromagnetic fields. Starting from Green's identity and adding the term $k^2 fg$ to both sides, one obtains

$$\oint_{\Sigma(V)} [f(\vec{r})\nabla g(\vec{r}) - g(\vec{r})\nabla f(\vec{r})]\cdot d\vec{\sigma} = \int_V f(\vec{r})\left(\nabla^2 + k^2\right) g(\vec{r})\, dV - \int_V g(\vec{r})\left(\nabla^2 + k^2\right) f(\vec{r})\, dV, \tag{5.21}$$

where f and g are scalar functions. Choosing $f = U(\vec{r}, \nu)$, a solution of the Helmholtz equation, and g as an arbitrary smooth function, the expression simplifies to

$$\oint_{\Sigma(V)} [U(\vec{r}, \nu)\nabla g(\vec{r}) - g(\vec{r})\nabla U(\vec{r}, \nu)] \cdot d\vec{\sigma} = \int_V U(\vec{r}, \nu)\left(\nabla^2 + k^2\right) g(\vec{r})\, dV. \tag{5.22}$$

The key lies in selecting g as the Green's function $G(\vec{r}, \vec{r}')$, defined by

$$\left(\nabla^2 + k^2\right) G(\vec{r}, \vec{r}') = 4\pi\delta(\vec{r} - \vec{r}'). \tag{5.23}$$

Remarkably, we may add an arbitrary number of Green's functions while keeping the solution invariant. What we can do is add an infinity of image charges, but *outside* the integration volume bounded by the surface $\Sigma(V)$, as shown in Figure 5.1.

In this spirit, equation (5.23) becomes

$$(\nabla^2 + k^2)G(\vec{r}, \vec{r}') = 4\pi\delta(\vec{r} - \vec{r}') + 4\pi\sum_{i=1}^{\infty}\delta(\vec{r} - \vec{r}_i''). \tag{5.24}$$

The explicit solution for a homogeneous medium of equation (5.23) is

$$G(\vec{r}, \vec{r}') = \frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r} - \vec{r}'\|}, \tag{5.25}$$

corresponding to an outgoing spherical wave centred at $\vec{r}'$. Substituting G into Green's identity, one derives

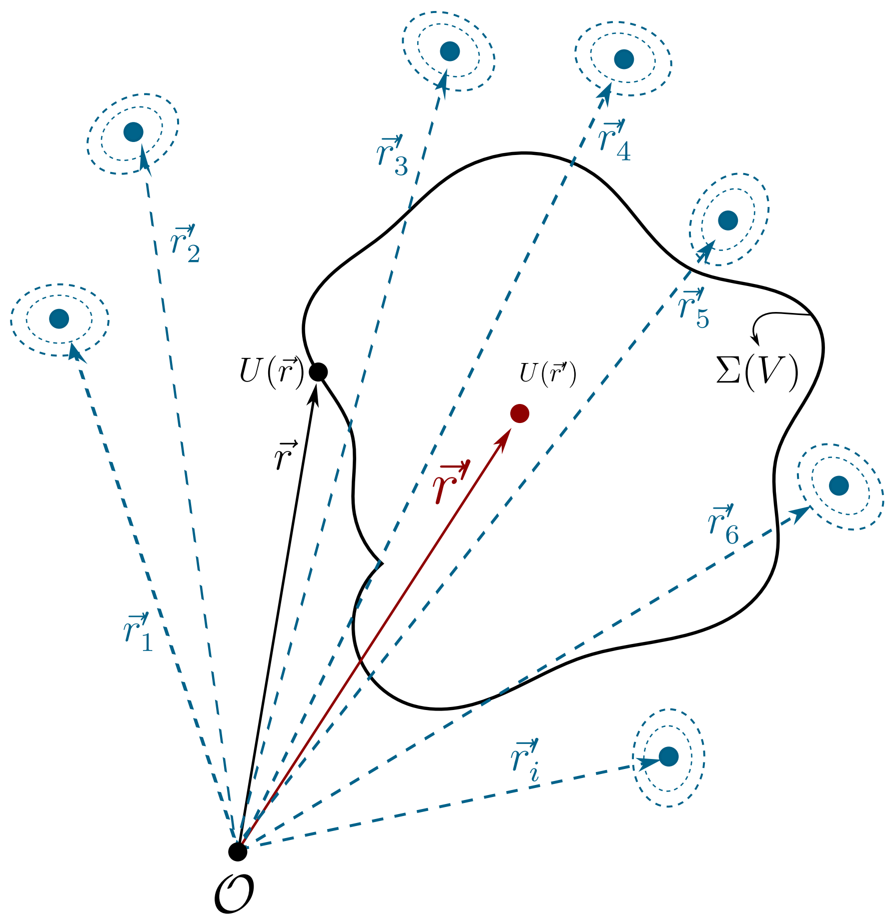


Figure 5.1: On the boundary $\Sigma(V)$ the contribution of the field components $U(\vec{r})$ is shown; inside the region the total field is built from the contribution of all the points on the boundary. One may add infinitely many point Green charges of arbitrary propagation nature; the condition for the invariance of equation (5.23) is that these image charges lie outside $\Sigma(V)$.

$$U(\vec{r}', \nu) = \frac{1}{4\pi} \oint_{\Sigma(V)} \left[U(\vec{r}, \nu) \nabla G(\vec{r}, \vec{r}') - G(\vec{r}, \vec{r}') \nabla U(\vec{r}, \nu) \right] \cdot d\vec{\sigma}, \tag{5.26}$$

known as the HelmholtzKirchhoff integral formula. This equation allows $U(\vec{r}', \nu)$ to be computed at any point of the volume V by means of an integral over its boundary $\Sigma(V)$.

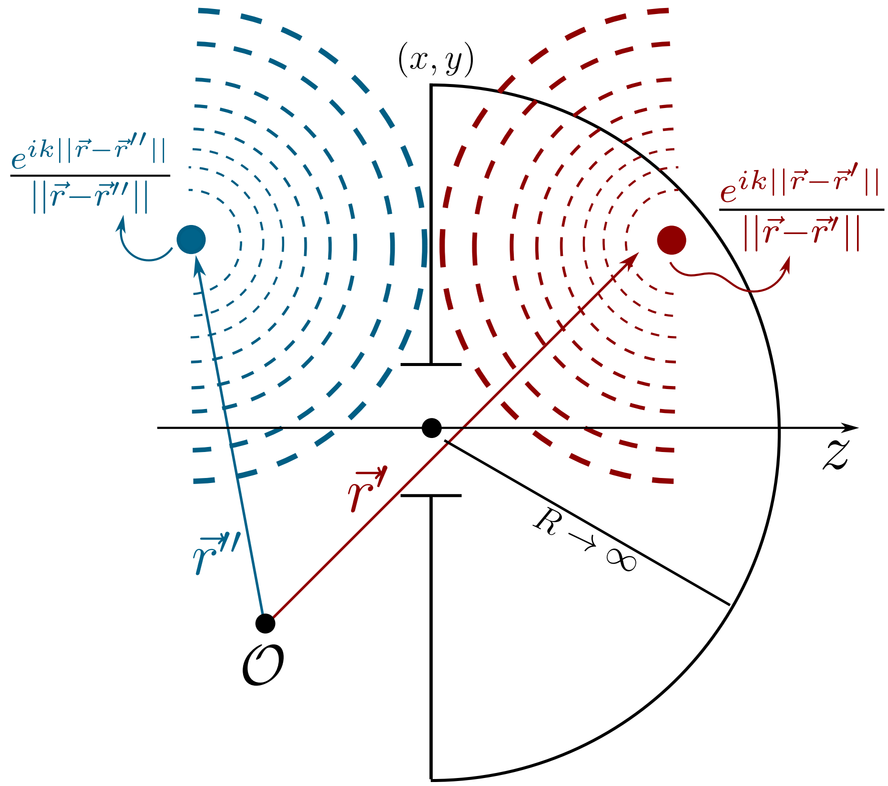


Figure 5.2: Representation of the Green's functions for point sources. Under these boundary conditions the Green's functions behave as point charges, one the mirror image of the other.

Before continuing, it is worth noting that the point-charge nature of the Green's functions $G(\vec{r}, \vec{r}')$ is an assumption taken for granted; there is, however, no restriction on their physical nature. As is well known in electromagnetic theory, radiation sources may have any volume and shape, so our problem (Figure 5.2) can be made more general, as shown in Figure 5.3, and the differential equation for Green's functions with volume would read

$$\left(\nabla^2 + k^2\right) G(\vec{r}, \vec{r}') = 4\pi\rho(\vec{r} - \vec{r}'_+) - 4\pi\rho(\vec{r} - \vec{r}'_-). \tag{5.27}$$

Let us continue with the development for point Green's functions. To simplify the evaluation of the integral, we consider a volume V bounded by an infinite plane Σ_1 ($z = 0$) and a hemisphere Σ_2

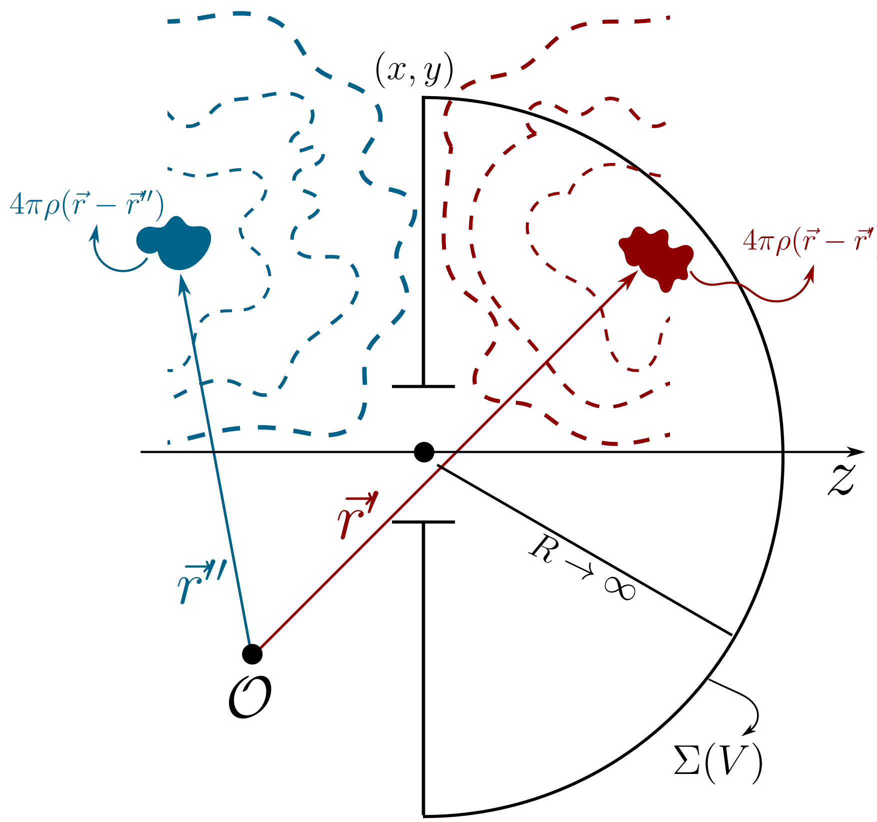


Figure 5.3: Representation of the Green's functions for non-point sources. Under these boundary conditions the Green's functions still behave as mirror images, but now with given volume and geometry; the only constraint on the emission of these sources is that, on the boundary $z = 0$, the total field must vanish, as required by the assumed Dirichlet conditions.

of radius R (see Figure 5.2). The contribution of Σ_2 vanishes as $R \to \infty$ (owing to the Sommerfeld radiation condition), reducing the expression to

$$U(\vec{r}', \nu) = \frac{1}{4\pi} \int_{\Sigma_1} \left[U(\vec{r}, \nu) \frac{\partial G}{\partial n} - G(\vec{r}, \vec{r}') \frac{\partial U}{\partial n} \right] d\sigma, \tag{5.28}$$

where $\frac{\partial}{\partial n} = \hat{n} \cdot \nabla$ denotes the normal derivative. However, this formulation requires specifying simultaneously U and $\partial U/\partial n$ on Σ_1, which over-determines the problem. This difficulty motivates the introduction of physically realistic boundary conditions, such as those of Kirchhoff, in order to resolve the ambiguity.

This problem is aggravated by a theorem of complex analysis, attributed to Riemann, which states that if a function and its normal derivative vanish on an open region of a plane, then the function must be identically zero throughout the domain. This restriction motivates the strategic need to eliminate one of the terms in the HelmholtzKirchhoff surface integral.

The key is to exploit the degrees of freedom inherent in the choice of the auxiliary Green's function G. Kirchhoff initially proposed the intuitive boundary conditions

- $U = 0$ outside the aperture on Σ (blocked field),
- $\partial U/\partial n = 0$ on opaque regions (vanishing normal derivative).

Although these conditions reproduced the experimental results of diffraction, they introduced mathematical inconsistencies by violating the CauchyRiemann uniqueness theorem. Poincaré showed that such assumptions generate contradictions in the mixed derivatives of the field. The resolution came with Sommerfeld, who redefined G by means of a double-layer (mirror) Green's function, ensuring that

- $G = 0$ on Σ (Dirichlet condition),
- $\partial G/\partial n$ remains non-singular,

thus eliminating the redundancy in the boundary conditions and preserving the self-consistency of the formalism.

5.4.3 The RayleighSommerfeld formulation

Returning to equation (5.26) and choosing the Dirichlet condition for the function G, we have

$$U(\vec{r}', \nu) = \frac{1}{4\pi} \int_{\Sigma_1} \left[U(\vec{r}, \nu) \nabla G \cdot \hat{n} \right] d\sigma. \tag{5.29}$$

Given the integration surface under consideration, we may express (5.25) as

$$G(\vec{r},\vec{r}') = \frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r}-\vec{r}'\|} - \frac{e^{ik\|\vec{r}-\vec{r}''\|}}{\|\vec{r}-\vec{r}''\|}, \tag{5.30}$$

where

$$\|\vec{r}-\vec{r}''\| = \|\vec{r}-\vec{r}'\|, \qquad x''=x', \quad y''=y', \quad z''=-z', \tag{5.31}$$

since $\vec{r}''$ is the mirror image of $\vec{r}'$ across the plane Σ_1. It follows that

$$\nabla G\cdot\hat{z} = 2\left[ik - \frac{1}{\|\vec{r}-\vec{r}'\|}\right]\frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r}-\vec{r}'\|}\cos(\vec{r}-\vec{r}',\hat{z}), \tag{5.32}$$

and with this, (5.29) becomes

$$U(\vec{r}',\nu) = \frac{1}{2\pi}\int_\Sigma U(\vec{r},\nu)\left[ik - \frac{1}{\|\vec{r}-\vec{r}'\|}\right]\frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r}-\vec{r}'\|}\cos(\vec{r}-\vec{r}',\hat{z})\,d\sigma. \tag{5.33}$$

We may now make the approximation appropriate to the RayleighSommerfeld regime, in which $\|\vec{r}-\vec{r}'\| \gg \lambda$, obtaining

$$U(\vec{r}',\nu) = \frac{ik}{2\pi}\int_\Sigma U(\vec{r},\nu)\frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r}-\vec{r}'\|}\cos(\vec{r}-\vec{r}',\hat{z})\,d\sigma, \tag{5.34}$$

a mathematical expression that embodies the HuygensFresnel principle: as can be appreciated, there is one spherical wave per point of the integration surface.

5.5 The Fresnel approximation

Starting from the field

$$U(\vec{r}') = \frac{ik}{2\pi}\int_\Sigma U(\vec{r})\frac{e^{ik\|\vec{r}-\vec{r}'\|}}{\|\vec{r}-\vec{r}'\|}\cos(\vec{r}-\vec{r}',\hat{z})\,\mathrm{d}\sigma, \tag{5.35}$$

it is possible to carry out the so-called mid-field or Fresnel approximation. To this end we take

$$\begin{aligned}\|\vec{r}-\vec{r}'\| &= \left[(x-x')^2+(y-y')^2+(z-z')^2\right]^{1/2}\\ &= z\left(1+\frac{(x-x')^2+(y-y')^2}{z^2}\right)^{1/2}\\ &\approx z+\frac{(x-x')^2+(y-y')^2}{2z}.\end{aligned} \tag{5.36}$$

It is clear that, for the factor $\dfrac{1}{\|\vec{r}-\vec{r}'\|}$, an expansion up to the first term of (5.36) suffices; this is not enough for the factor $e^{ik\|\vec{r}-\vec{r}'\|}$, however, since the latter has oscillatory behaviour over its domain, so both terms are required. We additionally take $\cos(\vec{r}-\vec{r}',\hat{z}) = \dfrac{z}{\|\vec{r}-\vec{r}'\|} \approx 1$.

In this spirit, we obtain an expression for the field of the form

$$U(\vec{r}') = \frac{ie^{ikz}}{\lambda z}\int U(\vec{r})\exp\left(\frac{ik}{2z}((x-x')^2+(y-y')^2)\right)d\sigma, \tag{5.37}$$

which describes the propagation or diffraction of the electromagnetic field in the Fresnel regime. It is worth stressing that equation (5.37) describes a parabolic wave, and although, at least conceptually, such propagation is valid for a mid-field, in practice it has proved more than sufficient for sufficiently short distances.

5.6 The Fraunhofer approximation

Finally, from expression (5.37) a second approximation may be carried out, corresponding to the far field or Fraunhofer regime, in which the observation distances are considerably larger than the area over which one wishes to study the propagated field (whether through a slit or another object). We focus on the factor

$$\exp\left(\frac{ik}{2z}((x-x')^2+(y-y')^2)\right) = \exp\left(\frac{ik}{2z}(x^2+y^2)\right)\exp\left(\frac{ik}{2z}(x'^2+y'^2)\right)\exp\left(-\frac{ik}{z}(xx'+yy')\right), \tag{5.38}$$

where we may make the far-field assumption $\exp\left(\frac{ik}{2z}(x^2+y^2)\right)\approx 1$, so that we obtain

$$U(\vec{r}') = \frac{e^{ikz}}{\lambda z}\exp\left(\frac{ik}{2z}(x'^2+y'^2)\right)\int U(\vec{r})\exp\left(-\frac{ik}{z}(xx'+yy')\right)d\sigma. \tag{5.39}$$

There we can still appreciate a quadratic phase term outside the integral, usually called the curvature factor, which shows precisely the dephasing induced by the difference of optical path between a plane called the observation screen and a spherical surface of radius z (the propagation distance of the field along the optical axis) known as the cardinal sphere. The introduction of this spherical surface is rather useful when one wishes to describe Fraunhofer diffraction on a screen as a Fresnel propagation through two confocal cardinal spheres, as shown in Figure 5.4.

This makes more sense once we analyse the effect of each propagation step. It is clear that, without imposing far-field approximations from the outset, propagating the field at $\mathcal{A}$ towards $\mathcal{F}$ with the Fresnel approximation, we would obtain

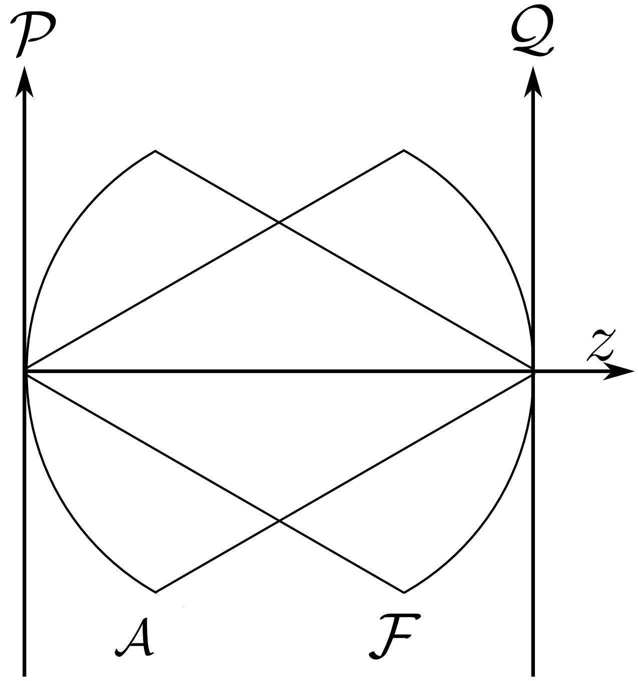


Figure 5.4: Scheme of the confocal cardinal spheres, with radii equal to the propagation distance z of the field.

$$U_{\mathcal{F}}(\vec{r}) = \frac{ie^{ikz}}{\lambda z} \exp\left(\frac{ik}{2z}(x^2+y^2)\right) \int u_{\mathcal{A}} \exp\left(\frac{ik}{2z}(x'^2+y'^2)\right) \exp\left(-\frac{ik}{z}(xx'+yy')\right) d\sigma', \quad (5.40)$$

where we may associate $u_{\mathcal{A}}$, i.e. the field on the sphere $\mathcal{A}$, with the field on $\mathcal{P}$ multiplied by its corresponding quadratic factor $u_{\mathcal{P}} = u_{\mathcal{A}}(\vec{r}') \exp\left(\frac{ik}{2z}(x'^2+y'^2)\right)$, while the term $\exp\left(\frac{ik}{2z}(x^2+y^2)\right)$ may be understood as the factor resulting from passing the field from the sphere $\mathcal{F}$ to the screen $\mathcal{Q}$, so that the field on $\mathcal{F}$ would correspond to

$$u_{\mathcal{Q}}(\vec{r}) = \frac{ie^{ikz}}{\lambda z} \int u_{\mathcal{P}} \exp\left(-\frac{ik}{z}(xx'+yy')\right) d\sigma', \quad (5.41)$$

thus recovering a Fraunhofer expression for diffraction on $\mathcal{Q}$. It should be noted that one must check whether the study of the propagated field is valid within a paraxial regime, where the field is confined to the neighbourhood of the optical axis and hence the curvature of the cardinal sphere does not differ much from the plane (the screen). In cases where the curvature factor ceases to be negligible, the paraxial regime loses validity and the problem must be addressed from a meta-axial standpoint.

Studying the propagation of the field through Fresnel diffraction is of great importance because, as will be discussed later, it bears a close relationship with what is known as the fractional Fourier transform, a tool that arises naturally when one chains paraxial propagation steps.

Chapter 6

Theory of Coherence

Coherence is the thread that weaves order out of the chaos of random light.

"Time advances in the direction in which correlations increase."
– attributed to Ludwig Boltzmann

6.1 A historical context of coherence theory

The history of coherence theory is an intricate and fascinating tale rooted in mathematics, statistics and physics, branching in classical and quantum directions with profound implications for our understanding of the world. To tell it, we must go back to the dawn of probability theory, where the idea of measuring relations between events and variables first took shape — a concept that, over the centuries, would evolve into the refined notions of correlation and, eventually, coherence as we understand it today.

It all began in the seventeenth century, when thinkers such as Blaise Pascal and Pierre de Fermat, in their correspondence on games of chance, laid the foundations of probability theory. Their reflections on expected value and combinatorics opened the way to Jacob Bernoulli, who in 1713, in his posthumous *Ars Conjectandi* Bernoulli (1713), formulated the law of large numbers, linking theoretical probability with observed frequency. These advances were consolidated by Pierre-Simon Laplace in 1812 with his *Théorie Analytique des Probabilités* Laplace (1812), where he unified scattered ideas and developed Bayesian inference, essential for understanding how variables relate in complex systems. Yet a mathematical language was still missing to quantify the interdependence between phenomena.

That language began to take form in the nineteenth century. Carl Friedrich Gauss, in his 1809 study of errors in astronomical observations (*Theoria Motus*) Gauss (1809), introduced the method of least squares, a tool that not only optimised measurements but also revealed hidden patterns in correlated data. But it was Francis Galton, in the 1880s, who coined the term "correlation" as we know it. Motivated by his studies of biological inheritance, Galton sought to measure how characteristics such as the height of parents and children varied together Galton (1888). His work inspired Karl Pearson, who in 1896 formalised the Pearson correlation coefficient Pearson (1896), a number between -1 and 1 quantifying the linear relationship between variables. This was a conceptual leap: no longer merely intuiting connections, but assigning them a rigorous numerical value.

In parallel, in the realm of physics, another kind of correlation — coherence — began to reveal itself through the study of light. In 1801 Thomas Young performed his celebrated double-slit experiment, showing that light interferes with itself, creating patterns of bright and dark fringes. This phenomenon, inexplicable under Newton's corpuscular theory, suggested that light was a wave whose phases could be correlated in space and time. Augustin-Jean Fresnel, between 1815 and 1820, developed a mathematical theory of light as a transverse wave, explaining not only interference but also polarisation. Coherence — the ability of waves to maintain a constant phase relationship — was, however, not yet understood in statistical terms.

The crucial step came in the first decades of the twentieth century, when optics met the need to describe realistic light sources, such as the Sun or incandescent lamps, whose emissions are partly random. In 1934 the Dutch physicist Pieter Hendrik van Cittert published a seminal work on the propagation of coherence from an incoherent source van Cittert (1934), followed in 1938 by Frits Zernike Zernike (1938), who formulated the theorem that bears both their names: the **van Cittert–Zernike theorem**. This result established that the light from an extended, incoherent source develops measurable spatial correlations as it propagates to a distant plane. It was the first time coherence had been quantified mathematically as a statistical property of optical fields.

Classical coherence theory reached maturity in the 1950s through Max Born and Emil Wolf, whose treatise *Principles of Optics* (1959) Born and Wolf (1999) became the bible of theoretical optics. In it they defined the **mutual coherence function**

$$\Gamma(\mathbf{r}_1, \mathbf{r}_2, \tau), \tag{6.1}$$

which measures the correlation between the electric field at two points in space ($\mathbf{r}_1$ and $\mathbf{r}_2$) at two instants separated by a delay τ. This function encapsulates both the temporal coherence (how long the light maintains a stable phase, related to its monochromaticity) and the spatial coherence (how correlated the vibrations are at different positions). Leonard Mandel and Emil Wolf, in the following decades, deepened these ideas in their treatise *Optical Coherence and Quantum Optics* Mandel and Wolf (1995), introducing concepts such as the coherence time and the coherence length, key for practical applications.

Coherence was not a mere abstraction: the Michelson interferometer, developed in the 1890s, exploited temporal coherence to measure wavelengths with unprecedented precision. But the technological milestone arrived in 1960 with the invention of the laser, whose light — highly coherent in time and space — is the product of stimulated emission, a phenomenon predicted by Einstein in 1917. Lasers not only confirmed the theories of coherence but opened new frontiers in communications, medicine and industry.

While classical coherence was being consolidated, a parallel revolution was beginning in the quantum world. In the 1960s quantum coherence found its theoretical framework. Roy J. Glauber, in a series of articles published in *Physical Review* (1963) Glauber (1963), reformulated optics from a quantum perspective. He introduced the **coherent states**, quantum analogues of classical waves with well-defined phase, and defined the quantum correlation functions $g^{(n)}$, which measure the probability of detecting several correlated photons. His work, awarded the Nobel Prize in 2005, not only explained phenomena such as photon antibunching but also laid the foundations of modern quantum optics. The mathematical backbone connecting the statistical description of fluctuations with their spectral content — the Wiener–Khinchin theorem Wiener (1930); Khinchin (1934) — remains, as we shall see, one of the central pillars of the whole edifice.

In the twenty-first century, coherence has acquired a new flavour: the idea that it is a quantifiable and exploitable **resource**, akin to energy or information Baumgratz et al. (2014), integrated into the broader scheme of quantum computation. In summary, the history of coherence is a journey from the abstract mathematics of statistical correlations to the engineering of fragile quantum states. In its classical guise it explains why stars twinkle, how lasers work and why holograms capture three-dimensional images; in its quantum facet it challenges our intuition. And although decoherence reminds us that these phenomena are ephemeral, it also underlines their value: in a world where noise and chance seem to reign, coherence — classical or quantum — is the thread that weaves order out of chaos.

6.2 How is statistical similarity quantified?

Suppose we have a completely random phenomenon — such as the electromagnetic radiation emitted by an incandescent bulb that is not in thermodynamic equilibrium, the number of cars in some capital city throughout the day, the motion of an ion in a plasma, or something as simple as throwing a ball into the air under non-ideal conditions.

When we wish to analyse statistically a single object or random variable, the familiar tools appear: the probability density, the mean value, the standard deviation, the kurtosis, and so on. What happens when we want to study at least two random variables and how they relate? When we wish to study the statistics of two or more random variables we must study the statistical similarity between them; but statistical similarity is a rather broad concept, since we may have similarity in the mean values, or in the standard deviations, or in the probability densities. To address this problem we return to linear algebra.

If we have two vectors $\vec{A}, \vec{B} \in \mathbb{R}^n$, the dot product $\vec{A} \cdot \vec{B}$ is interpreted as the magnitude of the projection of $\vec{A}$ onto $\vec{B}$ and vice versa; it can also be read as *how much of $\vec{A}$ lies along $\vec{B}$*. This behaviour can be extrapolated to continuous vector spaces, where one usually employs the notion of inner product: if f_A, f_B are two elements of a Hilbert space L^2 of dimension 1, one can define an inner product as

$$\langle f_A, f_B \rangle = \int_{\mathbb{R}} f_A(x) f_B(x) W(x)\, dx, \tag{6.2}$$

where $W(x)$ is some weight function.[1] The interpretation is still valid: this can be read as how much the function f_A over all the reals coincides with the function f_B over all the reals.

To reach the notion of correlation from linear algebra we use a bit of bra–ket notation, where the ket $|f_A\rangle$ represents a vector of an arbitrary Hilbert space L^2 and the bra $\langle f_A|$ represents an element of the dual of the same Hilbert space,[2] and the inner product between them $\langle f_A, f_A\rangle = \langle f_A|f_A\rangle$ equals the probability density of finding f_A in a state $q \pm \triangle q$, where q represents some generalised coordinate. Suppose that $f_A, f_B \in L^2$ have the form

$$|f_A\rangle = \hat{A}\,|\Psi_A\rangle\,, \quad |f_B\rangle = \hat{B}\,|\Psi_B\rangle\,, \tag{6.3}$$

where $\hat{A}, \hat{B}$ are the operators of the physical quantities of $|\Psi_A\rangle\,, |\Psi_B\rangle$, so that we may interpret $|f_A\rangle\,, |f_B\rangle$ as the wave functions steered by their expected value. Using the inner product (6.2) with negligible weight function, the inner product reads

[1] $W(x)$ receives this name because, as it says, it gives a weight to each value the integrand may take; there are no uniform contributions, but rather each possible value of the integrand within the domain contributes weighted by the value of W at that point.

[2] The Hilbert space is a particular case of the L^p spaces, the space of p-norm Lebesgue-integrable functionals; for Hilbert $p = 2$, and it has the property that the dual of L^2 is also L^2, which lets quantum mechanics arise naturally in this space.

$$\langle f_A | f_B \rangle = \int_{\mathbb{R}^\ltimes} \hat{A}\hat{B} \langle \Psi_A | \Psi_B \rangle d^n q = \int_{\mathbb{R}^\ltimes} \hat{A}\hat{B} P(A;B) d^n q, \tag{6.4}$$

where we see that the similarity between $|f_A\rangle$ and $|f_B\rangle$ can be viewed as the product of $\hat{A}$ with $\hat{B}$ times the probability density of having states A and B; that is, the similarity, or how *correlated* two variables are, can be seen as the integral of the product of the variables times their joint probability density.

With this analogy in mind we can understand correlation as a first measure of the similarity between one or more variables across space and time. If $x_1(\vec{r}, t_1), x_2(\vec{r}', t_2)$ are two random variables and $p_2(x_2, t'; x_1, t)$ is the joint probability density, then the correlation between x_1, x_2 is defined as

$$\Gamma(\vec{r}, \vec{r}'; t_1, t_2) = E\left\{ x_1(\vec{r}, t_1) x_2(\vec{r}', t_2) \right\}, \tag{6.5}$$

that is, as the ensemble average of $x_1 \cdot x_2$.

6.3 The coherence function

The concept of the coherence function is, at the mathematical level, identical to that of the correlation function given in equation (6.5); both represent correlations. However, coherence is a term widely used in the study of electromagnetic radiation; let us see a little of why this is so.

In Figure 6.1 we observe a component of the electric field, represented by $E(\vec{r}, t)$, seen across several observations or realisations. Behaving as a random variable, in each realisation it has a different behaviour in time at a point $\vec{r}$; in general, here we have indexed ${}^{(i)}E(\vec{r}, t)$ with integers, but in general there could be an uncountable set of possible states for the electric field. If we look at Figure 6.1 there are two temporal points, t, t', at which we isolate parts of these random oscillations; we could ask how correlated these oscillations are across the realisations over the interval $t' - t$, or in other words, how *coherent* the oscillations are across the realisations.

We speak of coherence when we want to analyse the correlation of electromagnetic radiation between space–time points. If we have two sources emitting an electric field (for the moment understood as scalar), their coherence function is given by

$$\Gamma(\vec{r}, \vec{r}'; t, t') = E\left\{ \mathcal{E}(\vec{r}, t) \mathcal{E}^*(\vec{r}', t') \right\}, \tag{6.6}$$

where we are computing the ensemble average of the components $\mathcal{E}$, which are complex numbers as already discussed in these notes. If we take $\vec{r} = \vec{r}'$ we obtain what is known as the *mutual coherence function*

$$\Gamma(\vec{r}'; t, t') = E\left\{ \mathcal{E}(\vec{r}', t) \mathcal{E}^*(\vec{r}', t') \right\}. \tag{6.7}$$

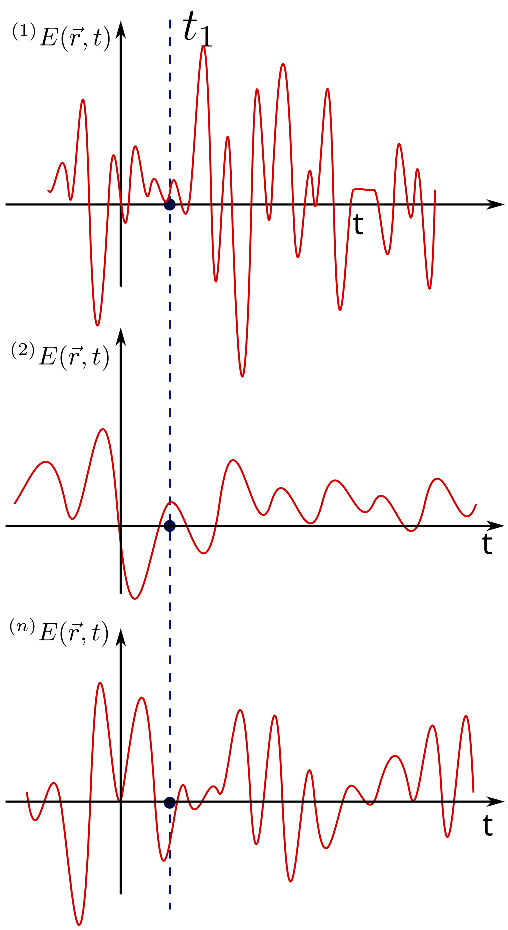


Figure 6.1: Representation of $E(\vec{r},t)$ as a random variable.

We can go even further by including the concept of *statistical stationarity*, but one must be careful since there are several kinds. Formally, a stochastic process $\{X_t\}$ is considered stationary when its statistical properties are invariant under time translations. This condition manifests in different degrees:

1. **Strict (strong) stationarity**: the most demanding form of stationarity requires that all joint probability distributions be time-invariant:

$$F_X(x_{t_1+\tau}, \ldots, x_{t_k+\tau}) = F_X(x_{t_1}, \ldots, x_{t_k}) \quad \forall \tau, k, t_i, \tag{6.8}$$

   where F_X is the joint distribution function. This implies that:

   - all moments (mean, variance, skewness, etc.) are constant;
   - the marginal distributions are identical at any instant;
   - it is a sufficient but not necessary condition for ARMA models.

2. **Weak (second-order) stationarity**: more practical in applications, it only requires invariance of the first two moments:

$$\begin{cases} \mathbb{E}[X_t] = \mu \quad \text{(constant)} \quad \forall t \\ \mathrm{Var}(X_t) = \sigma^2 \quad \text{(constant)} \quad \forall t \\ \mathrm{Cov}(X_t, X_{t+h}) = \gamma(h) \quad \text{(depends only on } h, \text{ not on } t) \end{cases} \tag{6.9}$$

   This is the definition used in most econometric models (ARIMA, GARCH) and in the Wiener–Khinchin spectral analysis (which we shall use later).

3. **Trend stationarity (deterministic stationarity)**: a non-stationary process that can be made stationary by removing deterministic components:

$$X_t = \alpha + \beta t + \epsilon_t, \tag{6.10}$$

   where ϵ_t is stationary. The non-stationarity here comes exclusively from the temporal trend βt.

We shall use a hypothesis of strong stationarity (although weak stationarity often suffices). In this spirit, if $\tau = t - t'$ we have

$$\Gamma(\vec{r}', t, t') = \Gamma(\vec{r}', t - t') = \Gamma(\vec{r}', \tau). \tag{6.11}$$

As we can see, the mutual coherence at a point $\vec{r}'$ under a stationarity hypothesis gives rise to an explicit dependence on the time difference τ in the ensemble average.

It is worth asking: how necessary is it to analyse the ensemble average; why not take the temporal average? Answering this is **very complicated**, since it touches on the problem of *ergodicity*. Ergodicity, in the sense of ergodic theory, requires that a dynamical system satisfy that the temporal averages coincide with the spatial averages for almost every initial condition. Its rigorous proof is highly non-trivial, to the point that the problems for which ergodicity has been proven with mathematical rigour are few.

From a phenomenological standpoint we shall assume, from now on, an *ergodic hypothesis* in which electromagnetic radiation has a stationary and ergodic character. Under these conditions the mutual coherence function coincides with the temporal average

$$\Gamma(\vec{r}';t,t') = \int_{\mathbb{R}} \mathcal{E}(\vec{r}',t')\mathcal{E}^*(\vec{r}',t-t')\,dt'. \tag{6.12}$$

The mutual coherence function has several properties, some of which are:

a) $\Gamma(\vec{r}',0) \geq 0$. This follows from the definition, since $\Gamma(\vec{r}',0) = \langle|\mathcal{E}(\vec{r}')|^2\rangle$, i.e. it corresponds to the intensity of the electric field and can only be zero when trivially the field is zero at every instant.

b) $\Gamma(\vec{r}',-\tau) = \Gamma^*(\vec{r}',\tau)$. This is known as the *Hermitian property* of the mutual coherence function, and follows from the fact that $\mathcal{E}(\vec{r}',t)$ is stationary, which allows arbitrary temporal translations from the origin. In this spirit,

$$\Gamma(\vec{r}',\tau) = \langle\mathcal{E}(\vec{r}')\mathcal{E}^*(\vec{r}',t+\tau)\rangle = \langle\mathcal{E}(\vec{r}',t-\tau)\mathcal{E}^*(\vec{r}')\rangle = \Gamma^*(\vec{r}',-\tau). \tag{6.13}$$

c) $|\Gamma(\vec{r}',\tau)| \leq \Gamma(\vec{r}',0)$. This property follows from the *Cauchy–Schwarz inequality*

$$|\Gamma(\vec{r}',\tau)|^2 = |\,\langle\mathcal{E}(\vec{r}')\mathcal{E}^*(\vec{r}',t+\tau)\rangle\,|^2 \leq \langle\mathcal{E}(\vec{r}',t)\mathcal{E}^*(\vec{r}',t)\rangle\,\langle\mathcal{E}(\vec{r}',t+\tau)\mathcal{E}^*(\vec{r}',t+\tau)\rangle = \Gamma^2(\vec{r}',0), \tag{6.14}$$

which implies that $|\Gamma(\vec{r}',\tau)|$ can never exceed its initial value $\Gamma(\vec{r}',0)$. (This property is, let us recall, exclusive to the mutual coherence function; when we consider the coherence between two sources we will see that it is not satisfied — on the contrary, it will increase.)

d) $\Gamma(\vec{r}',\tau)$ is non-negative. We can see this from the definition through the following, rather obvious, inequality

$$\left\langle\left|\int_{T_1}^{T_2} f(t)\mathcal{E}(t)\,dt\right|^2\right\rangle \geq 0, \tag{6.15}$$

where $f(t)$ is an arbitrary function and T_1, T_2 are arbitrary times. In this way we obtain the inequality

$$\int_{T_1}^{T_2}\int_{T_1}^{T_2} f^*(\vec{r}',t)f(\vec{r}',t')\Gamma(\vec{r}',t'-t)\,dt\,dt' \geq 0. \tag{6.16}$$

e) If the electric field is random, stationary and differentiable, then we can find a velocity $\xi(\vec{r}', t) = \dot{\mathcal{E}}(\vec{r}', t)$, which is stationary, random and of zero mean value, and whose mutual coherence function is

$$\Gamma_{\xi}(\vec{r}', \tau) = -\ddot{\Gamma}_{\mathcal{E}}(\vec{r}', \tau). \tag{6.17}$$

f) $\Gamma(\vec{r}', \tau)$ is continuous. This holds even when there are jump discontinuities.

Finally, it is worth mentioning that there is a quantity known as the complex degree of coherence, given by

$$\gamma(\vec{r}', \tau) = \frac{\Gamma(\vec{r}', \tau)}{\Gamma(\vec{r}', 0)}, \tag{6.18}$$

and from the Cauchy–Schwarz inequality we have $0 \leq |\gamma(\vec{r}', \tau)| \leq 1$.

6.3.1 Manifestations of coherence: the Wiener–Khinchin theorem

The mutual coherence function and the power spectral density are two sides of the same coin in the study of light and wave signals. Mutual coherence describes how the fluctuations of a luminous field are correlated in space and time, acting as a map that reveals to what extent the variations at one point of space are synchronised with those at another, even with a certain temporal delay. The power spectral density, in turn, decomposes the energy of the signal into its frequency components, showing which tones or colours — in the case of light — dominate its behaviour. The intimate relation between the two concepts lies in the fact that, through mathematical transformations, it is possible to convert the detailed information on the space–time correlations captured by mutual coherence into a precise description of how the energy is distributed in the frequency domain, and vice versa. This link allows researchers to jump between complementary perspectives: analysing interference patterns in an interferometer or identifying spectral signatures in a material, all from the same fundamental principles.

We are going to find one of the most important theorems in coherence theory, the *Wiener–Khinchin theorem*. Note that stationary random signals do not have a Fourier transform; hence we may define the following truncated random signals of the electric field

$${}^{\omega}\mathcal{E}_T(\vec{r}, t) = \begin{cases} {}^{\omega}\mathcal{E}(\vec{r}, t) & -\frac{T}{2} \leq t \leq \frac{T}{2} \\ 0 & \text{otherwise} \end{cases} \tag{6.19}$$

and we define the spectral power for a realisation ω as the squared magnitude of the Fourier transforms of the electric fields (6.19), that is,

$${}^{\omega}S_T(\vec{r}, \nu) = \left| \int_{\mathbb{R}}^{\omega} \mathcal{E}_T(\vec{r}, t) e^{i2\pi\nu t} dt \right|^2 = \left| \int_{-T/2}^{T/2} {}^{\omega}\mathcal{E}(\vec{r}, t) e^{i2\pi\nu t} dt \right|^2, \tag{6.20}$$

and the *power spectral density of the event* ω we compute as

$$\lim_{T\to\infty} \frac{{}^{\omega}S_T(\vec{r},\nu)}{T} = {}^{\omega}S(\vec{r},\nu). \tag{6.21}$$

Since we already possess a tool to study, in each event or realisation, the power spectral density, then, through its ensemble average, we can find the total density, that is,

$$S(\vec{r},\nu) = E\left\{{}^{\omega}S(\vec{r},\nu)\right\}, \tag{6.22}$$

introducing equations (6.20) and (6.21) we then have

$$S(\vec{r},\nu) = E\left\{\lim_{T\to\infty}\left|\frac{1}{T}\int_{-T/2}^{T/2} {}^{\omega}\mathcal{E}(\vec{r},t)e^{i2\pi\nu t}dt\right|^2\right\}, \tag{6.23}$$

$$S(\vec{r},\nu) = \lim_{T\to\infty}\frac{1}{T}\int_{-T/2}^{T/2}\int_{-T/2}^{T/2} E\left\{\mathcal{E}(\vec{r},t)\mathcal{E}^*(\vec{r}',t')\right\}e^{i2\pi\nu(t-t')}dt\,dt', \tag{6.24}$$

$$S(\vec{r},\nu) = \int_{\mathbb{R}} \Gamma(\vec{r},\vec{r}';\tau)e^{i2\pi\nu\tau}d\tau, \tag{6.25}$$

$$\boxed{S(\vec{r},\nu) = \mathcal{F}\left\{\Gamma(\vec{r},\vec{r}',\tau)\right\}}. \tag{6.26}$$

Result (6.26) reads: the Fourier transform of the coherence function corresponds to the power spectral density, and is known as the Wiener–Khinchin theorem. Its use is fundamental, since it tells us that by methods such as spectrometry we can find the coherence function of any phenomenon; we can even go further, since this result really stems from something more general — the correlation function — meaning that every stochastic phenomenon has an associated spectrum, and from the measurements or estimates of that spectrum we can compute the correlations in the system.

If we make a simple observation, the optical range of radiation — the visible — is a band of very low frequencies relative to the whole spectrum; the visible runs roughly from 400 nm to 700 nm, and we may say that all of optics is *narrowband*. This obeys the inequality

$$\frac{\triangle\nu}{\nu} \ll 1, \tag{6.27}$$

and, in this narrowband regime, the coherence function takes the form

$$\Gamma(\vec{r};\tau) = \tilde{\Gamma}(\vec{r})e^{i2\pi\tilde{\nu}\tau}, \tag{6.28}$$

which we can represent graphically as shown in Figure 6.2, where one can see the behaviour of the coherence function in the narrowband regime.[3] There one sees an envelope (the observable) enveloping all the fundamental frequencies.

[3] One usually finds in the literature a term similar but not equal to that of narrowband, known as the

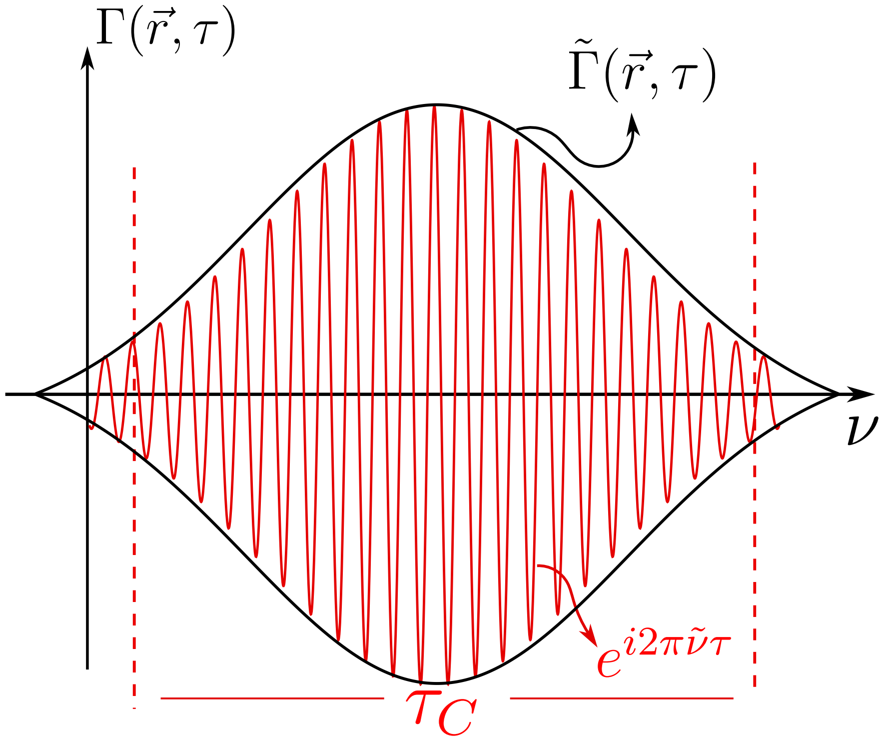


Figure 6.2: Graphical representation of the narrowband coherence function. In black, the envelope $\tilde{\Gamma}(\vec{r},\tau)$ is shown, and in red the oscillations $e^{i2\pi\tilde{\nu}\tau}$. These oscillations are confined within something known as the coherence time τ_C, which we shall see later.

6.3.2 The wave equation of coherence

Let us step a little out of the comfort zone and understand how correlations propagate through space and time. Let $\mathcal{E}^{(r)}(\vec{r},t)$ be some **real** component of the electric field, viewed as a random signal. As is well known from classical electromagnetic theory, it obeys Maxwell's equations and therefore has an associated wave equation, namely

$$\nabla^2\mathcal{E}^{(r)}(\vec{r},t) = \frac{1}{C^2}\frac{\partial^2\mathcal{E}^{(r)}(\vec{r},t)}{\partial t^2}, \tag{6.29}$$

where C is the speed of light in vacuum. We also have, by Fourier analysis,

$$\mathcal{E}(\vec{r},t) = \int_0^\infty \tilde{\mathcal{E}}(\vec{r},\nu)e^{-i2\pi\nu t}d\nu. \tag{6.30}$$

And if we apply the d'Alembertian operator $\Box = \nabla^2 - \frac{1}{C^2}\frac{\partial^2}{\partial t^2}$ to equation (6.30), we find that $\tilde{\mathcal{E}}(\vec{r},\nu)$ has the form of a Helmholtz-type equation, that is,

$$\nabla^2\mathcal{E}(\vec{r},t) - \frac{1}{C^2}\frac{\partial^2\mathcal{E}(\vec{r},t)}{\partial t^2} = \int_0^\infty \left[\nabla^2\tilde{\mathcal{E}}(\vec{r},\nu) + k^2\tilde{\mathcal{E}}(\vec{r},\nu)\right]e^{-i2\pi\nu t}d\nu, \tag{6.31}$$

which leads us to the fact that the electric field, as an analytic and complex signal, also obeys a wave-type equation

$$\nabla^2\mathcal{E}(\vec{r},t) = \frac{1}{C^2}\frac{\partial^2\mathcal{E}(\vec{r},t)}{\partial t^2}. \tag{6.32}$$

Taking the complex conjugate of the previous equation we have

$$\nabla^2\mathcal{E}^*(\vec{r},t) = \frac{1}{C^2}\frac{\partial^2\mathcal{E}^*(\vec{r},t)}{\partial t^2}. \tag{6.33}$$

If we multiply both sides of equation (6.33) by $\mathcal{E}(\vec{r}',t')$, and since we have taken the d'Alembertian with respect to the unprimed variables, using some sign rules we have

$$\nabla^2\left[\mathcal{E}^*(\vec{r},t)\mathcal{E}(\vec{r}',t')\right] = \frac{1}{C^2}\frac{\partial^2}{\partial t^2}\left[\mathcal{E}^*(\vec{r},t)\mathcal{E}(\vec{r}',t')\right]. \tag{6.34}$$

Taking the ensemble average of equation (6.34) over all realisations of the field, this wave equation becomes

quasi-monochromaticity condition, written as $\frac{\triangle\lambda}{\lambda} \ll 1$; but this bounds the wavelength so that it is as monochromatic as possible, while the narrowband condition admits several frequencies, provided they are narrow relative to the whole frequency spectrum.

$$\nabla^2\Gamma(\vec{r},\vec{r}';t,t') = \frac{1}{C^2}\frac{\partial^2}{\partial t^2}\Gamma(\vec{r},\vec{r}';t,t'), \tag{6.35}$$

and, similarly, using $\Box' = \nabla'^2 - \frac{1}{C^2}\frac{\partial^2}{\partial t'^2}$ we have another wave equation given by

$$\nabla'^2\Gamma(\vec{r},\vec{r}';t,t') = \frac{1}{C^2}\frac{\partial^2}{\partial t'^2}\Gamma(\vec{r},\vec{r}';t,t'). \tag{6.36}$$

Equations (6.35) and (6.36) tell us directly that the coherence functions between two space–time points evolve in time following a wave-type equation, separately, with respect to each pair of space–time coordinates — a result that is, without doubt, remarkable. We can take this result even further and study this propagation in the next section.

6.3.3 Propagation of coherence and the direction of time: the van Cittert–Zernike theorem

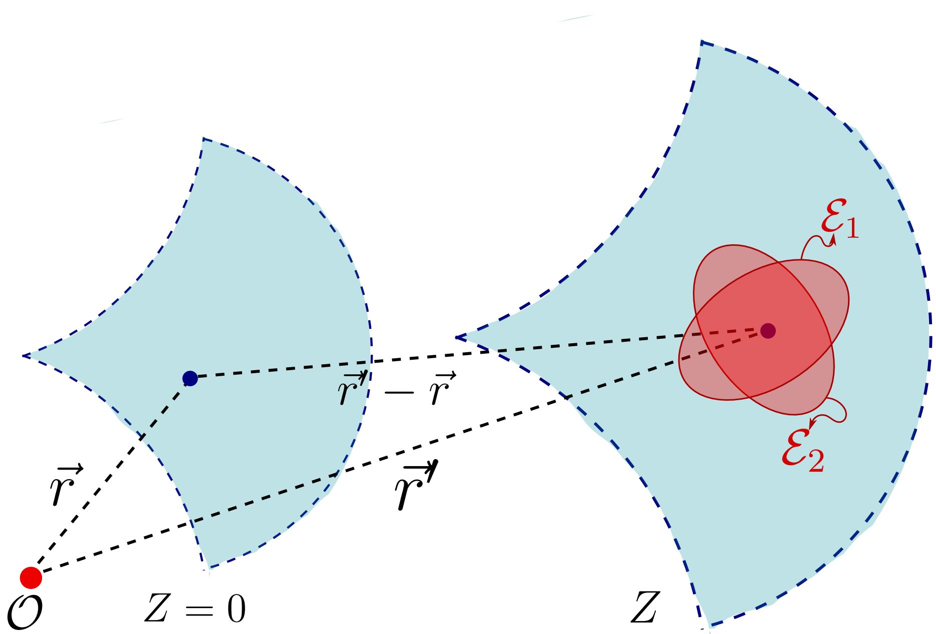


Figure 6.3: Propagation of the electric field from a plane $Z = 0$ to a plane Z, where each point of the wavefront carries its mean polarisation ellipse.

As we know, the electric field obeys a wave-type equation and the orientation of the fields is given by the polarisation of the field; in turn, as we saw in the diffraction chapter (see Section 5.2), at the classical level light fundamentally propagates as spherical waves, as illustrated in Figure 6.3, and at each point of the wavefront the components of the electric field oscillate randomly, but within the mean polarisation ellipse of their representative state. Mathematically we can represent this component as

$$\mathcal{E}_z(\vec{r}_1,\nu)=\frac{i}{\lambda}\int_{\Sigma}\mathcal{E}_0(\vec{r},\nu)\frac{e^{ik\|\vec{r}_1-\vec{r}\|}}{\|\vec{r}_1-\vec{r}\|}\cos(\hat{z};\vec{r}_1-\vec{r})\,d\sigma. \tag{6.37}$$

Using equation (6.30) we have

$$\mathcal{E}_z(\vec{r}_1,t)=\frac{1}{i}\int_{\mathbb{R}^+}\frac{\nu}{V}\int_{\Sigma}\mathcal{E}_0(\vec{r},\nu)\frac{\cos(\hat{z};\vec{r}_1-\vec{r})}{\|\vec{r}_1-\vec{r}\|}e^{i2\pi\nu\left(t+\frac{\|\vec{r}_1-\vec{r}\|}{V}\right)}d\sigma\,d\nu, \tag{6.38}$$

where V is the propagation velocity of the envelope. Rewriting a little more, we have

$$\mathcal{E}_z(\vec{r}_1,t)=\frac{1}{i}\int_{\Sigma}\frac{\cos(\hat{z};\vec{r}_1-\vec{r})}{\|\vec{r}_1-\vec{r}\|}\int_{\mathbb{R}^+}\frac{\nu}{V}\mathcal{E}_0(\vec{r},\nu)e^{i2\pi\nu\left(t+\frac{\|\vec{r}_1-\vec{r}\|}{V}\right)}d\nu\,d\sigma, \tag{6.39}$$

and, taking the narrowband approximation, we have

$$\mathcal{E}_z(\vec{r}_1,t)=\frac{1}{i\tilde{\lambda}}\int_{\Sigma}\mathcal{E}_0\left(\vec{r},t+\frac{\|\vec{r}_1-\vec{r}\|}{V}\right)\frac{\cos(\hat{z};\vec{r}_1-\vec{r})}{\|\vec{r}_1-\vec{r}\|}d\sigma. \tag{6.40}$$

We can visualise graphically how we intend to propagate in Figure 6.4, where we observe that we are starting from the coherence function between two points $\vec{r},\vec{r}'$ at instants t,t', respectively, and carrying it to a coherence function at the points $\vec{r}_1,\vec{r}_2$ at two instants t_1,t_2, respectively.

Starting from equation (6.40) we can compute the coherence function in its propagation as

$$\Gamma_z(\vec{r}_1,\vec{r}_2;\tau)=E\left\{\mathcal{E}_z(\vec{r}_1,t)\mathcal{E}_z^*(\vec{r}_2,t-\tau)\right\}, \tag{6.41}$$

$$\Gamma_z(\vec{r}_1,\vec{r}_2;\tau)=E\left\{\frac{1}{i\tilde{\lambda}}\int_{\Sigma}\mathcal{E}_0\left(\vec{r},t+\frac{\|\vec{r}_1-\vec{r}\|}{V}\right)\frac{\cos(\hat{z};\vec{r}_1-\vec{r})}{\|\vec{r}_1-\vec{r}\|}d\sigma\right. \tag{6.42}$$

$$\left.\cdot\frac{1}{-i\tilde{\lambda}}\int_{\Sigma'}\mathcal{E}_0^*\left(\vec{r}',t-\tau+\frac{\|\vec{r}_2-\vec{r}'\|}{V}\right)\frac{\cos(\hat{z};\vec{r}_2-\vec{r}')}{\|\vec{r}_2-\vec{r}'\|}d\sigma'\right\}, \tag{6.43}$$

$$\Gamma_z(\vec{r}_1,\vec{r}_2;\tau)=\frac{1}{\tilde{\lambda}^2}\int_{\Sigma}\int_{\Sigma'}E\left\{\mathcal{E}_0\left(\vec{r},t+\frac{\|\vec{r}_1-\vec{r}\|}{V}\right)\mathcal{E}_0^*\left(\vec{r}',t-\tau+\frac{\|\vec{r}_2-\vec{r}'\|}{V}\right)\right\}\frac{\cos(\hat{z};\vec{r}_1-\vec{r})\cos(\hat{z};\vec{r}_2-\vec{r}')}{\|\vec{r}_2-\vec{r}'\|\cdot\|\vec{r}_1-\vec{r}\|}d\sigma\,d\sigma' \tag{6.44}$$

$$\Gamma_z(\vec{r}_1,\vec{r}_2;\tau)=\frac{1}{\tilde{\lambda}^2}\int_{\Sigma}\int_{\Sigma'}\Gamma_0\left(\vec{r},\vec{r}';\tau+\frac{\|\vec{r}_1-\vec{r}\|+\|\vec{r}_2-\vec{r}'\|}{V}\right)\frac{\cos(\hat{z};\vec{r}_1-\vec{r})\cos(\hat{z};\vec{r}_2-\vec{r}')}{\|\vec{r}_2-\vec{r}'\|\cdot\|\vec{r}_1-\vec{r}\|}d\sigma\,d\sigma' \tag{6.45}$$

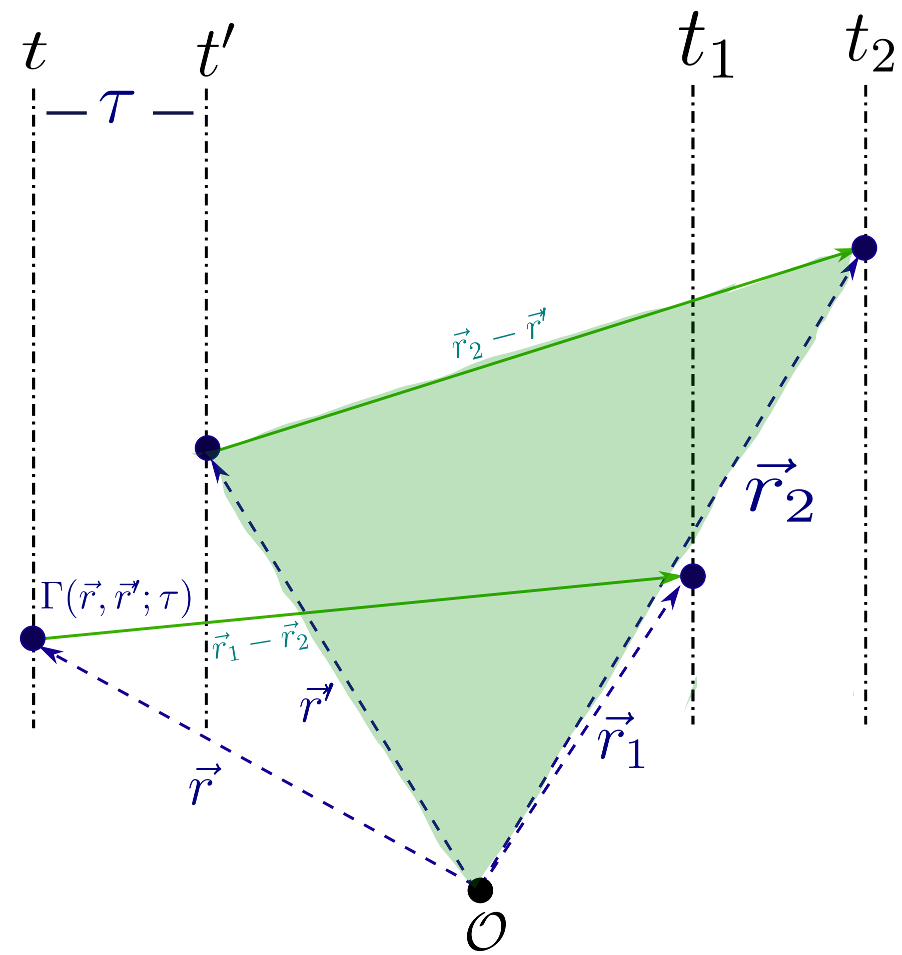


Figure 6.4: Propagation of the coherence function between two points $\vec{r}, \vec{r}'$ on the initial plane to the coherence between two points $\vec{r}_1, \vec{r}_2$ on the observation plane.

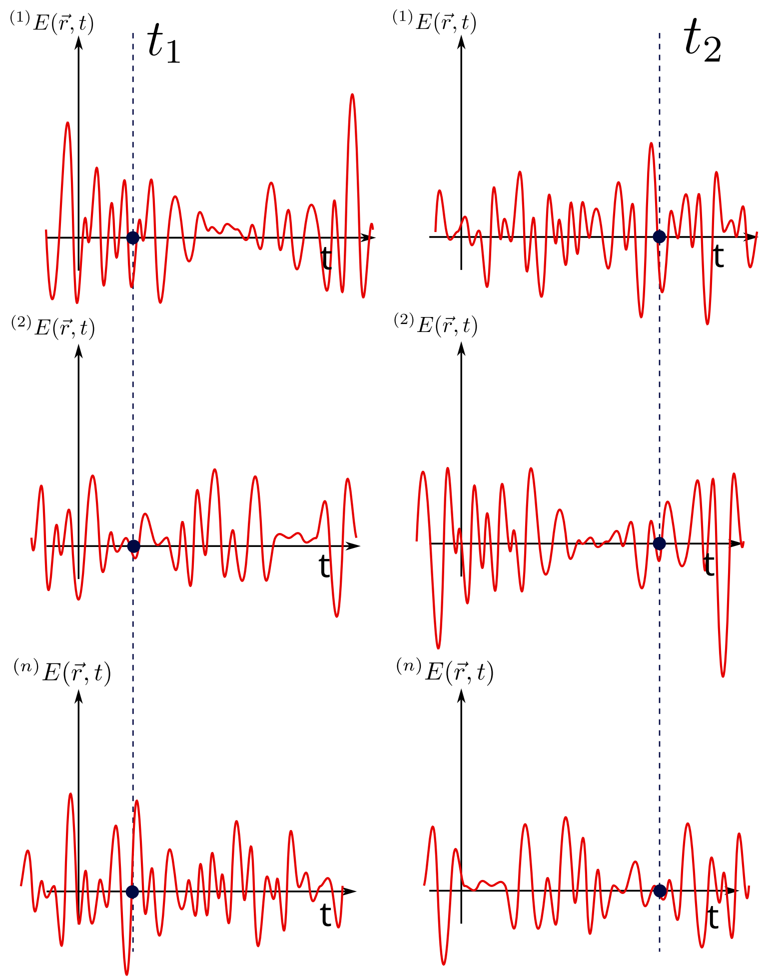


Figure 6.5: Evolution of the realisations upon propagating the electric field.

The propagation of the random oscillations across n realisations can be visualised in Figure 6.5.

We are going to impose the following condition on the coherence function

$$\Gamma(\vec{r},\vec{r}';\tau)=\Gamma(\vec{r}';\tau)\delta(\vec{r}-\vec{r}'), \tag{6.46}$$

which, in a few words, means that, spatially speaking, the distribution representing the coherence function is point-localised and weighted by the mutual coherence at the point $\vec{r}'$. Imposing this on equation (6.45) we have

$$\Gamma_z(\vec{r}_1,\vec{r}_2;\tau)=\frac{1}{\bar{\lambda}^2}\int_\Sigma \Gamma_0\left(\vec{r};\tau+\frac{\|\vec{r}-\vec{r}_1\|-\|\vec{r}-\vec{r}_2\|}{V}\right)\frac{\cos(\hat{z};\vec{r}_1-\vec{r})\cos(\hat{z};\vec{r}_2-\vec{r})}{\|\vec{r}_2-\vec{r}\|\cdot\|\vec{r}_1-\vec{r}\|}d\sigma. \tag{6.47}$$

In nature we will not always have narrowband sources; however, we can resort to another argument, namely that most instruments do not have an infinite or broad range but a short one — or, in other words, measurement instruments measure a narrow part of the spectrum. Therefore, including the narrowband condition we have

$$\Gamma_z(\vec{r}_1,\vec{r}_2)e^{i2\pi\bar{\nu}\tau}=\frac{1}{\bar{\lambda}^2}\int_\Sigma \Gamma_0(\vec{r})e^{i2\pi\bar{\nu}\left(\tau+\frac{\|\vec{r}-\vec{r}_1\|-\|\vec{r}-\vec{r}_2\|}{V}\right)}\frac{\cos(\hat{z};\vec{r}_1-\vec{r})\cos(\hat{z};\vec{r}_2-\vec{r})}{\|\vec{r}_2-\vec{r}\|\cdot\|\vec{r}_1-\vec{r}\|}d\sigma. \tag{6.48}$$

Given the coherence time defined by $\tau_C=\int_{\mathbb{R}^+}|\gamma(\vec{r},\tau)|^2d\tau$, and if we impose a condition of very small times, smaller than a coherence time $\tau\ll\tau_C$, we have

$$\Gamma_z(\vec{r}_1,\vec{r}_2)=\frac{1}{\bar{\lambda}^2}\int_\Sigma I_0(\vec{r})e^{i2\pi\left(\frac{\|\vec{r}-\vec{r}_1\|-\|\vec{r}-\vec{r}_2\|}{V/\bar{\nu}}\right)}\frac{\cos(\hat{z};\vec{r}_1-\vec{r})\cos(\hat{z};\vec{r}_2-\vec{r})}{\|\vec{r}_2-\vec{r}\|\cdot\|\vec{r}_1-\vec{r}\|}d\sigma. \tag{6.49}$$

Taking the far-field approximation we can simplify the problem as follows

$$\Gamma_z(\vec{r}_1,\vec{r}_2)=\frac{1}{\bar{\lambda}^2z^2}\int_\Sigma I_0(\vec{r})e^{i2\pi\left(\frac{\|\vec{r}-\vec{r}_1\|-\|\vec{r}-\vec{r}_2\|}{V/\bar{\nu}}\right)}d\sigma. \tag{6.50}$$

We may now make a Taylor-series expansion, the same one performed in the diffraction chapter. In this spirit the following terms become

$$\|\vec{r}-\vec{r}_1\|=\left(z^2+(x-x_1)^2+(y-y_1)^2\right)^{1/2}=z\left(1+\frac{(x-x_1)^2+(y-y_1)^2}{z^2}\right)^{1/2}\approx z+\frac{(x-x_1)^2+(y-y_1)^2}{2z}, \tag{6.51}$$

$$\|\vec{r}-\vec{r}_2\|\approx z+\frac{(x-x_2)^2+(y-y_2)^2}{2z}. \tag{6.52}$$

Introducing this into integral (6.50) we obtain

$$\Gamma_z(\vec{r}_1, \vec{r}_2) = \frac{1}{\tilde{\lambda}^2 z^2} \int_{\mathbb{R}^2} I_0(x, y) e^{-\frac{i2\pi}{\tilde{\lambda} z}(xx_1 + yy_1)} e^{\frac{i2\pi}{\tilde{\lambda} z}(xx_2 + yy_2)} e^{-\frac{i\pi}{\tilde{\lambda} z}(x_1^2 + y_1^2)} dx\, dy. \tag{6.53}$$

Let us define the vectors $\vec{S}_1 = (x_1, y_1)$, $\vec{S}_2 = (x_2, y_2)$ and $\vec{S} = (x, y)$. Then

$$\Gamma_z(\vec{S}_1, \vec{S}_2) = \frac{e^{i\frac{\pi}{\tilde{\lambda} z}(\|\vec{S}_1\|^2 - \|\vec{S}_2\|^2)}}{\tilde{\lambda}^2 z^2} \int_{\mathbb{R}^2} I_0(\vec{S}) e^{-i\frac{2\pi}{\tilde{\lambda} z}\vec{S}\cdot(\vec{S}_1 - \vec{S}_2)} dS. \tag{6.54}$$

If we set $\vec{S}_2 = \vec{0}$ we arrive at the Fourier integral

$$\boxed{\Gamma_z(\vec{S}_1) = \frac{e^{\frac{i\pi}{\tilde{\lambda} z}\|\vec{S}_1\|^2}}{\tilde{\lambda}^2 z^2} \int_{\mathbb{R}^2} \Gamma_0(\vec{S}) e^{-\frac{i2\pi}{\tilde{\lambda} z}\vec{S}\cdot\vec{S}_1} dS}, \tag{6.55}$$

which, basically, means that the correlations, under all the approximations we have made, propagate by means of a Fourier integral; or, what is the same, the van Cittert–Zernike theorem tells us that coherence diffracts — it propagates naturally as a diffraction integral (see Figure 6.6).

How can we tell in which direction time flows? This may sound like a question rather out of context for these notes, but if we could travel back to the golden age of Boltzmann, we would find a heated debate of the time on how statistical physics gives a direction to time, an *arrow*. Paraphrasing Boltzmann, it is said that *time advances in the direction in which correlations increase*. If we look at the van Cittert–Zernike theorem, we can start from two completely uncorrelated sources but, as they propagate, they gradually become correlated; time grows and so does coherence, until a point is reached at which the radiation of those sources is coherent. This is, without doubt, one of the strongest and most beautiful results we can find in statistical physics.

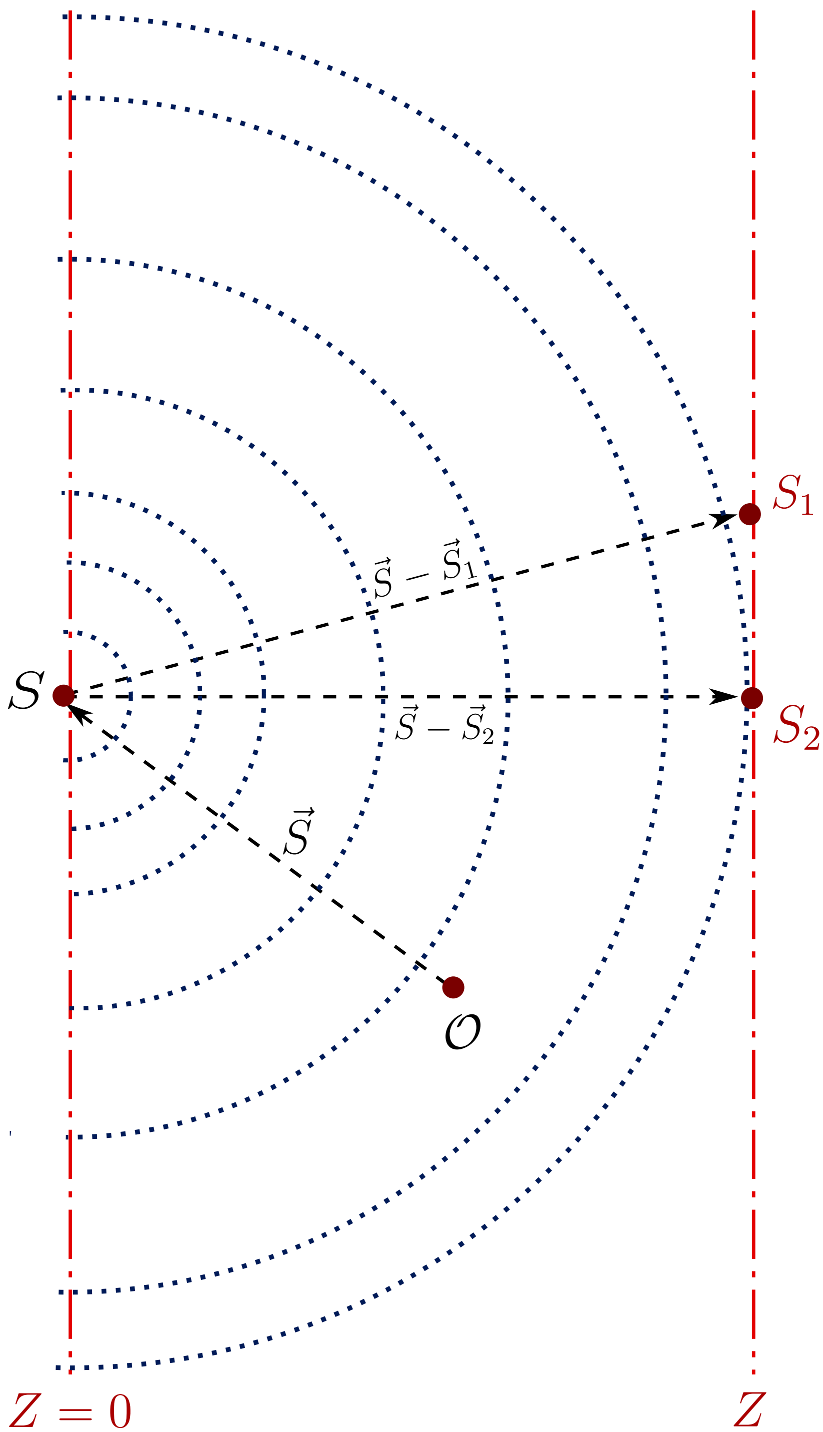


Figure 6.6: The mutual coherence propagated to the far field is the Fourier transform of the intensity distribution $\Gamma_0(\vec{S})$ of the source.

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